Quantum Reference Algebra for Information Flow

The physical interface

Complete Collected Volume

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Introduction

A positive control cost does not become a Lorentzian metric simply by being renamed. These three papers connect operator representation, actual propagation and physical actuation. The connection is conditional on the carrier and interaction: a per-species cone is not automatically a common cone, and a representation image is not the whole apparatus that realizes it.

A develops propagation and readout, B realizes protection and corrective control, and C follows representations into local contacts, material sources and finite autonomous execution. The separate Mass–Energy, Null Cluster and No View papers extend these questions; they remain companion papers rather than additional chapters in this volume.

Reading map

QR-A *Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass.* Local unitary propagation produces per-species Clifford readout and causal cones. Speed alignment requires divergent accumulated coupling in the mediator model, or decay of relative-speed perturbations near a supplied fixed point; connectivity alone is insufficient.

QR-B *Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost.* Persistent control is realized by physical couplings. Protection, calibration, active recovery and field energy are compared under explicit collection and reference permissions.

QR-C *Retained Quantum History, Physical Sources and Receiver Renewal in Q-RAIF.* One autonomous path performs finite repeated readings and receiver updates while retaining the body and history. Local linear-current control also implements neutral-sector stress flows; clock work and the limits of total-source matching are explicit.

The papers retain their complete proofs, appendices and references, and their own pagination. The contents list PDF page numbers; PDF bookmarks also provide section-level navigation.

Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass

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Abstract

A positive control metric cannot become Lorentzian by a change of variables. We instead derive the relative sign from the propagation of an internal quantum state. For a first-order Hermitian symbol with linearly independent spatial coefficients, every band lies on one quadratic shell if and only if the coefficients obey the corresponding Euclidean Clifford relations. Operator-norm bounds on deviations give a uniform squared-band certificate without asserting repair to an exact representation. The resulting form $\omega^2 - Q(q)$ has signature $(1, d)$; it describes the full characteristic cone when both frequency branches occur. The evolution parameter is supplied, while unitarity makes its conjugate frequency a hyperbolicity direction. The internal dimension bounds $d + [\mu \neq 0]$ by $2\nu_2(s) + 1$, so a qubit supports a massless shell in at most three spatial dimensions. On a lattice, the strict support cone, maximum group speed and low-energy read-out cone differ. For several species, off-diagonal mixing with fixed diagonal principal blocks cannot produce a single quadratic shell. Interactions can align speeds: in a one-loop mediator model, divergent accumulated coupling along every edge is sufficient, even when their active ranges of renormalization scale are disjoint. Near a supplied Lorentz-invariant conformal fixed point, additional conserved spin-2 currents mark the marginal exceptions to speed alignment. A finite record protocol for the same coined walk certifies its mass-setting angle, separating the ideal rest-and-curvature and group speeds from each other and from the supplied support speed. The geometric dictionary, actual noisy transport and physical inputs remain separate.

Keywords: Lorentzian signature, Clifford algebra, hyperbolic polynomials, quantum cellular automata, quantum walks, Weyl and Dirac equations, causal cone, spectral mass, Q-RAIF

1 Introduction

Programmes that reconstruct geometry from quantum data usually take the Lorentzian signature as given. Holographic entanglement formulas and thermodynamic derivations of the Einstein equation relate entropies to areas and energy fluxes in a spacetime whose causal

structure is already Lorentzian [2, 3]. The Holographic Alaya-Field Framework (HAFF) studies which distinctions a physical reduction can resolve and what geometric data can be reconstructed from them [42, 43]; a spacetime metric needs a further dictionary there as well. This paper asks where the indefinite direction of such a read-out can come from in a model whose dynamics is given explicitly.

Two facts fix the starting point. A positive definite form stays positive under every linear change of variables (Proposition 1), so no control metric, information metric or work form becomes Lorentzian by reparameterization. The companion paper on persistence makes the same point physically: its descent metric and its field-derived work form are both positive, and the Lorentzian form is a third structure [45]. Second, imposing Clifford relations $\{e_\mu, e_\nu\} = 2q_{\mu\nu}\mathbf{1}$ for a chosen Lorentzian q puts the signature into the premise; the universal property of $Cl(V, q)$ then supplies an encoding, rather than a signature-selection mechanism [4, 5].

The model class is local unitary propagation: a particle with an internal Hilbert space $\mathbb{C}^s$ hops on a lattice by a translation-invariant, finite-range unitary step, as in quantum walks and the one-particle sector of quantum cellular automata [9, 10, 11, 12, 13]. Its low-energy generator is a first-order Hermitian matrix symbol. In this setting four objects can be defined separately: a positive norm on generators, which plays the role of a control metric; the causal cones of the dynamics; the algebra of the symbol's coefficients; and the spectral mass. For a first-order Hermitian symbol, unitarity alone makes the characteristic polynomial hyperbolic in the frequency (Theorem 2). For linearly independent spatial coefficients, all bands lie on one quadratic shell if and only if the coefficients satisfy the shell relations of Theorem 4; the characteristic form is then ω^2 minus the Hilbert–Schmidt norm of the spatial symbol and has signature $(1, d)$, and the Lorentzian Clifford generators are $\gamma^0 = \beta$, $\gamma^i = \beta\alpha_i$ (Theorem 4, Corollary 5). The internal dimension bounds the spatial one, $d + [\mu \neq 0] \leq 2\nu_2(s) + 1$. For a qubit carrier three is the largest dimension with an isotropic massless cone, and the only one in which every internal rotation axis is coupled to a direction of motion (Theorem 8, Proposition 10).

On the lattice, group velocities lie in the convex hull of the neighbourhood, so the speed of any massless read-out cone is at most the inradius of that polytope per step (Proposition 12). Exact formulas for the split-step walk and computations for the Weyl automaton then show that the read-out map is not fixed by the dynamics: a band point, a momentum window and the observable that defines the speed all have to be chosen (Section 5).

The mathematical ingredients are standard: Dirac's square root of the Klein–Gordon operator [6], Gårding's hyperbolic polynomials [7], the representation theory of complex Clifford algebras [8], and the derivations of Weyl and Dirac equations from quantum walks and automata [9, 11, 13]. Hyperbolicity as the defining property of a physical dispersion relation is developed in Ref. [14], and an information-theoretic argument linking three spatial dimensions to the qubit is given in Ref. [15]. This paper separates the four objects in one model, identifies which inputs of a Lorentzian read-out follow from propagation, and calculates how the remaining read-out choices enter. The several-species analysis adds a quantitative fixed-block mixing obstruction and applies cut-balanced convergence to interacting speeds that need not couple simultaneously.

The argument proceeds from the symbol and its shell to dimension, lattice windows and several-species dynamics. A residual bound connects the exact shell criterion to calibrated

approximate symbols; a finite angle certificate then connects one such dictionary to actual records. The concluding sections distinguish these contributions from the physical selection of a carrier, an interacting fixed point and a complete source-and-return apparatus.

2 Local unitary propagation and four objects

2.1 The model

Let $\Lambda \subset \mathbb{R}^d$ be a lattice of physical positions and let the one-particle Hilbert space be $\ell^2(\Lambda) \otimes \mathbb{C}^s$. A translation-invariant step with finite neighbourhood $N \subset \Lambda$ acts as

$$(W\psi)(x) = \sum_{h \in N} A_h \, \psi(x - h), \qquad A_h \in M_s(\mathbb{C}), \tag{1}$$

and is unitary exactly when its symbol

$$W(k) = \sum_{h \in N} A_h \, e^{-ik \cdot h} \tag{2}$$

is unitary for every k in the Brillouin zone. One step takes time τ. The eigenvalues of $W(k)$ are $e^{-i\omega_b(k)\tau}$ with quasi-energies $\omega_b(k)$ defined modulo $2\pi/\tau$, and the group velocity of band b is $\nabla_k \omega_b$. With this convention the pure shift $A_h = \mathbf{1}$ for a single h moves every state by $+h$ per step.

Near a point k_* where a gap closes or nearly closes, the step is described by an effective generator. If a parameter θ controls the gap, we say that the walk has *first-order symbol*

$$H(q) = \sum_{i=1}^{d} q_i \alpha_i + \mu\beta, \qquad \alpha_i, \beta \in \mathrm{Herm}(s), \tag{3}$$

at k_* if $(i/\tau) \operatorname{Log} W_\theta(k_* + q) = H(q) + O(|q|^2 + |\mu|^2)$ with μ proportional to θ. This is the scaling limit studied for quantum walks and automata [11, 12, 13]: H generates the continuum dynamics $i\partial_t \psi = H(-i\nabla)\psi$ when lattice spacing and step time go to zero at fixed ratio. For example, the split-step walk of Section 5.2 has $W_\theta(k) = \mathrm{diag}(e^{-ika}, e^{ika}) R(\theta)$ with $R(\theta) = \cos\theta - i \sin\theta \, \sigma_y$, and at $k_* = 0$ its first-order symbol is $H(q) = (a/\tau) q \sigma_z + (\theta/\tau)\sigma_y$.

2.2 Four objects

A control metric. The generators of the internal space form the real vector space $\mathrm{Herm}(s)$, which carries the positive Hilbert–Schmidt norm

$$G_{\mathrm{HS}}(A) = \frac{1}{s} \operatorname{tr}(A^2). \tag{4}$$

Restricted to the span of the coefficients of H, it is the natural positive metric on the controls that the symbol couples to translations. Other positive forms of the same type appear in the series: the descent metric G of a gradient controller and the work form K derived from the stress of a field-coupled actuator [45], and monotone information metrics on states [29].

Causal cones. The step (1) has a strict causal region. After n steps a state supported in S is supported in $S + nN$, the n-fold Minkowski sum, so all influence lies in the cone over the polytope $C_N = \tau^{-1} \operatorname{conv} N$. A first-order symbol has a second cone, its characteristic cone, defined in Section 3. The two need not agree.

A representation. The coefficients α_i, β generate a real algebra inside $M_s(\mathbb{C})$. Whether they satisfy Clifford relations, and with which quadratic form, is a property of the dynamics, not a choice of encoding.

Spectral mass. For a selected logarithm and energy zero, the positive band's value at k_* is the rest energy $E_0 = \hbar|\omega(k_*)|$, and its curvature defines an inertial mass m_* through $\omega(k_* + q) \simeq \omega(k_*) + \hbar q^2/(2m_*)$. A measured phase difference alone does not fix an unknown common energy offset. A Lorentzian read-out identifies $E_0 = m_* c^2$, which defines a speed c; whether this c agrees with either causal cone is a question about the model.

Proposition 1 (Positive forms stay positive)**.** *Let G be a positive definite form on a real vector space V and $T : U \to V$ linear. Then T^*G, given by $(T^*G)(u, u') = G(Tu, Tu')$, is positive semidefinite, and positive definite if T is injective. In particular no positive definite form is congruent to a form with a negative direction.*

Proof. $(T^*G)(u, u) = G(Tu, Tu) \geq 0$, with equality only if $Tu = 0$. The last statement is Sylvester's law of inertia. □

The Lorentzian form of a read-out is therefore not a pullback or a reparameterization of G_{HS}, G or K. In the model it arises as a different kind of object, a form on frequency–momentum space built from the spectrum of the evolution.

3 Where the indefinite direction comes from

For a first-order symbol (3) write $P(\omega, q) = \det(\omega \mathbf{1} - H(q))$ for its characteristic polynomial and $P_0(\omega, q) = \det(\omega \mathbf{1} - \sum_i q_i \alpha_i)$ for the principal part. A homogeneous polynomial p on $\mathbb{R}^n$ is *hyperbolic with respect to e* if $p(e) \neq 0$ and $t \mapsto p(\xi + te)$ has only real roots for every ξ [7].

Theorem 2 (Hyperbolicity in the evolution frequency)**.** *For Hermitian α_i, β, every root in ω of $P(\omega, q)$ is real for every real q, and P_0 is hyperbolic with respect to the frequency direction $e = (1, 0, \ldots, 0)$. The propagation speed of the principal part in a unit direction u is the largest eigenvalue of $\sum_i u_i \alpha_i$.*

Proof. The roots of $P(\cdot, q)$ are the eigenvalues of the Hermitian matrix $H(q)$. For P_0, $P_0(e) = 1$ and the roots of $t \mapsto P_0(\xi_0 + t, \xi')$ are $-\xi_0$ plus the eigenvalues of $\sum_i \xi'_i \alpha_i$, which are real. The characteristic speeds of $i\partial_t \psi = \sum_i \alpha_i (-i\partial_i)\psi$ in direction u are the eigenvalues of $\sum_i u_i \alpha_i$. □

The evolution parameter is supplied by the model. Unitarity makes its conjugate frequency a hyperbolicity direction; it does not choose how many fundamental time parameters

a theory should have. The next lemma determines the signature when the characteristic form is quadratic.

Lemma 3 (Hyperbolic quadratic forms are Lorentzian)**.** *A real quadratic form p on $\mathbb{R}^n$ with $p(e) > 0$ is hyperbolic with respect to e if and only if it has exactly one positive eigenvalue. Then $p = \ell^2 - r$ with ℓ linear and r positive semidefinite, and if p is non-degenerate its signature is $(1, n-1)$.*

Proof. Decompose $\xi = se + \zeta$ with ζ in the p-orthogonal complement of e. Then $p(\xi + te) = p(e)(s+t)^2 + p(\zeta)$, which has real roots in t for every ξ if and only if $p(\zeta) \le 0$ on the complement. That is the statement that p has one positive direction and is negative semidefinite on a complementary hyperplane. □

Theorem 4 (Single shell, Clifford relations and the Lorentzian form)**.** *Let $\alpha_1, \dots, \alpha_d, \beta \in \mathrm{Herm}(s)$ with $\alpha_1, \dots, \alpha_d$ linearly independent, and fix $\mu \in \mathbb{R}$. The following are equivalent.*

(a) There is a quadratic form Q on $\mathbb{R}^d$ such that every eigenvalue ω of $H(q)$ satisfies $\omega^2 = Q(q) + \mu^2$ for all q.

(b) $H(q)^2 = (Q(q) + \mu^2)\mathbf{1}$ for all q.

(c) $\{\alpha_i, \alpha_j\} = 2Q_{ij}\mathbf{1}$, $\mu\{\alpha_i, \beta\} = 0$ and $\mu^2(\beta^2 - \mathbf{1}) = 0$.

If they hold, then:

1. *Q is positive definite and $Q(q) = G_{\mathrm{HS}}\big(\sum_i q_i \alpha_i\big)$;*
2. *the read-out form $\eta(\omega, q) = \omega^2 - Q(q)$ on frequency–momentum space has signature $(1, d)$, and the characteristic set of P_0 lies on its null cone; the sets coincide if $d \ge 2$, if $\mu \neq 0$, or if $d = 1$ and α_1 has both positive and negative eigenvalues;*
3. *if $d + [\mu \neq 0] \ge 2$, then s is even and $P(\omega, q) = (\omega^2 - Q(q) - \mu^2)^{s/2}$;*
4. *every group velocity $v = \nabla_q \omega$ satisfies $Q^{-1}(v, v) = Q(q)/(Q(q) + \mu^2) \le 1$, with equality only for $\mu = 0$.*

Proof. (a)⇒(b): $H(q)$ is Hermitian, hence diagonalizable, and all its eigenvalues square to $Q(q) + \mu^2$. (b)⇒(c): expanding,

$$H(q)^2 = \sum_{i,j} q_i q_j \tfrac{1}{2}\{\alpha_i, \alpha_j\} + \mu \sum_i q_i \{\alpha_i, \beta\} + \mu^2 \beta^2.$$

Comparing the parts of degree two, one and zero in q with $(Q(q) + \mu^2)\mathbf{1}$ gives the three relations. (c)⇒(a): the same expansion gives (b), and (b) gives (a).

For item 1, $Q(q)\mathbf{1} = (\sum_i q_i \alpha_i)^2 \ge 0$; if $Q(q) = 0$ the Hermitian matrix $\sum_i q_i \alpha_i$ squares to zero and so vanishes, and linear independence gives $q = 0$. Taking the normalized trace gives $Q(q) = G_{\mathrm{HS}}(\sum_i q_i \alpha_i)$. For item 2, η is ω^2 minus a positive definite form, and every root of $P_0(\cdot, q)$ belongs to $\{\pm\sqrt{Q(q)}\}$. When $d \ge 2$, the principal symbol at nonzero q anticommutes with an invertible symbol in a Q-orthogonal direction, pairing its two signs.

If $\mu \neq 0$, β provides the pairing. In the remaining one-dimensional case both signs occur exactly when α_1 has both signs. For item 3, choose coordinates with $Q = c^2\mathbf{1}$. The matrices α_i/c together with β (if $\mu \neq 0$) pairwise anticommute and square to $\mathbf{1}$. If at least two are present, each x anticommutes with an invertible g, giving $\operatorname{tr} x = \operatorname{tr}(g^{-1}xg) = -\operatorname{tr} x$. Thus $H(q)$ is traceless, and its eigenvalues $\pm\sqrt{Q+\mu^2}$ have equal multiplicity. For item 4, at a differentiable band point with $\omega \neq 0$, $\nabla_q \omega = Qq/\omega$ and $Q^{-1}(v,v) = Q(q)/\omega^2$. At a massless node the same statement applies to directional band limits. □

Corollary 5 (Lorentzian Clifford generators)**.** *Under Theorem 4 with $\mu \neq 0$, the matrices*

$$\gamma^0 = \beta, \qquad \gamma^i = \beta\alpha_i \tag{5}$$

satisfy $\{\gamma^0,\gamma^0\} = 2$, $\{\gamma^0,\gamma^i\} = 0$ and $\{\gamma^i,\gamma^j\} = -2Q_{ij}$, so they generate a representation of $Cl(1,d)$ on $\mathbb{C}^s$; γ^0 is Hermitian and the γ^i are anti-Hermitian. The eigenvalue equation $H(q)\psi = \omega\psi$ is equivalent to $(\omega\gamma^0 - \sum_i q_i\gamma^i - \mu)\psi = 0$, and $(\omega\gamma^0 - \sum_i q_i\gamma^i)^2 = \eta(\omega,q)\mathbf{1}$. For $\mu = 0$ and $d \geq 1$, the matrices

$$\Gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \qquad \Gamma^i = \begin{pmatrix} 0 & \alpha_i \\ -\alpha_i & 0 \end{pmatrix} \tag{6}$$

generate a representation of $Cl(1,d)$ on $\mathbb{C}^s \oplus \mathbb{C}^s$.

Proof. $(\beta\alpha_i)^2 = -\beta^2\alpha_i^2 = -Q_{ii}$ and $\{\beta,\beta\alpha_i\} = \alpha_i - \alpha_i = 0$ use $\{\alpha_i,\beta\} = 0$ and $\beta^2 = 1$; similarly $\{\beta\alpha_i,\beta\alpha_j\} = -\{\alpha_i,\alpha_j\}$. Multiplying $H(q) - \omega$ by β gives the second statement. The block matrices are checked by direct multiplication. □

What has been derived. In the model the indefinite direction of the read-out form is the frequency ω, the spectral variable of the one-parameter group of time steps. The positive part is the Hilbert–Schmidt norm of the spatial symbol, a control-type metric. The Lorentzian form is the difference $\omega^2 - G_{\mathrm{HS}}(\sum_i q_i\alpha_i)$, which by Proposition 1 could not be obtained from any positive form alone. Theorem 2 makes the supplied evolution frequency a hyperbolicity direction for every first-order Hermitian symbol; Lemma 3 says that a quadratic cone with one time direction has signature $(1,d)$; and Theorem 4 says that the cone is quadratic, with all internal states on it, exactly when the propagation coefficients satisfy Euclidean Clifford relations. The Lorentzian Clifford algebra of Corollary 5 is then constructed, not supplied: the time-like generator is the mass term β.

The signature is no longer an input. The Clifford relations are equivalent to an observable property, a single shell, which means no birefringence and no internal state that propagates differently from the others. That property, isotropy of Q (which can be absorbed into the choice of spatial coordinates for one species) and the existence of a band point with a linear symbol remain inputs.

Remark 6 (Two boundary cases). For $d = 1$ and $\mu = 0$, $\alpha_1^2 = Q_{11}\mathbf{1}$ allows unequal multiplicities of $\pm Q_{11}^{1/2}$, and for $s = 1$ the symbol $H(q) = cq$ has a single branch. Its characteristic set is one chiral branch, with one future null direction. This is the chiral field used as an actuator in Ref. [45]: it supplies one null direction, and a full $(1,1)$ cone needs both chiralities. In the other direction, a quadratic cone can occur without a single shell. The Maxwell

field in Riemann–Silberstein form has the symbol $c\,\mathbf{S}\cdot q$ with spin-one matrices; its characteristic polynomial is $\omega(\omega^2 - c^2|q|^2)$, and the flat factor is removed by the Gauss constraint. Theorem 4 concerns symbols for which every internal state lies on the cone.

3.1 A forward certificate for approximate shells

The exact operator identity has a finite-error consequence. It requires bounds on implemented coefficients or their calibration errors, not merely access to effects whose ideal algebra contains those coefficients.

Proposition 7 (Squared-band certificate)**.** *Choose a positive quadratic form Q and Hermitian coefficients α_i, β. Let*

$$R_{ij} = \tfrac{1}{2}\{\alpha_i, \alpha_j\} - Q_{ij} I, \qquad R_{i\beta} = \{\alpha_i, \beta\}, \qquad R_\beta = \beta^2 - I.$$

Suppose their operator norms are bounded by calibrated numbers $\epsilon_{ij}, \epsilon_{i\beta}, \epsilon_\beta$. If the implemented symbol is $\widetilde{H}(q) = H(q) + E(q)$ with Hermitian E and $\|E(q)\| \le e(q)$, put

$$\Delta(q) = \sum_{i,j} |q_i q_j| \epsilon_{ij} + |\mu| \sum_i |q_i| \epsilon_{i\beta} + \mu^2 \epsilon_\beta,$$
$$L(q) = \sum_i |q_i| \|\alpha_i\| + |\mu| \|\beta\|, \qquad B(q) = \Delta(q) + 2L(q)e(q) + e(q)^2.$$

Every eigenvalue $\widetilde{\omega}$ obeys

$$\left|\widetilde{\omega}^2 - Q(q) - \mu^2\right| \le B(q).$$

With $S = Q(q) + \mu^2 \ge 0$, this gives $\left||\widetilde{\omega}| - \sqrt{S}\right| \le \sqrt{B(q)}$. If $S > B(q)$, a sharper radius is $B(q)/(\sqrt{S} + \sqrt{S - B(q)})$. The bounds hold uniformly on any specified window where their input bounds hold.

Proof. Expanding $H^2 - SI$ gives $\sum q_i q_j R_{ij} + \mu \sum q_i R_{i\beta} + \mu^2 R_\beta$, whose norm is at most Δ. The additional square error is $HE + EH + E^2$, of norm at most $2Le + e^2$. Evaluate on an eigenvector of the Hermitian $\widetilde{H}$. Finally, $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for nonnegative x, y; away from zero use the difference of squares. □

This is a forward dispersion certificate. It does not say that nearly anticommuting coefficients can be repaired to an exact Clifford representation, that the two frequency multiplicities are balanced, or that a finite effect menu implements operator-norm tomography. A calibration valid on a restricted state set is not a norm bound on the entire carrier. The distinction matters: a Hilbert–Schmidt trace identity can hold exactly while different internal states retain different propagation speeds.

4 Spatial dimension and the internal space

Theorem 8 (Dimension bound)**.** *Suppose Theorem 4 holds on $\mathbb{C}^s$, and let $n = d + [\mu \ne 0]$. Write $s = 2^\nu m$ with m odd, $\nu = \nu_2(s)$. Then $n \le 2\nu + 1$. Conversely, for every s and every $n \le 2\nu_2(s) + 1$ there are coefficients satisfying Theorem 4.*

Proof. Choose spatial coordinates in which $Q = c^2\mathbf{1}$; the coefficients transform linearly, so the relations are preserved. Then $e_i = \alpha_i/c$ and, if $\mu \neq 0$, $e_{d+1} = \beta$ are n pairwise anticommuting Hermitian involutions, which define a representation of the complex Clifford algebra $\mathbb{C}l_n$ on $\mathbb{C}^s$. This algebra is isomorphic to $M_{2^{n/2}}(\mathbb{C})$ for even n and to $M_{2^{(n-1)/2}}(\mathbb{C}) \oplus M_{2^{(n-1)/2}}(\mathbb{C})$ for odd n [8]. Every finite-dimensional representation is a direct sum of irreducible ones of dimension $2^{\lfloor n/2 \rfloor}$, so $2^{\lfloor n/2 \rfloor}$ divides s and $\lfloor n/2 \rfloor \leq \nu$. For the converse take an irreducible representation of $\mathbb{C}l_{2\nu+1}$, which has dimension 2^ν, tensor it with $\mathbf{1}_m$, and use any n of its generators. □

Corollary 9 (Qubit carriers)**.** *For $s = 2$: an isotropic single-shell symbol exists for massless propagation in $d \leq 3$ and for massive propagation in $d \leq 2$. Massive single-shell propagation in $d = 3$ requires $4 \mid s$; the minimal case $s = 4$ is the Dirac symbol $\alpha_i = \sigma_x \otimes \sigma_i$, $\beta = \sigma_z \otimes \mathbf{1}$, a pair of Weyl qubits coupled by the mass term.*

For $s = 2$ and $d = 3$ the symbol is $H(q) = c\,\sigma \cdot q$ up to a unitary and a rotation, and

$$\det\bigl(\omega\mathbf{1} - c\,\sigma \cdot q\bigr) = \omega^2 - c^2|q|^2. \tag{7}$$

This is the Lorentzian form of signature $(1, 3)$, obtained from the Pauli relations $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$, which are Euclidean. No operator on $\mathbb{C}^2$ anticommutes with all three Pauli matrices, so $Cl(1, 3)$ has no representation on $\mathbb{C}^2$; the Lorentzian form exists anyway, as the characteristic form of the evolution.

Two forms on the same carrier. The identity $\det(x_0 I + x \cdot \sigma) = x_0^2 - |x|^2$ is preserved by determinant-one filtering, not by the positive Hilbert–Schmidt norm. For the massive Dirac carrier, the time-like generator in Corollary 5 is the mass coefficient, not a modular generator.

Proposition 10 (Full coupling of a qubit)**.** *Let $s = 2$, $\mu = 0$, and let the symbol satisfy Theorem 4 with traceless coefficients. The map $q \mapsto \sum_i q_i\alpha_i$ is an injective linear map from $\mathbb{R}^d$ into the three-dimensional space of traceless Hermitian 2×2 matrices. It is onto, so that every internal rotation generator is coupled to some direction of motion, if and only if $d = 3$.*

Proof. Injectivity is the linear independence assumed in Theorem 4; the target has dimension three. □

Proposition 10 is a precise version of a familiar observation: momentum space of a Weyl particle is identified with the Lie algebra of internal rotations of its qubit. It is a characterization under a stated premise, not a derivation of three dimensions. The premise, that no internal axis is left uncoupled, is a physical input. The following conjecture states what would have to be shown to turn it into a selection principle.

Conjecture 11 (Minimal carriers and three dimensions)**.** *In local unitary dynamics whose elementary internal carrier is a qubit and whose low-energy propagation is isotropic and single-shell, propagation that uses the full internal space selects $d = 3$, and massive propagation requires two carriers coupled by a chirality-exchanging term.*

Theorem 8 and Proposition 10 reduce this selection proposal to a physical reason for coupling every internal axis to momentum. Irreducibility alone does not supply it: $H = q_x\sigma_x + q_y\sigma_y$ generates the full matrix algebra, while the omitted σ_z is not a conserved charge. The three-dimensional Weyl automaton [13] and the minimal-directional-system argument of Ref. [15] motivate the proposal, but do not select three dimensions within the present class. A selection principle must distinguish the two-dimensional counterexample by stability, resources or formation, and must explain why a qubit is the relevant elementary carrier.

5 The lattice level: causal polytope, three speeds and nodes

5.1 Group velocities lie in the causal polytope

Proposition 12 (Group velocities and the neighbourhood)**.** *For the step* (1)*, let b be a band and k_0 a point where the eigenvalues of $W(k_0)$ are distinct. Then $\nabla_k\omega_b(k_0) \in \tau^{-1}\operatorname{conv} N$.*

Proof. Take $\tau = 1$. Let X be the position operator and ψ finitely supported. Then $\langle X\rangle_n = \langle W^n\psi, XW^n\psi\rangle$ satisfies $\langle X\rangle_n - \langle X\rangle_0 = \sum_{j<n}\langle W^j\psi, VW^j\psi\rangle$ with $V = W^*[X,W]$. From Eq. (1), $([X,W]\psi)(x) = \sum_h hA_h\psi(x-h)$, so V is bounded and translation invariant with symbol $V(k) = W(k)^*\, i\nabla_k W(k)$. Write $W(k) = \sum_b e^{-i\omega_b(k)}P_b(k)$ wherever the eigenvalues are distinct, which is almost everywhere once it holds at one point, since the discriminant of the characteristic polynomial is real-analytic in k. For each such k the Cesàro means $n^{-1}\sum_{j<n}W(k)^{-j}V(k)W(k)^j$ converge to $\sum_b P_bVP_b$, and differentiating $W = \sum_b e^{-i\omega_b}P_b$ with $P_b(\nabla P_b)P_b = 0$ gives $P_bVP_b = \nabla\omega_b\, P_b$. By bounded convergence,

$$\frac{\langle X\rangle_n - \langle X\rangle_0}{n} \longrightarrow \bar v(\psi) = \sum_b \int \nabla\omega_b(k)\, \langle\hat\psi(k), P_b(k)\hat\psi(k)\rangle\, \frac{dk}{|B|},$$

where B is the Brillouin zone. On the other hand $W^n\psi$ is supported in $\operatorname{supp}\psi + nN$, so for every unit vector u, $u\cdot\langle X\rangle_n \le \max_{x\in\operatorname{supp}\psi} u\cdot x + n\max_{h\in N} u\cdot h$. Hence $u\cdot\bar v(\psi) \le h_N(u) := \max_{h\in N} u\cdot h$. The map $\hat\psi \mapsto \bar v(\psi)$ is continuous in L^2 because $|\nabla\omega_b| \le \|V(k)\|$ is bounded, and finitely supported states are dense, so the inequality holds for all ψ. Taking $\hat\psi = P_b(k)\varphi$ with φ concentrating at k_0 gives $u\cdot\nabla\omega_b(k_0) \le h_N(u)$ for every u, which is membership in $\operatorname{conv} N$. □

The ballistic scaling used in the proof is the expectation-value form of the weak limit theorem for quantum walks [16].

Corollary 13 (Speed of a massless read-out cone)**.** *If a band has a first-order massless symbol at k_* whose cone is round with speed c, then $c \le \tau^{-1}r_{\mathrm{in}}(\operatorname{conv} N)$, where r_{in} is the inradius.*

Proof. Near k_* the group velocity tends to $c\,\hat q$ along every direction $\hat q$, so $c\hat q \in \tau^{-1}\operatorname{conv} N$ for all $\hat q$. □

5.2 The split-step walk

Consider the coined walk that applies a coin rotation and then a conditional shift,

$$W_\theta(k) = \begin{pmatrix} e^{-ika}\cos\theta & -e^{-ika}\sin\theta \\ e^{ika}\sin\theta & e^{ika}\cos\theta \end{pmatrix}, \qquad 0 < \theta < \frac{\pi}{2}, \tag{8}$$

with neighbourhood $N = \{a, -a\}$ and step time τ.

Proposition 14 (Three speeds of one walk)**.** *For the walk* (8)*:*

1. $\cos(\omega\tau) = \cos\theta\cos(ka)$*;*
2. *the support speed is* a/τ*, and the largest group velocity is* $(a/\tau)\cos\theta$*, attained at* $ka = \pm\pi/2$*;*
3. *near* $k = 0$*,* $\omega(k)^2\tau^2 = \theta^2 + \theta\cot\theta\,(ka)^2 + O((ka)^4)$*; the rest energy is* $E_0 = \hbar\theta/\tau$*, the inertial mass* $m_* = \hbar\tau\tan\theta/a^2$*, and the Lorentzian read-out speed defined by* $E_0 = m_* c_{\text{eff}}^2$ *is* $c_{\text{eff}} = (a/\tau)(\theta\cot\theta)^{1/2}$*;*
4. $\cos\theta < (\theta\cot\theta)^{1/2} < 1$*.*

Proof. Item 1: $\det W_\theta = 1$ and $\operatorname{tr} W_\theta = 2\cos\theta\cos(ka)$. Item 2: the support statement is Eq. (1) with $N = \{\pm a\}$. Differentiating item 1, $v = (a/\tau)\cos\theta\sin(ka)/\sin(\omega\tau)$, and $v^2 \le (a/\tau)^2\cos^2\theta$ is equivalent to $\cos^2\theta\cos^2(ka) \le \cos^2(ka)$, with equality only at $\cos(ka) = 0$. Item 3: differentiating item 1 twice at $k = 0$, where $\omega\tau = \theta$ and $\omega' = 0$, gives $\tau\sin\theta\,\omega'' = a^2\cos\theta$; then $\frac{1}{2}(\omega^2)'' = \omega\omega''$, $m_* = \hbar/\omega''$, and $E_0/m_* = \omega(0)\omega''(0)$. Item 4 is $\sin\theta\cos\theta < \theta < \tan\theta$. □

The Lorentzian speed read from the rest energy and the inertia of slow particles lies strictly between the largest transport speed and the support speed. No wave packet reaches it, because $v \le (a/\tau)\cos\theta < c_{\text{eff}}$; and the strict causal cone is wider. In the scaling limit $a, \tau \to 0$ with $a/\tau = c$ and $\theta/\tau = \mu$ fixed, all three speeds tend to c and the dispersion tends to $\omega^2 = c^2q^2 + \mu^2$ [11, 12]. At $\theta = 0$ the walk is a pair of exact shifts, with $\omega = \pm k$ and one speed. The deviations at finite mass are $1 - c_{\text{eff}}^2\tau^2/a^2 = \theta^2/3 + O(\theta^4)$ and $1 - \cos^2\theta = \theta^2 + O(\theta^4)$. The consequences for mass read-outs are developed in the companion paper on mass and energy [47].

5.3 The Weyl automaton on the body-centred cubic lattice

The minimal three-dimensional automaton of D'Ariano and Perinotti [13] has $s = 2$, neighbourhood $N = \{\ell(\pm1, \pm1, \pm1)/\sqrt{3}\}$ of eight vectors of length ℓ, and symbol

$$W(k) = e^{-i\kappa_x\sigma_x}\, e^{-i\kappa_y\sigma_y}\, e^{-i\kappa_z\sigma_z}, \qquad \kappa = \frac{\ell k}{\sqrt{3}}, \tag{9}$$

with $\cos(\omega\tau) = c_x c_y c_z - s_x s_y s_z$, where $c_i = \cos\kappa_i$, $s_i = \sin\kappa_i$. Expanding, $c_x c_y c_z = 1 - |\kappa|^2/2 + O(|\kappa|^4)$ and $s_x s_y s_z = \kappa_x\kappa_y\kappa_z + O(|\kappa|^5)$, so

$$\omega\tau = |\kappa| + \frac{\kappa_x\kappa_y\kappa_z}{|\kappa|} + O(|\kappa|^3). \tag{10}$$

At leading order $W(k) \simeq \mathbf{1} - i\sigma \cdot \kappa$, the symbol is $c\,\sigma \cdot k$ with $c = \ell/(\sqrt{3}\tau)$, and Eq. (7) gives a round Lorentzian cone. Isotropy fails at relative order $|\kappa|$ through the parity-odd term in Eq. (10). Table 1 collects analytic properties and numerical unitary wave-packet diagnostics. It is not a table of records obtained from a physical measuring instrument.

Property	**Value**
Causal polytope $\tau^{-1}\,\mathrm{conv}\,N$	cube $\lvert x_i\rvert \le \ell/(\sqrt{3}\tau)$ per unit time (analytic)
Inradius of the cube	$\ell/(\sqrt{3}\tau) = c$: Corollary 13 is saturated (analytic)
Group velocity components	at most c, attained on the cube faces
Largest group speed	$(3/2)^{1/2}c$; the cube corners, at speed $\sqrt{3}c$, are not reached by group velocities
Band-touching points in the Brillouin zone	four: at quasi-energy 0, $k = 0$ (chirality $+1$) and $\kappa = -\frac{\pi}{2}(1,1,1)$ (chirality -1); at quasi-energy π/τ, one of each chirality
One-site initial state	probability outside the cube: zero to machine precision; outside the round cone of radius ct: 0.50 after 16 steps, 0.47 after 24
Gaussian packets near $k = 0$, 24 steps	probability outside radius $ct + R_0$ (R_0 the 99% radius at $t = 0$): 0.084, 0.023, 0.008 for momentum widths $\sigma_\kappa = 0.4$, 0.25, 0.15

Table 1: Weyl automaton (9). Chirality is the sign of the determinant of the linear map $q \mapsto$ coefficient of $-i\sigma$ in $\pm W(k_0 + q)$. The packets carry a fixed spinor and therefore populate both bands.

5.4 What the read-out map adds

The lattice results show three ways in which the dynamics leaves the Lorentzian read-out underdetermined.

First, at finite mass the speed of the read-out depends on the observable that defines it. In the walk (8), rest energy and inertia give c_{eff}, the fastest wave packets give $(a/\tau)\cos\theta$, and the support front gives a/τ (Proposition 14).

Second, the Lorentzian cone governs low-momentum states only. In the Weyl automaton a state prepared on one site contains all momenta and leaves almost half of its probability outside the round cone, while it never leaves the cube; the leakage falls as the momentum width shrinks (Table 1).

Third, the Brillouin zone contains several band-touching points. In the Weyl automaton there are two at quasi-energy zero with opposite chiralities, as the doubling theorem for lattice Hamiltonians would lead one to expect [17], although that theorem does not apply directly to automata. A low-energy observer reads a Weyl cone of one chirality or the other depending on which point the prepared states are near. The dynamics does not choose; the preparation and the instruments do. In one dimension a single chiral cone is possible, since the shift by one site is a legitimate automaton; whether a strictly local isotropic automaton in three dimensions can carry a single node is a question we leave open.

So in this model the read-out map cannot be removed. What can be said precisely is what it consists of: a band point, a momentum window around it, and an observable defining the speed, each with computable consequences. This matches the treatment of access in HAFF, where what a physical reduction resolves depends on the channel and not only on the algebra it generates [42]. Here the Lorentzian geometry read by an observer is the characteristic geometry of the momentum window that its couplings reach.

5.5 Finite records certify one walk's dictionary

The choice of a read-out need not leave all its parameters inaccessible. The companion mass–energy paper realizes the same coin rotation by a fresh Gaussian oscillator, computes its exact open-channel linewidth, and reads its rest phase through controlled mixed-memory instruments [47]. On a supplied finite periodic ring, a normalized zero mode is preserved by a spatially uniform field coupling. Independent X and Z record trials at one and ten steps resolve the phase alias and certify θ. The ring, uniform support, axes, gain, clock, fresh ports and record archive are priced inputs.

Let that actual protocol give $\theta \in [\theta_-, \theta_+]$ at its stated confidence, with supplied a, τ. Monotonicity of $\tan\theta$, $\cos\theta$ and $C(\theta) = \theta\cot\theta$ gives simultaneous intervals

$$E_0 \in \frac{\hbar}{\tau}[\theta_-, \theta_+], \qquad m_* \in \frac{\hbar\tau}{a^2}[\tan\theta_-, \tan\theta_+],$$
$$c_{\text{eff}}^2 \in \frac{a^2}{\tau^2}[C(\theta_+), C(\theta_-)], \qquad v_{\max} \in \frac{a}{\tau}[\cos\theta_+, \cos\theta_-].$$

At $\theta_+ = \pi/2$ the inertial upper bound is infinite. With the priced angle radius $b = .004296823$ at 95% confidence derived in Ref. [47], true angle .3 and $a = \tau = 1$, every confidence interval on that event lies inside $[.3 - 2b, .3 + 2b]$. The implied intervals are

$$c_{\text{eff}} \in [.9838969, .9856636], \qquad v_{\max} \in [.9527616, .9578408], \qquad c_{\text{support}} = 1.$$

The support speed is supplied, not measured by the phase protocol. The other two are model implications of its angle certificate. The full preparation, pulse, read-memory and reference budget is given with that protocol.

These are parameter inferences for the prescribed ideal walk, informed by an actual open-drive experiment. They are not its noisy long-time transport velocities: nonzero-momentum transport requires the real Bloch channel and inter-momentum coherences. Nor does calibration prove that the dictionary is Lorentzian throughout the band. The companion finite-angle window theorem gives precisely where a Lorentzian mass reading becomes negative and must be rejected. Thus an observer's interface can be calibrated, can distinguish otherwise conflated speeds, and can fail at a testable boundary.

6 Several species: when do the cones become one?

Theorem 4 gives every single-shell species its own form Q. For isotropic species in one frame, a common light cone requires their speeds to agree. In a free model nothing makes them

agree: each speed is a parameter of the step. This section asks whether coupling the species can make the speeds equal, so that one cone is an output of the dynamics. The answer depends on the kind of coupling. Off-diagonal mixing cannot produce a single quadratic principal shell while its distinct diagonal shells are held fixed. Interactions can change those diagonal propagation laws: in the mediator model their speeds converge when each coupling has divergent accumulated strength toward low energy.

6.1 Mixing does not merge cones

Write the principal symbol of two species in a direction u as $A_1(u)$ on $\mathbb{C}^{s_1}$ and $A_2(u)$ on $\mathbb{C}^{s_2}$, both single-shell, $A_a(u)^2 = Q_a(u)\mathbf{1}$. A one-particle coupling adds a block off-diagonal term, which may itself depend linearly on q.

Theorem 15 (Mixing and the single shell)**.** *For every $s_1 \times s_2$ matrix V, the symbol $A = \begin{pmatrix} A_1 & V \\ V^* & A_2 \end{pmatrix}$ satisfies*

$$\min_{\lambda\in\mathbb{R}} \big\|A^2 - \lambda\mathbf{1}\big\|_{\mathrm{HS}}^2 \;\geq\; \tfrac{1}{2}\min(s_1, s_2)\,\big(Q_1(u) - Q_2(u)\big)^2. \tag{11}$$

Hence the coupled system has a single shell only if $Q_1 = Q_2$. Moreover the largest eigenvalue of A is at least the largest eigenvalue of either block and the smallest is at most the smallest of either block.

Proof. The diagonal blocks of $A^2 - \lambda$ are $Q_1 - \lambda + VV^*$ and $Q_2 - \lambda + V^*V$, and the Hilbert–Schmidt norm of a block matrix is at least that of its diagonal blocks. If V has rank r, then VV^* and V^*V share the nonzero eigenvalues $\sigma_1^2, \ldots, \sigma_r^2$ and have $s_1 - r$ and $s_2 - r$ zero eigenvalues. Using $(x + a)^2 + (x + b)^2 \geq \frac{1}{2}(a - b)^2$ for each shared eigenvalue, and for $k = \min(s_1, s_2) - r$ pairs of zeros, the squared norm is at least $\frac{1}{2}(r + k)(Q_1 - Q_2)^2$. The last statement is Cauchy interlacing for the principal submatrices A_1 and A_2. □

The bound (11) does not depend on the strength of the coupling. A mass-type coupling, independent of q, leaves the principal symbol untouched, but can change low-momentum band slopes. For example,

$$H(q) = \begin{pmatrix} v_1 q & g \\ g & v_2 q \end{pmatrix}, \qquad \omega_\pm(q) = \tfrac{1}{2}(v_1 + v_2)q \pm \sqrt{g^2 + \tfrac{1}{4}(v_1 - v_2)^2 q^2}.$$

For $g \neq 0$ both slopes at $q = 0$ equal $(v_1 + v_2)/2$, although the two distinct principal speeds remain v_1, v_2. Equal slopes at one band point are not a common quadratic characteristic cone. An on-site lattice coin can likewise change the effective logarithm of a step; the fixed-block theorem applies only after its principal blocks have been specified. For a chiral principal velocity matrix with fixed diagonal entries, the Schur–Horn theorem makes the eigenvalues majorize those entries, so off-diagonal mixing cannot reduce their variance. The interacting route below instead changes the propagation coefficients themselves.

6.2 Loops average speeds

Lemma 16 (Loops of one speed)**.** *Let a self-energy be a loop whose internal lines all have Lorentzian propagators of speed c and whose vertices are covariant under the Lorentz group of speed c. Then the coefficient of* $\ln\Lambda$ *in it is Lorentz covariant with speed c, for any fixed regular cutoff shape enclosing the origin and scaled by Λ.*

Proof. The substitution $k \mapsto (k_0, k/c)$ makes the integrand covariant under the standard Lorentz group. For a homogeneous integrand f of degree $-D$, the coefficient of $\ln\Lambda$ is the flux of the divergence-free field $xf(x)$ through a surface enclosing the origin, and it is the same for every such surface. □

The smallest interacting model is a real scalar of speed u coupled to N Dirac fermions of speed v by $g\phi\bar\psi\psi$ in $3+1$ dimensions. We integrate out a shell of spatial momenta with the frequency unrestricted; the logarithmic coefficients are scheme independent at this order. By Lemma 16 the fermion loop adds $\frac{Ng^2}{4\pi^2 v^3}(p_0^2+v^2p^2)\,dl$ to the inverse scalar propagator $p_0^2+u^2p^2$, where $l=\ln(\Lambda/E)$, so the squared scalar speed is replaced by a convex combination of u^2 and v^2. The fermion self-energy contains one fermion and one scalar line. Its frequency integrals are

$$J_0=\int\frac{t^2\,dt}{(t^2+v^2)(t^2+u^2)^2}=\frac{\pi}{2u(v+u)^2},\qquad J_s=\int\frac{dt}{(t^2+v^2)(t^2+u^2)^2}=\frac{\pi(v+2u)}{2vu^3(v+u)^2},\tag{12}$$

and the ratio of the spatial to the temporal coefficient gives the speed toward which v moves.

Proposition 17 (Speed flow in the Yukawa model)**.** *At one loop,*

$$\frac{dv}{dl}=a\Big(\frac{v+2u}{3}-v\Big),\quad a=\frac{g^2}{4\pi^2u(u+v)^2},\qquad \frac{du^2}{dl}=\frac{Ng^2}{4\pi^2v^3}\,(v^2-u^2),\tag{13}$$

and the coupling obeys $dg^2/dl=-(g^4/2\pi^2)\big[1/(vu(v+u))+N/(2v^3)+1/(u(v+u)^2)\big]$. At $u=v$ the linearized flow of $u-v$ has rate $(6N+1)g^2/(24\pi^2v^3)$, and $u+6Nv$ is conserved to linear order.

For $N=1$ these equations coincide with those of Anber and Donoghue [18], who derived them by dimensional regularization. The momentum-shell calculation gives the same logarithmic coefficients. Each speed moves toward a target that lies between the speeds of the lines in its loop: the fermion toward $(v+2u)/3$, the scalar toward v. The scalar moves $6N$ times faster, so the common speed lies close to the initial fermion speed. Since the Yukawa coupling is marginally irrelevant, $g^2\simeq 8\pi^2c_*^3/[(2N+3)l]$, where c_* is the common limiting speed and

$$\frac{|u-v|}{v}\;\propto\;\Big[\ln\frac{\Lambda}{E}\Big]^{-(6N+1)/(3(2N+3))},\tag{14}$$

with exponent $7/15$ for one Dirac fermion. Integrating the full flow to $l=10^9$ gives log–log slopes -0.4667 for $N=1$ and -0.7576 for $N=4$, as predicted.

Theorem 18 (Speeds converge through a common mediator)**.** *Let fermion species of speeds $v_1, \dots, v_n$ couple to one scalar of speed u with prescribed nonnegative, locally integrable functions $g_1^2(l), \dots, g_n^2(l)$, each fermion obeying the speed equations in* (13) *and u^2 receiving the sum of the fermion loops. The theorem is conditional on these coupling histories; a multi-species beta function must separately establish their accumulated strengths. Assume positive initial speeds. Let D be the difference between the largest and the smallest of $u, v_1, \dots, v_n$, and M_0 the initial largest speed. Then the largest speed never increases, the smallest never decreases, and*

$$D(l) \le D(0) \exp\Big(-\frac{1}{24\pi^2 M_0^3} \int_0^l \min_a g_a^2 \, dl'\Big). \tag{15}$$

If $\int g_a^2 \, dl < \infty$, the speed v_a changes in total by at most $D(0) \int g_a^2 \, dl/(24\pi^2 m_0^3)$, with m_0 the initial smallest speed.

Proof. Each v_a moves toward u at rate $\frac{2}{3}a_a$, and u^2 moves toward a weighted mean of the v_a^2, so u decreases when it is the largest speed and increases when it is the smallest. Consider the upper Dini derivative D^+ of the maximum minus the minimum. If both extremes are fermions $v_a \ge v_b$, then $D^+ \le \frac{2}{3}[a_a(u - v_a) - a_b(u - v_b)] \le -\frac{2}{3}(\min_c a_c)D$; if the maximum is u, then $D^+ \le -\frac{2}{3}a_b(u - v_b)$; the case of u minimal is symmetric. All speeds stay in $[m_0, M_0]$, where $\frac{2}{3}a_a \ge g_a^2/(24\pi^2 M_0^3)$, and Grönwall's inequality gives (15). The last statement follows from $|v_a'| \le \frac{2}{3}a_a D(0)$ and $a_a \le g_a^2/(16\pi^2 m_0^3)$. □

The estimate gives a rate when the weakest simultaneous coupling has divergent integral. Convergence itself needs less.

Corollary 19 (Intermittent interactions)**.** *In the one-loop star model of Theorem 18, suppose the nonnegative functions g_a^2 are locally integrable and initial speeds are positive. Every speed converges. Species connected to the mediator by edges with $\int_0^\infty g_a^2 \, dl = \infty$ have the same limiting speed as the mediator. In particular, all speeds converge to one value if each such integral diverges, even when the couplings are never active simultaneously.*

Proof. Write $v_a' = A_a(u - v_a)$ and $u' = \sum_a B_a(v_a - u)$, with

$$A_a = \frac{g_a^2}{6\pi^2 u(u + v_a)^2}, \qquad B_a = \frac{g_a^2(u + v_a)}{8\pi^2 u v_a^3}.$$

For an active edge, $B_a/A_a = \frac{3}{4}(1 + u/v_a)^3$. All speeds stay in $[m_0, M_0]$, so the two directional rates are uniformly comparable, and their integrals diverge exactly when $\int g_a^2$ does. This is a type-symmetric continuous-time consensus system. The cut-balanced convergence theorem [1], applied to the coefficients along the solution, gives convergence and equality of limits within each component of the infinite-integral interaction graph. The theorem allows state-dependent coefficients. It does not provide the same rate as (15) when $\min_a g_a^2$ vanishes. □

The accumulated interaction across renormalization scales decides the outcome. The variable $l = \ln(\Lambda/E)$ is scale, not laboratory time; intermittent edges here are active on different scale intervals. An identically uncoupled species keeps its cone; a finite integrated

coupling need not erase an initial difference. Marginally irrelevant couplings remove differences as a power of $\ln(\Lambda/E)$; near an interacting fixed point they fall as a power of E/Λ, with the one-loop exponent $7g_*^2/(24\pi^2 c_*^3)$ for $N = 1$. In $4-\varepsilon$ dimensions the Yukawa fixed point gives $(6N+1)\varepsilon/[3(2N+3)]$ at this order. The weak-coupling model must remain applicable along this flow. A mediator that breaks rotations can instead drive speeds apart, as in nematic criticality of d-wave superconductors [20].

The common speed lies between the smallest and the largest initial speed. Each initial speed of a massless round cone is at most the inradius of conv N per step (Corollary 13), so the emergent cone lies inside the causal polytope.

6.3 Fixed points and the number of cones

Near an interacting fixed point the question has an answer that does not rely on perturbation theory.

Theorem 20 (Speed splittings at a conformal fixed point)**.** *Let the infrared fixed point be a unitary conformal field theory in $D = d+1 \geq 3$ dimensions, Lorentz invariant for one cone, and consider local perturbations invariant under translations, spatial rotations and CPT. Up to total derivatives the Lorentz-violating perturbations are $\int d^D x\, O_{0\cdots 0}$ for symmetric traceless primaries of even spin $\ell \geq 2$. Their couplings satisfy $D-\Delta_O \leq 2-\ell \leq 0$; equality in the first inequality means that the current is conserved, while a marginal Lorentz-violating coupling requires a conserved spin-2 current. If the stress tensor is the only conserved spin-2 current, all independent spin-2 speed perturbations are irrelevant. When the excited non-conserved spin-2 modes have a smallest gap $\gamma_2 = \min(\Delta_O - D) > 0$, their slowest linearized decay is $(E/\Lambda)^{\gamma_2}$.*

Proof. Descendants integrate to zero. A primary transforms in a finite-dimensional irreducible representation of the Lorentz group, whose complexification is that of $SO(D)$, and because $(SO(D), SO(D-1))$ is a Gelfand pair only the symmetric traceless representations contain a rotation-invariant component, once each. CPT acts on a Hermitian rank-ℓ tensor by $(-1)^\ell$, which removes odd ℓ, and scalars are Lorentz invariant. The unitarity bound $\Delta \geq \ell + D - 2$ for $\ell \geq 1$ holds with equality exactly for conserved currents [21, 22]. The perturbation $\int T_{00}$ rescales time and changes no relative speed. □

The stability argument is established in Refs. [23, 24]; it assumes the Lorentz-invariant fixed point and does not prove that every microscopic dynamics reaches it. Additional conserved spin-2 currents are the marginal exceptions. In three dimensions, for exactly two such currents, a suitable basis generates separate translations and the two families of current correlators factorize [25, Section 8]. This is dynamical separation of the current sectors, not necessarily a tensor product of the full theories: a shared singlet constraint can remain. Nor does counting currents count sectors in general; two free scalar fields already have three conserved spin-2 currents. The result needed here is the local criterion for suppressing relative-speed perturbations, while identifying all independently propagating sectors requires further structure.

In two dimensions several propagation sectors can retain different speeds, as in spin–charge separation. A Sugawara decomposition gives an additional sector only when $T - T_J$

is nonzero. Under its positivity, spectrum and finite-signal-speed assumptions, the scale-to-Lorentz theorem permits either one Lorentz factor or several [26]; it does not forbid a common cone from being selected dynamically. Establishing one cone therefore requires information about the actual sectors and their interactions, rather than the dimensionality alone.

6.4 What the lattice adds

For number-conserving automata whose interaction preserves the supplied neighbourhood, each distinguishable particle stays in $x_j + nN$ after n steps. The centre-of-mass version of Proposition 12 therefore bounds every isolated bound-state band by $\operatorname{conv} N$ per step. Changing the neighbourhood changes that bound. Equalization of low-energy cones does not itself change the support cone.

For a loop dominated by one linear band point, the logarithmic coefficient carries that point's read-out speed by Lemma 16. Finite ultraviolet matching terms need not have the same ratio of temporal and spatial coefficients. A Lorentz-violating regulator can leave corrections of order $g^2/16\pi^2$ [28], subsequently reduced only at the infrared rates above.

Composites need not inherit a constituent's read-out cone. Take distinguishable walkers with $\theta_A = .3$, $\theta_B = .5$, $a = \tau = 1$, first apply both free coined steps, and then multiply the coincident-position sector by $e^{+.4i}$. With total momentum K conjugate to $(x_A + x_B)/2$, the isolated positive band at $K = 0$ has $\omega_0 = .7614121$ and $m_* = 1/\omega_0'' = .9023$, compared with the free threshold .8 and constituent mass sum .8556. Thus $c_{\text{eff}}^2 = \omega_0\omega_0'' = .8439$, below the constituent values $\theta_A \cot\theta_A = .9698184$ and $\theta_B \cot\theta_B = .9152439$. Periodic relative-coordinate calculations with 91 and 131 sites agree at the quoted precision. They also track a localized band beyond $|K| = .68$; no such endpoint or global maximum speed is inferred from this local mass calculation. Contact interactions can produce bound molecules in quantum walks [19]. In the example above, the separately computed quasienergy and curvature do not obey the relativistic binding-energy–inertia relation.

A common cone can also be kinematic. The minimal isotropic Weyl automaton is unique on the body-centred cubic lattice [13], so all massless species built from it share $c = \ell/(\sqrt{3}\tau)$; a Dirac automaton built from two of them has first-order speed $c(1 - m^2)^{1/2}$ in lattice units, and photons built from pairs of Weyl automata have the dispersion $2|n_{k/2}|$ and the same low-energy cone [27]. For an electron on a lattice of Planck spacing, $m \approx 4 \times 10^{-23}$ and the mass correction is of order 10^{-45}. Loop matching can generate relative-speed corrections of order $g^2/16\pi^2$ unless a symmetry forbids them. Contact binding also changes the composite cone in the example above, but its size is model dependent and carries no general loop suppression factor.

6.5 How much running is needed

With marginal couplings the logarithmic rate (14) is far too slow to produce the observed equality of speeds. Reducing a difference by a factor 10^{-13} requires $\ln(\Lambda/E) \approx 10^{28}$ for one Dirac fermion with $g^2 = 4\pi$ at the cutoff, and about 10^{12} even for a hundred fermions, while $\ln(M_{\text{Pl}}/m_e) = 51.5$. Anber and Donoghue reached the same qualitative conclusion [18].

Conjecture 21 (One cone needs a strongly coupled regime)**.** *In local unitary dynamics with several interacting gapless species, the relative speed difference of two sectors at scale E is $\delta c(E) \simeq \delta c(\Lambda) \exp(-\int_E^\Lambda \gamma \, d\ln\mu)$, where γ is the anomalous dimension of their relative stress tensor and $\delta c(\Lambda)$ is of order $g^2/16\pi^2$ from lattice matching even when the step is common. Equalities of order 10^{-15} then require $\int \gamma \, d\ln\mu \gtrsim 30$ above accessible energies, that is, a strongly coupled regime with γ of order one over many decades, or a symmetry that removes $\delta c(\Lambda)$. The largest residual differences sit between the most weakly coupled sectors.*

Theorems 18 and 20 tie equalization to interaction strength; lattice matching regenerates relative-speed differences, while marginal running removes them only logarithmically. The quantitative estimates above and the power-law emergence in strongly coupled holographic models [24] support the conjecture. Anber and Donoghue's observation that the weakest interactions leave the largest differences also makes gravitational waves against light a sensitive comparison [18].

A viable ultraviolet realization must place the strongly coupled range consistently with the observed weakly coupled particle description and experimental bounds. A composite realization is one possibility, not a necessary consequence of the conjecture. A symmetry route, such as non-relativistic supersymmetry protecting dimension-four terms, can avoid strong coupling but must survive symmetry breaking. Gravity is shared by all species yet is irrelevant at low energy; this flow cannot align its cone there. The relation of the gravitational-wave and matter cones therefore requires an ultraviolet mechanism or a protecting symmetry.

6.6 Consequences

The mediator model makes a common speed a conditional output of divergent accumulated interactions. The conformal argument establishes stability against relative-speed perturbations near a supplied Lorentz-invariant fixed point; it does not establish formation of that fixed point. This separates three inputs to the higher-dimensional Consciousness construction: each species' shell and signature, the alignment of their cones, and the actual infrared regime. Its boosted-ball family requires a common cone. In the two-dimensional HAFF gravity model [44], identifying the matter and Jackiw–Teitelboim cones remains an input; the two-dimensional sector theorem neither supplies nor forbids that identification.

7 Relations to other frameworks and papers

7.1 Algebraic frameworks

Sequential operations are represented here by linear maps under ordinary composition, which is associative by definition. Jordan and octonionic models need additional bracketing and composition rules before tensor products, subsystems or error correction can be transferred [30, 31, 32]; this is a statement of scope, not a no-go theorem. Von Neumann and C^*-algebras are compatible with everything above, since the step (1) and its symbol live in a concrete C^*-algebra; such algebras do not single out a quadratic form until a dynamics is

given. Clifford algebras are distinguished in this paper by a dynamical property, the single shell of Theorem 4, rather than by being chosen as an encoding.

7.2 Thermodynamic gravity, modular flow and spectral triples

Jacobson's derivation of the Einstein equation assumes local Lorentzian causal structure, local equilibrium and an entropy–area relation [3], and entanglement derivations of linearized gravity work on a Lorentzian background [33, 34]. The present model supplies local Lorentzian cones for the low-momentum sector of a local unitary dynamics. It supplies neither equilibrium nor the entropy–area relation, and Section 5.4 shows that its cones depend on the momentum window.

Local algebras in relativistic quantum field theory are of Type III [35, 36]. For the vacuum and wedge regions the Bisognano–Wichmann theorem identifies modular flow with the boost subgroup [37, 48]; the identification presupposes the Lorentzian structure it describes. Whether the modular data of states of a lattice automaton reproduce the boosts of its low-energy cone in the scaling limit is open and could be computed in the walk of Section 5.2.

In Connes' noncommutative geometry a Dirac operator with Euclidean Clifford symbol encodes Riemannian geometry [38], and Lorentzian versions are an active problem [39, 40, 41]. Theorem 4 says that using such a Dirac operator as a Hamiltonian, $i\partial_t\psi = D\psi$, produces the Lorentzian characteristic cone of signature $(1, d)$ on time times space. We have not worked out the relation to the Krein-space approaches cited.

7.3 The Q-RAIF papers and HAFF

The persistence paper of the series [45] distinguishes three quadratic structures: a designer's positive descent metric G, a positive work form K derived from field stress, and the Lorentzian read-out form. The present paper adds the source of the third, $\eta = \omega^2 - G_{\mathrm{HS}}$, and shows why neither G nor K could supply it. The actuator field of that paper is the chiral case of Remark 6; the stress it carries propagates along a single null ray, which is the matter source used in the HAFF paper on gravity [44].

The representation paper [46] asks for a quadratic-space morphism before signatures can be compared. Under the single-shell condition of Theorem 4, the spatial symbol $q \mapsto \sum_i q_i\alpha_i$ supplies an isometry from $(\mathbb{R}^d, Q)$ into the generators with G_{HS}. The converse fails: the trace identity $G_{\mathrm{HS}}(H(q)) = Q(q)$ does not imply the operator identity $H(q)^2 = Q(q)\mathbf{1}$. For example, the local unitary walk $W(k) = \mathrm{diag}(e^{-ik}, e^{-2ik})$ has $H(q) = q\,\mathrm{diag}(1, 2)$. Its symbol is an isometry for $Q(q) = 5q^2/2$, but its two branches have different speeds and do not share a quadratic shell. The operator identity, rather than its trace, is the condition that connects this metric to a common propagation cone.

The mass–energy paper [47] constructs a covariant positive-energy representation of the continuum single-shell symbol, closing the spectral interface used for its mass theorem. Selection of that sector remains physical input, and a finite lattice window is not invariant under arbitrary boosts. Its field pulse and actual phase protocol connect occupation, lifetime and finite records; Section 5.5 imports that certificate rather than treating an ideal band as a measurement.

8 Discussion

Ingredient of a Lorentzian read-out	**Status in local unitary propagation**
Evolution time	supplied; unitarity makes its frequency a hyperbolicity direction (Theorem 2)
Signature $(1, d)$	derived for any quadratic cone (Lemma 3)
Clifford relations	equivalent to a single shell (Theorem 4); Lorentzian generators constructed (Corollary 5)
Single shell, isotropy, linear band point	inputs; observable and testable in a given model
Spatial dimension d	input; bounded by the internal space (Theorem 8)
Speed c	a property of the dynamics; at finite mass it depends on the defining observable (Proposition 14)
Read-out map	input: band point, momentum window and speed observable (Section 5.4)
Common speed of several species	conditional output of the mediator model; stable against irrelevant relative-speed perturbations near a supplied Lorentz-invariant fixed point, whose formation is not derived (Theorems 18, 20)
Lattice, neighbourhood, step	inputs

Table 2: Inputs of a Lorentzian read-out in local unitary propagation.

Table 2 separates the derived signature from the remaining inputs: lattice data, absence of birefringence and the observer's read-out map. Deriving the first two requires a formation model; no property of the dynamics alone specifies the third.

The problems that follow have the most leverage. Section 6 answers the question of a common cone for several species under stated conditions; what remains there is the anomalous dimension γ_2 in a strongly coupled lattice model, which Conjecture 21 needs. Another is chirality: whether a strictly local, isotropic three-dimensional automaton can have a single node, which would remove one component of the read-out map. The mass-setting interface now has a nonempty field realization and finite certificate, but renewed drives, full noisy spatial transport and covariant support remain physical tasks. A fourth is the modular route: computing the modular flow of a lattice state and comparing it with the boosts of the low-energy cone. Position-dependent coins, which produce effective curved metrics, would connect the model to the gravitational questions of HAFF.

9 Conclusion

Propagation supplies the relative sign in $\omega^2 - G_{\mathrm{HS}}$; a positive control metric alone cannot. The single-shell criterion, dimension bound and calibrated residuals connect this form to a specified carrier. Finite angle records imply separated ideal dictionary-speed intervals, while

noisy transport remains a different observable. For several species, the mediator model aligns cones under accumulated-coupling conditions; the conformal argument tests their stability, not the creation of a fixed point. These are the respective interfaces to mass–energy read-out, the common-cone geometry of Consciousness II and the still-supplied matter cone of HAFF-D. Formation of the carrier, clock and accessible momentum window remains a dynamical question.

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Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost

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Persistent control has to be realized by an interaction, with an error and a cost. We construct one such actuator: a degenerate bit coupled to a mode of a free chiral field through $U = \exp[-i\lambda X(b+b^\dagger)]$ and a smooth switch g. For independent diagonal bit inputs, the same unitary fixes the flip error, added stress $(g')^2/4$, switching work $\lambda^2\bar{\omega}$ and a positive quadratic cost on switch profiles. A flip with failure ϵ needs mean drive occupation at least $\theta_\epsilon(1-2\epsilon)/[4\sqrt{\epsilon(1-\epsilon)}]-\frac{1}{2}$, where $\theta_\epsilon = \arccos(2\epsilon-1)$; the leading cost is $\pi/(4\sqrt{\epsilon})$. Displaced squeezed states nearly attain it, whereas classical drives need order ϵ^{-1}. A calibrated feedback task with scheduled fresh independent drives has an exact stationary loss, coupling rate and stability proof, with one acceptance condition joining calibration records to drive energy. A second bound prices response readout: if two branches differ only by probe delays bounded in difference by $\Delta\tau$, binary error ϵ requires $E\Delta\tau \geq \hbar[1-2\sqrt{\epsilon(1-\epsilon)}]/\zeta$, with $\zeta = \max_{x>0}(1-\cos x)/x$. Retained ancillas and joint decoding do not remove this bound, and a three-level construction saturates it across the allowed signed-delay families. A retained oscillator also supports a thermal branch: a coherent echo protects held quantum content, a mixed qubit port reads the coupling through one binary outcome, and a finite Gibbs bank restores the oscillator while retaining every spent output. These energy and memory accounts do not identify exported energy with removed entropy. The descent metric, physical work form and Lorentzian propagation form remain distinct.

I. INTRODUCTION

Interaction with a large environment drives a quantum system towards decoherence and thermalization [1, 6]. A subsystem that keeps a non-equilibrium steady state (NESS) does so through a control process paid for by its surroundings, and such processes can be modelled as feedback [7–10]. This paper asks what persistence requires of the controls. The question has two parts. One concerns the operator structure the control dynamics needs in order to remain Lyapunov stable. The other concerns how a control operation is physically realized, what error it carries and what it costs.

Persistence alone does not select a Clifford algebra or an indefinite metric. Gradient descent requires a positive control metric, and standard dissipative and coherent control stabilizes target states without Clifford structure [15–18]. Section III distinguishes associativity of operator composition, positivity of a descent metric and the optional Clifford encoding of a chosen quadratic form.

The physical part is the main subject. In an abstract control model a pulse is a free exact unitary. In a physical model it is an interaction with a field, and the field brings quantum fluctuations, energy and stress with it. We work out one such actuator exactly. A degenerate two-level system couples to the derivative of a free chiral scalar field through a smooth switch. The same unitary then fixes the flip channel and its failure probability, the outgoing stress and switching work, and a positive quadratic form on switch profiles that is derived from the field rather than chosen.

Three results follow from this construction. Theorem 6 bounds the energy cost of an accurate flip for every input state of the driving mode; squeezed Gaussian drives nearly attain the bound, while Proposition 8 shows that classical drives cannot approach it. Theorem 11 places the actuator in a calibrated feedback task and gives a complete closed-loop persistence theorem, including bath noise, reading noise and gate failure. Proposition 12 turns the task requirement into one condition on calibration records and drive energy together, so a shortage of either resource can make the task infeasible.

An interface can also be queried while it carries unknown quantum content. Section VII gives the corrected parity instrument and its collection-dependent damage. A coherent controlled-parity echo protects the body for arbitrary mode inputs and returns a finite meter; its mode access, control and readout are separate physical resources. Resonant contacts realize a retained receiver and a finite packet collector, with explicit occupation, pulse-duration and finite-area-error accounts. A positive-frequency finite-band field supplies a continuous counterpart: a finite prescribed drive prepares its matched packet, with certified collection loss and separately counted receiver renewal.

For the retained oscillator, Section VII C also supplies a thermal alternative to vacuum inquiry. A mixed qubit port produces an actual binary record after the echo, and a finite bank of Gibbs oscillators restores the active port without discarding its old correlations. The record gain, read backreaction and pulse durations are derived from this same protocol. Its bank and clocks are finite resources; it is a different implementation from travelling-field collection.

Control also needs the response to come back through an instrument. A bound based on mean energy, in the quantum-speed-limit tradition [34], prices a specified translation-only readout in Theorem 14. It permits mixed probes, retained ancillas and joint decoding. Its role is to constrain the physical response-to-program route of HAFF-D, not to identify every sensing channel with a time delay.

Within the Q-RAIF series this paper carries the control and persistence side of the physical interface. Its outgoing stress is the matter source whose gravitational response is studied in the HAFF paper on gravity [29], and its feedback task is the one analysed in the Imprint Chain [36], where the corrective flip was a supplied exact pulse. Nothing here makes an observer the author of geometry. Persistence is analysed inside a supplied field theory, and geometric reconstruction in HAFF depends on its own algebraic and dictionary inputs [28].

Section II states the feedback problem. Section III records the algebraic constraints. Section IV constructs the actuator, Section V proves the energy–accuracy results, and Section VI treats persistence, calibration, response readout and their resource accounts. Section VIII compares the quadratic structures, Section IX relates the results to other work, and Section X discusses what remains open.

II. PERSISTENCE AS A FEEDBACK PROBLEM

Consider $\mathcal{H}_{\mathrm{tot}} = \mathcal{H}_R \otimes \mathcal{H}_E$ with reduced dynamics

$$\dot{\rho}_R = -i[H_{\mathrm{eff}}, \rho_R] + \mathcal{D}[\rho_R]. \tag{1}$$

The controls must be specified in the composed system dynamics, rather than inferred from a formal stationarity condition on a protocol cost.

Standard quantum control writes $H_{\mathrm{ctrl}} = \sum_k u_k(t) G_k$ with generators of a Lie algebra $\mathfrak{g}$, $[G_i, G_j] = i f_{ijk} G_k$ [14]. The Lie algebra fixes the accessible directions. The size of a correction is quantified by a positive cost, norm or information metric supplied separately. This separation is standard and is not a defect of Lie-algebraic control. Section IV shows where such a cost can come from physically: from the stress of the field that implements the control.

III. WHAT THE OPERATOR ALGEBRA MUST SUPPLY

A. Associativity as a boundary of the model

Remark 1 (Associativity as a control-model boundary)**.** *The master-equation model represents sequential controls by linear maps under ordinary composition, which is associative by definition [1]. This is a declared scope condition, not a theorem excluding every non-associative dynamics. Jordan or octonionic models would need additional bracketing, subsystem and composition rules before a control calculation could be transferred [12, 13]. Q-RAIF Paper A [30] uses the same operator-composition boundary for propagation.*

B. A positive metric for Lyapunov descent

For an auxiliary gradient controller on parameters λ consider the descent rule

$$\dot{\lambda} = -\Gamma\, G^{-1} \nabla_\lambda V, \qquad \Gamma > 0. \tag{2}$$

Proposition 2 (Positive descent metric)**.** *The controller contribution to $\dot{V}$ under Eq. (2) is $-\Gamma(\nabla_\lambda V)^T G^{-1} (\nabla_\lambda V)$. It is nonpositive for every gradient exactly when G^{-1}, equivalently the invertible symmetric matrix G, is positive definite. If G has a negative direction, a gradient along it makes the contribution positive.*

Proof. The first statement is the chain rule. A symmetric quadratic form is nonnegative on all vectors precisely when it is positive semidefinite, and an invertible positive semidefinite matrix is positive definite. □

The descent mechanism therefore selects positive definiteness, not Lorentzian signature. Channel types such as unitary, dephasing and dissipative can be labelled by a discrete index, a direct-sum decomposition or supplied weights; their qualitative difference does not need a negative norm.

Parameter descent is not yet the physical state equation. With drift H_0, dissipator $\mathcal{D}$ and controls $H_{\mathrm{ctrl}}(\lambda) = \sum_i u_i(\lambda) G_i$, the closed loop is

$$\dot{\rho} = -i[H_0 + H_{\mathrm{ctrl}}(\lambda), \rho] + \mathcal{D}[\rho], \qquad \dot{\lambda} = -\Gamma G^{-1} \nabla_\lambda V, \tag{3}$$

and for differentiable $V(\rho, \lambda)$

$$\dot{V} = D_\rho V[-i[H_0 + H_{\rm ctrl}, \rho] + \mathcal{D}[\rho]] - \Gamma(\nabla_\lambda V)^T G^{-1}(\nabla_\lambda V). \tag{4}$$

A stability proof must control the first term as well. Setting a control amplitude equal to a gradient component gives no general sign.

Dissipative and coherent control can stabilize target states with positive costs and Lie or operator-algebraic controls [15–18]. Von Neumann and C^*-algebras are compatible with this model; neither supplies a preferred quadratic form without a dynamics. These examples also show why persistence alone imposes no indefinite Clifford structure.

C. Clifford algebra as a parameterization

If one chooses a quadratic space (V, q) and requires control generators to satisfy $v^2 = q(v)\mathbf{1}$, the universal property identifies $Cl(V, q)$ as the minimal algebra for that presentation [11]. This is a definitional fact. It does not show that persistence, channel discrimination or Lyapunov descent imposes those relations. Pauli controls of a qubit satisfy $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1}$, the relations of $Cl(3, 0)$, so the Clifford package is a convenient description of qubit controls. A particular feedback model may favour some representation because of its controls and costs, but that preference has to be shown in its closed-loop dynamics.

a. State stability. Equation (4) is the continuous-time obligation: a parameter descent controls only its own contribution, while drift, dissipator and controlled state evolution remain. For full-rank states and a fixed full-rank target, $\dot{D}(\rho\|\rho_*) = \mathrm{tr}[\dot{\rho}(\log\rho - \log\rho_*)]$. Theorem 11 establishes stability instead for a discrete composed channel with a physically realized, imperfect correction.

IV. A CONTROL FLIP REALIZED BY A FIELD COUPLING

A. The coupling and its channel

The field-coupling and drive bounds in this and the next section use $\hbar = 1$: $\bar{\omega}N$ and $\bar{\omega}\lambda^2$ are energies in these units. In physical units they are $\hbar\bar{\omega}N$ and $\hbar\bar{\omega}\lambda^2$; the field Hamiltonian and stress carry the same overall $\hbar$. Quadratures, occupation and pulse areas are dimensionless. The later delay, entropy and retained-oscillator accounts display $\hbar$ explicitly.

The controlled system is a two-level system with zero free Hamiltonian, Pauli operators X and Z. A bosonic mode with $[b, b^\dagger] = 1$ and quadratures $Q = b + b^\dagger$, $P = i(b^\dagger - b)$, $[Q, P] = 2i$, interacts through

$$U = \exp[-i\lambda XQ], \qquad \lambda > 0. \tag{5}$$

In the eigenbasis $X|x\rangle = x|x\rangle$, $U = \sum_x |x\rangle\langle x| \otimes e^{-i\lambda xQ}$, and $e^{-i\lambda xQ}$ is the displacement $D(-i\lambda x)$. Write $\chi_\sigma(k) = \mathrm{tr}(\sigma e^{-ikQ})$ for the characteristic function of Q in a mode state σ.

Proposition 3 (Channel of the coupling)**.** *Let the mode start in a state σ independent of the bit. The bit channel multiplies the matrix element $\rho_{xx'}$ in the X basis by $\chi_\sigma(\lambda(x - x'))$. For bit states diagonal in Z, its output Z statistics agree with a flip having failure probability*

$$\epsilon = \tfrac{1}{2}\big(1 + \mathrm{Re}\,\chi_\sigma(2\lambda)\big). \tag{6}$$

For a coherent input $|\alpha\rangle$, $\alpha > 0$, the factor is

$$e^{-\lambda^2(1 - xx') - 2i\alpha\lambda(x - x')}. \tag{7}$$

If $2\lambda\alpha = \pi/2$ and $N = \alpha^2$, the channel on every bit state is

$$\mathcal{F}_N(\rho) = s_N X\rho X + (1 - s_N)\rho, \qquad s_N = \frac{1 + e^{-2\lambda^2}}{2}, \qquad \lambda^2 = \frac{\pi^2}{16N}, \tag{8}$$

so that $\epsilon_N = 1 - s_N = \big(1 - e^{-\pi^2/(8N)}\big)/2$.

Proof. Tracing the mode gives the factor $\mathrm{tr}(\sigma\, e^{i\lambda x' Q} e^{-i\lambda x Q}) = \chi_\sigma(\lambda(x-x'))$. A Z-diagonal state $(I+zZ)/2$ has $\rho_{+-} = z/2$ in the X basis, so the output mean of Z is $z\,\mathrm{Re}\,\chi_\sigma(2\lambda)$, which is $(1-2s)z$ for a flip with success probability s. For the coherent input, $D(\beta)D(\alpha) = e^{(\beta\alpha^* - \beta^*\alpha)/2} D(\alpha+\beta)$ gives $e^{-i\lambda x Q}|\alpha\rangle = e^{-i\alpha\lambda x}|\alpha - i\lambda x\rangle$. The overlap is the coherent-state inner product $\langle \alpha - i\lambda x' | \alpha - i\lambda x\rangle = e^{-\lambda^2(1-xx') - i\alpha\lambda(x-x')}$ times the Weyl phase $e^{-i\alpha\lambda(x-x')}$ from composing the displacements; this is Eq. (7). At $2\lambda\alpha = \pi/2$ the off-diagonal factor is $-e^{-2\lambda^2}$, and $1 - 2s_N = -e^{-2\lambda^2}$ yields Eq. (8) on all inputs. □

The channel $\mathcal{F}_N$ differs from the ideal Ad_X by $\epsilon_N(\mathrm{id} - \mathrm{Ad}_X)$, so its half diamond-norm distance from the ideal flip is ϵ_N, and its average gate infidelity is $2\epsilon_N/3$. These are different error conventions. The coherent family above needs infinite occupation for an exact flip; Section V shows that this holds for every input state.

B. A local switch on a free field

Equation (5) becomes a local operation once the mode is defined by a switch. Take a right-moving free massless scalar in $1+1$ dimensions with null coordinate u,

$$\phi(u) = \int_0^\infty \frac{d\omega}{\sqrt{4\pi\omega}} \big(a_\omega e^{-i\omega u} + a_\omega^\dagger e^{i\omega u}\big), \qquad [a_\omega, a_{\omega'}^\dagger] = \delta(\omega - \omega'), \tag{9}$$

current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$. The current obeys $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$. For a real smooth compactly supported switch g define

$$Q_g = \int g(u)\, j(u)\, du = \lambda(b + b^\dagger), \qquad \lambda^2 = \int_0^\infty \frac{\omega}{4\pi} |\hat g(\omega)|^2 d\omega, \qquad \hat g(\omega) = \int g(u) e^{-i\omega u} du, \tag{10}$$

where $b = \int_0^\infty f^*(\omega) a_\omega d\omega$ with $f(\omega) = i\sqrt{\omega/(4\pi)}\,\hat g^*(\omega)/\lambda$ is a normalized mode. Its mean frequency is $\bar\omega = \int \omega |f|^2 d\omega$, and the current contains it as $j(u) = \psi(u) b + \psi^*(u) b^\dagger + j_\perp(u)$ with $\psi(u) = -i\int_0^\infty \sqrt{\omega/(4\pi)}\, f(\omega) e^{-i\omega u} d\omega$.

A probe at fixed position with interaction Hamiltonian $g(t) X j(t)$ evolves by $U = e^{i\vartheta} \exp(-iXQ_g)$ for a real phase ϑ. The reason is that $[g(t)Xj(t), g(t')Xj(t')]$ is a multiple of the identity, because $X^2 = 1$ and the current commutator is a c-number, so the Magnus series stops at second order and the second-order term is a phase. This uses three conditions: a single probe, a fixed coupling operator X, and a probe free Hamiltonian that is zero or commutes with X. Nonzero field frequencies are harmless. A probe Hamiltonian that does not commute with X makes $X(t)$ time dependent, and several probes keep relative phases; neither case is covered. A point-like limit of the switch needs regularization, so we use smooth profiles. This exact Weyl evolution of a degenerate detector is a known method [21, 22]; what we add is its use as a priced feedback actuator.

C. Stress, work and a derived quadratic cost

Proposition 4 (Stress of one encounter)**.** *For the unitary $U = \exp(-iXQ_g)$,*

$$U^\dagger j(u) U = j(u) + \tfrac{1}{2} X g'(u), \qquad U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u)\, X j(u) + \tfrac{1}{4} g'(u)^2. \tag{11}$$

The expected added stress is $g'(u)\langle X j(u)\rangle + g'(u)^2/4$. The cross term vanishes if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis. A Z-diagonal bit marginal alone is not sufficient.

Proof. $[Q_g, j(u)] = \int g(v)[j(v), j(u)] dv = -\frac{i}{2} g'(u)$ is a c-number, so $e^{iXQ_g} j(u) e^{-iXQ_g} = j(u) + \frac{1}{2} X g'(u)$ exactly. Squaring, using $X^2 = 1$, and noting that a c-number shift leaves normal ordering unchanged gives the stress identity. Under independence $\langle Xj\rangle = \mathrm{tr}(\rho X)\langle j\rangle = 0$, even when the input beam has $\langle j\rangle \neq 0$. If the joint state is block diagonal in the bit's Z basis, $X \otimes A$ has zero expectation for every field operator A. For the last claim, $(|\uparrow\rangle|0\rangle + |\downarrow\rangle|1_b\rangle)/\sqrt{2}$ has bit marginal $I/2$ but $\langle Xb\rangle = 1/2$, hence $\langle X j(u)\rangle = \mathrm{Re}\,\psi(u) \neq 0$; with a relative phase i the integrated cross term equals $-\lambda\bar\omega$ and changes the work below. □

Under the stated joint condition the added stress is

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2, \tag{12}$$

the second form following from $2\lambda \operatorname{Im}\psi = g'/2$. If the interaction is off before and after the encounter and the bit Hamiltonian is zero, the energy delivered by the external switching agent equals the change of field energy,

$$W_{\text{switch}} = \frac{1}{4}\int g'(u)^2 du = \frac{1}{4\pi}\int_0^\infty \omega^2 |\hat{g}(\omega)|^2 d\omega = \lambda^2 \bar{\omega}, \tag{13}$$

by Parseval's identity and $|f|^2 = \omega|\hat{g}|^2/(4\pi\lambda^2)$. This holds for every input state of the field that satisfies the joint condition.

Proposition 5 (Derived quadratic cost)**.** *For $g = \sum_i a_i g_i$ with real smooth compactly supported g_i, $W_{\text{switch}} = \sum_{ij} a_i K_{ij} a_j$ with $K_{ij} = \frac{1}{4}\int g_i' g_j' du$. If the g_i are linearly independent, K is positive definite.*

Proof. $a^T K a = \frac{1}{4}\int (\sum_i a_i g_i')^2 du$ vanishes only if $\sum_i a_i g_i$ is constant; compact support makes the constant zero, and linear independence gives $a = 0$. □

D. Coherent drive

Let the mode b carry a coherent state $|\alpha\rangle$, $\alpha = \sqrt{N}$, with all orthogonal modes in vacuum, independent of a Z-diagonal bit. The output field marginal is then

$$\rho_F^{\text{out}} = \tfrac{1}{2}|\alpha - i\lambda\rangle\langle\alpha - i\lambda| + \tfrac{1}{2}|\alpha + i\lambda\rangle\langle\alpha + i\lambda|, \tag{14}$$

independently of the bit's bias, although bit and field are correlated afterwards. The incoming field energy is $E_{\text{in}} = N\bar{\omega}$ and the outgoing energy is $(N+\lambda^2)\bar{\omega}$. A wave packet is not an eigenmode of the field Hamiltonian, so these are mean values; the field Hamiltonian is not $\bar{\omega} b^\dagger b$. The incoming stress is $4N[\operatorname{Re}\psi(u)]^2$ and the encounter adds F_g of Eq. (12). The gravitational use of these profiles is developed in Ref. [29].

V. THE ENERGY COST OF AN ACCURATE FLIP

How much drive is needed for a flip of given accuracy? The coherent family gives $N = \pi^2/(16\ell)$ with $\ell = -\frac{1}{2}\log(1-2\epsilon)$, about $\pi^2/(16\epsilon)$. The following theorem bounds every input state.

Theorem 6 (Energy–accuracy bound)**.** *Let σ be any state of the mode b with finite mean occupation $N = \operatorname{tr}(\sigma b^\dagger b)$, let $\lambda > 0$, and let the coupling* (5) *flip Z-diagonal bit states with failure probability $\epsilon \in (0, \frac{1}{2})$ as in Eq.* (6)*. With $\theta_\epsilon = \arccos(2\epsilon - 1) \in (\pi/2, \pi)$,*

$$N \geq \frac{\theta_\epsilon(1-2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}, \tag{15}$$

$$N + \lambda^2 \geq \frac{\theta_\epsilon}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}. \tag{16}$$

Both right-hand sides equal $\pi/(4\sqrt{\epsilon}) - 1 + O(\sqrt{\epsilon})$. The second inequality is a bound on an occupation-plus-activation budget. It equals the outgoing single-mode energy in units of $\bar{\omega}$ only with the independent zero-cross-term field inputs of Section IV C; correlations can change the actual switching work. An exact flip, $\epsilon = 0$, is impossible at finite occupation.

Proof. Write $C = \cos(2\lambda Q)$ and $U_2 = e^{-2i\lambda Q}$, so that $\langle C\rangle = \operatorname{Re}\chi_\sigma(2\lambda) = 2\epsilon - 1$ and $b^\dagger b = (Q^2 + P^2 - 2)/4$. Finite N makes both second moments finite.

Position quadrature. The function $G(c) = \arccos^2 c$ is convex on $[-1, 1]$: with $c = \cos y$, $y \in (0, \pi)$, one finds $G''(c) = 2(1 - y\cot y)/\sin^2 y \geq 0$, since $y\cot y \leq 1$. For every real q, $(2\lambda q)^2 \geq G(\cos 2\lambda q)$, with equality when $|2\lambda q| \leq \pi$ and strict inequality otherwise because $G \leq \pi^2$. Jensen's inequality for the spectral distribution of Q gives

$$\langle Q^2\rangle \geq \frac{G(\langle C\rangle)}{4\lambda^2} = \frac{\theta_\epsilon^2}{4\lambda^2}. \tag{17}$$

Momentum quadrature. Since $U_2^\dagger P U_2 = P - 4\lambda$, we have $[U_2, P] = 4\lambda U_2$ on the domain of P. Writing $\delta U_2 = U_2 - \langle U_2\rangle$, $\delta P = P - \langle P\rangle$ and applying Cauchy–Schwarz to the Hilbert–Schmidt products $\operatorname{tr}(\sigma\, \delta U_2\, \delta P)$ and $\operatorname{tr}(\sigma\, \delta P\, \delta U_2)$

gives $|\langle[U_2, P]\rangle| \le 2\,\Delta U_2\,\Delta P$ with $(\Delta U_2)^2 = 1 - |\langle U_2\rangle|^2$. Hence $\Delta P \ge 2\lambda|\chi|/\sqrt{1-|\chi|^2}$ with $\chi = \chi_\sigma(2\lambda)$. Because $|\chi| \ge |\operatorname{Re}\chi| = 1 - 2\epsilon$ and $x/\sqrt{1-x^2}$ increases,

$$\langle P^2\rangle \ge (\Delta P)^2 \ge \frac{\lambda^2(1-2\epsilon)^2}{\epsilon(1-\epsilon)}. \tag{18}$$

In particular $|\chi| < 1$ at finite N, which excludes $\epsilon = 0$.

Combination. Adding the two bounds,

$$N \ge \frac{1}{4}\left[\frac{\theta_\epsilon^2}{4\lambda^2} + \frac{\lambda^2(1-2\epsilon)^2}{\epsilon(1-\epsilon)}\right] - \frac{1}{2}, \tag{19}$$

and minimizing over λ by the arithmetic–geometric mean inequality gives Eq. (15). Adding λ^2 changes the coefficient of λ^2 to $\frac{1}{4}[(1-2\epsilon)^2/(\epsilon(1-\epsilon)) + 4] = 1/[4\epsilon(1-\epsilon)]$, and the same step gives Eq. (16). The expansion uses $\theta_\epsilon = \pi - 2\sqrt{\epsilon} + O(\epsilon^{3/2})$. □

The bound is nearly attained by Gaussian drives.

Proposition 7 (Coherent and Gaussian families)**.** *(a) The coherent family with $2\lambda\alpha = \pi/2$ needs $N = \pi^2/(16\ell)$, $\ell = -\frac{1}{2}\log(1-2\epsilon)$. (b) For a displaced pure Gaussian state whose Q has mean 2α and variance $V > 0$, $\chi(2\lambda) = e^{-4i\lambda\alpha}e^{-2\lambda^2 V}$ and $N = \alpha^2 + (V + V^{-1} - 2)/4$. At $2\lambda\alpha = \pi/2$ and failure ϵ,*

$$\min N = \frac{\sqrt{1+\pi^2/(4\ell)} - 1}{2}, \qquad \min(N+\lambda^2) = \frac{1}{2}\sqrt{\Big(1+\frac{\pi^2}{4\ell}\Big)(1+4\ell)} - \frac{1}{2}, \tag{20}$$

attained at $V = (1+\pi^2/(4\ell))^{-1/2}$ and $V = [(1+4\ell)/(1+\pi^2/(4\ell))]^{1/2}$ respectively. Both equal $\pi/(4\sqrt{\epsilon}) - \frac{1}{2} + O(\sqrt{\epsilon})$, so their gap to Theorem 6 tends to half a quantum.

Proof. The Gaussian characteristic function gives χ. The failure condition reads $\lambda^2 V = \ell$, so $\alpha^2 = \pi^2/(16\lambda^2) = aV$ with $a = \pi^2/(16\ell)$ and $N = (1+4a)V/4 + 1/(4V) - \frac{1}{2}$, minimized at $V = (1+4a)^{-1/2}$. Adding $\lambda^2 = \ell/V$ replaces $1/(4V)$ by $(1+4\ell)/(4V)$. □

Squeezing is what makes the better scaling possible.

Proposition 8 (Classical drives)**.** *If $\sigma = \int P(\beta)|\beta\rangle\langle\beta|\,d^2\beta$ with a probability density $P \ge 0$, then a flip with failure ϵ needs*

$$N \ge \frac{(\pi - \arccos(1-2\epsilon))^2}{16\ell} = \frac{\pi^2}{16\epsilon}\big(1 - 4\sqrt{\epsilon}/\pi + O(\epsilon)\big). \tag{21}$$

Proof. Each coherent component has $\chi_\beta(2\lambda) = e^{-2\lambda^2}e^{-4i\lambda\operatorname{Re}\beta}$, so $\int P\cos(4\lambda\operatorname{Re}\beta) = -(1-2\epsilon)e^{2\lambda^2}$. This must be at least -1, hence $\lambda^2 \le \ell$. The convexity argument of Theorem 6 applied to $y = 4\lambda\operatorname{Re}\beta$ gives $16\lambda^2\int P(\operatorname{Re}\beta)^2 \ge (\pi - \arccos[(1-2\epsilon)e^{2\lambda^2}])^2 \ge (\pi - \arccos(1-2\epsilon))^2$. Finally $N = \int P|\beta|^2 \ge \int P(\operatorname{Re}\beta)^2$ and $\lambda^2 \le \ell$. □

At fixed λ, a constrained minimum can be certified by a supporting ground-state problem. If ψ_μ is the ground state of $b^\dagger b + \mu\cos(2\lambda Q)$ and $\langle C\rangle_{\psi_\mu} = 2\epsilon - 1$, then every σ with the same $\langle C\rangle$ satisfies $\operatorname{tr}(\sigma b^\dagger b) \ge \langle b^\dagger b\rangle_{\psi_\mu}$, because $\operatorname{tr}\sigma(b^\dagger b + \mu C)$ is at least the ground energy. Table I lists numerical variational minima on a position grid, further minimized over λ, alongside the analytic bounds and the explicitly attainable Gaussian half-turn family. Halving the grid checks resolution stability, not the continuous global minimum. The computed states are close to displaced squeezed states, while the coherent half-turn family uses substantially more occupation.

Remark 9 (Scope of the occupation)**.** *(i) For a multimode field state, Cauchy–Schwarz applied to $b = \int f^* a_\omega d\omega$ gives $\langle b^\dagger b\rangle \le \omega_h^{-1}\langle H_F\rangle$ with $\omega_h^{-1} = \int |f|^2\omega^{-1}d\omega$, so the field energy is at least ω_h times the bound* (15)*; under the zero-cross-term condition of Section IV C, $\omega_h \le \bar\omega$ then makes field energy plus switching work at least ω_h times the bound* (16)*. (ii) The occupation is the drive level present at the encounter, not energy absorbed by the bit. The bit is degenerate; the beam leaves with energy $(N+\lambda^2)\bar\omega$ in the coherent case. Whether drive energy is lost depends on whether the outgoing beam is recovered. (iii) Squeezed inputs can have locally negative normal-ordered stress, and their preparation and phase reference are additional resources.*

Recovering the beam is not free of consequences. Two bits placed one after the other on the same chiral packet, with the same switch profile in u, experience $U = \exp[-i\lambda(X_1 + X_2)Q]$.

ϵ	Classical bound (21)	Coherent, $2\lambda\alpha = \pi/2$	Bound (15)	Grid variational	Gaussian half-turn (20)
0.1	3.496	5.529	1.165	1.847	1.904
0.05	8.589	11.709	2.278	2.921	2.958
0.01	53.53	61.07	6.742	7.314	7.330
0.001	591.7	616.2	23.80	24.32	24.33

TABLE I. Mean drive occupation needed for a flip with failure probability ϵ. Including one activation's switching work at $\epsilon = 0.01$: bound (16) 6.890, grid variational minimum 7.470, Gaussian half-turn 7.487, coherent 61.08. The grid minima change by less than 2×10^{-3} when the spacing is halved; this is not a certified global error bound.

Proposition 10 (Shared drive correlates errors)**.** *For a shared drive each bit's marginal channel is the single-encounter channel. For two bits starting at $Z = +1$, $\langle Z_1 Z_2 \rangle = \frac{1}{2}[1 + \mathrm{Re}\,\chi(4\lambda)]$. For the coherent family of Proposition 3 this equals $(1 + e^{-8\lambda^2})/2$, which exceeds the product $\langle Z_1 \rangle \langle Z_2 \rangle = e^{-4\lambda^2}$ obtained with independent drives.*

Proof. In the X basis the joint element with indices $(x_1 x_2), (x_1' x_2')$ is multiplied by $\chi(\lambda(x_1 + x_2 - x_1' - x_2'))$; tracing one bit recovers Proposition 3. $Z_1 Z_2$ connects (x_1, x_2) with $(-x_1, -x_2)$, which gives arguments $\pm 4\lambda$ and 0. For the coherent family $\chi(4\lambda) = e^{-8\lambda^2} e^{-2\pi i}$, and $(1 + y^2)/2 > y$ for $0 < y < 1$. □

At $N = 16$ the single failure probability is 0.0371, and the probability that both flips fail is 0.00393, against 0.00138 for independent drives. Drive energy can be shared only at the price of correlated failures. The persistence analysis below uses a fresh drive at each opportunity.

VI. PERSISTENCE WITH A FINITE-ENERGY ACTUATOR

A. The task

We use the feedback task of the Imprint Chain [36] with the corrective flip replaced by the coupling of Section IV. The persistent subsystem is a bit restricted to Z-diagonal states, with value $W = \pm 1$ and target $W = +1$. A bath step flips W with probability $d \in (0, \frac{1}{2})$; put $c = 1 - 2d$. A reading prepares a fresh probe, copies W by a CNOT, rotates the probe by U_a with $a \in \{0, 1\}$, and measures it with effects $E_{y|s} = \frac{1}{2}(I + y\nu\sigma_s)$, $\sigma_0 = Z$, $\sigma_1 = X$, where the instrument label S is hidden and fixed. Then

$$\Pr(Y = y \mid W = w, S = s, a) = \begin{cases} (1 + yw\nu)/2, & a = s, \\ 1/2, & a \neq s. \end{cases} \tag{22}$$

A calibration stage selects a; its mismatch probability $q = \Pr(a \neq S)$ is fixed along a trajectory. In each task step the system reads Y, accepts feedback with an independent probability $h \in (0, 1]$, switches on the coupling if accepted and $Y = -1$, and then the bath acts. The loss is the occupation probability $L_t = \Pr(W_t = -1)$.

Two declarations fix the resource accounting. Each opportunity has its own drive, initially independent of the bit and of earlier records, and the outgoing field never returns to the controller's reading access. The drive is scheduled: the same incoming beam arrives at every opportunity and the command only decides whether g is switched on. On-demand emission of a whole packet would need a router and a timing scheme that we do not model.

B. Closed-loop stability

Theorem 11 (Persistence with a finite-energy actuator)**.** *Let each coupling act on the bit as the flip channel $sX\rho X + (1 - s)\rho$, which is the case exactly when $\chi_\sigma(2\lambda)$ is real, for instance for the coherent and Gaussian families at $2\lambda\alpha = \pi/2$. Put $\Lambda = c(1 - hs)$ and $B = chs\nu$, and let the task start from W uniform.*

(a) Conditional on $a = S$, the bit evolves by a fixed channel $\mathcal{T}$ on Z-diagonal states, $m \mapsto \Lambda m + B$ for the mean m of W. Conditional on $a \neq S$, $m \mapsto \Lambda m$.

(b) $\mathcal{T}$ has the unique fixed point $\rho_* = (I + m_* Z)/2$, $m_* = B/(1 - \Lambda)$. The trace distance to ρ_* contracts by exactly Λ per step, and $D(\rho_t \| \rho_*)$ is nonincreasing and tends to zero.

(c) The stationary loss is

$$L_\infty(q) = \frac{1}{2} - \frac{(1-q)B}{2(1-\Lambda)}. \tag{23}$$

(d) The stationary rate of actual couplings is

$$p(q) = \frac{h}{2}\left[1 - \frac{(1-q)\nu B}{1-\Lambda}\right] = h\nu L_\infty(q) + \frac{h(1-\nu)}{2}, \tag{24}$$

of which a fraction s succeed. Over a window of T steps the expected number of couplings is $\frac{h}{2}\big[T - (1-q)\nu\frac{B}{1-\Lambda}\big(T - \frac{1-\Lambda^T}{1-\Lambda}\big)\big]$.

(e) For two calibrations with $q' < q$, $L_\infty(q) - L_\infty(q') = (q-q')B/[2(1-\Lambda)]$ and $p(q) - p(q') = h\nu[L_\infty(q) - L_\infty(q')]$.

Proof. With $a = S$ a reading has $\mathbb{E}[WY] = \nu$, and the coupling is attempted with probability $h(1-Y)/2$ and flips with probability s. Thus the mean after feedback is $m - 2s\,\mathbb{E}[Wh(1-Y)/2] = (1-hs)m + hs\nu$, and the bath multiplies it by c. With $a \neq S$, Y is independent of W and the mean after feedback is $(1-hs)m$. Fresh probes and fresh drives make each step the same channel on the bit marginal. Trace distance between Z-diagonal states is $|m-m'|/2$, which the affine map multiplies by $\Lambda < 1$. Data processing gives $D(\mathcal{T}\rho\|\rho_*) \le D(\rho\|\rho_*)$, and $m_t \to m_*$ gives convergence. Starting from $m_0 = 0$, the branch means are $m_*(1-\Lambda^t)$ and 0, weighted by $1-q$ and q; this gives (c). The attempt probability is $h(1-\nu m)/2$ for $a = S$ and $h/2$ otherwise, which gives (d) after averaging and using $(1-q)m_* = 1 - 2L_\infty$. Part (e) follows from (c) and (d). □

Gate failure enters the recursion through hs, while the attempt rate keeps h: a failed gate uses the whole encounter. Stability here does not rely on a gradient rule or a metric G. It follows from the contraction of the composed channel, and the relative entropy to the stationary state is a Lyapunov function of the full closed loop, which is what Section III C 0 a required of a proof. Whether persistence is useful is a separate matter: the stationary polarization is positive only on the calibrated branch and grows with s.

Table II uses $\nu = 0.8$, $c = 0.9$, $h = 0.4$ and coherent drives. The calibrated column uses 32 passive two-read blocks, $q_{32} = 0.0404888$ (Section VI C). With an exact flip the calibrated loss would be 0.199631 [36].

N	ϵ_N	Loss, $q = \frac{1}{2}$	Loss, q_{32}	Rate p, $q = \frac{1}{2}$	Rate p, q_{32}	Added energy per opportunity
1	0.354394	0.360165	0.231654	0.155253	0.114129	0.070401
4	0.132699	0.348517	0.209300	0.151525	0.106976	0.016497
16	0.037104	0.344779	0.202127	0.150329	0.104680	0.004036
64	0.009546	0.343806	0.200259	0.150018	0.104083	0.001003

TABLE II. Stationary loss and coupling rate with a coherent drive of occupation N. The last column is $p\lambda^2$ at q_{32} in units of $\bar{\omega}$; the incoming energy $N\bar{\omega}$ is paid at every opportunity in addition. Reading, probe and calibration costs are not included.

C. Calibration with the same imperfect gate

The Imprint Chain separates calibration blocks by an exact randomization $\mathcal{R}(\rho) = \frac{1}{2}(\rho + X\rho X)$. With an imperfect gate, triggering it with probability $\frac{1}{2}$ leaves a fraction $1-s$ of the bit's mean. Triggering with probability $1/(2s)$, which is at most one because $s \ge \frac{1}{2}$, and discarding the coin gives

$$\Big(1 - \frac{1}{2s}\Big)\rho + \frac{1}{2s}\big(sX\rho X + (1-s)\rho\big) = \tfrac{1}{2}(\rho + X\rho X) \tag{25}$$

exactly. This presumes that s is known. If the trigger probability is rounded to k binary digits, each randomization has flip probability within $s2^{-k-1}$ of $\frac{1}{2}$. Since $\mathrm{id} - \mathrm{Ad}_X$ has diamond norm 2, the record distributions of a protocol with $n+1$ randomizations differ from the exact ones by at most $(n+1)s2^{-k-1}$ in total variation, or 1.52×10^{-5} for $N = 16$, $n = 32$, $k = 20$.

Passive two-read blocks randomize, read, let the bath act and read again, without a feedback flip. With the compensated randomization they keep the Imprint Chain likelihood: the product of the two readings is positive with probability $p_{\mathrm{cal}} = (1 + c\nu^2)/2$ if $a = S$ and $\frac{1}{2}$ otherwise. After n blocks the optimal mismatch probability is

$$q_n = \frac{1}{2} \sum_{k=0}^{n} \binom{n}{k} \min\{p_{\mathrm{cal}}^k (1 - p_{\mathrm{cal}})^{n-k}, 2^{-n}\}. \tag{26}$$

Active blocks, with a feedback flip after the first reading, change once the flip is imperfect. For $a = S$,

$$\Pr(y_1, y_2) = \tfrac{1}{4}\big[1 + c\nu^2 y_2\big(s + (1 - s)y_1\big)\big], \tag{27}$$

and the pair is uniform for $a \neq S$. This law is obtained from the passive pair by an S-independent post-processing that keeps the first reading and relabels with probability s, so gate error creates no calibration information about the sensor. The success probability s itself is readable from the closed loop: in the persistence task the lag-one correlation of readings on gated steps differs from that on ungated steps by a term proportional to s, so the agent's own records determine s, the bath contraction and the squared gain without references, and never the sign [33]. Calibration with n blocks uses $2n$ readings and $n + 1$ randomizations with an expected $(n + 1)/(2s)$ couplings.

D. Records and drive in one acceptance condition

Proposition 12 (Joint record–drive condition)**.** *If $chs\nu > 0$, the stationary loss satisfies $L_\infty(q) \leq L_*$ exactly when*

$$q \leq 1 - \frac{(1 - 2L_*)(1 - c + chs)}{chs\nu}. \tag{28}$$

If the right-hand side is negative, no amount of calibration achieves L_* with that actuator.

Proof. Rearrange Eq. (23) using $1 - \Lambda = 1 - c + chs$. □

Take $L_* = 0.2$. A coherent drive with $N = 1$ fails even with a known interface, whose loss is 0.220330. For $N = 4$, 16 and 64 the minimal numbers of passive blocks are 57, 36 and 33; since q_n is nonincreasing, checking that $n - 1$ blocks fail and n succeed establishes the minimum. With 32 blocks fixed, Eq. (28) requires $s \geq 0.9943781$, that is $\epsilon \leq 0.0056219$. Coherent half-turn drives then need $N \geq 109.10$, classical drives need $N \geq 98.93$ by Proposition 8, and by Theorem 6 every single-mode drive acting as a flip channel needs $N \geq 9.390$ (and $N + \lambda^2 \geq 9.503$). The Gaussian half-turn family reaches 9.957 and the computed grid variational minimum is 9.945. These minima refer to the declared block protocol and feedback rule, not to every possible experiment.

Corollary 13 (Scaling of the persistence cost)**.** *Among drives acting as flip channels, holding the stationary loss within δ of its exact-flip value at a known interface requires a drive occupation per activation that grows as $\delta^{-1/2}$ for optimal drives and as δ^{-1} for classical drives.*

Proof. At $q = 0$, L_∞ is a smooth decreasing function of s with nonzero derivative at $s = 1$, so δ is proportional to ϵ to first order. Theorem 6 and Proposition 7 give occupation of order $\epsilon^{-1/2}$ for optimal drives, and Proposition 8 together with the coherent family gives order ϵ^{-1} for classical drives. □

E. Resource accounting

The resources of the task are kept as a vector. Each task opportunity uses one reading, one probe and CNOT, a gate coin, a bath interval and one incoming drive of occupation N, prepared whether or not it is used. Each actual coupling adds switching work $\lambda^2\bar{\omega}$ at rate p. For $N = 16$ and 32 blocks, calibration uses 64 readings and 33 randomization opportunities carrying incoming energy $528\bar{\omega}$, with an expected 17.14 couplings and switching work $0.6606\bar{\omega}$. Over a task window of 60 steps the expected number of couplings is 6.494. The occupations 61.07 and 7.33 at $\epsilon = 0.01$ are input levels, and 7.49 minimizes input plus one activation's work; none of these is the total energy of an experiment, which would also include reading, preparation, records and the switching apparatus.

F. Finite-energy response readout

An outgoing stress can change a boundary trajectory, which can in turn change signal delays. Consider the particular readout in which these delays are the only branch-dependent signal. Use one calibrated proper clock, nonnegative probe Hamiltonians H_j on separate ports, and a joint probe–ancilla state independent of the unknown branch. The ancilla may be correlated with the probes but is not acted on by the signal. In branch $k = 0, 1$ the probes undergo $U_k = \exp[-i \sum_j H_j \tau_{kj}/\hbar]$. Preparation and final joint measurement can be arbitrary. The charged probe input is $E = \sum_j \operatorname{Tr}(\rho H_j)$, above the declared zero levels.

Theorem 14 (Sharp signed-delay energy bound)**.** *Let* $\Delta\tau = \max_j |\tau_{1j} - \tau_{0j}| > 0$ *and define*

$$\zeta = \max_{x>0} \frac{1 - \cos x}{x} = 0.724611\ldots, \qquad z = \frac{\zeta E \Delta\tau}{\hbar}. \tag{29}$$

The trace distance of the two output states, including the retained ancilla, is at most $\sqrt{z(2-z)}$ *for* $0 \le z \le 1$*, and at most one otherwise. For equal priors, error at most* $\epsilon < 1/2$ *therefore requires*

$$E \ge \frac{\hbar}{\zeta \Delta\tau} \left[1 - 2\sqrt{\epsilon(1-\epsilon)}\right]. \tag{30}$$

This lower bound is sharp across the permitted probe families, including oppositely signed delay differences. It need not be sharp for a specified Hamiltonian or a readout with all delay differences of the same sign. If $\Delta\tau = 0$*, no such protocol distinguishes the branches.*

Proof. The scalar maximum occurs at the unique $x_* \in (0, \pi)$ satisfying $\tan(x_*/2) = x_*$; for $x \ge \pi$, $(1-\cos x)/x \le 2/\pi$, whereas the value at $2\pi/3$ gives $\zeta \ge 9/(4\pi) > 2/\pi$. Thus $1 - \cos x \le \zeta |x|$ for every real x. Purify the probe–ancilla input. The overlap of the two resulting pure states is

$$A = \operatorname{Tr} \rho \exp(-iK), \qquad K = \sum_j H_j (\tau_{1j} - \tau_{0j})/\hbar.$$

The Hamiltonians commute strongly, and their positivity gives $|K| \le (\Delta\tau/\hbar) \sum_j H_j$ by joint spectral calculus. Consequently $\operatorname{Re} A \ge 1 - z$. For $z \le 1$, the pure-state trace distance is $\sqrt{1 - |A|^2} \le \sqrt{z(2-z)}$; discarding the purifying reference cannot increase it. Equal-prior error is at least $(1 - D)/2$ for trace distance D. Solving this inequality gives Eq. (30).

For saturation, take two probes with equal excited energy $e = x_* \hbar/\Delta\tau$, and delay differences $+\Delta\tau, -\Delta\tau$. On the three states $|00\rangle, |e0\rangle, |0e\rangle$ prepare

$$|\psi\rangle = \sqrt{1-p}\,|00\rangle + \sqrt{p/2}\,(|e0\rangle + |0e\rangle), \qquad p = \frac{1 - 2\sqrt{\epsilon(1-\epsilon)}}{1 - \cos x_*}.$$

Here $0 \le p < 1$, $E = pe$, and the branch overlap is exactly $1 - p(1 - \cos x_*) = 2\sqrt{\epsilon(1-\epsilon)}$. The optimal binary measurement attains error ϵ and Eq. (30). □

Mean-energy quantum speed limits constrain distinguishable evolution, including arbitrary-fidelity targets [34, 35]. Here the relevant resource is the sum of the physical probe energies, not the signed generator's ground-state shift; this accounts for opposite delay differences without assigning a negative probe cost. A readable branch marker, changed output frequency or amplitude, branch-dependent preparation, or a signal interaction other than translation falls outside the theorem. For example, an unpriced orthogonal marker would distinguish the branches at zero probe energy. Receiver, clock and preparation costs remain additional resources.

In HAFF-D, three echoes determine the coefficients of a supplied JT solution. Its source-compensated inverse permits known stresses in both null directions and propagates timing and source-calibration errors to the next coupled projection [29]. This connects a selection-dependent source, its response and a subsequent operation, conditional on the declared instrument. For its compact-pulse example the largest proper-delay difference is 0.05001338; a translation-only instrument with error 0.1 requires input energy at least $11.0374\hbar$ in that clock's units. The geometric inverse does not construct such a probe or its reflecting boundary. Its supplied bulk source-compensation test is not an energy–resolution demonstration. A joint apparatus must meet the readout bound, the actuator bound and its actual stress and entropy accounts without counting the same outgoing energy twice.

G. Finite-rate entropy export

Drive occupation and exported heat obey different bounds. If the maintained subsystem and its finite reused controller remove a net dimensionless entropy current $\dot{s} > 0$, and all of it leaves through a chiral conformal exhaust of capacity $c_{\rm ex}$ above a thermal input at temperature T, HAFF-D gives [29]

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\rm ex}} \dot{s}^2.$$

The current must include retained side information and controller consumption; the number of attempted flips alone does not determine it. The bound assumes an extensive long-window entropy-current account, zero chemical potential or the matched conserved-charge account, and one common clock. It is not an exact bound for every short packet. It remains positive at $T = 0$ for a thermal input, where ordinary Landauer heat is zero. For independent parallel exhausts at one temperature, $c_{\rm ex}$ is their total capacity.

A nonthermal drive changes this comparison. If its incoming energy and dimensionless entropy currents are $e_{\rm in}$ and $\sigma_{\rm in}$, the same Gibbs argument gives

$$\dot{Q} + e_{\rm in} - E_{\rm th}(\sigma_{\rm in}) \geq k_B T_\sigma \dot{s} + \frac{3\hbar}{\pi c_{\rm ex}} \dot{s}^2, \qquad E_{\rm th}(\sigma) = \frac{3\hbar\sigma^2}{\pi c_{\rm ex}}, \quad T_\sigma = \frac{6\hbar\sigma_{\rm in}}{\pi c_{\rm ex} k_B}.$$

The incoming energy above its entropy-matched thermal value can pay for export. In particular, a coherent beam with zero entropy is not a zero-energy reservoir. Its drive occupation and the additional heat must be accounted for separately.

This heat is neither the incoming occupation $N\bar{\omega}$ nor an extra charge to add to an outgoing energy already counted. It is a constraint on that energy when it carries the removed entropy. The scheduled drive calculation above does not establish its exhaust-capacity assumptions. A complete apparatus must price reading, recovery, sensing and entropy transport in the same cycle.

VII. CALIBRATING AN INTERFACE WHILE IT CARRIES QUANTUM CONTENT

Preparing test states before storage is not the only way to learn an interface. The outgoing field can carry information about the coupling without carrying the logical content that the apparatus is trying to keep. We give the instrument, its loss under incomplete collection, and a coherent implementation.

On the two-qubit code $\mathcal{C} = \mathrm{span}\{|01\rangle, |10\rangle\}$, let Z_L have eigenvalues $+1, -1$. The coupling $Z_1 + \alpha a Z_2$ restricts to mZ_L, $m = 1 - \alpha a$, with unknown $\alpha > 0$ and chosen command a. Write

$$Q_f = \sqrt{k}(b + b^\dagger), \qquad U_m = e^{-imZ_L Q_f}, \qquad k > 0, \quad \Pi = (-1)^{b^\dagger b}.$$

The independently prepared incoming mode is vacuum. A beam splitter collects fraction ξ of this mode, leaving fraction $1 - \xi$ inaccessible.

Proposition 15 (A content-preserving calibration instrument)**.** *Measure the collected mode's parity and apply Z_L for an odd outcome. The corrected instrument, including an arbitrary retained reference, is*

$$\mathcal{I}_\pm(\rho) = p_\pm(a)\mathcal{D}_{\xi,m}(\rho), \qquad p_\pm(a) = \frac{1 \pm e^{-2k\xi m^2}}{2}, \tag{31}$$

where $\mathcal{D}_{\xi,m}$ fixes the populations and multiplies logical coherence by $e^{-2k(1-\xi)m^2}$. At full collection each outcome is a scalar multiple of the identity channel. Fresh-port queries chosen from earlier records can therefore calibrate the interface without replacing its held state.

Proof. The accessible displacement has the expansion $I_L \cos(m\sqrt{\xi}Q_A) - iZ_L \sin(m\sqrt{\xi}Q_A)$. On the vacuum the two field vectors have opposite parity, with squared norms p_+, p_-. The conditional pulse makes both logical factors I_L. The independent inaccessible displacement commutes with this correction and leaves $\mathcal{D}_{\xi,m}$. Each branch is thus Eq. (31), not merely an equality of its unconditional channel. Composition proves the adaptive statement; the same computation tensors with any reference. □

Environment-assisted correction of unitary Kraus branches is a known general mechanism [3, Proposition 2]. Here its records are also the data for learning the same interaction's gain and collection fraction. Repeated ancilla-based

parity measurements have been implemented for a cavity mode [5]; collecting the full mode of a travelling chiral field is an additional requirement, not supplied by that experiment.

Partial collection prices the calibration itself. For N queries at each of $a = \pm h$, with fixed α, ξ, the held coherence is multiplied by

$$\exp[-4Nk(1-\xi)(1+\alpha^2 h^2)]. \tag{32}$$

This loss occurs before any subsequent operating benefit. With $k = .2$, $\alpha = 1.1$, $h = .5$ and $N = 4096$, keeping the calibration's dephasing error below .01 needs $1 - \xi \le 4.7335 \times 10^{-6}$. At $\xi = .99$ its error is essentially $1/2$. More records are not automatically better for held content.

A. A resonant collection port

The collection fraction can itself be realized by a finite-duration interaction. This construction starts with a retained source oscillator b and a receiver oscillator c of the same frequency ω, with c initially in vacuum. It does not assume that a broadband travelling mode has already been trapped by this interaction. In the interaction picture put

$$H_{\mathrm{cap}}(t) = \hbar g(t) C, \quad C = b^\dagger c + c^\dagger b, \quad \theta = \int g(t)dt, \quad U_{\mathrm{cap}} = e^{-i\theta C}.$$

Proposition 16 (Finite resonant collection)**.** *For $0 \le \theta \le \pi/2$, the receiver collects $\xi = \sin^2\theta$ of the source mode. Corrected receiver-parity readout realizes Eq.* (31) *with this ξ. If $|g| \le g_{\max} < \omega$, the total Hamiltonian $\hbar\omega(n_b + n_c) + H_{\mathrm{cap}}$ is bounded below and the pulse needs duration at least $\theta/g_{\max}$. A pulse starting and ending at zero does no net prescribed-switch work in this isolated two-mode model. For the independently prepared vacuum encounter, the mean occupations after collection are ξkm^2 and $(1-\xi)km^2$. Their sum is km^2.*

A pulse-area error δ changes ξ by at most $|\delta|$. For any input with $\langle n_b + n_c\rangle \le \bar{n}$, including a retained reference, its half-trace error is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$.

Proof. The Heisenberg transforms are $b \mapsto b\cos\theta - ic\sin\theta$ and $c \mapsto c\cos\theta - ib\sin\theta$. The conditional coherent states produced by U_m therefore split into receiver and remainder amplitudes with squared fractions $\xi, 1-\xi$. The receiver's known phase does not change parity. Applying Proposition 15 proves the instrument and the occupation account, for any logical input.

The normal modes $(b \pm c)/\sqrt{2}$ have instantaneous frequencies $\omega \pm g$, which proves the lower bound. The amplitude bound proves the duration statement. Both C and total occupation commute with the Hamiltonian, so $\int \langle \partial_t H_{\mathrm{cap}}\rangle dt = \hbar\langle C\rangle[g(t_{\mathrm{end}}) - g(t_{\mathrm{start}})] = 0$. Finally $|\sin^2(\theta+\delta) - \sin^2\theta| \le |\delta|$ and $|C| \le n_b + n_c$. The purified-vector estimate $2\langle 1 - \cos(\delta C)\rangle \le 2|\delta|\bar{n}$ proves the energy-constrained error bound. □

This account includes transferred energy and a gate clock. Receiver parity control, reading and retaining a syndrome, preparing the next empty receiver, and supplying the switching apparatus remain separate resources. Emptying the receiver exports energy; it is not included in the zero net switch work above. Restoring a degraded collection setting also needs a stated physical permission and price. Brain Anchor [4] uses such a permission in a finite model where the same content survives inquiry, operation and maintenance. Resonant port capture, local capture of a continuous field and sustained thermal export are three different implementation questions.

Whole-mode control can be reduced to successive port contacts in a finite packet model. Take n source oscillators of frequency ω and one vacuum receiver c. For real $f_j \ge 0$, $\sum_j f_j^2 = 1$, form the packet mode $b_f = \sum_j f_j b_j$. Sequential source interactions $e^{-im\sqrt{k} f_j Z_L (b_j + b_j^\dagger)}$ produce the same packet displacement as $e^{-im\sqrt{k} Z_L (b_f + b_f^\dagger)}$. After contacting port j, couple only that port to c through the resonant Hamiltonian above, with area $0 \le \theta_j \le \theta_{\max} \le \pi/2$. Released ports are inaccessible. The receiver's packet amplitude, up to its known common phase, obeys

$$y_0 = 0, \qquad y_j = y_{j-1}\cos\theta_j + f_j \sin\theta_j.$$

Proposition 17 (Sequential local-port collection)**.** *The receiver collects fraction $\xi = y_n^2$ of the packet, and its corrected parity instrument is Eq.* (31)*. Among these permitted real resonant contacts, final collection is maximized by*

$$\theta_j = \min\{\theta_{\max}, \arctan(f_j / y_{j-1})\},$$

with the first angle $\pi/2$ before applying the cap when $y_{j-1} = 0 < f_j$. When $y_{j-1} = f_j = 0$, choose $\theta_j = 0$. For a uniform packet and $\theta_{\max} = .5$, the first five contacts are capped and the remaining ones are unconstrained. For $n \ge 5$ the exact optimum is

$$\xi = 1 - \frac{d}{n}, \qquad d = 5 - \cot^2(.25)(1 - \cos^5(.5))^2 = 1.47392688\ldots.$$

Thus 148 uniform ports suffice for collection at least .99, and fewer do not under the same cap. With no cap below $\pi/2$, the angles $\theta_1 = \pi/2$, $\theta_j = \arctan(1/\sqrt{j-1})$ collect a uniform finite packet completely.

Proof. Passive mode mixing sends the packet's conditional coherent displacement to receiver amplitude $-iy_n$ and a remainder of squared norm $1 - y_n^2$. Orthogonal input modes are vacuum. Their overlaps prove the instrument. Each step maximizes $y\cos\theta + f_j \sin\theta$ at the stated angle. The maximum is nondecreasing in $y \geq 0$, because all permitted cosines are nonnegative. Induction therefore makes this stepwise optimum a final optimum. For a uniform packet, the first capped steps give $y_j = n^{-1/2}\cot(.25)(1 - \cos^j(.5))$. After the fifth, the interior optimum has $y_j^2 = y_{j-1}^2 + 1/n$. This proves the formula and threshold. The uncapped recursion gives $y_j = \sqrt{j/n}$. □

Each contact has a Hamiltonian bounded below when $g_{\max} < \omega$, needs duration at least $\theta_j/g_{\max}$ and conserves total occupation. The source prepares mean total occupation km^2, the receiver holds ξkm^2 and the released ports hold $(1-\xi)km^2$. Parity control is needed only on the retained receiver. For contact-area errors δ_j, the half-trace error is at most $\min\{1, \sum_j \sqrt{2|\delta_j|\bar{n}}\}$ when total occupation is bounded in mean by $\bar{n}$; telescope the preceding energy-constrained bound. Source switching, receiver reuse and detector errors require their own account. These are local contacts in a finite collision architecture, not a proof that a specified relativistic wave packet admits that architecture or that its clocks and references can be sustained indefinitely.

B. Positive-frequency continuous collection

A specified continuous field can also supply the collection port. Let $\omega = \Omega + \delta$, $|\delta| \leq B < \Omega$, and couple it to a retained oscillator c. With $[a_\delta, a_{\delta'}^\dagger] = \delta(\delta - \delta')$, take a real even form factor g, $J = g^2$ and $G = \|g\|_2$. Put

$$H(h) = \hbar\Omega n_c + \hbar\int_{-B}^{B} (\Omega+\delta) a_\delta^\dagger a_\delta d\delta + \hbar h(t)\int_{-B}^{B} g(\delta)(c^\dagger a_\delta + a_\delta^\dagger c)d\delta, \qquad |h| \leq 1. \tag{33}$$

This is an excitation-preserving effective Hamiltonian with a specified band-smeared coupling. Its frequency form factor gives spatial tails; it is not a strictly point-local relativistic detector. No negative physical frequencies or Markov extension of the band are used. Matched pulse absorption by a tunable resonator is an established mechanism [2]; the question here is how its finite-band realization enters the held-content instrument and source account.

In the frame rotating at Ω, the one-particle operator is

$$A = \begin{pmatrix} 0 & \langle g| \\ |g\rangle & \delta \end{pmatrix}, \quad S = e^{-iAT}, \quad r(T) = \langle c|S|c\rangle.$$

The capture pulse has $h = 1$ on $[0, T]$ and is otherwise off. Deviations from this prescribed pulse are bounded below.

Proposition 18 (Matched finite-band capture)**.** *If $\Omega > B + G$, every Hamiltonian in Eq. (33) is bounded below by $\hbar(\Omega - B - G)N$. Write $\xi_0 = 1 - |r(T)|^2 > 0$. Among normalized incoming field modes, the mode*

$$f = -P_{\text{field}} S^\dagger |c\rangle / \sqrt{\xi_0}$$

maximizes collection and is collected with fraction ξ_0. It can be prepared from independent vacuum by a finite prescribed body-controlled linear drive with envelope $u(s) = r(T-s)$, $0 \leq s \leq T$. For even J, r and this envelope are real. The prepared field amplitudes are

$$m\beta w_u(\delta), \qquad w_u(\delta) = -ig(\delta)\int_0^T u(s)e^{i\delta s}ds, \quad w_u = -P_{\text{field}} S^\dagger|c\rangle, \quad \beta = \sqrt{k/\xi_0}. \tag{34}$$

For an arbitrary prepared envelope and $w_u \neq 0$, replace k by $k_{\text{eff}} = \beta^2\|w_u\|^2$ and use $\xi = |\langle c|S(0, w_u)\rangle|^2/\|w_u\|^2$. Corrected receiver-parity readout then realizes Eq. (31) with these actual parameters.

Proof. $\|A\| \leq B + G$ and second quantization prove the lower bound. The norm of the field part of the receiver row is $\sqrt{\xi_0}$; Cauchy–Schwarz proves the optimum. Its orthogonal complement has zero collection, so the statement is not uniform over unknown packets.

Eliminating the field component of $S^\dagger|c\rangle$ gives Eq. (34). Reflection $\delta \mapsto -\delta$, field sign reversal and complex conjugation define an antiunitary taking A to $-A$ and leaving its propagator invariant. Hence r is real. One explicit preparation places the band-smeared source at $x = -T$ and the receiver at $x = 0$, with propagation speed one. Drive

the source during $t \in [-T, 0]$ with envelope $u(t+T)$ and a prescribed carrier phase Ω. Explicitly its interaction-picture Hamiltonian is

$$H_s^I(t) = \hbar m \beta Z_L u(t+T) \int_{-B}^{B} g(\delta)[a_\delta^\dagger e^{i\delta(t+T)} + a_\delta e^{-i\delta(t+T)}]d\delta.$$

Its displacement is $m\beta w_u$; the receiver is coupled only after this drive. The source Hamiltonian is linear in the band field and proportional to mZ_L. Its time-ordering commutator is a scalar times Z_L^2, so it changes only an overall phase. Completing the square bounds its instantaneous energy below since $\Omega - B > 0$ and $\int J/\omega$ is finite.

Passive field evolution splits the conditional coherent amplitudes into a receiver component of squared fraction ξ and a remainder of fraction $1-\xi$. Their vacuum overlaps give the claimed instrument for arbitrary logical input and retained reference. Initial independence and parameters fixed within an encounter are required. □

The propagation prescription is within the band-smeared model: a strictly band-limited field operator has no compact spatial support. The construction supplies a drive and a receiver, not their formation from unknown interactions. For real envelopes and even g, the free input and output energies are

$$E_{\text{source}} = \hbar\Omega k_{\text{eff}} m^2, \qquad E_c = \hbar\Omega\xi k_{\text{eff}} m^2, \qquad E_{\text{free,out}} = \hbar\Omega(1-\xi)k_{\text{eff}} m^2. \tag{35}$$

The same reflection symmetry makes the mean detuning energy zero before and after collection. Consequently the net prescribed capture-switch work is zero in this family. The complete source drive supplies E_{source}. For other packets the capture work need not vanish: at mean occupation $\bar{n}$, its magnitude is at most $2\hbar B\bar{n}$. Detector, carrier, clock and support costs are separate from these accounts.

Proposition 19 (A finite waveform certificate)**.** *Let $J(\delta) \le J_0$. Approximate $r(T-s)$ by linear interpolation of node values on L uniform intervals, each rounded on a prescribed grid of spacing q. Include a further uniform envelope error η_u. With $\nu = \sqrt{\xi_0}$,*

$$\sigma = \sqrt{2\pi J_0 T}\left[\frac{G^2(T/L)^2}{8} + \frac{q}{2} + \eta_u\right]$$

bounds the prepared field-vector error relative to Eq. (34)*. An interval enclosure error for node values is added inside the brackets. Let duration and arrival errors be bounded by d_T, d_t, and the capture pulse have $\int |h_{\text{actual}} - h|dt \le d_h$. Put $\varepsilon = (B+G)d_T + Bd_t + Gd_h$. If $\sigma < \nu$, then*

$$1 - \xi \le \left[\frac{|r|\nu + \sigma + \varepsilon(\nu+\sigma)}{\nu - \sigma}\right]^2, \qquad \left(1 - \frac{\sigma}{\nu}\right)^2 \le \frac{k_{\text{eff}}}{k} \le \left(1 + \frac{\sigma}{\nu}\right)^2. \tag{36}$$

Arrival and duration bounds here concern collection and the logical parity instrument, which are insensitive to the common carrier phase.

Proof. $|r''(s)| = |\langle Ac|e^{-iAs}|Ac\rangle| \le G^2$ gives the interpolation bound. Plancherel over the entire frequency line, followed by restriction to the band, gives $\|w_u - w_{\text{ideal}}\| \le \sigma$. Duhamel bounds the carrier-referenced propagator error by ε. The ideal outgoing field remainder has norm $|r|\nu$. Adding preparation and propagator errors to this norm and dividing by the actual input norm, at least $\nu - \sigma$, proves the first bound. The second follows by the same input-norm bounds. □

For a concrete continuum certificate use

$$J(\delta) = \frac{1}{2\pi}\left[1 - (\delta/B)^2\right]^2, \quad \Omega = 20, \quad B = 5, \quad T = 16.$$

Frequencies are measured in units of $2\pi J_0$ and times in its reciprocal. Then $G^2 = 8/(3\pi)$. The exact even bath moments are $\mu_{2j} = 16J_0B^{2j+1}/[(2j+1)(2j+3)(2j+5)]$. If $M_n = \langle c|A^n|c\rangle$, then

$$M_0 = 1, \qquad M_{2n} = \sum_{j=0}^{n-1} \mu_{2j} M_{2(n-1-j)}, \qquad r(t) = \sum_{n=0}^{\infty} \frac{(-1)^n M_{2n} t^{2n}}{(2n)!}.$$

Decomposing each departure and return to c proves the recurrence. The degree-P Taylor remainder is at most $[(B+G)T]^{P+1}/(P+1)!$ by self-adjoint functional calculus. Directed intervals at 120 decimal digits and $P = 400$ give $r(16) = 0.000061743299929498335 73\ldots$ with remainder below 1.53×10^{-79} and $\xi_0 > 0.99999999618$.

Choose $L = 512$, $q = 1/65536$, $\eta_u = d_T = d_t = d_h = 10^{-5}$. The 513 rounded node values fit the grid $[-1/64, 1]$, whose 66561 labels need 17 bits each. The fixed table uses 8721 logical bits, apart from its decoder, interpolator, carrier and clock. A separately encoded normalization amplitude rounded on a 2^{-24} grid contributes less than 10^{-7} relative error at nominal $k = .2$; it fits within η_u and uses 24 more label bits. Eq. (36) gives $1 - \xi < 4.42935 \times 10^{-7}$ and $k_{\text{eff}}/k \in [.999030268, 1.000970203]$. For $\alpha = 1.1$, $h = .5$, $N = 4096$ probes at each of $\pm h$ and nominal $k = .2$, the calibration's collection-dependent dephasing error is below .000946. This contrasts with the loss of content at $\xi = .99$ above; it does not include parity-gate or detector errors.

A second numerical representation replaces frequency multiplication by cell midpoints while retaining g itself. The invariant vector in each cell is $g\mathbf{1}_{\text{cell}}/\|g\mathbf{1}_{\text{cell}}\|$, with coupling $\sqrt{\int_{\text{cell}} J}$. Its finite matrix approximates the continuum propagator with operator error at most TB/M for M cells. These cells are a numerical resolution, not fresh physical ports. Dense and sparse matrix calculations agree with the moment enclosure and the prepared waveform. The recurrence and rounding rule specify the waveform without a fitted infinite-precision function.

After an encounter, swapping c with an independent same-frequency vacuum oscillator returns c exactly to vacuum, even when its old state is correlated with a reference. The swap exports E_c and has the finite duration and occupation account of Proposition 16. It supplies a reusable receiver for a finite bank of encounters, not a free thermal sink. Repeating the query law needs independent field preparations as well as independent vacuum exhausts. Independent field sectors selected one at a time give a finite-bank realization whose Hamiltonian remains bounded below. 8192 queries consume 8192 such preparations and exhausts, while sharing the waveform program. In this calibration example the nominal source energy is $E_{\text{probes}}/[\hbar(2\pi J_0)] = 4096(.2)(20)[(1 - .55)^2 + (1 + .55)^2] = 42680.32$. It is predominantly collected and later exported, not eliminated by the small content-damage bound. Serial preparation and capture take $8192(2T) = 262144$ time units before parity control and receiver reset. Time windows in one band-limited waveguide are not automatically independent; that alternative requires a return/reset bound. Thermal exhaust resets c to a thermal state and changes the readout law. For example, for the exactly matched mode, a thermal receiver of mean occupation n_T gives even-parity probability

$$\frac{1}{2}\left[1 + \frac{e^{-2\xi k m^2/[1+2(1-\xi)n_T]}}{1 + 2(1-\xi)n_T}\right]$$

after mixing with the vacuum source packet, rather than the vacuum formula. For an unmatched packet, the thermal occupation term instead uses $|\langle c|S|c\rangle|^2 n_T$, which need not equal $(1 - \xi)n_T$.

Proposition 20 (Coherent parity echo)**.** *Suppose the relevant mode remains available for joint control. The bounded involution*

$$R = |0_L\rangle\langle 0_L| \otimes I + |1_L\rangle\langle 1_L| \otimes \Pi = I_L \otimes P_{\text{even}} + Z_L \otimes P_{\text{odd}}$$

satisfies, on the quadrature domain,

$$R(Z_L \otimes Q_f)R = I_L \otimes Q_f, \qquad RU_m R = I_L \otimes e^{-imQ_f}. \tag{37}$$

This protects the body's marginal for every mode input and even for an initially correlated body–mode state. For an independent vacuum input, the first R acts trivially; the final R alone disentangles the body and mode.

Proof. $\Pi Q_f \Pi = -Q_f$. In the $+1$ logical block the conjugated generator is Q_f, and in the -1 block it is $-\Pi Q_f \Pi = Q_f$. Functional calculus gives the unitary identity. A unitary on the mode alone cannot change the body's reduced state. On vacuum $\Pi|0\rangle = |0\rangle$. □

One qubit meter implements R coherently. Prepare it in $|0\rangle_C$, apply

$$M = \exp[-i(\pi/2)P_{\text{odd}} \otimes Y_C], \qquad C_Z = I_L \otimes |0\rangle\langle 0|_C + Z_L \otimes |1\rangle\langle 1|_C,$$

and then apply $M^\dagger C_Z M$, with the tensor identities understood. The meter computes parity, controls the body phase and uncomputes parity; it returns exactly to $|0\rangle_C$. Bounded Hamiltonians $g(t)P_{\text{odd}} \otimes Y_C$ and $\gamma(t)(I_L - Z_L)/2 \otimes |1\rangle\langle 1|_C$ suffice, with pulse areas $\pi/2$ and π. Amplitude bounds impose corresponding nonzero durations. No classical syndrome bit has to be erased in this coherent correction. Producing a retained classical calibration record is a separate operation. Pulse area is not a heat bill: the control field, capture, clock and support still need their physical energy account. Half-diamond errors of supplied primitive channels accumulate at most by their sum, by telescoping compositions; such a bounded-channel error alone does not bound an unbounded stress observable.

A retained oscillator offers a direct alternative to the spectral meter. In its interaction picture, with $n = b^\dagger b$ and $P_1 = |1_L\rangle\langle 1_L|$, the dispersive Hamiltonian $H_R(t) = \hbar\chi(t)P_1 \otimes n$ implements R when $\int \chi(t)dt = \pi$. Such number-controlled parity gates are established cavity operations [5, pp. 2–3]. For a degenerate logical body, $\chi \geq 0$ and

oscillator energy $\hbar\omega n$, the total Hamiltonian is bounded below. If χ starts and ends at zero and the gate is isolated, $P_1 n$ is conserved, so its net prescribed-switch work is exactly zero:

$$W_R = \hbar \int \dot{\chi}\, \langle P_1 n\rangle dt = 0.$$

The oscillator's stored energy, gate duration and controller are not zero. The integrated expected interaction energy is $\pi\hbar\langle P_1 n\rangle$; $\chi \le \chi_{\max}$ gives duration at least $\pi/\chi_{\max}$. If the pulse area has error δ, its half-trace error, with any reference and an input occupation bound $\langle n\rangle \le \bar{n}$, is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$. Indeed the relative unitary is $e^{-i\delta P_1 n}$, and the purified vector error squared is bounded by $2|\delta|\langle P_1 n\rangle$, using $1-\cos x \le |x|$. This is an energy-constrained error, not an unrestricted diamond bound. It gives an explicit retained-oscillator gate account; it does not price trapping a broadband travelling mode or operating the classical switch.

C. Thermal inquiry with a binary record and a finite bank

The echo gives an actual thermal inquiry protocol on the retained oscillator. Let its input be the centered Gibbs state τ_T of $H_b = \hbar\omega n$, with $n_T = (e^{\hbar\omega/(k_B T)} - 1)^{-1}$ and $\nu_T = 2n_T + 1$. This input is independent of the held body, its old reference and earlier records. Both parity gates in $RU_m R$ are used: the first is generally consequential on a thermal input. The result is $\rho_{L\mathrm{old}} \otimes D_m \tau_T D_m^\dagger$, where $D_m = e^{-im\sqrt{k}Q}$. Thus the body and its entire old reference are preserved, while the oscillator has

$$\langle P\rangle = -2m\sqrt{k}, \qquad \mathrm{Var}(P) = \nu_T, \qquad E_b - E_{b,\mathrm{in}} = \hbar\omega k m^2.$$

The displaced Gibbs state has the same entropy as τ_T.

A quadrature mean is not yet a saved record. Prepare a fresh degenerate qubit C in $\eta_v = (I + vZ_C)/2$, $0 < v \le 1$, independently of the oscillator and all earlier outputs. Couple it through

$$V_\kappa = e^{-i\kappa P \otimes Y_C}, \qquad \kappa > 0,$$

then measure X_C and store the outcome $Y = \pm 1$. The preparation, orientation, measurement and storage are declared operations.

Proposition 21 (Thermal binary inquiry)**.** *For this protocol the instrument on the held body and its old reference is*

$$\mathcal{J}_y(\rho_{L\mathrm{old}}) = \frac{1 + y\mu_T(m)}{2}\rho_{L\mathrm{old}}, \qquad \mu_T(m) = -ve^{-2\kappa^2\nu_T}\sin(4\kappa m\sqrt{k}). \tag{38}$$

The read interaction adds oscillator energy $\hbar\omega\kappa^2$ before conditioning on the record. If $m = \pm m_0$ with known $m_0 > 0$, $0 < \kappa \le \pi/(8m_0\sqrt{k})$ gives sign gain $\gamma_T = ve^{-2\kappa^2\nu_T}\sin(4\kappa m_0\sqrt{k}) > 0$. For J fresh trials, the sign estimate from $-\sum_j Y_j$ has error at most $e^{-J\gamma_T^2/2}$, including ties as errors.

Proof. The echo acts only on the independent oscillator, so all subsequent oscillator–port operations factor from $\rho_{L\mathrm{old}}$. Conjugation gives $V_\kappa^\dagger X_C V_\kappa = \cos(2\kappa P)X_C + \sin(2\kappa P)Z_C$. The Gibbs characteristic function is $\langle e^{itP}\rangle = e^{-t^2\nu_T/2}$, and displacement changes it by $e^{-2itm\sqrt{k}}$. Taking the qubit expectation proves Eq. (38). Also $V_\kappa^\dagger Q V_\kappa = Q + 2\kappa Y_C$, while P is fixed. Since the independent port has $\langle Y_C\rangle = 0$ and $Y_C^2 = I$, $n = (Q^2 + P^2 - 2)/4$ increases in mean by κ^2. Each sign-aligned binary variable has mean γ_T; Hoeffding's inequality gives the stated bound. Independence here follows from fresh inputs and the renewal below, rather than from reusing a disturbed mode. □

This record reads a sign that parity cannot resolve. Its orientation reference is an additional permission. The sine is periodic, so one setting does not identify arbitrary continuous m; an unknown initial quadrature mean also confounds it. If the mode initially shares correlations with the body, the echo still preserves the unconditional body state, but measuring the mode may steer it. That case does not give Eq. (38).

The pulse can have bounded amplitude and finite duration. In the laboratory frame choose a smooth real $g(t)$ of compact time support, with $\int g(t)dt = \kappa$, and set

$$H_{\mathrm{read}}(t) = \hbar\omega n + \hbar g(t)[P\cos(\omega t) + Q\sin(\omega t)] \otimes Y_C. \tag{39}$$

Its interaction-picture generator is exactly $\hbar g(t) P \otimes Y_C$. The final laboratory evolution includes a known free rotation; subsequent pulses and read axes use the same clock. Completing the square gives $H_{\mathrm{read}}(t) \ge -\hbar g(t)^2/\omega$. Adding the

scalar $\hbar g(t)^2/\omega$ makes it nonnegative and changes only an overall phase; the scalar vanishes at both endpoints. For $|g| \le g_{\max}$ the duration is at least $\kappa/g_{\max}$. The net prescribed-switch work is the endpoint free-Hamiltonian energy increase $\hbar\omega\kappa^2$ in this isolated, degenerate-port model.

The echo's displacement admits the same finite pulse construction, with laboratory force $\hbar m\sqrt{k}f(t)[Q\cos(\omega t) - P\sin(\omega t)]Z_L$ and $\int f(t)dt = 1$. Completing the square bounds its Hamiltonian below by $-\hbar k m^2 f(t)^2/\omega$; a finite command range keeps this bound uniform. The source pulse supplies $\hbar\omega k m^2$ on the Gibbs input. The two parity gates have the finite dispersive realization above. These rotating profiles supply phase and clock access explicitly; their implementation by an autonomous controller is a further physical question.

Proposition 22 (Finite thermal renewal with retained outputs)**.** *Take a bank of J independent same-frequency oscillators in τ_T, independent mixed read ports, and storage for J binary records. After each echo and read, couple b to the next fresh bank oscillator through $\hbar h(t)(b^\dagger c + c^\dagger b)$ with area $\pi/2$. The active oscillator returns exactly to τ_T, independent of all earlier retained outputs. Its entire post-read state and correlations are transferred to the spent bank oscillator, up to known phase rotations. After summing over the new outcome, the entire old body/reference/history state is preserved, including for commands selected from earlier records. If $|h| \le h_{\max} < \omega$, the total contact Hamiltonian is bounded below and its duration is at least $\pi/(2h_{\max})$. A pulse off at both endpoints has zero net prescribed-switch work. For fixed m and κ, the spent oscillator's mean excess energy above its incoming Gibbs value is*

$$\Delta E_{\text{spent}} = \hbar\omega(km^2 + \kappa^2). \tag{40}$$

This is a finite J-trial realization of Eq. (38)*.*

Proof. The contact sends $b \mapsto -ic$, $c \mapsto -ib$. For any joint post-read state ρ_{bA}, where A contains the held body, old reference, port and records, the output is $\tau_{T,b} \otimes \rho'_{cA}$. Gibbs states are invariant under the known phase rotation. This identity transfers correlations instead of tracing them out. The two instantaneous normal frequencies are $\omega \pm h$. Total occupation and $b^\dagger c + c^\dagger b$ are conserved during the contact, which proves the stability and zero switch-work statements as in Proposition 16. The preceding two displacement accounts prove Eq. (40) after averaging over the retained record. For an old classical history r, the new record appends the kernel $p(y \mid m_r)$ to its existing conditional body/reference state. Summing over y leaves each old block, and hence their full joint state, unchanged. Induction proves renewal and this nonselective preservation for adaptive commands. Conditioning only on a new outcome can reweight old history blocks and change the reader's body/reference state; it need not preserve that conditional state. For a fixed unknown sign and fixed settings the new inputs give independent trials. □

For example, $k = .2$, $m_0 = .7$, $v = .6$ and $\kappa = \min\{\pi/(8m_0\sqrt{k}), 1/(2\sqrt{\nu_T})\}$ give sufficient trial counts 178, 247 and 593 for error .05 at $n_T = .2$, .5 and 2. The respective excess oscillator energies per trial, in units of $\hbar\omega$, are .276571, .223 and .148. Higher temperature weakens the record even though this choice reduces the read-pulse energy. A bank consumes one fresh Gibbs oscillator and one mixed read port per trial, retains the spent oscillator and a binary record, and pays both parity gates, the source/read profiles and the resonant contact duration. The initial active Gibbs oscillator is needed in addition to the J fresh bank inputs. Preparation, transport, readout, storage, orientation and the clock/support must be charged in addition.

Neither term in Eq. (40) is an entropy-export certificate. The source displacement preserves entropy; the read operation changes correlations with its port and record. Replenishing the bank or erasing spent outputs requires their joint state, including side information. This construction supplies repeated thermal inquiry for each finite bank. It does not supply the chiral capacity of Section VI, an infinite thermal line or the total covariant stress of a controller.

For a centered independent input the original outgoing source is $m^2 F_f$, $F_f = (f')^2/4$. The coherent echo leaves this coupling-added stress in the field while preserving content. Its implementation requires an available mode and controlled parity; it supplies neither a local capture apparatus for a whole continuous field nor a sustained thermal exhaust. The finite oscillator identity and the moving-field apparatus are different levels of the construction.

VIII. THREE QUADRATIC STRUCTURES

The analysis now contains three quadratic objects, and they should not be identified.

The descent metric G of Proposition 2 is a positive form on changes of controller parameters. A designer of a gradient controller chooses it, and its positivity is what descent requires. The work form K of Proposition 5 is a positive form on switch profiles. It is derived from the field stress and prices a control setting. The Lorentzian form used in Q-RAIF Paper A [30] represents spacetime read-out; it is indefinite and derived from a single-shell propagation symbol with independently supplied evolution time and spatial carrier.

K and G live on different spaces: K on profiles, G on tangent vectors of the parameter space. If the controller parameters are the amplitudes a of $g = \sum_i a_i g_i$, the parameter space is linear and one may choose $G = K$. The descent

direction $-K^{-1}\nabla V$ then maximizes the first-order decrease $-\nabla V \cdot v$ among parameter changes with $v^T K v = 1$, by Cauchy–Schwarz in the K inner product. This is a design option that measures parameter changes in the units that price settings. Persistence does not force it, and the feedback of Section VI is not a gradient controller: its stability was proved without any G.

Positive quadratic drive charges with supplied coefficients appear in other Q-RAIF protocols. The field coupling shows how such a coefficient can be fixed by the stress of the field that carries the control. Both G and K are positive, so the actuator does not supply the negative direction of a Lorentzian form.

IX. RELATIONS TO OTHER WORK

A. Q-RAIF Papers A and C

Paper A [30] looks for the relative sign in translation-invariant local unitary propagation. When all bands share one quadratic shell, the propagation symbol obeys Euclidean Clifford relations. Its form $\eta = \omega^2 - Q(q)$ has one positive frequency direction and d negative spatial directions; the mass term together with the spatial generators yields $Cl(1, d)$. Reversing the overall metric convention reverses these signs, not their relative opposition. By Sylvester's law neither positive G nor K of Section VIII can be brought to that indefinite form.

The two papers use different quadratic structures, and Section VIII keeps them apart. Paper C [31] asks what a coupling supplies in place of an assumed representation map. For the coupling XQ_g of this paper the algebra represented exactly is the commutant of X, for every drive state, and the defect on the remaining controls reproduces the energy–accuracy bound of Theorem 6.

The mass–energy paper [32] now uses the same quadrature interaction to realize a quantum-walk coin. With a fresh Gaussian drive each step, its mean angle fixes the rest resonance, while drive fluctuations set its decay rate. At input occupation N the smallest Gaussian rate is $\theta^2/[2\tau N(N+1)]$, attained by a displaced squeezed state. This connects the price of control to the persistence of a spectral readout. The rate is an open-channel linewidth, not a new spectral mass, and shift, preparation and port reset costs are separate.

B. Quantum control theory

Lie-algebraic control characterizes reachability through the algebra generated by drift and control Hamiltonians [14]. Stabilization against perturbations also needs magnitudes and costs. In the realized actuator these come from the physical coupling: the rotation angle from the drive amplitude, the error from the drive's quadrature fluctuations, and the cost from the field stress. A Clifford quadratic form is not needed to supply them.

C. Decoherence-free subspaces

Decoherence-free subspaces [19, 20] are protected passively by symmetry. The feedback task gives the active counterpart. The held-code instrument, coherent echo and finite collection protocols also join passive encoding to priced inquiry and correction: their exact identities protect a supplied code, while collection loss and gate error determine what survives. Brain Anchor [4] uses records of this interface for finite-window inquiry and maintenance decisions. Forming the code and its physical permission from a prior history is a further mechanism, not a consequence of encoding it.

D. The Imprint Chain and HAFF

The Imprint Chain [36] supplies the instrument, the calibration and the task, with an exact supplied flip. Replacing that flip by the field coupling raises the calibrated loss from 0.199631 to 0.202127 at $N = 16$ and makes the drive energy part of the feasibility condition. Each coupling also adds the stress F_g to the outgoing field. The HAFF gravity paper [29] uses this outgoing field as a matter source in a Jackiw–Teitelboim model; there, calibration changes the mean source through the coupling rate p of Eq. (24). Selecting the coupled projection changes its represented algebra and source even at fixed activation schedule. A boundary readout can then update the next projection, with known probe stresses and bounded errors included in the inverse. Section VI F supplies one energy constraint on a specified realization of that return.

E. Quantized control fields

The effect of field quantization on gate accuracy has a substantial literature. For Jaynes–Cummings-type couplings, conservation of excitation number leads to error bounds controlled by photon-number fluctuations, of order $1/\bar{n}$ for coherent fields [24]; coherent-field control limits for such models were analysed in Ref. [25], and simultaneous driving of several atoms by one quantized pulse in Ref. [26]. The coupling studied here is a quadrature coupling of a degenerate bit and conserves no excitation number, which is why squeezing changes the scaling from ϵ^{-1} to $\epsilon^{-1/2}$. Theorem 6 gives the quadrature-specific bound and explicit near-attainment under these assumptions. A covariant stress tensor for a finite detector, including its localization, is constructed in Ref. [23]; it shows why the apparatus stress cannot simply be omitted.

X. DISCUSSION

For gradient-type feedback the algebra must compose associatively and the descent metric must be positive. When control is realized by a field, the structure that limits persistence is the canonical commutation relation of the drive quadratures: $[Q, P] = 2i$ fixes the trade-off between flip accuracy and drive occupation in Theorem 6, and it is the reason squeezed drives do better than classical ones. The Pauli and Clifford relations of the controlled bit label the operation; they do not set its cost. Persistence has a closed-loop proof in this model, with an explicit Lyapunov function, and the drive energy and calibration records enter one feasibility condition.

Several inputs remain supplied. The free chiral field, the degenerate bit, the clock and switch, the drive ports and the instrument family are given. The feedback task also supplies the success probability for compensated randomization and scheduled beams. Its analysis concerns Z-diagonal bit states and flip channels; complex $\chi_\sigma(2\lambda)$ instead rotates Z into Y. The held-content construction concerns arbitrary logical inputs. Its finite-band realization prepares and collects a specified packet, with a finite waveform, clock tolerances and independent vacuum exhausts. It returns a coherent meter; the carrier, detector, port selection and vacuum supply remain separate inputs. The stress of the switch and its support is not computed; outgoing-field stress is used outside the apparatus.

The thermal bank identifies what renewal has to preserve. A fresh input restores the oscillator only because the previous state, including its record correlations, moves to a spent output. Bank replenishment and memory erasure must therefore be priced on that joint output, as in the information thermodynamic account [9, 10]. The chiral capacity bound constrains a specified entropy flow; the finite oscillator construction gives an implementable inquiry and transfer, rather than that flow's attainability.

Repeated propagation in one field remains a stronger problem. Proposition 10 proves that shared drives can preserve each marginal channel while correlating failures. A persistence calculation on that joint history may therefore have a different energy scaling from the fresh-drive task. The other unresolved connection is formation: which interactions and retained records produce the code, access permissions and controller used here? T-DOME, Imprint and Brain Anchor address that mechanism; the present results constrain the resources of the operations it must supply.

XI. CONCLUSION

Persistence of an open subsystem needs controls, and controls are physical. At the algebraic level, associativity and a positive descent metric are what gradient feedback requires, and Clifford algebra remains an optional parameterization. At the physical level, a flip realized by a field coupling has an error, an outgoing stress, a switching work and a derived quadratic cost. Every drive state needs occupation of order $\pi/(4\sqrt{\epsilon})$ for failure ϵ, squeezed drives nearly reach this, and classical drives need order $\pi^2/(16\epsilon)$. In a calibrated feedback task the same actuator gives a complete stability proof, exact loss and coupling rates, and a single condition that ties calibration records to drive energy. Returning a response through a translation-only instrument has a separate sharp mean-energy bound, even with retained ancillas and joint decoding. Together these results price both directions of a control interface; the finite thermal oscillator branch realizes content-preserving inquiry and physical port renewal, but does not supply the chiral exhaust capacity or a reusable geometric readout. The work form, the descent metric and the Lorentzian read-out form stay distinct.

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Retained Quantum History, Physical Sources and Receiver Renewal in Q-RAIF

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Abstract

Acquired quantum records can drive, sense and act again on the same complete material without replacing its old history by independent baths. We construct a positive autonomous Gaussian interface with physical energy-moment and smooth interacting-source bounds, retaining its clock and all old references. A paid Gibbs bank renews a specified receiver. Acquisition, decision, force, renewal and the next reading share one finite cursor path with arbitrarily small joint error as clock resources grow. Exact algebraic transport is a separate question: a reduced coupling supplies a completely positive map whose multiplicative domain is its exactly represented algebra. Discarded non-atomic ports limit recovery; retained oscillator loops and closed propagation permit exact operations.

A full-mode fermion-ring realization renews the same body to uniform per-cycle tolerance through four thermal outlets, then reinstalls the mixed receiver and current purifier by neutral pumps. Physical gates compress noisy multi-cycle records, load overlapping packets and sort the actual spectrum into a paid finite-volume thermal carrier. With every atypical label and old purification retained, the complete output Gibbs deficit per unit time can approach zero in a prescribed finite layout. A separate massive-outside body supplies full-angle local gates and weighted physical-source control.

A spatial work-clock field executes a finite smooth schedule using the actual retained field state. Matrix Ward equations and the delayed incoming kernel close at the same derivative order; physical energy moments control the complete joint gate/reset output. Gap work, clock baseline stress and its finite work correction remain charged. This uses static endpoints and a supplied material frame, with horizon-dependent clock resources. The distinct receiver, ring and massive-body constructions do not yet supply one dynamically supported, covariant apparatus with sustained fixed-capacity sharp exhaust.

Keywords: Q-RAIF, multiplicative domain, quantum records, autonomous control, material response, local stress, thermal renewal

1 Introduction

A physical interface must say what happens to the degrees of freedom that acquire its records. Are they lost, coherently returned, or used in another interaction? Q-RAIF A supplies Lorentzian read-out under its propagation-symbol conditions; Q-RAIF B gives actuation, energy and record permissions; the mass–energy paper identifies momentum and mass read-outs [25, 26, 27]. Here the task is retained-history control and physical renewal, including the source produced by those operations.

The main argument starts with discarded versus retained ports. One oscillator closes exact Weyl loops; one continuum has a compact-time closure obstruction but admits finite adaptive return with its old correlations retained. The multiplicative-domain and propagation results in Appendices A–D state precisely which algebra a coupling transports exactly [29, 30, 31]. This is not a universal conservation-law no-go: Wigner–Araki–Yanase bounds address different specified measurement architectures [34, 35, 36].

Sections 5–6 build the material interface: charged fields and their spin sectors, a finite mode with its remainder, same-Hamiltonian state/source certificates, and actual acquisition–force–sensing–action under one positive autonomous Hamiltonian. The record remains an operator in the old joint history. Its connected fluctuations contribute to the source even when mean trajectories coincide with a classical mixture. The interacting Gibbs reference is defined on that same Hamiltonian, but equilibrium alone supplies no running cursor or blank records.

Section 7 separates port replacement, later free-field thermal windows and whole interacting-equilibrium preparation. Its full-ring construction adds uniform body renewal, neutral block compression, a finite-width packet interface and charge-sector thermal sorting on the actual carrier. The covariance and energy need not commute. Complete-output sharpness retains every bad label; it does not prepare an indefinitely renewed inlet or controller. Section 8 supplies the retained-state field–clock evolution and full joint-state error for a finite gate/reset schedule. The massive-outside gate is a separate physical layout, not the massless ring's reset with a different name.

Three clock/resource statements must be kept distinct. Corollary 16 holds clock mean energy and norm fixed by taking $J = A/m$, while receiver-renewal duration and space grow. Proposition 17 holds J and duration fixed while clock energy mJ grows, placing all finite rounds on one cursor. Proposition 24 evolves a spatial work field χ, retaining its stress and finite work, with static endpoints/frame and inertia chosen for the finite horizon. None establishes an indefinitely running fixed-resource apparatus.

The auxiliary battery, outlet, thermal-preparation, recoil, diagnostic and stress-flow proofs remain in the appendices. Their carriers and supports are explicitly distinct. Section 9 finally supplies a dynamical-support/clock action and identifies the quantum gate and total-source matching still needed. A finite cursor is not thereby a transported relativistic controller, nor is a free thermal window the interacting material's Gibbs state.

Unless stated otherwise, $\hbar = c = k_B = 1$. Energies expressed as frequencies acquire a factor $\hbar$ when units are restored, times and pulse areas use the corresponding frequency units, and entropy is in nats. The combinations $b + b^\dagger$, $i(b^\dagger - b)$ and occupation numbers are dimensionless; material smearings use the normalizations specified with their Hamiltonians below. All joint-state assertions include the stated old references; a preserved nonselective

marginal need not remain unchanged after conditioning on a new record.

2 Discarded ports and coherent return

A discarded record port and a retained environment permit different controls. Write $\chi(t) = \mathrm{Tr}(\sigma_E e^{-itE})$ for the port's spectral characteristic function. Appendix B derives the single-coupling channel and its multiplicative domain; the following bound survives arbitrary later controls that cannot reach the lost port.

Proposition 1 (A fresh port leaves an irreversible defect)**.** *Suppose a finite-dimensional system first undergoes ideal unitary controls and then one coupling $e^{-iX\otimes E}$ to an initially independent port, with X having at least two distinct eigenvalues and the environment's spectral distribution non-atomic. Let F collect every auxiliary port accessible later; just before this coupling the joint state is $\rho_S \otimes \sigma_E \otimes \tau_F$ for every system input. If E is thereafter inaccessible, no subsequent system channel can make the complete channel unitary on the full system algebra. More precisely, let Λ include all the preceding and subsequent controls, and let $\mathcal{V}$ be any unitary channel. For every pair of distinct eigenvalues x, x',*

$$\frac{1}{2}\|\Lambda - \mathcal{V}\|_\diamond \;\geq\; \frac{1 - |\chi(x' - x)|}{2}. \tag{1}$$

Proof. Choose eigenvectors $|x\rangle, |x'\rangle$ and the orthogonal states $|\pm\rangle = (|x\rangle \pm |x'\rangle)/\sqrt{2}$. Their trace distance is one before the coupling and $|\chi(x' - x)| < 1$ afterwards. All subsequent operations induce a system channel and cannot increase this distance. Pull the two inputs back through the preceding unitaries and call them $\rho_\pm$. If $e_\pm = D(\Lambda\rho_\pm, \mathcal{V}\rho_\pm)$, the triangle inequality gives $1 \leq e_+ + |\chi(x' - x)| + e_-$. Thus $\max e_\pm \geq (1 - |\chi|)/2$, which also bounds half the diamond norm from below. □

The independence concerns the entire later-accessible auxiliary system jointly, including any purification of the discarded port. A later register correlated with that port can reveal which phase was applied and allow its reversal; the system's reduced channel alone then does not describe the available recovery operation. For a qubit and no later noise, the bound is attained: remove the phase of χ, and the remaining dephasing is the mixture $(1 + |\chi|)\mathcal{I}/2 + (1 - |\chi|)\mathcal{Z}/2$. Its half diamond distance from $\mathcal{I}$ is $(1 - |\chi|)/2$.

Proposition 2 (An exact interface from a retained oscillator)**.** *Let $Q = b + b^\dagger$, $P = i(b^\dagger - b)$ satisfy the Weyl relations, so $[Q, P] = 2i$, and let X be Hermitian with finite spectrum. Then*

$$e^{-i\lambda XQ} e^{-i\mu XP} e^{i\lambda XQ} e^{i\mu XP} = e^{-2i\lambda\mu X^2} \otimes I. \tag{2}$$

Every initial oscillator state is returned exactly. The induced system map has multiplicative domain $M_s(\mathbb{C})$, including operators that do not commute with the coupling observable X.

Proof. In an X eigenblock, the two pulse generators have scalar commutator

$$[-i\lambda x Q, -i\mu x P] = -2i\lambda\mu x^2.$$

The Weyl group commutator is therefore that scalar phase. Summing the spectral blocks gives (2). □

For a qubit choose $X = (I + \sigma_x)/2$. With $\lambda\mu = \pi/2$, the loop is $e^{-i\pi X} = -\sigma_x$: an exact corrective flip, even with a vacuum or thermal oscillator input. The intermediate system–oscillator correlations are returned, not discarded. Conditional displacement loops of this kind are established geometric gates [24]; the point here is their implication for the interface. Continuous spectrum alone cannot prohibit exact control of a retained port.

Corollary 3 (One retained oscillator gives universal qubit control)**.** *Suppose the four pulses of Proposition 2 can be applied through either $P_z = (I + \sigma_z)/2$ or $P_x = (I + \sigma_x)/2$, with arbitrary real pulse areas. Every qubit unitary, up to global phase, is a sequence of at most three closed loops, and the oscillator returns after each loop.*

Proof. A loop with $2\lambda\mu = \alpha$ implements $e^{-i\alpha P_z} = e^{-i\alpha/2}e^{-i\alpha\sigma_z/2}$, and similarly for P_x. The z–x–z Euler decomposition of a qubit unitary therefore gives three loops. For example the three angles $\pi/2, \pi/2, \pi/2$ give a Hadamard up to global phase. Each identity holds on the full system–oscillator tensor product, so composition returns even an oscillator initially correlated with another system. □

For a larger system the same construction supplies the one-parameter groups e^{-itP_j}. Full unitary control follows when the real Lie algebra generated by iP_j is $\mathfrak{u}(s)$; generation of the associative matrix algebra alone is not the required controllability condition. The pulse permissions, including the choice between noncommuting projectors, remain physical inputs.

The loop is not costless. For a centred input and a mode of frequency ω, the largest conditional occupation increase along the four corners is $(\lambda^2 + \mu^2)x^2$. Minimizing it for the qubit flip gives $\lambda = \mu = \sqrt{\pi/2}$ and peak mean stored energy $\pi\omega\langle X\rangle \leq \pi\omega$. The mode's final energy is unchanged; an ideal reversible switch can recover the intermediate work. This is a stored-energy statement, not a total-device dissipation estimate. It assumes access to both quadratures, their controlled pulse order, and coherent retention until closure. A travelling field that has left the apparatus need not supply these permissions. An implementation with smooth local spacetime couplings must establish them separately.

Thus a record carried away by an inaccessible port constrains future control, while a coherently returned record can disappear without thermal erasure. Retaining an interface can replace a preparation-accuracy requirement by a closure requirement. The oscillator, its routing, clock and switching controls are part of that retained interface; they are not supplied by an abstract algebra homomorphism.

3 Finite return through one thermal field

The oscillator loop closes at one frequency. A propagating field has an interval of frequencies, and a sequence timed for its carrier does not close at every detuning. We now compute what remains, and how a single thermal line can be used over a finite adaptive protocol without resetting it between operations. The field theory, phase controls and reading instruments are still inputs.

3.1 Finite-width forces and the outgoing record

Let $\omega = \Omega + \delta$, $|\delta| \le B < \Omega$, and let g be real, even, square-integrable, and nonzero on an interval. In units $\hbar = 1$, the field Hamiltonian is $H_F = \int (\Omega + \delta) a_\delta^\dagger a_\delta \, d\delta$. For a system projector P with both eigenvalues $0, 1$, apply

$$H_s^I(t) = iP \int g(\delta) \left[u(t) e^{i\delta t} a_\delta^\dagger - \overline{u(t)} e^{-i\delta t} a_\delta \right] d\delta. \tag{3}$$

This is the interaction picture of a prescribed carrier-modulated force. It supplies two field quadratures and a clock; it is band-smeared, not compactly localized in space. At finite $u(t)$ the laboratory Hamiltonian is bounded below by $-|u(t)|^2 \int g^2/(\Omega+\delta)\, d\delta$. A scalar shift makes it nonnegative without altering the gate.

Use four disjoint rectangular pulses of width $\sigma < \tau$, centred at $j\tau$, with complex labels $z_j = (1, i, -1, -i)$ and heights $r z_j/(\sigma\sqrt{Z})$, where

$$Z = \int g(\delta)^2 \operatorname{sinc}^2(\delta\sigma/2) d\delta, \qquad f(\delta) = g(\delta) \operatorname{sinc}(\delta\sigma/2)/\sqrt{Z}, \quad \operatorname{sinc} x = \sin x / x.$$

Thus $\|f\|_2 = 1$. The individual field displacements are $\alpha_j = r z_j f e^{ij\delta\tau}$. Put

$$M(t) = \int |f|^2 \cos(\delta t) d\delta, \qquad C = 3M(\tau) - M(3\tau), \qquad F = rf(1 - e^{2i\delta\tau})(1 + ie^{i\delta\tau}).$$

Proposition 4 (Continuum loop and its thermal defect)**.** *The complete four-pulse unitary is*

$$U_{\text{loop}} = e^{ir^2 CP} D(PF), \qquad D(PF) = \exp \int P(Fa^\dagger - \overline{F}a) d\delta. \tag{4}$$

If $C > 0$, choose $r^2 = \theta/C$ for a target phase $e^{i\theta P}$, $\theta > 0$. For an initially independent thermal field with inverse temperature β, define

$$D_\beta = \int |F(\delta)|^2 \coth[\beta(\Omega + \delta)/2] \, d\delta. \tag{5}$$

The thermal state is specified by its Weyl characteristic; a continuum KMS state need not be a trace-class density matrix on vacuum Fock space. After removal of the target phase the system channel is dephasing between the two P blocks, with coherence factor $e^{-D_\beta/2}$. Its half diamond distance from the target unitary is $(1 - e^{-D_\beta/2})/2$, and its representation defect norm on a unit-norm off-block Pauli control is $1 - e^{-D_\beta}$.

No nonzero compact-time force through this single profile can close the continuum exactly. This last statement concerns the source (3)*, not all multimode control architectures.*

Proof. The source commutator is a scalar field commutator times $P^2 = P$, so the time-ordered exponential is a conditional displacement and a commuting P phase. Within each constant-phase pulse the self-phase vanishes after integration over the even band: its sine kernel is odd in δ. The six inter-pulse Weyl phases add to $r^2[3M(\tau) - M(3\tau)]$. Summing the four displacement vectors gives F. This derivation includes finite pulse width; the pulses are not delta kicks.

The thermal Weyl characteristic is $\langle D(F)\rangle_\beta = e^{-D_\beta/2}$. Tracing the field gives the stated channel, a mixture of identity and conjugation by $2P - I$ after the target phase. Orthogonal off-block superpositions attain the half diamond distance; multiplying its Heisenberg off-block factor gives the defect.

For a general integrable compact-time u the displacement is $g(\delta)\int u(t)e^{i\delta t}dt$. Exact unitary control after discarding the thermal field requires this displacement to vanish. The Fourier transform of compact-time u is entire. Vanishing on an interval where $g \neq 0$ forces it to vanish identically, and Fourier uniqueness gives $u = 0$. At one oscillator frequency only a single zero is required, which is why Proposition 2 has no such obstruction. □

For $\theta = \pi$ and $P = (I + X)/2$, the target is a corrective flip up to phase. With $\mu_2 = \int \delta^2 |f|^2 d\delta$,

$$C \geq 2 - \tfrac{3}{2}\tau^2\mu_2, \tag{6}$$

$$\|F\|_2^2 \leq 8r^2\tau^2(1 + B\tau)\mu_2, \tag{7}$$

$$D_\beta \leq \coth[\beta(\Omega - B)/2]\|F\|_2^2. \tag{8}$$

The first uses $M(\tau) \geq 1 - \tau^2\mu_2/2$ and $M(3\tau) \leq 1$. For the second, $|1 - e^{2i\delta\tau}| \leq 2|\delta|\tau$ and $|1 + ie^{i\delta\tau}|^2 = 2 - 2\sin(\delta\tau) \leq 2(1 + B\tau)$. Temperature multiplies the residual record; it does not remove it.

3.2 One line throughout an adaptive sequence

Multiplying the reduced one-loop channels would assume a fresh field. That is not the evolution we consider. The same field and its correlations remain present, and subsequent projectors need not commute.

Proposition 5 (Finite reuse without independence)**.** *Start with one thermal line independent of system, controller and retained reference. A protocol uses at most L loops, interspersed with arbitrary system–controller instruments whose dilations do not act on the line. Projectors may be selected from acquired records and may differ between loops. Suppose every permitted branch has $D_\beta \leq D_{\max}$ and its target phase is implemented as in* (4)*. The complete actual process, including final records, system and field, is at operational state distance at most*

$$\min\{1, L\sqrt{D_{\max}}\}$$

from the process with ideal target gates and the same initial line. Every terminal loss in $[0, 1]$ *has an expectation difference within this bound, as does the total variation of the complete record laws. No reset or independent-error hypothesis is used. Here operational distance is half the norm distance of the states on their joint observable algebra; it is half trace distance whenever a density-matrix description applies.*

Proof. Purify the initial thermal state, or use its thermal GNS cyclic vector. For a loop acting on the ideal-prefix state,

$$\|(D(PF) - I)|\Psi\rangle\|^2 = 2\langle P\rangle(1 - e^{-D_\beta/2}) \leq D_\beta.$$

This is uniform over system inputs, retained entanglement and the record-controlled choice of projector. In the ideal prefix the line remains thermal and independent; the actual prefix need not. Telescope the actual unitary dilations against the ideal dilations, placing the ideal prefix on each difference and leaving the actual suffix as a norm-preserving factor. The vector difference is at most $L\sqrt{D_{\max}}$. The norm distance of vector states, followed by restriction to physical observables or processing records, gives the stated operational bound. The proof does not replace any actual correlated prefix by a reset state. □

For constant per-pulse relative amplitude error at most e_a, centre error d_t and carrier-phase error e_φ, with unchanged pulse widths and still disjoint pulses, each displacement changes in vacuum norm by at most

$$e = r[e_a + (1 + e_a)(Bd_t + e_\varphi)].$$

The six Weyl phases change by at most $6(2re + e^2)$. The same proof therefore gives the finite-setting bound

$$L\left[\sqrt{D_{\max}} + 4\sqrt{\coth[\beta(\Omega - B)/2]}\, e + 6(2re + e^2)\right]. \tag{9}$$

It concerns these specified setting errors, not arbitrary time-dependent source noise or heating added during the force.

As a concrete family take $g^2 = 15[1 - (\delta/B)^2]^2/(16B)$, $\Omega = 10$, $B = 1$, $\beta = .2$ and $\sigma = \tau/2$. The unfiltered second moment is $B^2/7$; since $\operatorname{sinc}^2(\delta\sigma/2) \le 1$, $\mu_2 \le B^2/(7Z)$ and $Z \ge \operatorname{sinc}^2(B\sigma/2)$ in the stated small-width range. At $\tau = 5 \times 10^{-5}$, $e_a = e_\varphi = 10^{-8}$ and $d_t = 10^{-9}$, conservative rational bounds in (9) put an entire 64-loop adaptive protocol below .00512 operational distance. Tight numerical integration gives one-loop error 1.48×10^{-9}. The sequence statement comes from the joint bound, not from treating those tiny errors as independent. The clock's frequency units are supplied by B; these numbers are not unlabelled laboratory seconds.

3.3 Source energy and control resources

For one loop and an independent zero-mean thermal input, the excess free-field energy is

$$\Delta E_F = \hbar\langle P\rangle \int (\Omega + \delta)|F(\delta)|^2 d\delta. \tag{10}$$

For the chiral current normalization of Paper B, its corresponding free outgoing excess stress is $4\langle P\rangle[\operatorname{Re}\psi_F(u)]^2$, where $\psi_F(u) = \int -i\sqrt{\omega/(4\pi)}F(\omega)e^{-i\omega u}d\omega$. The chosen g vanishes at band edges and gives finite first two time moments. This is the free field source, not the stress of its support, phase reference or switching apparatus.

A reused line has extra cross terms. Let $E_1 = \int \omega|F|^2 d\omega$ bound the unit-projector displacement energy of each loop, and let $n_{\max} = [e^{\beta(\Omega - B)} - 1]^{-1}$. For loops sharing the stated profile, with arbitrary projector choices,

$$|\Delta E_F^{(L)}| \le \hbar\left[2L\sqrt{(\Omega + B)n_{\max}E_1} + L^2E_1\right]. \tag{11}$$

Indeed the final annihilation field is its initial value plus $\sum_j F_j P_j^{\mathrm{past}}$, with $\|P_j^{\mathrm{past}}\| \le 1$. The quadratic term is at most $L^2 E_1$ by the weighted L^2 triangle inequality. Each initial thermal cross term is at most $\sqrt{(\Omega + B) n_{\max} E_1}$ by Cauchy–Schwarz. Thus the single-loop energy cannot simply be multiplied by L. This bound prices an excess over the supplied thermal background, not that background itself.

Small final radiation does not mean small driving demands. With $\|g\|_2 = 1$, the four rectangular sources have

$$\int |u(t)|^2 dt = \frac{4r^2}{\sigma Z}, \qquad |H_{\min}| \le \frac{\hbar r^2}{\sigma^2 Z(\Omega - B)}$$

during a pulse. The first is a declared quadratic envelope charge, not heat; the second is the magnitude of a lower energy bound, not consumed work. Narrowing the loop reduces its escaped record but increases these source and clock requirements. At the 64-loop settings the single-loop envelope charge is about 2.51×10^5, although its unit-projector radiated energy is about $4.49 \times 10^{-8}\hbar$. Finite pulse heights are allowed; ideal switching of the external apparatus is still a supplied control.

For fixed band, temperature and target phase, $D_\beta = O(\tau^2)$ as $\tau \to 0$. Taking $\sigma = \tau/2$ and choosing τ proportional to ε/L therefore suffices for joint error ε, with setting errors also reduced to order ε/L. Within this family the total quadratic envelope charge scales as L^2/ε. This is a constructive resource scaling, not a lower bound for every field-control architecture.

Finite reuse removes the independent-port premise; the initially thermal line is never reset. Stored records, controller renewal, exhaust and total source/geometric response still require a compatible apparatus, not merely a small reduced-channel error.

4 What must be retained by a material interface

A material interface must let the next operation use the record without replacing the held system or its old references. The appendices supply a guarded battery and clock (E), local export (F), actual thermal sources with switching work (G), and contact reaction/recoil (H). Their lattice, co-propagating and bidirectional carriers are distinct.

The main construction now fixes the complete material K. Its mode projection retains the omitted reaction, its local drive uses the actual initial state, and its source certificate controls an unbounded observable. These inputs support actual quantum acquisition, force and later sensing on the same Hamiltonian.

5 From a charged material to a certified finite interface

A material action can retain the charged fields rather than replace them by neutral oscillators. The operator boson–fermion correspondence gives a convenient construction on a circle of circumference L. For one Dirac flavor with antiperiodic spatial fermions, its current oscillators and integer chiral charges have free energy

$$H_0 = \frac{\pi}{L}(N_R^2 + N_L^2) + \sum_{k \ge 1} \frac{2\pi k}{L}(m_{R,k} + m_{L,k}).$$

The current descendants of the charge vacua form a complete orthonormal basis of the fermion Hilbert space [7, Sections 2.5–2.6]. The correspondence includes Klein factors and charged odd operators; it is stronger than equality of partition functions.

For a scalar ϕ of period $\sqrt{\pi}$, put $w = N_R + N_L$ and $n = N_R - N_L$. Its zero-mode description is

$$\bigoplus_{w\in\mathbb{Z}} L^2\big([0,\sqrt{\pi}],\, \psi(\phi_0+\sqrt{\pi}) = (-1)^w\psi(\phi_0)\big) \otimes \mathcal{F}_{\mathrm{osc}}.$$

Here $p_0 = \sqrt{\pi}n$, $n \equiv w \pmod 2$, and the charge energy is $\pi(w^2+n^2)/(2L)$. The state $w = 1, n = 0$ would have $N_R = N_L = 1/2$ and is excluded. An ordinary compact scalar with periodic zero-mode wavefunctions has different sectors. Fixed-spin bosonization requires this additional spin information [8, Section 3.5].

Take two such flavors and real material fields Z, θ, y. The local classical coframe is $d\phi_a + \lambda_a\theta f(Z)dZ$, together with $m_Z dZ, \sqrt{I}\,d\theta, \sqrt{J}\,dy$, where $m_Z^2, I, J > 0$. Its positive canonical energy is

$$H_{\mathrm{mat}} = \frac{1}{2}\int\left[\sum_a \Pi_a^2 + \frac{(\Pi_Z - \sum_a \lambda_a\theta f(Z)\Pi_a)^2}{m_Z^2} + \frac{\Pi_\theta^2}{I} + \frac{\Pi_y^2}{J}\right]dx$$
$$+ \frac{1}{2}\int\left[\sum_a (\phi_a' + \lambda_a\theta f(Z)Z')^2 + m_Z^2 Z'^2 + I\theta'^2 + Jy'^2 + 2V\right]dx.$$

Assume smooth f and a smooth nonnegative V independent of the scalar zero modes. A Fourier regulator retains modes $0 < |k| \le K$ of the interaction and material fields, all winding and charge sectors, and the complete free oscillator tail above K. The real Fourier coordinates are not separately wrapped angles.

Proposition 6 (Positive charged material at finite resolution)**.** *For each finite K, the displayed ordered sum of squares has a nonnegative self-adjoint Friedrichs realization. With the free tail retained, the known boson–fermion unitary transports it to the full four-chiral-field CAR Hilbert space tensor the material space. It strongly commutes with all four canonical chiral charges. No small-coupling bound on the finite λ_a is needed.*

Proof. Work first in each winding sector on smooth zero-mode line-bundle sections and compactly supported smooth wavefunctions of the finite real coordinates. The kinetic factor

$$D_Z(x) = \Pi_Z(x) - \sum_a \lambda_a\theta(x)f(Z(x))\Pi_a(x)$$

is symmetric: its extra coefficient is independent of every ϕ_a coordinate, so its formal divergence vanishes. The other momentum factors are symmetric and the spatial factors are real multiplication operators. Their integrated squares and V define a densely defined nonnegative symmetric operator on the common core. Its Friedrichs extension exists. Finite sums of winding sectors give a dense core for their direct sum. Add the positive free-tail Hamiltonian and use the complete current/charge basis to transport this operator to the fermion space. The smeared fermion operators themselves are transported, not replaced by cutoff vertex exponentials, so their CAR are retained. Translations of each $\phi_{a,0}$ leave the core and closed form invariant, as do winding phases. The associated charge unitaries therefore commute with the self-adjoint realization. □

Low-mode canonical squares retain their common zero-point constant; the free tail uses one fixed normal ordering. No coupling-dependent vacuum subtraction is made. The material momenta, including the shift of Π_Z, remain in the operator. Fourier resolution acts on the coupling, not on the carrier's charges or occupation numbers. It is spatially nonlocal at finite K. Removing this cutoff while controlling propagation, renormalized stress and interacting thermal preparations is a separate limit.

The local current also fixes which coupling is intended. With signature $(-,+)$ and $\epsilon^{01}=1$, let $j^\mu=\epsilon^{\mu\nu}\partial_\nu\phi/\sqrt{\pi}$ and $a_\mu=\lambda\theta f(Z)\partial_\mu Z$. Then

$$-a_\mu\partial^\mu\phi=A_\mu j^\mu,\qquad A_\mu=\sqrt{\pi}\,\epsilon_{\mu\nu}a^\nu.$$

For a static texture, $A_0=-\sqrt{\pi}\lambda\theta f(Z)Z'$ and $A_1=0$, giving the required density potential. The coframe's $-a_\mu a^\mu/2$ term and momentum shifts accompany it. The local dictionary does not by itself fix global holonomies or spin weights [9, Eq. (9) and Section III]; the construction above uses the specified operator correspondence.

Rotating (ϕ_1,ϕ_2) to active and spectator Gaussian modes must also rotate their charge lattice. For instance $(w_1,w_2)=(1,0)$ becomes $(1/\sqrt{2},1/\sqrt{2})$, not two independent integer windings. This scalar normal-mode change is different from rotating the actual fermion flavors within each direction. A Gaussian diagonalization cannot discard their spin constraint or their charged operators.

This material also admits finite autonomous record-controlled operation. Extend its Hamiltonian H_K trivially over finite degenerate record registers. For a bounded self-adjoint permitted operator C and real $h\in C_c^\infty$, set

$$A=\int h(s)e^{isH_K}Ce^{-isH_K}ds,\qquad V=e^{i\ell A}.$$

Integration by parts gives $\|[H_K,A]\|\le\|h'\|_1\|C\|$ and $\|[H_K,[H_K,A]]\|\le\|h''\|_1\|C\|$. The energy-change estimates preceding Proposition 52 now apply with the *interacting material* H_K, so finite products meet Proposition 49. Record-controlled C are finite block sums of the permitted operators; all previous reference systems remain in its input purification.

For $0\le g\in C^\infty(S^1)$, weighting the regulated coupled-sector energy density above by g gives a positive form Q_g with $0\le Q_g\le\|g\|_\infty H_K$. It is a closable sum of the same weighted differential and multiplication squares. Its actual expectation during autonomous execution obeys Eq. (114), with $a=\|g\|_\infty$ and $d=0$. This supplies a state and energy certificate on the charged material itself, without a hard physical energy cutoff. It concerns the regulated energy test: a renormalized covariant stress, spatially local history clock and interacting thermal preparation need their own estimates. Gate permission and the finite-rms initial state are specified inputs, not products of this smoothing argument.

5.1 A finite material mode and its remainder

The finite mechanical coordinates need not arise from fields uniform over the whole line. A local binding potential supplies normalizable angle and storage modes, with the discarded modes still present. Here we specify a fixed-texture Gaussian realization; its state, source

and thermal estimates are not an equivalence theorem for the preceding nonlinear charged regulator. In normalized fields $u = (\sqrt{I}\theta, \sqrt{J}y)$, let

$$A_{\mathrm{m}} = \begin{pmatrix} k/I & -k/\sqrt{IJ} \\ -k/\sqrt{IJ} & (k+K)/J \end{pmatrix}, \qquad \alpha = \lambda_{\min}(A_{\mathrm{m}}) > 0.$$

Choose $0 < s < 1$ and $\mu > 0$ with $\alpha > s\mu^2$. The material operator

$$L_{\mathrm{m}} = -\partial_x^2 + A_{\mathrm{m}} + s^2\mu^2 I - s(s+1)\mu^2 \operatorname{sech}^2(\mu x) I$$

has a strictly positive pointwise mass matrix. For a supplied kink $Z_0 = z_0 \tanh(\mu x)$, the well is $1 - Z_0^2/z_0^2$. Using that polynomial in the material potential keeps it nonnegative for all real Z, since $1 - Z^2/z_0^2 \le 1$. The well can therefore belong to the same positive local action; the kink's preparation and position fluctuations remain physical inputs.

Proposition 7 (Finite modes and their reaction bound)**.** *The normalized $b(x) = N \operatorname{sech}^s(\mu x)$ gives two material modes with squared-frequency matrix A_{m}. The orthogonal material space has gap $m_{\mathrm{gap}}^2 = \alpha + s^2\mu^2$. For compact real $a(x)$, the complete Gaussian form is*

$$q[\phi, u] = \|\phi' + a u_\Theta\|^2 + \langle u, L_{\mathrm{m}} u\rangle.$$

Retaining the carrier and $u = b q_{\mathrm{m}}$ gives

$$q_P[\phi, q_{\mathrm{m}}] = \|\phi' + ab q_\Theta\|^2 + q_{\mathrm{m}}^T A_{\mathrm{m}} q_{\mathrm{m}}.$$

For $z < m_{\mathrm{gap}}^2$, elimination of the orthogonal material space has self-energy $\Sigma(z) \ge 0$, with

$$\langle \chi, \Sigma(z)\chi\rangle \le \frac{\|Q(a\phi' + a^2 b q_\Theta, 0)\|^2}{m_{\mathrm{gap}}^2 - z}.$$

Here Q projects off b in both material components, and the exact retained operator is $K_{PP} - \Sigma(z)$.

Proof. For $D_s = \partial_x + s\mu \tanh(\mu x)$, $L_{\mathrm{m}} = D_s^* D_s + A_{\mathrm{m}}$ and $D_s b = 0$. Its partner is

$$D_s D_s^* = -\partial_x^2 + s^2\mu^2 + s(1-s)\mu^2 \operatorname{sech}^2(\mu x) \ge s^2\mu^2.$$

The nonzero spectra of the two products agree; thus b is the only spatial bound profile, and its orthogonal space has the stated gap. The two retained frequencies need not both lie below that gap. Normalization gives $N^2 = \mu\Gamma(s+1/2)/(\sqrt{\pi}\Gamma(s))$ and $\|b'\|^2 = s^2\mu^2/(2s+1)$, so its gradient cost is finite and retained in A_{m}. Projection gives q_P exactly. The omitted block is $K_{QQ} = Q[L_{\mathrm{m}} + \operatorname{diag}(a^2, 0)]Q \ge m_{\mathrm{gap}}^2 Q$, and $K_{QP}\chi = Q(a\phi' + a^2 b q_\Theta, 0)$. Solving its equation gives $\Sigma(z) = K_{PQ}(K_{QQ} - z)^{-1} K_{QP}$ and the bound. □

The projected contact has profile ab and its complete square. The original k, K, I, J are recovered, rather than fitted after the projection. The remaining material field has not been reset or removed. For a response bound put

$$d_{\mathrm{eff}} = \|ab\|^2, \qquad \beta_a = 4\|a\|_\infty^2(1 + d_{\mathrm{eff}}/\alpha) + 2\|a^2 b\|^2/\alpha.$$

The inequality $\|\phi'\|^2 \le 2(1 + d_{\text{eff}}/\alpha)q_P$ yields $\Sigma(z) \le \beta_a K_{PP}/(m_{\text{gap}}^2 - z)$ as forms. For $\sigma > 0$, let $\epsilon = \beta_a/(m_{\text{gap}}^2 + \sigma^2) < 1$, $R_{\text{ex}} = P(K + \sigma^2)^{-1}P$ on the retained space and $R_P = (K_{PP}+\sigma^2)^{-1}$. Normalizing the Schur complement by $K_{PP}+\sigma^2$ gives $I-D$, $0 \le D \le \epsilon I$, hence

$$\|R_{\text{ex}} - R_P\| \le \frac{\epsilon}{(1-\epsilon)\sigma^2}.$$

This certifies the exponentially weighted causal response: $\int_0^\infty e^{-\sigma t}\sin(t\sqrt{K})/\sqrt{K}\,dt = (K + \sigma^2)^{-1}$.

For finite-time linear-response forcing $F(t) = K_{QP}\chi(t)$ and zero omitted response displacement and velocity, its exact reaction is

$$u_Q(t) = -\int_0^t K_{QQ}^{-1/2}\sin((t-r)\sqrt{K_{QQ}})F(r)\,dr, \qquad \sqrt{2E_Q(t)} \le \int_0^t \|F(r)\|dr.$$

Here E_Q is the energy of that response, not its vacuum fluctuations. The force uses the actual retained history, including backreaction. An initial omitted excitation adds its homogeneous evolution. Near the material threshold, or near a retained resonance, a small projection alone does not bound unfiltered scattering or rapid gates. The full local Gaussian propagator retains its light cone; a single collective coordinate generally does not preserve that sharp front. The estimates concern linear response and omitted energy, not a complete quantum-state or interacting thermal equivalence. They give a finite-body reduction with an explicit remainder while the physical clock, nonlinear recoil and thermal renewal are still to be matched.

5.2 A local record drive on the same quantum material

A finite record can drive the material without replacing its vacuum by that of the projected oscillator. Keep the complete Gaussian operator K of Section 5.1, let $\omega = \sqrt{K}$ and $H = d\Gamma(\omega)$, and use one common material vacuum calibration. For a real material smearing $h = (0, h_\Theta, h_Y)$ define

$$Q(h) = \frac{a(\omega^{-1/2}h) + a^*(\omega^{-1/2}h)}{\sqrt{2}}.$$

The bound $q \ge \alpha\|u\|^2$ gives $\langle h, K^{-1}h\rangle \le \|h\|^2/\alpha$ by the inverse-form variational formula. Spectral Cauchy–Schwarz then gives $\|K^{-1/4}h\|^2 \le \|h\|^2/\sqrt{\alpha}$. Thus these material smearings are defined even when the carrier has no mass gap. No static infrared dressing unitary is assumed.

Let $f \in C_c^\infty(0,T)$ be real and $F_1 = \int |f|$. An acquired classical bit $r \in \{0,1\}$ selects the drive $-rf(t)Q(h)$. In the interaction picture its exact unitary is

$$U_{I,r} = e^{ir^2\Phi_h}W(rz_h), \qquad z_h = \frac{i}{\sqrt{2}}\omega^{-1/2}\int_0^T f(t)e^{it\omega}h\,dt,$$
$$\Phi_h = \frac{1}{2}\int_{s>t} f(s)f(t)\langle h, \omega^{-1}\sin((s-t)\omega)h\rangle\,dt\,ds. \tag{12}$$

The commutator is a scalar, so the Magnus expansion stops here. The Weyl convention and coherent-state overlap are standard [6]. In the Schrödinger picture the mean displacement is the full retarded response

$$\xi_h(t) = r \int_0^t \frac{\sin((t-s)\omega)}{\omega} h f(s)\, ds.$$

For centred Gaussian input, its added energy is

$$E_h = \|\omega^{1/2} r z_h\|^2 = \frac{r^2}{2} \left\| \int f(t) e^{it\omega} h \, dt \right\|^2 \le \frac{r^2 F_1^2}{2} \|h\|^2.$$

It equals the actual force work $\int r f \langle h, \dot{\xi}_h \rangle dt$. The instantaneous forced form is bounded below by $-r^2 f(t)^2 \langle h, K^{-1} h \rangle / 2$. Adding the opposite scalar makes it nonnegative; that offset is not net work, heat or a construction of the driver.

To address the angle mode, choose smooth $0 \le \chi_R \le 1$, equal to one on $|x| \le R$ and zero outside a finite larger interval. Set

$$A_R = \int b^2 \chi_R, \quad h_R = (0, b\chi_R / A_R, 0), \quad h_b = (0, b, 0), \quad d_R = \|h_R - h_b\|.$$

Both have retained force $\langle b, h_\Theta \rangle = 1$. Only h_R is the permitted compact drive; h_b is a comparison smearing in the same full model. If $t_R = \int_{|x|>R} b^2$, then $d_R^2 \le t_R/(1-t_R)$ and $\|h_R\|^2 \le (1-t_R)^{-1}$. For $s = 1/2$, $t_R = 1 - (2/\pi)\arctan(\sinh(\mu R))$. Reducing this tail grows the actual drive support.

Proposition 8 (Local actuation with the initial state retained)**.** *Let $\Delta z = z_{h_R} - z_{h_b}$, with both drives evolved under full K. Then*

$$\|\Delta z\|^2 \le \frac{F_1^2 d_R^2}{2\sqrt{\alpha}}, \qquad \Delta E := \|\omega^{1/2} \Delta z\|^2 \le \frac{F_1^2 d_R^2}{2}.$$

For an actual initial field marginal σ_r with its reference retained, the half trace distance of the two complete outputs obeys

$$D_r \le \sqrt{1 - |\operatorname{Tr} \sigma_r W(r\Delta z)|^2}.$$

A classical distribution of records gives joint distance at most $\sum_r p_r D_r$. On the same Gaussian ground state this becomes $\sqrt{1 - e^{-r^2\|\Delta z\|^2}}$. On a finite Dirichlet interval, for the actual full-K Gibbs state $\sigma_\beta = e^{-\beta H}/Z$, the two displaced field states satisfy

$$S(\sigma_{h_R} \| \sigma_{h_b}) = \beta r^2 \Delta E, \qquad D_{\mathrm{field},r} \le \sqrt{\beta r^2 \Delta E / 2}.$$

Proof. The bound on $K^{-1/4}$ and the spectral integral in Eq. (12) prove the first two inequalities. Writing $B_R = \int b^2 \chi_R^2 \le A_R$ gives $d_R^2 = B_R/A_R^2 - 1 \le A_R^{-1} - 1 \le t_R/(1-t_R)$. Purify the actual field and reference state. The overlap of the two outputs is its Weyl characteristic function, up to a scalar phase. The pure-state trace-distance formula and partial-trace contraction give the retained-reference bound. For the Gibbs identity, conjugate both outputs by $W(-r z_{h_b})$. Displacement preserves entropy, $\log \sigma_\beta = -\beta H - \log Z$, and the initial mean is zero. Its energy increment is $r^2 \Delta E$; Pinsker gives the field-only bound. □

The Gibbs energy bound concerns the reduced field states. A retained thermal purification still uses the characteristic-function bound; its covariance can make that distance larger. Here ΔE is the energy of the difference response, not $E_{h_R} - E_{h_b}$, which has a cross term. No independent ground state of K_{PP} is substituted. Its dynamic reduction still has the reaction error of Proposition 7. An actual old field state must use its own characteristic function; initial correlation is not replaced by a fresh thermal packet. Coherent record controls also retain the Magnus and Weyl phases and are not covered by the classical-record distance bound.

For a centred Gaussian input the covariance is unchanged. A quadratic source, renormalized with the same fixed vacuum subtraction in both experiments, therefore changes by the classical value $T[\xi_h]$. Nonzero initial mean adds its bilinear cross term. The local energy balance has force input $rf(t)h \cdot \dot{\xi}_u$; the driver must supply that work. It uses the complete response, including omitted material modes, not merely the isolated mode's energy. This is the fixed-texture Gaussian source, not the full moving quantum apparatus's stress. The emitted carrier is determined by that material motion. Its equation is $(\partial_t^2 - \partial_x^2)\xi_\phi = \partial_x(a\xi_\Theta)$. For zero past displacement, outside the compact contact the right and left outgoing profiles are

$$\xi_R(u) = -\tfrac{1}{2}\int a(y)\xi_\Theta(y, u+y)dy, \qquad \xi_L(v) = \tfrac{1}{2}\int a(y)\xi_\Theta(y, v-y)dy, \quad u = t-x,\ v = t+x.$$

Integrating the one-dimensional retarded Green function against $\partial_y(a\xi_\Theta)$ gives these endpoint integrals directly. The extra stresses of centred Gaussian input are $(\xi_R')^2$ and $(\xi_L')^2$. They use the complete material response, not an independent emitter. With no incoming drive-induced radiation, the total emitted carrier energy is at most the force work E_h; the remaining material energy is retained. A zero contact $a = 0$ emits no carrier source despite possible stored material energy. Opposite drives also have equal quadratic stress on centred input, so a command label alone does not identify the emitted source.

A finite menu of these record-selected unitaries admits the bounded prepared-bit dilation of Appendix I.1. For compact smooth h_R on the Gaussian ground state, its Weyl amplitudes have all positive frequency moments, so the required twice differentiable H graph orbits exist. Smoothing and autonomous return then use this same H. Other initial states need their own graph-moment certificate. The extra bit, clock, battery, program and retained outputs remain resources. The record must reach the entire drive support by physical transport before the operation; the dilation does not prove that transport or the clock's covariant stress. Approximating a smooth local source during autonomous execution also needs its relative source bound, not trace distance alone. The following estimate provides it for finite-interval output readings of this material.

5.3 A source certificate for the quantum material

Fix a Dirichlet interval of length L and keep all modes of the complete Gaussian K. Write its pointwise mass matrix as $mI \le M(x) \le M_{\max}I$, $m > 0$. For $X = (\phi, \Theta, Y)$ its spatial energy form is

$$q[X] = \|\phi' + a\Theta\|^2 + \|u'\|^2 + \langle u, Mu\rangle, \qquad u = (\Theta, Y).$$

With $C_g = 2 + 2\|a\|_\infty^2/m$ and $V = 2\|a\|_\infty^2 + M_{\max}$, Cauchy–Schwarz gives $\|X'\|^2 \le C_g q[X]$ and $q[X] \le 2\|X'\|^2 + V\|X\|^2$. Min–max on three Dirichlet components bounds the ordered frequencies:

$$c_- j \le \omega_j \le c_+ j, \qquad c_- = \frac{\pi}{3L\sqrt{C_g}}, \quad c_+ = \sqrt{2\pi^2/L^2 + V}.$$

The full tower, not a finite Fock cutoff, enters the next statement.

Let $\mathcal{Y} = (\Pi, \phi' + a\Theta, u', \sqrt{M}u)$ be the energy-factor fields, so their squared norm divided by two is the energy density. For a real symmetric bounded spatial matrix $B(x)$, $\|B(x)\| \le G$, and real $h \in C_c^\infty$, set

$$O = \frac{1}{2}\int h(t) \int : \mathcal{Y}(t,x)^T B(x) \mathcal{Y}(t,x) : \, dx\, dt.$$

Normal ordering uses the same K vacuum in both compared states. $B = gI$ tests the local energy. In a free exterior carrier region a rank-one choice tests $g(\Pi_\phi \mp \phi')^2/4$, the corresponding null stress. These tests may have negative quantum expectations.

Proposition 9 (A same-Hamiltonian quadratic-source bound)**.** *Put $H = d\Gamma(\omega)$, $H_N = \|h^{(N)}\|_1$, $N \ge 2$. The source is a Hermitian form on* $\operatorname{Dom} H^{1/2}$*,*

$$O = d\Gamma(A) + P + P^*, \qquad P = \sum_{ij} b_{ij} a_i a_j.$$

It obeys $|d\Gamma(A)| \le a_0 H$, $a_0 = G\|h\|_1$. Define

$$\kappa^2 = \sum_{ij} \frac{|b_{ij}|^2}{\omega_i \omega_j}, \qquad \nu^2 = \sum_{ij} \frac{|b_{ij}|^2}{\omega_i}.$$

Both converge, with the explicit sufficient bounds

$$\kappa \le \frac{G H_N}{2c_-^N} \sqrt{\zeta(2N-1) - \zeta(2N)},$$
$$\nu^2 \le \frac{G^2 H_N^2 c_+}{8 c_-^{2N}} [\zeta(2N-2) - \zeta(2N-1)]. \tag{13}$$

For any reference energy $\Lambda > 0$, let $C_O = a_0 + 2\kappa + \nu/\sqrt{\Lambda}$. Then $(H+\Lambda)^{-1/2} O (H+\Lambda)^{-1/2}$ is a bounded Hermitian form of norm at most C_O.

Proof. Choose real eigenmodes e_j of K. If $L_s^* L_s = K$ is its spatial energy factor, the annihilation coefficients of $\mathcal{Y}$ are

$$v_j = (-i\sqrt{\omega_j/2}\, e_j, L_s e_j/\sqrt{2\omega_j}), \qquad \langle v_i, v_j \rangle = \omega_j \delta_{ij}.$$

Spatial multiplication by B is bounded by G on this factor space. Time evolution preserves the energy norm; integration against h proves the number-form estimate. The pair coefficient is $b_{ij} = \frac{1}{2}\int v_i^T B v_j dx\, \widehat{h}(-\omega_i - \omega_j)$, where $\widehat{h}(\xi) = \int h(t) e^{i\xi t} dt$. Integrating N times by parts gives

$$|b_{ij}| \le \frac{G\sqrt{\omega_i \omega_j} H_N}{2(\omega_i + \omega_j)^N}.$$

Summing over $i + j = k$ uses $k - 1$ pairs and $\sum_{j=1}^{k-1} j = k(k-1)/2$; the frequency bounds give Eq. (13).

For a unit finite-particle spectral vector of energy E, Cauchy–Schwarz applied to $\sqrt{\omega_i}a_i\psi$ and $\sum_j \overline{b_{ij}}a_j^*\psi/\sqrt{\omega_i}$ gives

$$|\langle P\rangle| \le \sqrt{E(\kappa^2 E + \nu^2)} \le \kappa E + \nu\sqrt{E}.$$

The creation-vector square is a vacuum term ν^2 plus a number form bounded by $\kappa^2 H$: the matrix $\omega^{-1/2} b \omega^{-1/2}$ has Hilbert–Schmidt norm κ. Thus $|\langle O\rangle| \le (a_0 + 2\kappa)E + 2\nu\sqrt{E} \le C_O(E + \Lambda)$. Pair tails converge in these two sums; the uniformly bounded number forms converge on the finite-energy core. Their normalized forms therefore extend to the stated bounded Hermitian form. □

This estimate concerns a spacetime-smeared source form, not an unsmeared density. Its constants depend on L; they do not prove a uniform real-line limit. Compact $W^{2,1}$ time tests also satisfy the $N = 2$ proof. No free chiral energy inequality has been assigned to the coupled material.

Corollary 10 (Output source error with references retained)**.** *For normalized actual and target vectors $\psi, \varphi \in \operatorname{Dom} H$, including their actual references, let $e = \|\psi - \varphi\|$ and $g_H = \|H(\psi - \varphi)\|$. Then*

$$\begin{aligned} |\langle O\rangle_\psi - \langle O\rangle_\varphi| &\le C_O\sqrt{\Lambda e^2 + e g_H} \\ &\quad \times \left[\sqrt{\Lambda + \|H\psi\|} + \sqrt{\Lambda + \|H\varphi\|}\right]. \end{aligned} \tag{14}$$

Proof. Tensor the bounded normalized form with the reference identity and polarize. Its difference is bounded by C_O times $\|(H + \Lambda)^{1/2}(\psi - \varphi)\|$ times the sum of the two weighted norms. The first squared norm is at most $\Lambda e^2 + e g_H$; each remaining squared norm is at most $\Lambda + \|H\psi\|$ or $\Lambda + \|H\varphi\|$. □

The energy-regular return theorem supplies e, g_H for a specified same-H graph-domain operation. It consequently certifies these output source readings under common subsequent H evolution, not the missing driver or clock stress during its interaction. In fact its uniform physical rms bounds M_a, M_i suffice: $g_H \le M_a + M_i$ in Eq. (14). The resulting source error tends to zero with the autonomous state error even as the history clock grows; its physical rms bounds do not grow with that clock. This does not add a permanent halt or decouple an always-running controller for free. A mixed input uses its retained purification. On this finite interval a full-K Gibbs purification has finite energy moments: with $q_\beta = e^{-\beta c_-}$,

$$\langle H\rangle \le \frac{c_+ q_\beta}{(1 - q_\beta)^3}, \qquad \operatorname{Var} H \le \frac{c_+^2 q_\beta(1 + q_\beta)}{(1 - q_\beta)^5}.$$

The bosonic product partition function converges. Bounding occupations by $q_\beta^j/(1 - q_\beta)$ and summing $j q_\beta^j$ and $j^2 q_\beta^j$ proves the two estimates. All higher polynomial moments are finite as well. This does not prepare the Gibbs state, prove a particular gate's graph orbit, or renew a correlated reservoir.

For the vacuum-driven outputs $|z\rangle, |w\rangle$ of two actual forces on this same material, put $c = e^{-\|z-w\|^2/2}$, $E_z = \|\omega^{1/2} z\|^2$ and $\Delta E = \|\omega^{1/2}(z-w)\|^2$. Choosing only a comparison phase to make their overlap c gives the exact identity

$$\|(H+\Lambda)^{1/2}(|z\rangle - |w\rangle)\|^2 = c\Delta E + (1-c)(E_z + E_w + 2\Lambda).$$

Indeed $\langle z|H|w\rangle = \langle z|w\rangle\langle z, \omega w\rangle$. Their source difference is at most C_O times the square root of this expression times $\sqrt{E_z+\Lambda} + \sqrt{E_w+\Lambda}$. Proposition 8 bounds its two difference terms for compact forces. This comparison keeps full K; changing the box or using the projected oscillator changes the problem. For coherently controlled records the actual relative phases remain, rather than being physically aligned branch by branch.

Trace distance alone cannot replace these estimates: a trace-small admixture can have arbitrarily large energy. Nor does matching mean energy suffice. A small vacuum–two-particle amplitude changes a pair-source reading to first order while its energy changes to second order. The source certificate retains both the state and its energy norm. Matching this finite-interval Gaussian readout to a charged nonlinear continuum and the complete moving apparatus remains a separate physical construction.

6 Acquired quantum records on the complete material

The preceding force calculation conserved a classical record during its drive. An acquired record need not be classical in the full old history. Keep the complete finite-interval K and $H_F = d\Gamma(\omega)$, $\omega = \sqrt{K}$, of Section 5.3. Let the finite controller include probes, record cells and program cursor. Its nonnegative Hamiltonian H_D need not commute with the force amplitudes. The following construction therefore includes acquisition and a later record-controlled response within one autonomous dynamics.

6.1 A positive common Hamiltonian and a uniform physical moment

Let B_a be finitely many commuting bounded self-adjoint controller operators and h_a real smooth compact material smearings. Define

$$F = \sum_a B_a h_a, \qquad C_{\min} = \frac{1}{2}\sum_{ab} B_a B_b \langle h_a, K^{-1} h_b\rangle,$$
$$H_{\mathrm{tot}} = H_D + H_F - \sum_a B_a Q(h_a) + C_{\min}. \tag{15}$$

The full canonical material fields obey $[Q(h), P(k)] = i\langle h, k\rangle$, with $Q(h) = [a(\omega^{-1/2}h) + a^*(\omega^{-1/2}h)]/\sqrt{2}$. This is the canonical normalization of the material Hamiltonian, distinct from the earlier dimensionless $[b+b^\dagger, i(b^\dagger - b)] = 2i$ convention. The amplitudes commute at equal Schrödinger time, not necessarily at different Heisenberg times. In particular a cursor-controlled record force and a probe force can commute while acquisition edges in H_D change those same records.

Proposition 11 (Positive quantum interface and physical rms bound)**.** *The Hamiltonian* (15) *is self-adjoint and nonnegative. Its domain is equivalent to* $\operatorname{Dom} H_F$*. For a normalized initial vector* ψ *in that domain, including all old references, put*

$$L_F = \sqrt{2}\sum_a \|B_a\|\|\omega^{-1}h_a\|, \qquad c_F = \frac{1}{\sqrt{2}}\sum_a \|B_a\|\|\omega^{-1/2}h_a\|,$$

$$A_F = \|H_{\rm tot}\psi\| + \|H_D\| + \|C_{\rm min}\| + c_F, \qquad M_F = \left[\frac{L_F + \sqrt{L_F^2 + 4A_F}}{2}\right]^2.$$

Then $\|H_F e^{-itH_{\rm tot}}\psi\| \le M_F$ *for every* t*. If the initial cursor lies in a force-idle region and* $\|H_D\|$*, coupling norms and initial physical rms stay fixed, this certificate does not grow with the finite program length.*

Proof. Resolve the common spectrum of the B_a into projections P_b and write $f_b = \sum_a b_a h_a$, $z_b = -\omega^{-3/2} f_b/\sqrt{2}$. The finite interval has a gap; hence the finite controlled Weyl unitary $D = \sum_b P_b \otimes W(z_b)$ exists and $D^* H_F D = H_F - \sum B_a Q(h_a) + C_{\rm min}$. It follows that $H_{\rm tot} = D^* H_F D + H_D \ge 0$. The linear field bound

$$\|Q(h)\zeta\| \le \sqrt{2}\|\omega^{-1}h\|\|H_F^{1/2}\zeta\| + \|\omega^{-1/2}h\|\|\zeta\|/\sqrt{2}$$

also makes the force infinitesimally H_F-bounded, giving the graph domain statement by the closed operator realization. For $\psi_t = e^{-itH_{\rm tot}}\psi$ and $x = \|H_F\psi_t\|$, conservation of $\|H_{\rm tot}\psi_t\|$ gives $x \le A_F + L_F\sqrt{x}$, since $\langle H_F\rangle_t \le x$. Solving this quadratic inequality proves the bound. Spectral-core approximation extends the calculation to the stated domain. In a force-idle initial region, $\|H_{\rm tot}\psi\| \le \|H_F\psi\| + \|H_D\|$. Every step tensors with the actual reference identity. □

The displacement D proves positivity; it is not a free preparation of a dressed vacuum. This finite-interval argument is not an infinite-line static-dressing theorem. A positive nearest-neighbor Feynman cursor with bounded edge gates can have $\|H_D\| \le 4J + \|H_{\rm data}\|$ independently of its site count [4]. This removes a program-length factor from the physical moment bound, not from cursor space, execution time or preparation cost. For the uniform-J cursor, Foundation III's lemma "Wholly idle input and a finite completion window" in the appendix "A Finite Arrival Bound at Fixed Hopping Energy" bounds arrival for a packet supported wholly in the input idle region of the pure gate-dressed hopping comparison [50]. Cell gaps and forces require the subsequent Duhamel control; this is neither the spin-clock nor the spatial work-clock estimate below. Linear finite-system/field Hamiltonians and their domain analysis are established tools [5]; the issue here is their common retained-record interface.

The field obeys the operator retarded equation

$$Q(t) = \cos(\omega t)Q(0) + \omega^{-1}\sin(\omega t)P(0) + \int_0^t \omega^{-1}\sin[\omega(t-s)]F(s)\,ds. \tag{16}$$

It follows by solving $\dot{Q} = P$, $\dot{P} = -KQ + F(t)$ as weak smeared identities on the energy core. The $B_a(s)$ in $F(s)$ are the actual joint Heisenberg operators. Their expectations are not an

externally supplied force history. With material-only force, the exterior carrier formula of Section 5.2 holds with the complete operator $\Delta\Theta$:

$$\Delta\phi_R(u) = -\tfrac{1}{2}\int a(y)\Delta\Theta(y, u+y)\,dy, \qquad \Delta\phi_L(v) = \tfrac{1}{2}\int a(y)\Delta\Theta(y, v-y)\,dy.$$

An additional carrier-smeared probe contributes its own retarded source. Quadratic stress uses correlations of these operators, not only the square of their mean.

6.2 Actual acquisition, sensing and the next action

For a probe P and a blank record R, the unitary $C_Y = P_-^Y \otimes I_R + P_+^Y \otimes X_R$ acquires the probe's Y reading. Retain P, R and their old references. During a subsequent block that conserves R, the force of Section 5.2 acts as $V = \sum_{r=0,1} |r\rangle\langle r| \otimes e^{ir^2\Phi_h} W(rz_h)$. For actual initial material/reference blocks $\sigma_{rr'}$, the output is

$$\sum_{rr'} |r\rangle\langle r'| \otimes e^{i(r^2-r'^2)\Phi_h} W(rz_h)\sigma_{rr'}W(r'z_h)^*. \tag{17}$$

This is not a sampled classical force. For a pure complete input with both record branches present, dephasing R changes the full retained state by half trace distance $\sqrt{p(1-p)}$, where $p = \Pr(R=1)$. Indeed the deleted off-diagonal block connects two orthogonal vectors of norms $\sqrt{p}, \sqrt{1-p}$ and has two eigenvalues $\pm\sqrt{p(1-p)}$.

Now let a fresh sensor S start in $|+x\rangle$, couple by $e^{igZ_S Q(b)}$ to the material after its actual propagation, and acquire its Y value into a cell E. The latter starts with thermal occupation ν, independently prepared with its purifier retained. For a centered Gaussian material initially independent of the first probe, let μ_r be the actual conditional mean of $Q(b)$ and V_b its conditional variance after the force. Then

$$\Pr(E=1 \mid R=r) = \frac{1}{2} - \frac{1-2\nu}{2} e^{-2g^2V_b}\sin(2g\mu_r). \tag{18}$$

The sensor's Y mean is $-\operatorname{Im}\langle e^{2igQ(b)}\rangle$; the Gaussian characteristic function and the mixed cell's visibility $1-2\nu$ give the formula. The acquisition acts on the same disturbed material, and the actually acquired n_E can control the next force. Its energy and source are part of that dynamics. For a correlated or non-Gaussian input use its actual characteristic function, not this Gaussian specialization.

Before sensing, a conserved source record and centered product input give $\langle\Delta j\rangle = pj_h$ but $\Delta\langle{:}\, j^2 \,{:}\rangle = pj_h^2$, where j_h is the complete driven carrier current. The connected contribution is $p(1-p)j_h^2$, not an extra energy charge. Similarly the full added material/carrier energy is pE_h, of which p^2E_h is the energy of the mean displacement. Record dephasing leaves this immediate field marginal unchanged while changing the old joint history. Equal mean source therefore does not establish equal later control.

The formulas describe the sequential block. Under the autonomous Hamiltonian (15), acquisition, force and sensing instead occupy cursor regions that a dispersive cursor can revisit. Its deadline statistics require the actual joint evolution, without postselection or assignment of the sequential probabilities.

A one-mode check retains the old probe reference and the second cell's purifier: $p = .2766071$, $\Pr(E = 1 \mid R = 1) = .4267590$, dephasing distance .4473205, and first-acquisition probe/reference change below 4.1×10^{-16}. Cutoffs $6, 10, 14$ reduce Gaussian-law error from 1.04×10^{-7} to 1.11×10^{-16}. Field energy .0418368 splits into mean .0115724 and connected .0302645; sensing adds .04805 $= g^2/2$ at $g = .31$. This checks the sequence, not the full spatial K theorem.

A separate 31-site positive cursor diagnostic executes actual first acquisition, record force, sensing, second acquisition, receiver/bank SWAP and a later force controlled by the spent cell's actual record. With both cell purifiers retained, its unpostselected deadline $t = 40$ has field energy .0300724, physical rms at most .186211 against the uniform bound 3.01330, and energy residual below 1.9×10^{-13}. But only .857524 of its cursor probability is after the SWAP region; its receiver occupation is .157712, exceeding the prepared bank value .13 by .027712. Unfinished and returning cursor branches retain pre-SWAP data and can revisit acquisition and force regions. The program therefore fails an exact deadline-renewal claim. The force-idle window condition and its completion error must be certified separately, as done below; the sequential probability formula is not assigned to this autonomous output.

6.3 The local field source during interaction

The free-output qualification in Corollary 10 is unnecessary for a specified smooth test under the same H_{tot}. Let $B(x)$ be a real symmetric smooth matrix, supported strictly inside the Dirichlet interval, with $\|B\|_\infty = G$; let $w \in C_c^\infty$ be real and supported in $[0, T]$. Use the same free-K vacuum subtraction and energy factors $\mathcal{Y} = (P, L_s Q)$, $L_s^* L_s = K$. Define

$$O_{\mathrm{int}} = \int w(t) \int [\mathcal{Y}(t,x)^T B(x) \mathcal{Y}(t,x)/2 - \text{free vacuum}_K]\, dx\, dt.$$

This is the material/carrier field source, not the controller's stress. By (16),

$$\mathcal{Y}(t) = \mathcal{Y}_0(t) + \Delta\mathcal{Y}(t), \qquad \Delta\mathcal{Y}(t) = \sum_a \int_0^t M_{t-s} h_a B_a(s)\, ds,$$

where $M_\tau h = (\cos(\omega\tau)h, L_s \omega^{-1} \sin(\omega\tau) h)$ and $\|M_\tau h\| = \|h\|$. As a vector of joint-state operators, $\|\Delta\mathcal{Y}(t)\psi\| \le W(t)\|\psi\|$, $W(t) = t \sum_a \|B_a\| \|h_a\|$.

Proposition 12 (An interacting-window field-source bound)**.** *Assume smooth coefficients of K and the smooth compact h_a above. For each a, define finite constants*

$$D_{P,a} = \sup_{0 \le \tau \le T} \|\omega^2 [B M_\tau h_a]_P\|,$$
$$D_{S,a} = \sup_{0 \le \tau \le T} \|\omega^2 L_s^* [B M_\tau h_a]_S\|.$$

With $\omega_i \ge c_i$ as in Section 5.3, set

$$C_0(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\| [D_{P,a} c_-^{-3/2} \zeta(3/2) + D_{S,a} c_-^{-5/2} \zeta(5/2)],$$
$$C_1(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\| [D_{P,a} c_-^{-2} \zeta(2) + D_{S,a} c_-^{-3} \zeta(3)].$$

For any $\Lambda > 0$, O_{int} is a Hermitian form with

$$\|(H_F + \Lambda)^{-1/2} O_{\mathrm{int}} (H_F + \Lambda)^{-1/2}\| \le C_{\mathrm{int}}, \tag{19}$$

where

$$C_{\mathrm{int}} = C_O + \int |w(t)| \left[\frac{C_1(t)}{\sqrt{\Lambda}} + \frac{C_0(t) + GW(t)^2/2}{\Lambda} \right] dt.$$

Here C_O is the free source constant with time test w. All references and the finite controller are retained. For two initial joint vectors of rms at most M_F and vector distance e, their actual interacting-window source difference is at most

$$2C_{\mathrm{int}} \sqrt{\Lambda e^2 + 2M_F e} \sqrt{\Lambda + M_F}. \tag{20}$$

Proof. The free part is Proposition 9. The response-square term is bounded by $GW(t)^2/2$ without assuming commuting response components. For the symmetrized cross term, let $v_i(t,x)$ be the free annihilation coefficient of mode i and $C_i(t) = \int v_i(t,x)^T B(x) \Delta\mathcal{Y}(t,x)\, dx$. Projection of the two response components onto e_i and insertion of ω^2 give

$$\|C_i(t)\| \le \frac{t}{\sqrt{2}} \sum_a \|B_a\| [D_{P,a} \omega_i^{-3/2} + D_{S,a} \omega_i^{-5/2}].$$

Interior multiplication by smooth B removes boundary compatibility conditions. Iterating the elliptic operator $K = -I\partial_x^2 +$ smooth lower-order terms controls the necessary Sobolev orders of propagated compact h_a, proving finiteness of the displayed graph norms uniformly over the compact time interval. Thus $\sum_i \|C_i(t)\| \le C_0(t)$ and $\sum_i \|C_i(t)\|/\sqrt{\omega_i} \le C_1(t)$.

For $E = \langle H_F \rangle_\psi$, $\|a_i \psi\| \le \sqrt{E/\omega_i}$ and $\|a_i^* \psi\| \le \sqrt{E/\omega_i} + 1$. Bounding $\langle a_i C_i \rangle$ by $\|a_i^* \psi\| \|C_i \psi\|$, and $\langle a_i^* C_i^* \rangle$ by $\|a_i \psi\| \|C_i^* \psi\|$, bounds the cross expectation by $2C_1(t)\sqrt{E} + C_0(t)$. No commutation between C_i and a_i has been used. The C_0 term is the creation contribution; treating the record as a c-number would incorrectly remove it. Apply $2\sqrt{E} \le (E + \Lambda)/\sqrt{\Lambda}$ and add the free and response bounds. Absolute convergence of the cross terms, the bounded response vector and the established free form give spectral-core convergence. Polarization proves (19). The weighted-form difference estimate of Corollary 10, with $\|H_F(\psi - \varphi)\| \le 2M_F$, proves (20). □

The constants do not depend on cursor site count if its coupling norms stay fixed. They do depend on the box, smooth test, profiles and time window. Rough spatial multipliers, boundary distributions and unsmeared stress are not covered. This result compares retained inputs under one common H_{tot}; it does not compare arbitrary different apparatus Hamiltonians or supply the controller's missing stress.

6.4 A local contact energy and the interacting thermal reference

The inverse-K counterterm in (15) is useful for proof but is not a local physical density. For smooth compact material profiles $w_a(x)$ choose $h_a = (0, M(x) w_a(x))$ and replace it by

$$C_{\mathrm{loc}} = \frac{1}{2} \sum_{ab} B_a B_b \int w_a^T M w_b \, dx.$$

The actual mass-potential density becomes

$$V_{\mathrm{loc}}(x) = \tfrac{1}{2}[u - \sum_a B_a w_a]^T M[u - \sum_a B_a w_a]. \tag{21}$$

All gradient and current squares of the complete material remain. For each joint spectral branch $w = \sum_a b_a w_a$, the inverse-form variational formula gives

$$\langle (0, Mw), K^{-1}(0, Mw)\rangle = \sup_X \{2\langle w, Mu\rangle - q[X]\} \le \langle w, Mw\rangle,$$

because $q[X] \ge \langle u, Mu\rangle$. Consequently $C_{\mathrm{loc}} - C_{\min} \ge 0$ is a bounded controller operator. The actual local-contact Hamiltonian is $D^* H_F D + H'_D$ with $H'_D = H_D + C_{\mathrm{loc}} - C_{\min} \ge 0$. The positivity, rms and source proofs therefore apply to this physical contact, using its actual C_{loc} and the adjusted controller norm. No local stress is assigned to a nonlocal inverse operator.

Proposition 13 (Interacting equilibrium and retained record capacity)**.** *For either contact, put $d = \dim \mathcal{H}_D$, $d_0 = \|H'_D\|$, and $Z_F(\beta) = \operatorname{Tr} e^{-\beta H_F}$. Then the actual interacting Gibbs state exists and*

$$\begin{aligned} d e^{-\beta d_0} Z_F(\beta) \le Z_{\mathrm{tot}}(\beta) \le d Z_F(\beta), \\ \langle H_{\mathrm{tot}}^2 \rangle_\beta \le \frac{16 e^{\beta d_0}}{e^2 \beta^2} \frac{Z_F(\beta/2)}{Z_F(\beta)}. \end{aligned} \tag{22}$$

If independent uniform input bits $X_1, \dots, X_N$ are retained in the accessible controller and material with field mean energy at most M, and each is later readable with error $\epsilon_j \le 1/2$, then

$$\sum_j [\log 2 - h_2(\epsilon_j)] \le \log d + \inf_{\beta > 0} [\beta M + \log Z_F(\beta)]. \tag{23}$$

Any decoder-accessible outside reference is an additional resource.

Proof. The eigenvalues of $D^* H_F D + H'_D$ lie between those of $D^* H_F D$ and those eigenvalues plus d_0, by min–max. Summing scalar exponentials proves the partition bounds; operator monotonicity of the exponential is neither available nor needed. For $x \ge 0$, $x^2 e^{-\beta x/2} \le 16/(e^2\beta^2)$. The partition ratio then proves the moment bound; its controller multiplicity cancels. Subadditivity and the field Gibbs variational inequality give $S(\rho) \le \log d + \beta M + \log Z_F(\beta)$. Binary Fano, conditional entropy subadditivity for the independent classical inputs, and data processing give $\sum_j [\log 2 - h_2(\epsilon_j)] \le I(X^N : \text{apparatus}) \le S(\rho)$. Take the infimum. The frequency bound also yields $\log Z_F(\beta) \le q/(1-q)^2$, $q = e^{-\beta c_-}$, so the capacity is finite for fixed resources. □

The same-H Gibbs purification is a graph-domain input, not the product Gaussian in (18). Stationary equilibrium supplies neither directed cursor nor blank records; $\beta^{-1} D(\rho_{\mathrm{prep}} \| \rho_\beta)$ is a finite running state's free-energy excess, not its preparation protocol. At fixed energy, cursor dimension increases record capacity only logarithmically. Any outside history bank or entropy channel remains a resource.

7 Finite renewal and continuing return

7.1 Replacing a port is not renewing the interacting material

There are three distinct thermal operations below: replacement of a specified finite port; local thermal restriction on a free-carrier window; and preparation of the whole interacting Gibbs state. Only the first two have the stated finite operating protocols.

Proposition 14 (Universal held-content preservation forces a product extension)**.** *Let a channel $\mathcal{E}$ append a finite-dimensional port B to a held d-dimensional system A, whose marginal is the identity channel on every input, including all reference extensions. Then $\mathcal{E}(\rho) = \rho \otimes \tau$ for one fixed port state τ. Thus such a universal nondisturbing preparation cannot reproduce a correlated interacting Gibbs state. For $H = JZ_AZ_B$, $x = \beta J$, its Gibbs state is $\gamma = (I - \tanh(x)Z_AZ_B)/4$. At held input $I/2$ every such extension has half trace distance from γ at least $|\tanh x|/2$, attained by $\tau = I/2$.*

If instead only the classical Z_A label is conserved while conditional port Gibbs states $\tau_\pm = (I \mp \tanh(x)Z_B)/2$ are installed, the largest retained coherence factor is $\operatorname{sech} x$*. After the best held-system phase correction, an input $|+\rangle$ therefore suffers half trace distance at least $(1 - \operatorname{sech} x)/2$.*

Proof. Dilate the channel and purify its fixed ancilla. A pure held input must have a pure unchanged held marginal; hence the isometry sends $|\psi\rangle$ to $|\psi\rangle|e_\psi\rangle$. Applying this to basis states and their superpositions forces one common environment vector. Tracing the environment gives the product extension. Equivalently the identity channel has rank-one Choi state. For the Gibbs example the observable Z_AZ_B has norm one, expectation zero in $(I/2)\otimes\tau$ and $-\tanh x$ in γ, giving the distance bound; direct diagonal subtraction proves attainment. In a Z-nondemolition conditional preparation the two environment/port vectors purify τ_+, τ_-. Their overlap is at most their root fidelity, $\sum_z \sqrt{p_+(z)p_-(z)} = \operatorname{sech} x$, with equal Schmidt bases attaining it. That overlap multiplies the held off-diagonal matrix element. The qubit trace-distance formula proves the last assertion. □

This is a universal identity-channel condition, not a prohibition on preparing one specified state or maintaining a selected classical content. It supplies no universal constant for arbitrary approximate identity channels. An entire-body SWAP can prepare a fresh correlated copy, but it moves the old body's quantum content and correlations to the spent copy; it is not that identity channel.

For an attainable finite update, specify instead one receiver cell E with $H_E = \delta n_E$ and a fresh bank cell B with the same gap, Gibbs state η_β, and retained purifier. All held material, earlier records and old references form $\mathcal{O}$; the previous state of E may be correlated with all of them. The true S_{EB} commutes with $H_E + H_B$. The positive pulse $g(t)(I - S_{EB})$, $g \geq 0$, $\int g = \pi/2$, implements the true SWAP at finite duration, at least $\pi/(2g_{\max})$. At its endpoint E is η_β independently of the old $\mathcal{O}E$ state. That complete old state now belongs to $\mathcal{O}B$, with its correlations retained. The new cell remains correlated with its own Gibbs purifier. The $\mathcal{O}$ marginal is unchanged nonselectively; old records stored in E have moved to the spent slot rather than being erased. For an isolated prescribed pulse its switch work is zero because the bare cell energy is conserved and the pulse vanishes at the endpoints. This is not zero total controller or preparation cost.

Such a SWAP is also a bounded edge gate in H_D of the positive common material Hamiltonian. The cell must be force-idle while it is replaced. If n_E is still coupled to the material, $[S_{EB}, n_E Q(h)] \neq 0$; the matched gap alone gives no exact reset for that interacting contact. A finite-window estimate can retain this imperfection explicitly.

Proposition 15 (Finite-window port renewal with a live material)**.** *Compare the actual autonomous interface with an ideal renewal evolution having the same H_D, matched cell gaps and SWAP edge gate, but no material force in its renewal window $[0, T_R]$. Suppose their difference on the reference trajectory is $-\sum_a \Delta B_a Q(h_a) + \Delta C$, where $\Delta B_a = P_{\rm leak}\Delta B_a$, $\Delta C = P_{\rm leak}\Delta C$ and these controller operators commute with H_F. Here $P_{\rm leak}$ is a controller orthogonal projection; the differences are self-adjoint, so their displayed one-sided support is also two-sided support. Write $b_a = \|\Delta B_a\|$, $c = \|\Delta C\|$, $p(s) = \|P_{\rm leak}\psi_s^i\|^2$, and assume $\|H_F\psi_s^i\| \leq M$. Define*

$$L_\Delta = \sqrt{2}\sum_a b_a \|\omega^{-1} h_a\|, \qquad c_\Delta = c + \sum_a b_a \|\omega^{-1/2} h_a\|/\sqrt{2}.$$

Including the bank and every old reference, the vector error is at most

$$\epsilon_R \leq \int_0^{T_R} [L_\Delta \sqrt{M}\, p(s)^{1/4} + c_\Delta \sqrt{p(s)}]\, ds. \tag{24}$$

Any specified incomplete ideal-SWAP or clock error is added separately. The same bound controls half trace distance of the renewed cell and held-reference marginal from their ideal outputs.

Proof. Duhamel bounds the vector difference by the integrated norm of the Hamiltonian difference on ψ_s^i. Apply the linear-field estimate to $P_{\rm leak}\psi_s^i$. Its norm is $\sqrt{p}$ and

$$\|H_F^{1/2} P_{\rm leak}\psi_s^i\|^2 \leq \|P_{\rm leak}\psi_s^i\| \|H_F P_{\rm leak}\psi_s^i\| \leq M\sqrt{p}.$$

The field and counterterm contributions give (24). Pure-vector distance bounds half trace distance, which contracts under the stated restrictions. □

Corollary 16 (A constructive finite receiver-renewal family)**.** *Fix $A > 0$, $0 < p < 1/2$, a matched receiver/bank pair and a material smearing h. For even m, use the positive spin-history clock with $J = A/m$, a true receiver/bank SWAP on the first edge and identities thereafter. Put $\chi_m = \mathbf{1}_{\{j \geq m/2\}}$ and*

$$H_m = H_{\rm hist} + H_E + H_B + H_F - \chi_m n_E Q(h) + \tfrac{1}{2}\chi_m n_E \langle h, K^{-1} h\rangle.$$

Its initial cursor is $j = 0$; the material/old receiver state and all old references may be correlated, with initial physical rms $M < \infty$. The independent bank is Gibbs with its own purifier retained. This is a positive autonomous Hamiltonian. At $t_R = (m/A)\arcsin\sqrt{p}$ its joint half trace distance from a completed receiver renewal, with an explicitly specified comparison clock, is at most

$$2(1-p)^{m/2} + \frac{m \arcsin\sqrt{p}}{A}[L_\Delta \sqrt{M}\, e^{-mD/4} + c_\Delta e^{-mD/2}], \quad D = D_{\rm Ber}(1/2\|p) > 0. \tag{25}$$

It tends to zero while the clock mean energy A, its norm bound $2A$ and the common-H physical rms certificate remain bounded independently of m. Clock space and duration grow; the bank is not cycled for free.

Proof. The field/contact terms are the positive controlled-displacement Hamiltonian of Proposition 11; the history clock and matched cell energies are nonnegative. For the force-free reference, each prefix is either the true SWAP or identity and commutes with $H_E + H_B + H_F$. Prefix conjugation therefore gives the same binomial cursor law for every correlated input. For $s \le t_R$ its binomial parameter is at most p; Chernoff gives $\Pr(j \ge m/2) \le e^{-mD}$. The ideal physical rms is the unchanged M. Proposition 15 proves the second term. The no-SWAP probability is $(1-p)^m$. As comparison use the same untrimmed force-free cursor with the completed SWAP and freely evolved material/reference state. Only its $j = 0$ branch differs from the ideal joint state; the retained-purification vector error is at most $2\sqrt{(1-p)^m}$. Half trace distance is no larger. No clock outcome is postselected. Triangle inequality proves (25). Finally $\|H_{\mathrm{hist}}\| \le 2A$ and its initial mean is A. The cell norms and force/counterterm norms are fixed, so the initial total rms and the physical bound of Proposition 11 are uniform. □

This condition has a finite nonempty realization. Take the positive spin-history clock with $m = 64$, $J = .02$, the matched SWAP on its first edge, and a material force $-.2\,\mathbf{1}_{\{j\ge 32\}} n_E Q(h)$ with $\omega = 1$ and unit one-mode smearing. At $T_R = \arcsin\sqrt{.1}/J = 16.08753$ the force-free reference cursor has $\mathrm{Bin}(64, .1)$ law; throughout this window its source tail is at most 7.04652×10^{-16}. Its uncompleted first-edge probability is $.9^{64} = .00117902$. For vacuum material input $M = 0$, (24) is below 6.9×10^{-8}. Completing the ideal SWAP on the same untrimmed cursor costs at most $2\sqrt{.9^{64}} = .068674$ in vector norm. The clock's mean energy is 1.28. Direct sparse evolution of the full force-on Hamiltonian at Fock cutoffs $6, 10, 14$, retaining a correlated held-body/receiver input, its old reference and the fresh bank purifier, gives vector difference 6.54×10^{-10} from the same force-free reference. The receiver's half trace distance from its bank state is .00043624; the held-body/reference marginal changes by less than 2.5×10^{-13} and energy conservation residual is below 4.1×10^{-13}. These are finite-regulator diagnostics of the nonempty renewal window, not a continuum proof. This is a paid receiver-renewal block with given correlated input, not an assertion that a previous program produces its cursor for free.

Actual leakage and completion require paid force-idle padding. The receiver certificate preserves held material and old content, but neither prepares its interacting Gibbs state nor implements an entropy outlet. A J-cell bank retains J spent slots, outputs and purifiers; each isolated replacement transfers excess energy $\delta(\langle n_E\rangle - \nu)$ of either sign. Preparation, clock, storage and removal remain charged. Q-RAIF B's oscillator-bank formulas [26] concern its different material. Repeated noisy readings of one command also do not justify $J \log 2$ in the independent-bit capacity bound (23).

7.2 Several rounds on one running material

The next reading can be installed on the same cursor that performs replacement. Fix a finite sequence of permitted data gates U_b, including receiver–bank swaps, and a nonnegative data energy H_0 with $[U_b, H_0] = 0$. Degenerate logic and matched-cell exchanges meet this condition. Keep the body, all banks, their purifiers, previous records and reversible workspace in the input. Between gates use the linear material forces and completing terms of Eq. (15).

Choose rational $0 < a_1 < \cdots < a_q < 1$ and integers m for which $k_b = ma_b$ are integral. Put U_b on edge k_b of the spin clock; other edges are identities. Each cursor interval has its own fixed force amplitudes, commuting within that interval. Orthogonal cursor supports make the full amplitudes commute. With J fixed, the resulting $H_m \geq 0$ is time independent and has common domain $\operatorname{Dom} H_F$. Its comparison schedule is

$$p(t) = \sin^2(Jt), \qquad t_b = J^{-1} \arcsin \sqrt{a_b}, \qquad T = \pi/(2J).$$

It applies the specified gates at t_b while the same complete material evolves throughout. Only the comparison uses these switching times; the actual apparatus has its gates and forces fixed in space along its cursor.

Proposition 17 (A finite sequence with its history retained)**.** *Let the smooth force profiles satisfy $h_\ell, \omega^{1/2} h_\ell \in L^2$, and fix an input purification with $\|H_F \psi\| \leq M_0$, including arbitrary old references. The actual H_m approximates the scheduled sequence jointly with its complete, untrimmed cursor, at every fixed checkpoint away from the t_b. At T all data gates have been included. The joint vector error tends to zero as $m \to \infty$, uniformly on this input class. Both actual and comparison material rms energies are bounded by a constant independent of m.*

More explicitly, set

$$L_P = \sqrt{2} \sum_\ell \|B_\ell\| \|h_\ell\|, \qquad c_P = \frac{1}{\sqrt{2}} \sum_\ell \|B_\ell\| \|\omega^{1/2} h_\ell\|.$$

For any energy $E_* > 0$ a common rms bound is

$$M_T = \left[\sqrt{M_0 + E_*} + \frac{T}{2} \left(L_P + \frac{c_P}{\sqrt{E_*}} \right) \right]^2 - E_*. \tag{26}$$

In the prefix gauge, let H_b be the material/data Hamiltonian in interval b, and choose K_b so that $\|(H_b - H_{b-1})\zeta\| \leq K_b$ for normalized vectors with $\|H_F \zeta\| \leq M_T$. These constants are finite by the linear field bound. Then the gauge-state error satisfies

$$e_m(t) \leq \sum_{b=1}^{q} K_b \int_0^t e^{-m[p(s) - a_b]^2} \, ds. \tag{27}$$

At a physical checkpoint add $2 \sum_b e^{-m[p(t) - a_b]^2}$; this additional term is zero at T. For $0 < \alpha_b < \min(a_b, 1 - a_b)$ and $v_b = 2J \sqrt{(a_b - \alpha_b)(1 - a_b - \alpha_b)}$,

$$e_m(T) \leq \sum_b K_b \left[\frac{\sqrt{\pi}}{v_b \sqrt{m}} + T e^{-m\alpha_b^2} \right]. \tag{28}$$

Proof. The prefix conjugation used for the history clock transforms its hopping term to $mJ + 2JJ_x$ and each force to the corresponding prefix-conjugated data amplitude. The transformed material Hamiltonian is piecewise constant in j. Compare it with the product

of the untrimmed spin-clock state and the data/material sequence H_b. The clock distribution is $\mathrm{Bin}(m, p(t))$. Writing both piecewise Hamiltonians as sums of their jumps gives

$$H_m^{\mathrm{gauge}} - H_{\mathrm{cmp}}(t) = \sum_b (\mathbf{1}_{\{j \geq k_b\}} - \mathbf{1}_{\{t \geq t_b\}})(H_b - H_{b-1}).$$

The norm of each clock factor on the comparison state is the square root of its wrong-side probability. Hoeffding's bound is $e^{-m[p(t)-a_b]^2}$. Duhamel's formula therefore proves (27). On $|p - a_b| \leq \alpha_b$ one has $p' \geq v_b$; integrate the Gaussian there and bound its complement by $Te^{-m\alpha_b^2}$ to obtain (28). Undoing the gauge at a checkpoint costs at most twice the square root of the probability of an incorrect prefix. At T the reference cursor is exactly $|m\rangle$, giving no prefix error.

For the energy estimate, all controller terms commute with H_F, even though their norms grow with m. On the energy core, $[H_F, Q(h)] = -iP(h)$ and

$$\left\| \sum_\ell B_\ell P(h_\ell) \zeta \right\| \leq L_P \| H_F^{1/2} \zeta \| + c_P \| \zeta \|.$$

For $x(t) = \|H_F \psi_t\|$ this gives $D^+ x \leq L_P \sqrt{x} + c_P$. Hence $D^+ \sqrt{x + E_*} \leq (L_P + c_P/\sqrt{E_*})/2$, which proves (26). The data gates commute with H_F and cause no jump in this bound. Finite-mode energy-core regularization gives the same estimate before removing the cutoff; the common-domain propagator limit and closed-operator lower semicontinuity extend it to $\mathrm{Dom}\, H_F$. The linear field bound then supplies the K_b used above. Every estimate acts on the full purification and tensors with the old reference identity. □

For prescribed finite N, compile acquisition, decision, record force and bank swap on this path. Proposition 57 supplies one decision circuit from actual meter records and retained posterior weights; its repair flag can drive the force. Registers and thermal pairs are prepared initially, and spent correlations remain. At checkpoints, (20) and (26) certify a later reading and source test under the same subsequent H_m.

As a finite diagnostic, two probe readings of one oscillator, an intervening receiver–bank swap and a later force controlled by both records give endpoint joint vector errors $.0701594, .0355814, .0192136, .0107206, .00393182$ at $m = 16, 32, 64, 128, 512$, with $J = 1$. An old probe reference and the spent bank are retained. At $p = 7/16$, after the swap and before the second probe interval, receiver excitation is .0321012 for $m = 128$ and .00089552 for $m = 512$; endpoint accuracy alone would miss this timing error. The $m = 128$ final coordinate is .0393364 against the scheduled .0393304; removing the record-controlled forces gives -2.89×10^{-6} against zero. Fock cutoffs 8 and 10 change the endpoint vector error by 1.1×10^{-11}. These are regulator checks of the construction.

The clock uses $m + 1$ states, mean energy mJ, variance mJ^2, norm $2mJ$ and maximal edge coefficient at most $J(m+1)/2$. Fixed duration thus costs growing energy; fixed-energy renewal instead costs growing duration. Neither is indefinite fixed-capacity operation, and this material-source bound excludes controller covariant stress.

Appendix K gives a further use of this program: local linear-current pulses implement stress flows in the neutral $c = 1$ carrier, uniformly on bounded mean-energy inputs. The

construction supplies a physical permission used in finite modular-geometry tasks, with explicit control and coherence conditions.

The comparison clock need not have the actual clock's energy. For one oscillator take $H_F = \omega a^\dagger a$, $Q = (a + a^\dagger)/\sqrt{2}$, $V = -bQ + C$ and $C = b^2/(2\omega)$. Use identity clock edges and turn on V for cursor sites $j \geq m/2$, with even m. The positive Hamiltonian is $H_m = H_{\mathrm{clock,m}} + H_F + \mathbf{1}_{j \geq m/2} V$. Starting in $|0\rangle_{\mathrm{clock}} |0\rangle_F$, its conserved total energy is mJ. The scheduled endpoint is $|m\rangle \otimes \zeta$ up to phase, where $\zeta = e^{-i(T/2)(H_F+V)}|0\rangle$ and $\langle H_F + V\rangle_\zeta = C$. Consequently

$$\langle H_m \rangle_{\mathrm{scheduled}} - \langle H_m \rangle_{\mathrm{actual}} = C \quad \text{for every even } m. \tag{29}$$

The joint-state error still tends to zero. The finite work is paid by the actual clock and missed by its unperturbed product comparison.

This defect can be repaired for the total-energy observable. Rotate the terminal spin toward negative S_x by an angle θ. Its clock energy is $mJ - mJ \sin\theta$, and its vector distance from $|m\rangle$ is at most $|\theta|\sqrt{m}/2$. If $\epsilon_m(\theta)$ is its probability below $m/2$, the product state's total-energy excess is $-mJ \sin\theta + C - \epsilon_m(\theta)\langle V\rangle_\zeta$. It equals C at zero and is negative at $\theta = \arcsin[2C/(mJ)]$ for sufficiently large m: the last probability is at most $e^{-3m/8}$ once $\theta \leq \pi/6$. Continuity therefore supplies an exactly energy-matched comparison state with an additional $O(m^{-1/2})$ vector error. This is a physical comparison vector, not an extra gate supplied by the original dynamics. Matching one energy expectation does not match an extended stress distribution. HAFF-D uses the mass-work identity of its common action to keep this debit in a finite source comparison [23].

7.3 Local thermal windows and a continuing diagnostic return

Port replacement is distinct from using a later region of a correlated field. Appendix I.1 executes a local thermal preparation with a paid finite autonomous clock. Appendix I.2 proves its record-selected same-field conditional error from vacuum mutual information and controls the source with second moments. The complete outside purification stays present; a spatially separate window is not declared globally fresh. These free-CAR certificates require a new thermal matching proof before use on the nonlinear interacting material.

Appendix J.1 gives a separate finite-horizon return in which the actual meter word, not an inaccessible fault label, selects repair of the held body and a thermal exchange. Appendix J.2 supplies an actual terminal record and replacement of its specified probe. The full input bank, posterior arithmetic, reversible garbage, clock energy and retained output history belong to that horizon. The actual mixed-port quantum-material update above does not inherit their different deadline or mean-support guarantees.

7.4 Repeated renewal of a complete ring body

Use a metric-graph carrier with four complex chiral body fields on rings of length w, each coupled to its own thermal line over an arc $0 < \ell < w$. All spatial modes remain. Their charges are $(+1, -1, +1, -1)$; the first pair is the receiver, the second its current purifier. Two colocated, copropagating rails on each arc obey

$$i(\partial_t + \partial_x)\Psi = h(x)W(t - x)\Psi, \quad h \in C_c^\infty(0, \ell), \quad \int h = 1. \tag{30}$$

The exact ray transfer is $e^{-iW(u)}$, $u = t - x$. Take $W(u) = \theta(u)\sigma_y$ and include a physical ring connection with round-trip phase $e^{-i\Delta w}$. The overlap transit is included in w; the voltages and their switching work are supplied controls.

Choose $\tau = Kw$, $K \in \mathbb{N}$, a real $\lambda \in C_c^\infty(0,\tau)$, and $\theta = \sqrt{w}\lambda$, $|\theta| < \pi/4$. Iteration gives

$$A_w(u) = \prod_{j=0}^{K-1} \cos[\sqrt{w}\lambda(u + jw)], \qquad 0 \le u < w, \tag{31}$$

$$F_w(s) = \frac{\sin[\sqrt{w}\lambda(s)]}{\sqrt{w}} e^{i\Delta s} \prod_{j\ge 1:\, s+jw<\tau} \cos[\sqrt{w}\lambda(s + jw)]. \tag{32}$$

The old-ring multiplier is A_w; F_w is the captured zero-mode envelope. Put $\eta^2 = w^{-1}\int_0^w |A_w|^2$. Retaining the full complementary outgoing field gives $\|F_w\|_2^2 = 1 - \eta^2$ by CAR. Endpoint flatness makes the last product smooth. With $\Omega = 2\pi/w$, the captured nth mode has carrier $\Delta + n\Omega$.

The reference sea fills $n < 0$ and leaves $n \ge 0$ empty. Sum over all four species in

$$N_{\mathrm{ex}} = \sum_{n\ge 0} d_n^\dagger d_n + \sum_{n<0} d_n d_n^\dagger, \tag{33}$$

$$P = \Omega\left(\sum_{n\ge 0} n d_n^\dagger d_n + \sum_{n<0} |n| d_n d_n^\dagger\right), \qquad Q = N_{\mathrm{ex}} + P/\Omega. \tag{34}$$

Let $r(E)$ be the occupation of the normalized captured envelope with carrier E in the actual temperature-T KMS line. Its real envelope gives $r(-E) = 1 - r(E)$. For $A_w(w\xi) = \sum_k a_k e^{2\pi i k\xi}$ set

$$\alpha = \sum_k \sqrt{1 + |k|}\, |a_k|, \tag{35}$$

$$V_A = \tfrac{1}{2}\sum_{k>0} k(k+1)|a_k|^2 + \tfrac{1}{2}\sum_{k<0} |k|(|k|+3)|a_k|^2, \tag{36}$$

$$S_Q = (1-\eta^2)\left[r(\Delta) + \sum_{n\ge 1}(1+n)\{r(\Delta + n\Omega) + r(n\Omega - \Delta)\}\right], \tag{37}$$

$$b = 2(\sqrt{V_A} + \sqrt{S_Q}). \tag{38}$$

Proposition 18 (Uniform renewal with retained history)**.** *Suppose $\langle Q\rangle < \infty$, $0 < \Delta < \Omega$, and the next incoming supports have their unconditional KMS marginals. Correlations with the body and all earlier records are allowed. The actual reset satisfies*

$$\sqrt{\langle Q\rangle_{\mathrm{reset}}} \le \alpha\sqrt{\langle Q\rangle_{\mathrm{old}}} + b. \tag{39}$$

Between resets allow the translation-invariant pumps below and supplied logical noise on the four zero modes. For post-reset $x_j = \sqrt{\langle Q\rangle_j}$, if $\sqrt{2}\alpha < 1$, then

$$x_{j+1} \le \sqrt{2}\alpha x_j + 2\alpha + b, \quad \sup_j x_j \le \max\left\{x_0, \frac{2\alpha + b}{1 - \sqrt{2}\alpha}\right\}. \tag{40}$$

Against the actual outside marginal, including all old references,

$$\tfrac{1}{2}\|\rho_{\mathrm{body},R} - |\mathrm{sea}\rangle\langle\mathrm{sea}| \otimes \rho_R\|_1 \le 2x_j. \tag{41}$$

At fixed cycle duration and T, one finite choice of pulse strength, gap and ring length makes this error arbitrarily small uniformly in the number of cycles, with four active outlet species.

Proof. Apply all weighted wrong-occupation annihilators to a purification of the actual joint state. Weighted convolution, using $1 + |n| \le (1 + |m|)(1 + |n - m|)$, bounds their old excitation-annihilation part by $\alpha\sqrt{\langle Q\rangle}$. CAR bounds the creation covariance by the identity; its squared norm is at most V_A per species. The incoming vector has squared norm S_Q. Minkowski's inequality on the four-species direct sum proves (39), without factoring the old state from the line. Pumps and logical noise preserve P, and $N_{\mathrm{ex}} \le P/\Omega + 4$, so $Q_{\text{after gate}} \le 2Q_{\text{before gate}} + 4$ in expectation. This gives (40). The nonsea projection is bounded by $N_{\mathrm{ex}} \le Q$. Projection of a joint purification onto the pure sea and trace-distance contraction give (41).

The full sums are finite before imposing any numerical cutoff. Set $\Gamma = \int \lambda^2$ and $g_w = \log\cos(\sqrt{w}\lambda)$, extended by zero. For each $m \ge 3$, $G_m = \sup_{w\le w_0} w^{-1}\|g_w^{(m)}\|_1 < \infty$ if $\sqrt{w_0}\|\lambda\|_\infty < \pi/4$. Poisson summation bounds the nonconstant Fourier coefficients of $\log A_w$ by $G_m w^m/(2\pi|k|)^m$. Weighted Wiener exponentiation therefore gives

$$\alpha \le e^{-\Gamma/2+b_m}, \quad V_A \le 2e^{-\Gamma}(e^{b_m} - 1)^2, \quad b_m = \frac{2G_m w^m}{(2\pi)^m}[\zeta(m) + \zeta(m-1)]. \tag{42}$$

Write $F_w = e^{i\Delta s} f_w$. Every derivative of its logarithmic product is a sum of $O(1/w)$ terms of size $O(w)$, with flat endpoints. Thus $D_m = \sup_{w\le w_0} \|f_w^{(m)}\|_2^2 < \infty$. Splitting the Fourier integral at $E/2$ bounds the exact Fermi occupation by

$$r(E) \le e^{-E/(2T)} + \frac{(2/E)^{2m} D_m}{1 - \eta^2}, \qquad E > 0. \tag{43}$$

For $\Omega \ge 2\Delta$ and $z = e^{-\Omega/(4T)}$,

$$\begin{aligned} S_Q \le{}& (1 - \eta^2) r(\Delta) + 2(1 - \eta^2)\frac{z(2 - z)}{(1 - z)^2} \\ &+ 2D_m(4/\Omega)^{2m}[\zeta(2m) + \zeta(2m - 1)]. \end{aligned} \tag{44}$$

Choose Γ large, then Δ large, then $w = \tau/K$ small. All errors become small simultaneously. A finite initial reset puts x_0 at the same tolerance. Earlier nonselective even operations on disjoint line characteristic supports leave the next unconditional KMS marginal unchanged; guards longer than the overlap transit provide that separation. □

The mixed target is restored by a local neutral quartet pump

$$V_4 = \int_0^w (D_1^\dagger D_2^\dagger D_3^\dagger D_4^\dagger + \mathrm{h.c.})\, dx. \tag{45}$$

It commutes with P. On the sea, creation requires four nonnegative mode indices summing to zero: only the four zero modes occur. The reverse transition obeys the same argument. Their two-dimensional subspace is exactly invariant, with matrix element $1/w$.

For the full operator, bosonize the periodic fields with $q_i \in \mathbb{Z}+1/2$. The charge coordinate $\phi_c = \sum_i \phi_i/2$ gives the weight-two vertex $e^{2i\phi_c}$. At fixed transverse charges, $q_c = q_0 + 2\mathbb{Z}$ with $q_0 \in \frac{1}{2}\mathbb{Z}$, a fixed parity sector of a charge fermion; the other three bosons and the original Klein phases remain. The weight-two vertex splits this stress into two positive $c = 1/2$ Virasoro stresses with velocities $1 \pm \kappa/\pi$ [1]. The four Ramond ground weights sum to $1/2$, hence

$$\|V_4\psi\| \le \pi^{-1}\|(P + \Omega/2)\psi\|. \tag{46}$$

Finite-dimensional P eigenspaces are invariant; their direct sum defines the self-adjoint pump. The pair pump $B = \int (D_1^\dagger D_2^\dagger + \text{h.c.})dx$ obeys $|B| \le P/\Omega + 1$: every active nonzero pair costs at least Ω in P. For sea-normal-ordered number $N \ge -P/\Omega$,

$$P + \Delta N + \kappa V_4 + b_2 B \ge \left(1 - \frac{\Delta}{\Omega} - \frac{|\kappa|}{\pi} - \frac{|b_2|}{\Omega}\right) P - \frac{|\kappa|}{w} - |b_2|. \tag{47}$$

Small enough pump peaks give a common finite physical ground offset and positive Hamiltonian. This offset is part of the apparatus source.

A quartet pulse with phase $e^{-4i\Delta t}$ and area $w^{-1}\int \kappa(t)dt = \arcsin\sqrt{\epsilon}$ prepares

$$|\Phi_\epsilon\rangle = \sqrt{1-\epsilon}|00\rangle_A|00\rangle_R + \sqrt{\epsilon}|11\rangle_A|11\rangle_R. \tag{48}$$

The pair pump selects logical Z or X, retaining spectrum $(1-\epsilon, \epsilon)$ and the actual current purifier. Unitary contraction preserves (41); the next reset exports that purifier and its correlations through the same lines. Fixed gate duration requires $\kappa_{\text{peak}} = O(w)$, compatible with (47) and the ordered reset parameters.

The body has finite energy-constrained storage capacity rather than finite Hilbert dimension. Its positive-energy Gibbs partition function is finite for every inverse temperature $\beta > 0$, giving $S \le \beta\langle H\rangle + \log Z(\beta)$. Four complex outlets, any anomaly-cancelling partners, gap energy, drive work and stored outputs are charged. Fixed per-cycle error does not imply vanishing error for the whole growing history; that stronger task requires decreasing error per cycle and increasing resources. Autonomous covariant driving is treated below. Direct renewal exports the current purifier's two correlated occupation records into different species. Against a thermal reference that factorizes across species, their redundancy gives a positive complete-output relative-entropy rate whenever the reset and full-joint clock errors are small enough [23, Proposition "Output deficit and direct renewal"]. Improving this reset alone therefore does not approach the sharp heat-current limit. That task requires a physical output converter with its own work, memory and charge account.

Proposition 19 (Neutral block compression by physical gates)**.** *Buffer L original noisy A/R quartets in L equal-circumference slots of each physical charge species $(+,-,+,-)$. Permit local current phases, the pair and quartet pumps above, and same-charge quadratic mixers between connected slots. These prescribed gates can approximate a reversible code which confines their 4^L-dimensional support to m_L active zero modes and fixes the remaining*

modes, retaining arbitrary old references and intercycle correlations, where

$$m_L = \min\left\{ r : \binom{r}{\lfloor r/2 \rfloor} \geq 4^L, \quad 0 \leq 2L - \lfloor r/2 \rfloor \leq 4L - r \right\} = 2L + \tfrac{1}{2}\log_2 L + O(1). \quad (49)$$

No additional pure working register or outgoing species is used. The buffer, initialization, control, offsets and eventual discharge are physical resources.

Proof. Relabel only the existing negative-charge zero modes by $f_i = d_i^\dagger$ and the positive-charge ones by $f_i = d_i$. Neutrality is $\sum f_i^\dagger f_i = 2L$ on $4L$ modes; this relabeling leaves the full sea and original charge representation unchanged. Pair pumps and same-charge mixers become one-body rotations. Connected rotations and currents generate $U(4L)$. The quartet becomes, with its calibrated Klein phase, $Q_{ab;cd} = f_a^\dagger f_b^\dagger f_c f_d + \text{h.c.}$. Permutations move it to any four distinct labels.

Set $n_i = f_i^\dagger f_i$. Balanced rotations in $(a, c), (b, d)$, followed by averaging over their sixteen current sign phases, give $-\frac{1}{2}(n_a - n_c)(n_b - n_d)$. This average is implemented by a deterministic Trotter word, without a discarded random record. Quadrilateral differences span all two-body diagonal terms modulo currents on the fixed-number sector: their missing coefficient arrays are $\beta_{ij} = u_i + u_j$, for which $\sum_{i<j} \beta_{ij} n_i n_j = (2L - 1) \sum_i u_i n_i$. Choose a diagonal $H_D = \sum \alpha_i n_i + \sum \beta_{ij} n_i n_j$ whose finitely many hopping transition frequencies are nonzero and distinct up to sign. Such coefficients avoid finitely many proper hyperplanes. Vandermonde interpolation in iterated commutators with H_D isolates the real and imaginary generators of each individual transition between occupied $2L$-subsets. These subsets form a connected Johnson graph. Its edge generators give the full special unitary group on the neutral sector; finite group-commutator and Trotter words approximate each required Givens rotation to the stated tolerance.

Enumerate the original 4^L pair-code basis states and inject them into the first 4^L occupation strings on m_L active modes with $\lfloor m_L/2 \rfloor$ particles. Fix the inactive pattern to $2L - \lfloor m_L/2 \rfloor$ particles and extend the injection to a neutral-sector unitary. Its action is $U_L \otimes I$ on every old reference. Stirling's formula gives (49). For the physical embedding, the translation-invariant equal-ring integrals commute with P. Its complete $P = 0$ sector is precisely the zero-mode Fock space tensored with the same spectator sea; the control word never projects out the full field tower. The previous relative-energy margins, with finite offsets and small primitive amplitudes, define the full-field gates. Unitary contraction carries a certified input-reset error through the word. □

For example, two quartets occupy eight zero modes but need only six active modes after this encoding. The fixed occupations result from reversible redistribution, not from a recurring purity supply. Two paid banks can hold old batches while new states arrive; both must subsequently be discharged and reused through the same four lines. The compiler may be very long and expensive. It proves finite prescribed controllability, not a sustainable autonomous clock or an efficient fixed-rate layout. It also preserves the data spectrum: concentrating it does not prepare thermal occupations of the outlet translation generator. The next two propositions supply finite physical loading, sorting and return; their clock, source and repeated-inlet accounts remain distinct.

Proposition 20 (A finite-width packet interface)**.** *Fix a physical charge species, m paid equal-ring memories and one of its original outlet lines. Let $v(t)$ be a smooth row of m overlapping envelopes, flat outside an interior guard of $(0,\tau)$, with $\int v^\dagger v\,dt = (1-\eta^2)I_m$, $0<\eta<1$. A local Hermitian mixer on a finite common arc emits these memory columns with $O(w)$ packet error and terminal amplitude $\eta I_m + O(w)$ as the ring round-trip time w decreases. An analogous mixer captures any such incoming overlapping packet span. All original body modes and the complementary outgoing field remain in the canonical map.*

For a finite physical carrier of length V with no return during the window, let the joint input have physical excitation energy at most E, retaining arbitrary old references. With $\Omega = 2\pi/w$, $M_\Lambda = O(V\Lambda) + m$ low modes and any fixed sufficiently large smoothness order r, the actual and comparison full outputs have half trace distance at most

$$C\left(\sqrt{E/\Lambda} + \sqrt{E/\Omega}\right) + C_{m,V,\tau,\eta,r}\left[(M_\Lambda+1)w(1+\Lambda)^2 + (1+\Lambda)^{1-r} + w^{r-1}\right]. \tag{50}$$

The constants include the supplied envelope norms. First choose Λ, then w. This is a finite-carrier, prescribed-control result; it is not a global trace theorem for an infinite KMS rail.

Proof. Set $K(t) = I - \int_0^t v^\dagger v\,ds$, $\Phi = K^{1/2}$ and

$$\lambda^\dagger = v\Phi^{-1}, \qquad A = \dot\Phi\Phi^{-1}, \qquad H = i\left(A + \tfrac{1}{2}\lambda\lambda^\dagger\right).$$

Since $K \geq \eta^2 I$ and $A + A^\dagger = \Phi^{-1}\dot K\Phi^{-1} = -\lambda\lambda^\dagger$, H is smooth and Hermitian. The comparison equations

$$\dot d = Ad - \lambda\psi_{\mathrm{in}}, \qquad \psi_{\mathrm{out}} = \psi_{\mathrm{in}} + \lambda^\dagger d$$

have homogeneous propagator $\Phi(t)$ and emitted columns $\lambda^\dagger\Phi = v$. For capture, choose a column F with $\int FF^\dagger = (1-\eta^2)I$, and use

$$R(t) = \left[I - \int_t^\tau FF^\dagger\,ds\right]^{1/2}, \quad \lambda = R^{-1}F, \quad A = -R^{-1}\dot R.$$

The same identity gives Hermitian H; the future propagator is $G(\tau,t) = R(t)$, so the capture column is $-F(t)$. Its sign is an ordinary current phase, not reversed chiral transport.

In units of propagation speed one, choose a smooth arc profile $h_w(x) = w^{-1}h_0(x/w)$ of integral one and use the actual local potential

$$\mathcal{F}_w(t,x) = h_w(x)W_w(t-x), \qquad W_w = \begin{pmatrix} wH & -i\sqrt{w}\lambda \\ i\sqrt{w}\lambda^\dagger & 0\end{pmatrix}. \tag{51}$$

The internal gap is lowered for this window, with its work charged. Along a characteristic the exact arc transfer is $S_w = e^{-iW_w} = \left(\begin{smallmatrix} B_w & C_w \\ D_w & E_w\end{smallmatrix}\right)$. Thus the full fields obey

$$b(t) = B_w(t-w)b(t-w) + C_w(t-w)\psi_{\mathrm{in}}(t-w), \qquad \psi_{\mathrm{out}}(t) = D_w(t)b(t) + E_w(t)\psi_{\mathrm{in}}(t).$$

In each fixed derivative norm, $B_w = I + wA + O(w^2)$, $C_w = -\sqrt{w}\lambda + O(w^{3/2})$ and $D_w = \sqrt{w}\lambda^\dagger + O(w^{3/2})$. Telescoping the noncommuting ordered products over τ/w returns gives

the $O(w)$ comparison. The complete star is unitary, so its emitted Gram matrix plus its residual-body Gram matrix is exactly I_m.

The full tower also contributes. Each positive derivative of B_w is $O(w)$; differentiating an ordered product sums $O((\tau/w)^j)$ terms with j differentiated factors, each supplying $O(w^j)$. Its derivatives are therefore bounded uniformly in small w. Flat guards make the residual multiplier periodic on the ring. Its nonconstant Fourier coefficients satisfy $\|a_k\|_{\mathrm{HS}} \le C_r w^r/|k|^r$. The emitted and captured envelopes have the corresponding uniform smooth bounds; their nth body aliases retain the exact factors $e^{\pm in\Omega t}$.

The complementary line kernel is the direct term $E_w(t)\delta(t-s)$ plus its causal delayed returns

$$\sum_{j\ge 1} D_w(t)B_w(t-w)\cdots B_w(t-(j-1)w)C_w(t-jw)\delta(s-t+jw).$$

Each coefficient and each fixed center derivative is $O(w)$, with $O(\tau/w)$ terms. The center derivative $\partial_t+\partial_s$ leaves both these delayed diagonals and the comparison Volterra factor $\Theta(t-s)$ unchanged. Fourier integration by parts therefore bounds its nonidentity kernel by $C_r(1+|\omega-\omega'|)^{-r}$, uniformly in w. Across opposite sea polarizations, $|\omega-\omega'| = |\omega|+|\omega'|$; the squared tail above Λ is $O((1+\Lambda)^{2-2r})$. On a fixed low-frequency subspace, the causal sum is a Riemann approximation with one-particle error $Cw(1+\Lambda)^2$. Together with the body alias bounds this controls the relative transformation's opposite-polarization Hilbert–Schmidt block.

For the state estimate, a proof-only energy cutoff excludes field excitations above Λ and nonzero body excitations with probability at most $E/\Lambda + E/\Omega$. Gentle projection works with the old reference retained. The remaining finite excitation space has M_Λ modes, so its Fock lift costs at most M_Λ times the one-particle error. The canonical pair-rotation decomposition bounds the vacuum-lift error by the opposite-polarization Hilbert–Schmidt norm [48, Section 7.4, Theorem 31]. The actual normal-ordered lifts may differ by an overall phase, irrelevant for this prescribed channel. Combining the cutoff, low-band and full-tower estimates gives (50). □

Capture retains the actual covariance $C_{\mathrm{cap}} = \int F_w(t)C_{\mathrm{in}}(t,s)F_w(s)^\dagger\,dt\,ds$, including off-diagonal packet correlations and old body–line terms. Orthogonal wavefunctions need not have independent thermal occupations. A quantum program coherently selecting different loaders also requires their actual relative lift phases to be calibrated and charged. For the same finite graph, the source comes from its full covariance and local point splitting, and the work is $\int\langle\partial_t H_{\mathrm{graph}}\rangle dt$, including gap changes, contact energy and physical ground offsets. With $C_h = \int h_0^2$, the local quadratic scale satisfies

$$\int \operatorname{tr}\mathcal{F}_w^2 dx = C_h\left[w\operatorname{tr}H^2 + 2\|\lambda\|^2\right].$$

This does not bound all higher spatial derivatives or clock resources uniformly. Nor does (50) alone control heat: an old spectator occupied with probability E/Ω can export energy E while disappearing in trace norm. Heat or source convergence requires propagated uniform energy integrability, such as a controlled higher energy moment or explicit weighted output tails. The packet map supplies the loading step for the following thermal converter. Its

energy estimate uses the actual carrier rather than packet occupation numbers assigned independently of it.

Proposition 21 (Charge-sector thermal sorting on a retained carrier)**.** *Use the original four charge species and neutral gates, and retain four physical carrier loops of round-trip time $V = L/\nu$ in their complete finite-volume Gibbs state $\tau_{T,V}$, $T > 0$. Include its purification and all old references. Supply L independently realized noisy original quartets with marginal $\rho_S = \rho_0^{\otimes L}$, where*

$$\rho_0 = \begin{pmatrix} (1-q)(1-\epsilon) & 0 & 0 & (1-2q)\sqrt{\epsilon(1-\epsilon)} \\ 0 & q\epsilon & 0 & 0 \\ 0 & 0 & q(1-\epsilon) & 0 \\ (1-2q)\sqrt{\epsilon(1-\epsilon)} & 0 & 0 & (1-q)\epsilon \end{pmatrix}, \qquad h_\lambda = S(\rho_0).$$

The source and complete carrier initially have a product marginal; their external purifications are retained. There are prescribed finite words of the original smooth gates that return the source near its original blank and achieve, in natural units,

$$\lim_{\Lambda\to\infty} \limsup_{L\to\infty} \frac{D(\rho_{B,\mathrm{out}} \| \tau_{T_*,V})}{V} = 0, \qquad T_* = T + \frac{6\nu h_\lambda}{\pi c_{\mathrm{ex}}}, \quad c_{\mathrm{ex}} = 4. \tag{52}$$

Implementation error can be made arbitrarily small after choosing each finite block and cutoff Λ, with physical energy tails controlled. The entire output B, its uncaptured complement and every atypical label are present in this relative entropy. Finite paid internal pipeline stages permit the specified throughput without adding external outlets. This is a finite-carrier converter; an actual stationary inlet, repeated preparation and an autonomous reused controller require further physical constructions.

Proof. On each loop let $D = -i\partial_x$, P_Λ its Fourier subspace with $|\omega| \le \Lambda$, and χ a smooth cutoff flat zero in endpoint guards of length g. If $P_\Lambda \chi^2 P_\Lambda u_j = g_j u_j$, retain $g_j \ge 1 - e$ and set $f_j = \chi u_j/\sqrt{g_j}$. These are orthonormal smooth envelopes for Proposition 20. For their isometry F, the actual captured matrices are

$$C_T = F^*(1 + e^{D/T})^{-1} F, \qquad h_F = F^* D F.$$

They need not commute. A physical one-body rotation diagonalizes C_T for its spectral probabilities, whereas the final occupation states use the eigenbasis of h_F. Relabel existing negative-charge modes by $f = d^\dagger$ as in the neutral code, retaining their energy $-h_F^T$ and the physical sea constant.

Here is an exact finite count. In total charge sector Q, let $E_{\beta,Q}, S_{\beta,Q}, \sigma^2_{\beta,Q}$ be the energy, entropy and energy variance of the actual carrier-sector Gibbs matrix. For $b_Q > \sigma_{\beta,Q}$ there are at least

$$\left(1 - \frac{\sigma^2_{\beta,Q}}{b_Q^2}\right) e^{S_{\beta,Q} - \beta b_Q} \tag{53}$$

occupation states below $E_{\beta,Q} + b_Q$. Chebyshev gives this shell's mass; each state in it has probability at most $e^{-S_{\beta,Q}+\beta b_Q}$. Map the charge-resolved good input eigenlabels injectively

into these low-energy states with the source blank, then complete the injection to a sector bijection. Every bad label occupies a remaining physical state. If their probability is $p_{\rm bad}$,

$$\Pr(S_{\rm out} \neq b) \leq p_{\rm bad}, \qquad S(S_{\rm out}) \leq h(p_{\rm bad}) + 4Lp_{\rm bad} \log 2.$$

For the depolarizing source specified in the proposition, exact blank restoration would violate the full-rank finite input's rank count. No label is postselected away.

At fixed sufficiently large Λ, the source eigenvalue strings and the Gaussian carrier's actual spectral probabilities have information variance $O(L)$. An information margin $L^{9/10}$ and charge window $|Q| \leq L^{5/8}$ lose $o(1)$ probability. Choose the finite-bank target entropy larger than the input entropy by $3L^{19/20}$ and put $b_Q = L^{17/20}$. For these charges exponential tilting of the bounded occupation Bernoullis gives $\log \Pr_\beta(Q) = -O(Q^2/L + \log L)$. Fourier inversion bounds the tilted probability at its mean below by $cL^{-1/2}$: its characteristic function is bounded by $e^{-cL\sin^2(\theta/2)}$, and its scaled central part converges to a nonzero Gaussian integral. Conditioning its $O(L)$ energy variance therefore gives $\sigma^2_{\beta,Q} = O(L^{3/2})$ and $|E_{\beta,Q} - E_\beta| = O(|Q| + L^{3/4})$. The count loss in (53) is smaller than the entropy margin. Hence $p_{\rm bad} \to 0$, source entropy is $o(L)$, and the carrier energy excess is $o(L)$.

The bijection uses the original gates. For a hop $a \to b$, a quadrilateral interaction $(n_a - n_c)(n_b - n_d)$ has difference $n_d - n_c$. Two distinct spectator subsets have the same differences for every c, d only when one is empty and the other full. Generic current and quadrilateral coefficients therefore separate all other oriented edges; finite Vandermonde filtering and the connected occupation graphs give independent special-unitary actions in sectors $2 \leq k \leq m-2$. The linked $k = 1, m-1$ sectors are unnecessary: the original source has $k_S = 2L$ on $4L$ modes, so every occupied source–carrier sector is interior. Spectral diagonalization and the chosen bijection are finite commutator/Givens words. Their phases are retained.

Capture and return act on the same complete carrier as $U^*_{\rm load} U_{\rm sort} U_{\rm load}$. For commensurate loops $U_0(V) = 1$ up to its declared sea phase. Thus the inverse of a free interval δ is the positive wait $V - \delta$. Smooth signed current kicks invert the local mixing steps: evolution with $H_0 + (a/\varepsilon)J(F)$ for time ε approaches $e^{-iaJ(F)}$ on the smooth energy core, with error at most $\varepsilon \int_0^1 \|H_0 e^{-isaJ(F)}\psi\| ds$. Higher weighted versions follow from smooth multiplication and the summable opposite-sea tails. Reverse a finite Trotter word using these waits and kicks. Their work, peak ground shifts, relative phases and contact sources are paid. Its G whole-loop waits require G internal storage stages; narrowing a contact does not remove that latency. All stages have the original species and only the original four external outlets.

To pass from the bank to the complete physical carrier, let d be the number of discarded Fourier vectors and M the bank size. Then $d \leq C(\Lambda g + 1)/e$. The retained Fourier span need not be invariant under D; its discarded cross block contributes $C\Lambda d$, in addition to the cutoff error $CM(\Lambda\sqrt{e} + 1/g)$. Taking $g = V^{1/2}$ and $e = V^{-1/4}$ makes these and the entropy compression error $o(V)$ at fixed Λ. The unselected thermal tails have energy and entropy per V at most $C(1+\Lambda)^2 e^{-\Lambda/T_*}$. The ideal conjugation preserves the entropy of source plus full carrier, so Araki–Lieb and subadditivity give

$$|S(B_{\rm out}) - S(\tau_{T,V}) - Lh_\lambda| \leq S(S_{\rm out}).$$

Together with the physical energy estimate and Gibbs identity, this bounds the deficit in (52) by $o_\Lambda(1) + C(1+\Lambda)^2 e^{-\Lambda/T_*}$. Full finite-carrier Gibbs moments and the smooth gate

graphs control implementation tails; Gibbs energy-constrained entropy continuity then gives the stipulated finite accuracy. Trace error alone would not give this energy or entropy conclusion. □

Remark 22 (Other supplied neutral source spectra). The proof of Proposition 21 also applies to a product source $\bigotimes_{j=1}^{L} \sigma_j$ from a fixed finite alphabet of known states in the same neutral quartet code, with the same joint source–carrier product marginal, paid gates and controlled physical energy tails. Joint operations include the receiver and its current, resettable purifier; external old references remain untouched. Put $s_L = \sum_j S(\sigma_j)$ and suppose $s_L/L \to s$ and the source information variance is $O(L)$. Replace h_λ by s in T_* and Lh_λ by s_L in the entropy identity. The finite-bank target uses the actual s_L plus the same subextensive entropy margin; its target temperature tends to T_*. No rate of convergence of s_L/L is needed. Eigenlabel concentration uses this variance and entropy. The source still has $k_S = 2L$, so the neutral-sector compiler remains in its interior sectors. The final full-carrier entropy bound uses s_L; thermal counts, energy tails and paid loading/unloading are unchanged. HAFF-D's pointer-aligned dephasing supplies such rank-two sources at fixed noise parameters [23]. This is supplied-state renewal, not recovery of arbitrary unknown content against a forbidden purifier or a proof of the product marginal for an adaptive source.

The source debit is the complete quartet entropy h_λ. For $0 < q, \epsilon < 1/2$, its difference from the bare receiver change is the correlation lost in the logical depolarizing step:

$$h_\lambda - [h(\epsilon_q) - h(\epsilon)] = I(A:R)_{\text{before}} - I(A:R)_{\text{after}} \geq 4q\, h(\epsilon) > 0.$$

The equality uses the pure corrected logical quartet and its noisy marginals; joint convexity for the depolarizing mixture gives the bound. With a conserved classical axis P, the source entropy conditional on P is Lh_λ. Its near-blank return removes $s^{PS} = Lh_\lambda - o(L)$ from the complete source–program system. Entropy conservation with the independent thermal carrier gives

$$S(B_{\text{out}}|P) - S(\tau_{T,V}) = s^{PS} + I(S_{\text{out}} : B_{\text{out}}|P).$$

No noise purifier is granted as usable side information. An autonomous controller's sensing and renewal must also enter its complete ledger. HAFF-D gives the corresponding conditional deficit account [23].

The fixed Z and X noisy quartets are isospectral. Their entropy-optimal outlet heat agrees to leading order, although their retained correlations and recovered work can differ. The direct source contrast below belongs to its specified unsorted reset. Its causal intervention remains valid, but does not certify persistent heat gain for the sharp converter.

Corollary 23 (The selected axis changes the maintenance source)**.** *Fix* $0 < \epsilon < 1/2$ *and* $0 \leq q < 1/2$. *On the same complete post-reset state, compare the two physical axis preparations above, followed by the logical depolarizing channel with* $\epsilon_q = \epsilon + q(1-2\epsilon)$ *and the same reset. Do not undo the* X *rotation. There are fixed finite reset parameters for which, uniformly after the initial preparation cycles,*

$$\int \zeta(t)(\langle T_{\text{out}}\rangle_X - \langle T_{\text{out}}\rangle_Z)dt \geq \frac{\Delta(1-\eta^2)}{2}(1-2q)(1-2\epsilon) > 0, \tag{54}$$

where $\zeta \geq 0$ is smooth and equals one on the emitted pulse. This is a paired causal intervention on one pre-preparation state, not a comparison of histories selected by conditioning on different acquired outcomes.

Proof. The ideal total zero occupations are $2(\epsilon_q + \epsilon)$ and $1 + 2\epsilon$. Their difference is $D = (1 - 2q)(1 - 2\epsilon)$. Let v_w be the real emitted zero-mode envelope. The emitted nth mode is $v_n = e^{-i(\Delta+n\Omega)t}v_w$, and $J_k = \int \zeta v_w^2 e^{-ik\Omega t}$, $J_0 = 1 - \eta^2$. The exact old-body matrix of the smeared stress is

$$B_{mn} = [\Delta + \Omega(m + n)/2]J_{n-m}.$$

Its diagonal is $J_0(P + \Delta N_s + \Delta N_0)$ after sea normal ordering. Both interventions preserve P; $|N_s| \leq P/\Omega$ and $\|N_0\| = 4$. If d bounds the half-trace reset error and M_E bounds $\langle P/\Omega\rangle$, their diagonal contrast is at least $\Delta J_0(D - 16d - 2M_E)$.

These errors can be controlled independently of the zero occupation. Separating the old zero mode in the excitation-vector proof gives

$$\sqrt{\langle P/\Omega\rangle_{j+1}} \leq \alpha\sqrt{\langle P/\Omega\rangle_j} + 2(\sqrt{Z_A} + \sqrt{V_P} + \sqrt{S_P}),$$

where $Z_A = \sum_{n\neq 0} |n||a_n|^2$, $V_P \leq V_A$ is the nonzero-mode creation leakage, and S_P is S_Q with its zero term removed and weight n instead of $1 + n$. Thus $M_E \to 0$ as $w \to 0$ at fixed Γ, Δ. Before reset take $M_Q = 4 + 2M_E$ as a bound on $\langle Q\rangle$.

Set $R_2 = \sum_{k\neq 0}(1 + |k|)^2|J_k|$. Weighted Schur estimates for equal sea polarizations and a Hilbert–Schmidt estimate across the sea give a two-branch off-diagonal error at most $2(\Delta + \Omega)R_2(M_Q + 4\sqrt{M_Q})$. For $m \geq 5$, integration by parts gives

$$R_2 \leq 2\|(\zeta v_w^2)^{(m)}\|_1\Omega^{-m}[\zeta_R(m) + 2\zeta_R(m - 1) + \zeta_R(m - 2)],$$

where ζ_R denotes the Riemann zeta function. The envelope derivatives are uniformly bounded as w decreases.

Old body–line cross terms also remain. Let S_f be the exact line-to-line scattering operator and $g_n = S_f^*(i\zeta\partial_t + i\zeta'/2)v_n$. Its delay kernel contains shifts kw. Since $e^{-in\Omega kw} = 1$, direct substitution gives $g_n = e^{-iE_n t}(E_n a_w + b_w)$, $E_n = \Delta + n\Omega$, with all derivatives of a_w, b_w uniformly bounded independently of n, Δ. The wrong KMS occupation of this unnormalized mode is bounded by

$$2(E_n^2\|a_w\|_2^2 + \|b_w\|_2^2)e^{-|E_n|/(2T)} + 2(2/|E_n|)^{2m}(E_n^2\|a_w^{(m)}\|_2^2 + \|b_w^{(m)}\|_2^2).$$

Sum this over all integers n and the four species to obtain G_T. The sum converges for $m \geq 2$. Its zero term tends to zero with Δ; its nonzero terms tend to zero with $\Omega \geq 2\Delta$. CAR and Cauchy–Schwarz bound the difference of the actual cross stresses by $4\sqrt{4 + M_E}\sqrt{G_T}$, without a product-state premise. The incoming-only term cancels because body-only interventions leave the same incoming marginal unchanged. The full lower bound is therefore

$$\begin{aligned}\Delta J_0(D - 16d - 2M_E) - 2(\Delta + \Omega)R_2(M_Q + 4\sqrt{M_Q}) \\ - 4\sqrt{4 + M_E}\sqrt{G_T}.\end{aligned} \tag{55}$$

Choose Γ large, then Δ large, then w small. The reset bounds and the displayed convergent tails make the error less than $\Delta J_0 D/2$. These choices remain compatible with the pump positivity condition (47). □

The energy difference is paid by preparing and maintaining different physical axes at a fixed gap. It is emitted by the reset itself, not by an additional program-tagged source. Conditional histories can have different incoming backgrounds; the causal result makes no claim that those observed conditional means obey the same formula.

8 A spatial work clock for the retained thermal body

The waveforms above can be generated by a physical clock field over a specified finite horizon. Its source depends on the retained quantum state, including the body–line correlations created by earlier gates. For this construction supply a static material frame (U, V) and reflecting endpoints. Let

$$S_\chi = \frac{I}{2} \int \sqrt{-g}[(U \cdot \nabla\chi)^2 - C(V \cdot \nabla\chi)^2], \qquad 0 < C < 1, \tag{56}$$

with Neumann data, initially $\chi = 0$, $\chi_t = 1$, $\chi_x = 0$. Gate profiles are smooth functions of χ and the material coordinate. Use flat guard intervals between a zero-gap, fixed real quartet quadrature and the quadratic gap, pair and mixer gates. The actual gap-lowering and restoration work, offsets and finite normal-order terms are included in the Hamiltonian. Internally mixed components have a common positive free speed v_0.

Proposition 24 (Retained-state field–clock evolution)**.** *Fix a finite list of such windows and a finite horizon H. Prepare the original mixed four-mode target and its purifier, the empty-zero body reference and an actual finite-volume Gibbs reservoir; retain every subsequent state and reference. Assume smooth stress and kernel remainders, sufficiently many finite energy moments, and the strict quartet inequalities* (60) *below. Choose the reservoir long enough to prevent a return before H. Then sufficiently large finite physical inertia I_H gives a timelike work clock and the same sequence of gate types throughout H. Its force is that of the actual evolved full-CAR state. The quadratic windows and their delayed matrix boundary close at the same derivative order. The quartet window admits a reflected symmetrizer at that order. In each specified lower source norm the field and the paid clock-work source have a comparison error $O_H(I^{-1/2})$; a quadratic window with identical incoming data has $O_H(I^{-1})$ error. These estimates retain the clock's $O(I)$ baseline stress and its finite work contribution.*

Proof. During a quadratic window the Heisenberg multiplet obeys $(\partial_t + \sigma v_0 \partial_x)\Psi = -iF\Psi$, where $F = F(\chi, a)$ is Hermitian and $\sigma = \pm 1$ is chirality. Pair gates use constrained Majorana components; their doubling adds no physical outlets. Subtract the fixed symmetric spatial singularity $-i\sigma\kappa_J/(x-y)$, $\kappa_J = 1/(2\pi)$, from $\Gamma(x,y) = \langle \Psi^\dagger(y)\Psi(x)\rangle$. Let $J = \Gamma_{\rm reg}(x,x)$ and $K = -i\sigma(\partial_x - \partial_y)\Gamma_{\rm reg}/2|_{y=x}$. Point splitting gives the full matrix equations

$$\begin{aligned} J_t + \sigma v_0 J_x &= -i[F, J] - \sigma\kappa_J F_x, \\ K_t + \sigma v_0 K_x &= -i[F, K] - \frac{\sigma}{2}\{F_x, J\}, \qquad F_x = F_\chi \chi_x + F_a a_x. \end{aligned} \tag{57}$$

The force in (56) is $-\sum_\sigma \langle F_\chi, J_\sigma\rangle - \partial_\chi E_o$. For complex components the contraction is a trace; for constrained Majoranas it is half the trace. Finite sea shifts are retained: the

periodic empty-zero convention has $[L_n^{\rm sea}, J_m^{\rm sea}] = -mJ_{n+m}^{\rm sea} - (n/2)\delta_{n+m,0}$. Passing to canonical currents uses $J_n = J_n^{\rm sea} - \delta_{n0}/2$ and $L_n = L_n^{\rm sea} + \delta_{n0}/12$, with the corresponding fixed Hamiltonian subtraction. It leaves the physical zero occupation unchanged.

For a mixing arc of length $\ell < w$, set $d_a = \ell/v_0$, $d_{\rm out} = (w-\ell)/v_0$ and $d = w/v_0$. The arc Wilson matrix $G(u)$ maps entry time u to exit $u + d_a$. With its body/line blocks G_{BB}, G_{BF} and the actual outside phase Q, the incoming body satisfies

$$b(u) = A(u)b(u-d) + B(u)f(u-d), \quad A(u) = QG_{BB}(u-d), \quad B(u) = QG_{BF}(u-d). \tag{58}$$

Finite iteration expresses its cross kernel as

$$\Gamma_{BF}(u,v) = R_M \Gamma_{B_0F}(u - Md, v) + \sum_{j=0}^{M-1} R_j B(u - jd)\Gamma_{FF}(u - (j+1)d, v),$$

where $R_j = A(u)\cdots A(u-(j-1)d)$ and $R_0 = 1$. These are kernels of the original joint state. At $v = u$ every reservoir separation is at least d; all clock-dependent incoming data precede u by at least $d_{\rm out} > 0$.

The incoming cross stress requires covariant relative time differentiation. For constant exterior connections $F_{0,B}, F_{0,F}$,

$$K_{BF} = \frac{i}{2v_0}(\partial_u - \partial_v)\Gamma_{BF}|_{v=u} - \frac{F_{0,B}\Gamma_{BF} + \Gamma_{BF}F_{0,F}}{2v_0}.$$

For the full right-moving transfer put $S = (iG'G^\dagger + GF_{0,\rm in}G^\dagger - F_{0,\rm out})/v_0$. The complete spatial point-split conversion is

$$\begin{aligned} J_{\rm out} &= GJ_{\rm in}G^\dagger + \kappa_J S, \\ K_{\rm out} &= GK_{\rm in}G^\dagger + \tfrac{1}{2}\{S, GJ_{\rm in}G^\dagger\} + \tfrac{\kappa_J}{2}S^2. \end{aligned} \tag{59}$$

Distinct body/line finite-circle backgrounds are added before and subtracted after this conversion. Differentiating unitarity twice in the symmetric split cancels the anti-Hermitian second derivative, including noncommuting connection terms. A scalar total stress would lose the BF data in this equation.

Differentiate the arc Wilson matrix by Duhamel's formula. Its $(m+1)$st entry-time derivative needs only derivatives through m of χ_t, a first-order clock variable. The initial two-time kernel remainders and their required derivatives are bounded away from the declared diagonal singularity. Finite history products therefore give a locally Lipschitz C^m boundary functional in (χ_t, χ_x, J, K), $m \geq 3$. On time steps shorter than $d_{\rm out}$ its incoming data are already fixed. Characteristic iteration of (57) and the acoustic-clock wave equation uses these same jets and yields the quadratic-window estimate. The constants depend on H/d and the initial norms.

For a quartet window write $v_s = v_0 + sa(\chi)$, $g_s = sa_\chi$, $s = \pm$, and $\gamma_s = 1/(48\pi)$. The actual chiral stress satisfies $T_{\sigma s,t} + \sigma v_s T_{\sigma s,x} + 2\sigma v_{s,x} T_{\sigma s} = \sigma\gamma_s v_{s,xxx}$. Use $Y_{\sigma s} = (1-\partial_x^2)^{-1}T_{\sigma s}$ and $Z_{\sigma s} = \partial_x Y_{\sigma s}$, retaining their means. For $r = \chi_t$, $p = \chi_x$, the principal matrix has $A_{rp} = -C$, $A_{rZ} = -g_s/I$, $A_{pr} = -1$, $A_{Zp} = \sigma\gamma_s g_s$, $A_{ZZ} = \sigma v_s$. Its symmetrizer has $H_{rr} = I$, $H_{pp} = IC$, $H_{pZ} = g_s$, $H_{ZZ} = v_s/\gamma_s$. Direct Schur elimination gives

$$IC > 2\gamma_s a_\chi^2 \sum_s v_s^{-1}, \qquad I(1-C) > 2\gamma_s a_\chi^2 \sum_s \frac{v_s}{1-v_s^2}. \tag{60}$$

Thus H and $H \pm HA$ are positive. The exact remainder follows by commuting $1-\partial_x^2$ through the Ward equation; its terms with $v_{s,xx}Z$ gain one derivative under $\partial_x(1-\partial_x^2)^{-1}$. They are locally Lipschitz at the same Sobolev order. At a reflected endpoint $p = 0$, $Y_{Rs} = Y_{Ls}$, $Z_{Rs} = -Z_{Ls}$, so the energy flux vanishes. Smooth higher classical energies recover the desired lower-norm actual stress. Increasing finite I preserves the flat guard intervals and a timelike clock over the fixed finite list.

For the solved smooth profiles, the full-tower propagator carries the actual state through each seam. Its distributions satisfy these same linear Ward and kernel equations; uniqueness identifies the mean force with that retained state. Finitely higher initial graph and classical norms can be selected for the finite list. In a gate controlled by a conserved record projector use unnormalized weighted kernels; each c-number anomaly carries that projector's actual weight. The controlled unitary also retains off-diagonal record coherence. Register updates occupy their separately paid slots.

Finally the actual clock work is

$$\frac{d}{dt}\frac{I}{2}\int (r^2 + Cp^2) = -\int r\,[\sum_{\sigma}\langle F_\chi, J_\sigma\rangle + \partial_\chi E_o],$$

with the quartet force replacing the bracket during its slot. Write $\chi = t + \eta_I/I$ and solve $\eta_{0,tt} - C\eta_{0,xx} = -\text{force}_{\text{ref}}$ with the same corrected initial data. The source estimate controls $\eta_I - \eta_0$; the finite retarded work term and the $O(I^{-1})$ quadratic clock remainder remain in the comparison. No growing clock baseline multiplies an uncontrolled state error. □

Corollary 25 (Full joint accuracy of the finite schedule)**.** *Use the original charge-glued field space, its finite reservoir and all retained quantum registers and references in Proposition 24. In fixed energy units put $G_{\text{en}} = 1 + \sum_{\text{rails}} v_0 P_{\text{rail}}$, the positive physical excitation energy, including holes and every spectator. The finite zero-mode multiplicity is retained. Suppose the joint initial state has $M_1^2 = \operatorname{Tr}(\rho_0 G_{\text{en}}^2) < \infty$, with the finitely higher smooth input norms required for the schedule. Compare its solved clock history with the prescribed history on the same field space and in the same initial state. If*

$$a_I(t) = \|(H_I(t) - H_0(t))G_{\text{en}}^{-1}\|, \qquad b(t) = \|[G_{\text{en}}, H_0(t)]G_{\text{en}}^{-1}\|,$$

then, for every $t \le H$, the half trace distance of their full joint quantum outputs is bounded by

$$e_H = M_1 \int_0^H a_I(s) \exp\left(\int_0^s b(r)\,dr\right) ds = O_H(I^{-1/2}). \tag{61}$$

At a reset checkpoint whose prescribed-history whole-body error is d_{ring} in (41), the actual output satisfies

$$\tfrac{1}{2}\|\rho^I_{\text{body},R} - |\text{sea}\rangle\langle\text{sea}| \otimes \rho^I_R\|_1 \le d_{\text{ring}} + 2e_H. \tag{62}$$

Here R is the actual outside system, including every old reference. The same joint bound applies after preparing the original mixed receiver with its current purifier and before its next specified readout. It is an energy-constrained state estimate, not an unconstrained diamond-norm assertion.

Proof. The local quartet is a stress smearing in its original sector representation. The positive-energy bound [3, Sections 2.2–2.3]

$$\|L_n\psi\| \le \sqrt{1+c/12}(1+|n|^{3/2})\|(1+L_0)\psi\|$$

and Fourier Cauchy–Schwarz control a profile difference by its H^m norm, $m > 2$. The energy commutator needs one further derivative. For a quadratic current, pair or mixer, separate the same-sea number part from the pair part. The first is bounded relative to excitation number and hence to G_{en} on the fixed finite hardware. A smooth compact-arc profile gives pair matrix entries bounded by $C_s(1+E_n+E_m)^{-s}$. A sum-frequency shell has $O(q)$ entries: absolute summation requires $s > 2$, and the energy commutator requires $s > 3$. Three and four integrable spatial derivatives, respectively, are safe integer choices. Different circumferences enter through their physical energies and finite spacing constants; an unweighted sum of L_0's would not supply this mixer commutator. The estimates act across the original CAR sectors, including record coherences and their scalar phases. They use no Gaussian replacement at a seam.

The smooth propagators preserve the common energy core. For a purification Ψ_0 of the actual initial state, the commutator estimate gives $\|G_{\mathrm{en}}U_0(s)\Psi_0\| \le M_1\exp(\int_0^s b)$. Duhamel's formula then bounds $\|(U_I(t)-U_0(t))\Psi_0\|$ by (61); every reference carries the identity. The clock comparison at the chosen lower coefficient norm gives $\int_0^H a_I \le C_H I^{-1/2}$, after supplying the required higher norms. Its existence interval already covers the entire schedule. Triangle inequality with the prescribed reset costs e_H; replacing its outside marginal by the actual one costs another e_H. Re-preparation and the same subsequent instrument contract this error. □

The delay in reset bounds is the round-trip time d, whereas w is spatial length. For example, $v_0 = .5$, $w = 10^{-8}$, $\Delta = 10^7$, $T = \tau = 1$ and $\Gamma = 32$ admit a smooth flat reset pulse with $D_0 < 9.9\times10^8$ and $r_d < 4\times10^{-5}$. The four-mode joint reset bound is 0.026503. Finite additional clock accuracy is obtained by increasing I_H; this argument does not provide its numerical value. Restoring the gap preserves the original X/Z work difference $\Delta(1-2q)(1-2\epsilon)$. For a constant arc background $F_0 = \mathrm{diag}(\Delta, 0)$ the actual local connection

$$F = F_0 + v_0 h(a)e^{-iF_0a/v_0}W(\chi - a/v_0)e^{iF_0a/v_0}, \qquad \int h\,da = 1,$$

gives the full circuit phase $e^{-i\Delta d}$, retaining its spatial force and emitted carrier energy.

The evolving work clock still has a supplied frame/endpoints; traction and geometric reaction need their own equations. Equal mixer speeds, smooth inputs and separated quartet/current slots are required. Horizon-dependent inertia and reservoir length leave fixed-capacity sharp exhaust open.

8.1 A local gate on a massive body

A spatial body need not have the massless ring's energy graph. Pay instead for a finite physical circle with a mass-zero central interval of length ℓ and a positive smooth Dirac mass throughout the rest of its return arc. Its four thermal outlet fields remain separate,

with the original charges and capacity. Both mass layers, the exterior polarization, support pressure and initial preparation belong to the apparatus. This layout replaces a compact barrier followed by a long gapless body exterior; it does not give that old layout a gap by assumption.

Proposition 26 (A guarded local gate with full physical energy control)**.** *Let $h_M = -iv\sigma_z\partial_x + M(x)\sigma_x$ on this circle. There is a nonempty smooth mass-profile class with a simple lowest positive mode E_0, no zero mode and next positive energy $E_1 > E_0$. Set $H_M = d\Gamma(|h_M|)$, with particles and holes above the actual filled mass sea, and $A = 1 + H_M$. Let $h \geq 0$ be smooth, supported strictly inside the mass-zero region. On the original charge-glued space define the local vertex*

$$Q_+(h) = \int h(x)[R_1^\dagger R_2^\dagger R_3^\dagger R_4^\dagger + L_1^\dagger L_2^\dagger L_3^\dagger L_4^\dagger]dx, \qquad Q = Q_+ + Q_+^\dagger.$$

All modes and old references are retained. For each fixed graph order s, sufficiently weak smooth resonant drives

$$H_{\rm lab}(t) = H_M + \lambda b(\lambda t)[e^{-4iE_0t}Q_+(h) + e^{4iE_0t}Q_+(h)^\dagger] \tag{63}$$

implement a prescribed full logical angle on the four-ground-mode code P_0, in physical time $T = U/\lambda$, with

$$\|A^s(U_{\rm lab}(T) - U_{\rm target,lab}(T))P_0\| \leq C_s\lambda. \tag{64}$$

The target includes its calibrated laboratory phase. The error is uniform in external old references, not an unrestricted-energy diamond bound. Smooth physical mass-force and stress tests satisfy $\|A^{-4}S_fA^{-4}\| < \infty$, so weighted state errors also control their actual renormalized expectations at this fixed mass.

Proof. The original orthogonal boson rotation and re-fermionization give, for each chirality, the operator identities

$$Q_\sigma(h) = \pi^{-1}(T_{\sigma,1}(h) - T_{\sigma,2}(h)), \qquad T_{\sigma,1} + T_{\sigma,2} + \sum_{a=1}^{3} T_{\sigma,a} = \sum_{i=1}^{4} T_{\sigma,{\rm Dirac},i}.$$

The individual rotated stresses need not be quadratic in the massive species. Their total on the right is quadratic. The commuting sector stresses have positive shifted forms by the chiral QEI [12, Theorem 4.1 and Remarks 2–3]. Their joint spectral calculus therefore bounds $\|Q(h)\psi\|$ by $C\|(K_h + C_h)\psi\|$, where K_h is the actual original localized kinetic stress. Global charge gluing is retained; no independently prepared sector vacua are supplied.

In the physical mass polarization its one-particle operator is

$$D_h = -iv\sigma_z(h\partial_x + h'/2) = hh_M - iv\sigma_z h'/2,$$

since $hM = 0$. For a number block b and $\omega = |h_M|$, $b^2 \leq C^2\omega^2$ and hence $|b| \leq C\omega$. On an n-particle tensor space, different-particle factors commute, so

$$(\sum_i b_i)^2 \leq \sum_i b_i^2 + 2\sum_{i<j}|b_i||b_j| \leq C^2(\sum_i \omega_i)^2.$$

This gives a uniform Fock bound without a spurious $\sqrt{n}$. The number-block closure follows from the commutator criterion with $1+\omega^2$ on the smooth elliptic core. On the guard all mass jets vanish. The complete symbols of the spectral projections there agree with the massless chiral projections; consequently $P_+ D_h P_-$ is smoothing. Its actual mass-eigenbasis coefficients decay faster than every fixed power of $E_n + E_m$. Number and pair estimates give QA^{-1} bounded. For signed derivatives use a wider guarded positive cutoff, not a bound on h'/h. Local translation gives $\mathrm{ad}^j_{H_M} Q(h) = (-iv)^j Q_R(h^{(j)}) + (iv)^j Q_L(h^{(j)})$. Thus every fixed required energy commutator is relatively bounded and the smooth weak-drive propagator preserves the corresponding graphs over its entire gate time.

The physical spectral class can be constructed, rather than postulated. For a step reference with mass m on the exterior arc of length d, the subthreshold parity equations are

$$\tan\frac{E\ell}{2v} = \sqrt{\frac{m-E}{m+E}}\tanh\frac{d\sqrt{m^2-E^2}}{2v} \quad (+),$$

with the negative reciprocal square-root on the right for parity $(-)$. If $m > 3\pi v/(2\ell)$, the first positive root is simple, $0 < E_0 < \pi v/(2\ell)$, while $E_1 > \pi v/\ell$. At zero the transfer is $\exp((\int M/v)\sigma_y)$ and has no unit eigenvalue. Smooth the two steps just outside the body. Transfer Duhamel bounds the perturbation on fixed complex contours by $C\|M_{\mathrm{smooth}} - M_{\mathrm{step}}\|_1$; choosing it below the positive minimum of the explicit step characteristic function preserves these root counts and a strict gap. This gives a real smooth class, without treating a sampled eigengrid as its proof. In the body its normalized ground mode has $R = e^{iE_0x/v}/\sqrt{2Z}$ and $L = e^{-iE_0x/v}/\sqrt{2Z}$. For $\operatorname{supp} h \subset (-\ell/8, \ell/8)$,

$$g_4 = \frac{1}{2Z^2}\int h\cos(4E_0x/v)dx > 0, \qquad g_2 = \frac{1}{Z}\int h\cos(2E_0x/v)dx > 0.$$

Let N be signed flavour number, particles minus holes. It is not excitation number or total electric charge. Then $K_0 = H_M - E_0 N \geq 0$ has the sixteen-dimensional kernel P_0 and gap $\delta = \min(E_1 - E_0, 2E_0) > 0$; its graph is equivalent to A. Because $[N, Q_+] = 4Q_+$, e^{-iE_0tN} transforms (63) exactly to $K_0 + \lambda b(\lambda t)Q$. This is a proof coordinate, not a physical global-number pulse. The compression is $P_0QP_0 = g_4C_4^\dagger + \overline{g}_4C_4$; all off-code terms remain in the laboratory Hamiltonian.

On $|z| = \delta/2$ the weighted resolvent norms $\|A^jQ(K_0 - z)^{-1}A^{-j}\|$ are finite. Choose $\lambda\|b\|_\infty$ within their Neumann tube and the physical relative-bound margin. The contour projection and its Kato transport give an effective code matrix $aP_0QP_0 + O_s(a^2)$, $a = \lambda b$. Its integrated remainder is $O_s(\lambda)$ because $\int_0^T a^2dt = \lambda\int_0^U b^2du$. The transport residual costs $\int_0^T |\dot{a}|dt = \lambda\,\mathrm{TV}(b)$. Energy propagation costs $\exp(C_s\int|b|du)$, independent of λ. Choosing $\int bdu = \theta/g_4$ proves (64). Pair resonance at $2E_0$ and guarded currents supply the paired-code rotations and phase. For $\theta = \arcsin\sqrt{\epsilon}$ the vacuum preparation is the original mixed A/R target, at final energy $4E_0\epsilon$ up to the weighted error, supplied by the pump.

Finally a smeared physical mass bilinear has order zero and a metric stress order one. A safe pair coefficient bound is $C_f(1 + E_n + E_m)^d$, $d \leq 1$. Weighting both sides by A^{-4} bounds each creation term by $(1 + E_n + E_m)^{-4}$. One-dimensional energy-shell counting has $O(q)$ pairs, so the absolute sum converges. Number terms obey the previous estimate. With one fixed Hadamard subtraction scheme, for $d_4 = \frac{1}{2}\|A^4(\rho - \sigma)A^4\|_1$ this yields $|\operatorname{Tr} S_f(\rho - \sigma)| \leq$

$2\|A^{-4}S_f A^{-4}\|d_4$. Counterterm changes between backgrounds are not cancelled by this fixed-background comparison. □

This gate acts on the full massive carrier, with the actual pump phase, work and wall source. It does not use a hardware projector onto four smeared modes. Its source certificate is for smooth tests at a fixed mass, not pointwise source jets on moving walls. The massive carrier's outlet mixer and whole-sea renewal must be computed on its own spectral generator; the earlier massless ring's reset coefficients cannot be reused. Nor are its ground modes already the massless short-ring storage modes in Proposition 21. Physical transfer, moving support and a reused work clock must join these layouts before they form one sustained device.

9 A common covariant ledger: controller, clock and support

The local potential (21) has potential stress $-g_{\mu\nu}V_{\rm loc}$ if its variables are Lorentz scalars. That term alone omits the mechanism which moves its profile and changes its amplitude. A common local action makes this omission explicit and offers a candidate completion at the classical action and regulated canonical levels.

Take dynamical scalars $X = (\phi_1, \phi_2, Z, \theta, y, \chi, r_1, \dots, r_d)$ on a $(1+1)$-dimensional background of signature $(-,+)$. Here Z is a support texture, χ a clock coordinate, and r_j memory/control coordinates. Specify the positive target coframe

$$\begin{aligned} e_{\phi_a} &= d\phi_a + \lambda_a \theta f(Z) dZ, & e_Z &= m_Z dZ, \\ e_\theta &= \sqrt{I}\, d\theta, & e_y &= \sqrt{J}\, dy, \\ e_\chi &= \sqrt{I_\chi}\, d\chi, & e_{r_j} &= \sqrt{I_j}\, dr_j. \end{aligned}$$

with all diagonal coefficients strictly positive, and put $G = \sum e \otimes e$. Let $b_a(\chi, r)$ be smooth bounded functions, $w_a(Z)$ smooth profiles compact in the texture value, and $M(Z)$ a smooth strictly positive material matrix. With $u = (\sqrt{I}\theta, \sqrt{J}y)$ and $v = u - \sum_a b_a(\chi, r) w_a(Z)$, choose

$$S = -\int \sqrt{-g}\, [\tfrac{1}{2} G_{AB}(X) g^{\mu\nu} \partial_\mu X^A \partial_\nu X^B + V(X)]\, d^2x, \tag{65}$$

$$V = V_Z(Z) + V_{\rm mem}(r) + V_{\rm clock}(\chi) + \tfrac{1}{2} v^T M(Z) v \geq 0. \tag{66}$$

The last square replaces the material mass potential rather than adding a second copy of it. Z, χ, r are actual fields in this candidate, not prescribed positions, waveforms or cursor amplitudes. A prepared finite-energy texture can localize the contact; localization is not produced by writing $w_a(Z)$.

Proposition 27 (Common-action stress and clock reaction)**.** *For smooth finite-energy classical data of* (65)*, metric variation gives*

$$T_{\mu\nu} = G_{AB} \partial_\mu X^A \partial_\nu X^B - g_{\mu\nu} [\tfrac{1}{2} G_{AB} \partial X^A \cdot \partial X^B + V]. \tag{67}$$

The Euler–Lagrange equations are

$$\nabla_\mu(G_{AB}\partial^\mu X^B) - \tfrac{1}{2}(\partial_A G_{BC})\partial X^B \cdot \partial X^C - \partial_A V = 0. \tag{68}$$

They imply $\nabla^\mu T_{\mu\nu} = 0$ for the complete matter action. In specified Minkowski coordinates its canonical energy is

$$H = \int [\tfrac{1}{2}\Pi^T G^{-1}\Pi + \tfrac{1}{2}X'^T G X' + V]\, dx \geq 0.$$

At finite Fourier resolution, the following specified canonical ordering has a nonnegative Friedrichs realization on the supplied charge sectors and finite real coordinate space.

Proof. Vary the inverse metric with all scalar variables held fixed and use $T_{\mu\nu} = -2(\delta S/\delta g^{\mu\nu})/\sqrt{-g}$. Integration by parts in a field variation gives (68); contraction with $\partial_\nu X^A$ gives the stress divergence. In Minkowski coordinates $\Pi_A = G_{AB}\dot{X}^B$, so the Legendre transform has the displayed positive squares. The coframe is nonsingular and triangular. For a spatial Fourier regulator, let q denote its finite real coordinates and compact scalar zero modes. The reduced Lagrangian has positive kinetic matrix $G_K(q)$; put $A_K = G_K^{-1}$ and let $W_K(q) \geq 0$ be its integrated gradient and potential energy. Use Lebesgue measure on the real coordinates and the scalar zero-mode line bundles of Proposition 6, with their charge/winding direct sum. On smooth compactly supported sections choose the divergence-form kinetic ordering

$$\mathfrak{q}_K[\psi] = \tfrac{1}{2}\int (D_i\psi)^*(A_K)^{ij}(D_j\psi)\, dq + \int W_K|\psi|^2\, dq.$$

The D_i are coordinate derivatives with the stated zero-mode boundary conditions. Smooth locally elliptic coefficients make this densely defined nonnegative form closable; its Friedrichs realization is the specified operator. If the interaction acts only on retained low modes, the complete free carrier tail is added positively, as in the charged regulator. This choice of canonical measure and ordering is not a claim of regulator-independent covariant quantization. □

In particular,

$$I_\chi \Box\chi = \partial_\chi V = V'_{\text{clock}} - \sum_a (\partial_\chi b_a) w_a^T M v.$$

The amplitude change reacts on the clock. The memory equations have the corresponding $\partial_{r_j} b_a$ force, and the support equation includes derivatives of w_a, M and the target coframe. Dropping these terms returns an externally driven model. For a free massless clock initially having $\partial_t\chi = \Omega$ over a proper-length interval ℓ, its initial kinetic energy is at least $I_\chi\Omega^2\ell/2$. Smooth spatial tapers and contact reaction add costs and change that free trajectory. A locally timelike clock gradient, texture, memory preparation and boundary conditions are specified initial data; they are not guaranteed throughout an arbitrary run.

The action includes clock/support reaction; canonical positivity is only in the chosen coordinates and finite regulator. Curved-space quantum positivity, renormalized continuum stress and causal propagation after cutoff removal require further proofs. Matching (15)

requires quantum acquisition and transported gates, the prepared texture/charge sectors, and a common finite-cursor/source estimate. The Gaussian K needs static supplied Z and controlled freezing or small perturbations of r, χ; dynamical cross terms and backreaction otherwise change H. That approximation needs an error bound, as does matching HAFF D's dilaton coordinates and full supported quantum state.

10 Discussion

An abstract representation, an exact reduced operation and a renewable physical interface are different requirements. Image closure follows from a homomorphism. A coupling supplies a completely positive map, whose exact algebra is its multiplicative domain. A discarded non-atomic control port therefore limits exactness, but a retained oscillator or closed propagation does not obey that discarded-port premise.

Acquisition is a physical change of the old history, not a supplied source label. The actual record drives the same complete material and produces another reading. Its connected fluctuations contribute to the source; dephasing can preserve an immediate field marginal while changing later joint control. Physical moments and smooth-source forms certify that return without replacing an unbounded stress by a trace-distance estimate.

Renewal depends on what is held. Universal identity-channel preservation cannot append a correlated interacting equilibrium. A specified receiver can nevertheless be replaced by a paid Gibbs bank, with spent correlations retained; the full ring can be reset and its mixed target reinstalled through four thermal outlets. Neutral compression, packet loading and spectral sorting supply a complete-output converter, not merely thermal mean energy. Isospectral axes need not retain a heat-source contrast after optimal sorting, and accurate direct reset alone leaves its purifier redundancy. Persistent sharp-exhaust gain remains a joint implementation question.

The clock distinctions in the introduction fix the resource claims: bounded-energy slow receiver renewal, fixed-duration energy-growing finite rounds, and a spatial work field with horizon-dependent inertia. The full joint-state estimate for the latter retains every old reference and charges finite clock work in the source comparison. Static frame/endpoints and dynamical support are still different problems. The neutral-current stress compiler gives another finite physical control permission, with coherence and precision budgets; it is not noise-protected control over every charged sector.

Foundation III supplies a complementary finite semiclassical mean-action scalar/JT return at common initial constraints: a learned frame protects old content, its record changes the material source, and a later X-basis reading of the old probe carries the geometric response into the spent cell [50]. Its first-order signal arises after acquisition; the cursor and old purifications remain. Matching that run to the complete quantum material here requires a physical state/source bridge. HAFF D's rest-frame mean-stress proposition identifies one excess-radiation sector under fixed covariance and linear contact assumptions, not the entire acquired-record apparatus. Neither construction treats a spent port as a fresh bath.

A world model must retain the carrier and conserved total source; an interface must specify actual records and executable access; a continuing observer must maintain the content and resources those records serve. The common-action candidate connects these respon-

Interface question	Established construction	Required input or remaining bridge
Exact operations	Multiplicative domain; propagation symbol; retained loops	Spectral law, access and quadratic-space conditions
Actual quantum return	Acquisition, full material response and second record under one positive H	Prepared ports, allowed gates, physical transport and finite cursor
Local source	Same-H smooth interacting field bound; local positive contact potential	Smooth interior tests and moments; controller/support stress is separate
Finite maintenance	Receiver-bank SWAP; correlated-window and diagnostic protocols	Paid bank, storage and clock; not full interacting-material Gibbs renewal
Repeated body renewal	Full ring contraction, mixed-target preparation, neutral code, packet loading and finite-carrier sorting	Four thermal outlets, paid buffer, prescribed gates and energy-constrained capacity
Massive spatial body	Guarded local full-angle gate and weighted physical stress tests	Paid mass profile and state; actual outlet renewal and moving support remain
Field–clock evolution	Retained-state finite schedule with matrix Ward and delayed boundary	Static walls/frame; smooth input norms and horizon-dependent inertia
Geometric flow tasks	Local linear-current synthesis of neutral $c = 1$ stress gates	Circle recurrence, coherent control and growing precision budgets
Covariant apparatus	Common local action with dynamic clock and support reaction	Quantum gate matching, charged continuum and renormalized total stress

Table 1: The main interfaces and their distinct burdens. Auxiliary apparatus proofs are collected in the appendices.

sibilities at the classical and regulated canonical levels. Its quantum acquisition, charged continuum, moving supports and total renormalized stress still need matching. Neither exact multiplicativity nor finite repair derives experience or public geometric agreement. Conservation-law and multimode bounds [36, 38, 39] do not settle all these architectures at once. Split/modular inclusions and virtual subsystems [43, 44, 42] organize retained access; formation and maintenance of its permissions remain questions for T-DOME and Imprint.

A The algebraic statement

Let (V, q) be a finite-dimensional real vector space with a non-degenerate quadratic form and $Cl(V, q)$ the real Clifford algebra generated by V with $v^2 = q(v)\mathbf{1}$. When generators are to act as operators, those with $q(e) > 0$ are taken Hermitian and those with $q(e) < 0$ anti-Hermitian; this fixes an involution with $e^* = \operatorname{sign}(q(e))\, e$ on an orthogonal basis.

Definition 28 (Representation input)**.** A *representation* of a control algebra $\mathcal{O}_{\text{int}} \cong Cl(V, q)$ in an environmental C^*-algebra $\mathcal{A}_{\text{ext}}$ is a unital $*$-homomorphism $\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}}$.

Lemma 29 (Semisimplicity)**.** *$Cl(V, q)$ is semisimple. If $\dim V$ is even it is simple, and every nonzero algebra homomorphism out of it is injective.*

Proof. By the classification of real Clifford algebras [40, 41], $Cl(V, q)$ is a matrix algebra over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$, or a direct sum of two such algebras, according to signature and dimension modulo eight. For even $\dim V$ it is a single matrix algebra. The kernel of a homomorphism is a two-sided ideal, and a simple algebra has none besides 0 and itself. □

Proposition 30 (Image closure)**.** *Let ϕ be as in Definition 28. Then $\operatorname{Im}\phi$ is a $*$-subalgebra of $\mathcal{A}_{\text{ext}}$ in which the Clifford relations hold. If $Cl(V, q)$ is simple, in particular when $\dim V$ is even, ϕ is injective. If it is the sum of two simple algebras, its image is isomorphic to the full algebra or to one of the two summands.*

Proof. Images of $*$-homomorphisms are $*$-subalgebras and relations descend. Simplicity gives injectivity. The only proper ideals of a sum of two simple algebras are zero and either summand; a unital representation has a proper kernel. □

Non-degeneracy is needed: for $q(x, y) = x^2$ on $\mathbb{R}^2$, the map $e_1 \mapsto \operatorname{diag}(1, -1)$, $e_2 \mapsto 0$ is a nonzero unital homomorphism into $M_2(\mathbb{R})$ that is not injective. An algebra homomorphism also does not compare signatures. That requires a quadratic-space morphism $j : (V, q) \to (W, Q)$ with $Q(jv) = q(v)$, and, where relevant, preservation of the involution and grading by the induced map of Clifford algebras. Proposition 30 provides neither the map ϕ nor such a j. Sections B–D ask where they can come from.

B Couplings give channels

Let a finite-dimensional system S interact with an environment $\mathcal{E}$ through a unitary U on $\mathcal{H}_S \otimes \mathcal{H}_\mathcal{E}$, with the environment in a normal state σ. The induced Heisenberg map on system observables is

$$\Phi^*(A) = \operatorname{tr}_\mathcal{E}\big[(\mathbf{1} \otimes \sigma)\, U^*(A \otimes \mathbf{1})U\big]. \tag{69}$$

It is unital and completely positive, and by Stinespring's theorem every such map arises in this way [29]. It is in general not multiplicative. Its *multiplicative domain*

$$\mathcal{M}_\Phi = \{A :\ \Phi^*(A^*A) = \Phi^*(A)^*\Phi^*(A),\ \Phi^*(AA^*) = \Phi^*(A)\Phi^*(A)^*\} \tag{70}$$

is a C^*-subalgebra on which Φ^* is a $*$-homomorphism, and $\Phi^*(AB) = \Phi^*(A)\Phi^*(B)$ whenever $A \in \mathcal{M}_\Phi$ [30, 31]. It is the largest part of the system's algebra that the coupling represents exactly. For other elements we measure the failure by the *defect*

$$\delta(A) = \Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A) \ \geq 0. \tag{71}$$

Theorem 31 (Exactly represented algebra of a dephasing coupling)**.** *Let $X = \sum_x xP_x$ have finite spectrum, let E be self-adjoint on $\mathcal{H}_\mathcal{E}$, and $U = \exp(-iX \otimes E)$. Write $\chi(t) = \operatorname{tr}(\sigma e^{-itE})$ and $A_{xx'} = P_x A P_{x'}$. Then*

$$\Phi^*(A) = \sum_{x,x'} \chi(x'-x)\, A_{xx'}, \tag{72}$$

the relation $x \sim x'$ defined by $|\chi(x'-x)| = 1$ is an equivalence relation on the spectrum of X, and

$$\mathcal{M}_\Phi = \{A :\ A_{xx'} = 0 \text{ whenever } |\chi(x'-x)| < 1\}. \tag{73}$$

On $\mathcal{M}_\Phi$, Φ^* is an inner automorphism $A \mapsto W^*AW$ with $W = \sum_x u_x P_x$ for phases u_x. If $|\chi(t)| < 1$ for every nonzero difference t of eigenvalues of X, then $\mathcal{M}_\Phi = \{X\}'$ and Φ^* is the identity on it.

Proof. $U = \sum_x P_x \otimes e^{-ixE}$, so $U^*(A \otimes \mathbf{1})U = \sum_{x,x'} A_{xx'} \otimes e^{ixE}e^{-ix'E}$, and taking the expectation in σ gives Eq. (72).

If $|\operatorname{tr}(\sigma e^{-itE})| = 1$, write $\sigma = \sum_i p_i |\psi_i\rangle\langle\psi_i|$; each $|\langle\psi_i|e^{-itE}|\psi_i\rangle| \leq 1$, so equality forces every ψ_i to be an eigenvector of e^{-itE} with one common eigenvalue $\chi(t)$. Hence e^{-itE} acts as the scalar $\chi(t)$ on the support of σ. If this holds for t_1 and t_2, it holds for $t_1 + t_2$ with $\chi(t_1+t_2) = \chi(t_1)\chi(t_2)$. Together with $\chi(0) = 1$ and $\chi(-t) = \overline{\chi(t)}$ this shows that $\sim$ is an equivalence relation, and that within a class with base point x_0 the phases $u_x = \chi(x - x_0)$ satisfy $\chi(x'-x) = \overline{u_x}u_{x'}$.

The diagonal blocks of the defect are

$$P_y\big[\Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A)\big]P_y = \sum_x \big(1 - |\chi(y-x)|^2\big) A_{xy}^* A_{xy}, \tag{74}$$

a sum of positive operators with nonnegative coefficients. It vanishes for all y only if $A_{xy} = 0$ whenever $|\chi(y-x)| < 1$; the condition with AA^* is the same with the roles of rows and columns exchanged. Conversely, if A has this block pattern, Eq. (72) gives $\Phi^*(A) = \sum \overline{u_x}u_{x'}A_{xx'} = W^*AW$, and these A form a $*$-algebra on which Φ^* is multiplicative. If no nonzero difference has $|\chi| = 1$, the classes are single eigenvalues, the algebra is $\bigoplus_x P_x B(\mathcal{H}_S) P_x = \{X\}'$, and W is scalar on each block. □

Corollary 32 (Continuous spectrum forbids exact coupled control)**.** *If the spectral distribution of E in σ has no atoms, then $|\chi(t)| < 1$ for every $t \neq 0$, so $\mathcal{M}_\Phi = \{X\}'$. In particular this holds for every normal state when E has purely continuous spectrum. Every exactly represented element A commutes with X, and hence with the interaction on its natural domain. No nonzero rescaling of the coupling changes this conclusion.*

Proof. The equality $|\int e^{-ite}\,d\mu(e)| = 1$ requires the phase e^{-ite} to be constant μ-almost everywhere. For $t \neq 0$ its level sets are translates of the countable lattice $(2\pi/t)\mathbb{Z}$, and a non-atomic probability measure gives each such set measure zero. Absolute continuity is sufficient but unnecessary. □

A field quadrature, or the smeared current Q_g of Paper B, has purely absolutely continuous spectrum, so Corollary 32 applies to every normal drive state, whatever its energy. For other couplings, $|\chi(t)| = 1$ requires support in an eigenspace of e^{-itE}, not necessarily an eigenstate of E. Periodic spectra can give exact revivals: with $X = \pi Z$, $E = \mathrm{diag}(0, 1)$ and $\sigma = |+\rangle\langle+|$, the interaction unitary is $I \otimes \mathrm{diag}(1, -1)$ and the system channel is identity, although σ is not an E eigenstate. A Fock state in a number-type coupling gives another finite-energy possibility. The quadrature obstruction concerns the specified single coupling; extending it to composite local controls requires a separate argument.

C The field-coupled flip

Paper B couples a degenerate bit to one mode of a free chiral field through $U = \exp(-i\lambda XQ)$, $\lambda > 0$, with $Q = b + b^\dagger$, $P = i(b^\dagger - b)$ and $[Q, P] = 2i$ [26]. Here $E = \lambda Q$ and the eigenvalues of X are ± 1. Write $\chi = \mathrm{tr}(\sigma e^{-2i\lambda Q})$ and $\sigma_\pm = e^{\mp i\lambda Q}\sigma e^{\pm i\lambda Q}$ for the field states conditioned on $X = \pm 1$. A flip of Z-diagonal states fails with probability $\epsilon = (1 + \mathrm{Re}\,\chi)/2$.

Theorem 33 (Representation defect of the field-coupled flip)**.** *Let the drive mode be in a normal state σ with mean occupation $N = \mathrm{tr}(\sigma b^\dagger b) < \infty$.*

1. *In the X eigenbasis, Φ^* multiplies the two off-diagonal entries by χ and $\overline{\chi}$, and $\mathcal{M}_\Phi = \{X\}'$, the algebra generated by X.*

2. *For every A with vanishing diagonal in the X basis, in particular Z and Y, $\delta(A) = (1 - |\chi|^2)\,\mathrm{diag}(|A_{-+}|^2, |A_{+-}|^2)$; thus $\delta(Z) = \delta(Y) = (1 - |\chi|^2)\mathbf{1}$.*

3. $1 - |\chi|^2 \;\geq\; \dfrac{\arccos^2(\mathrm{Re}\,\chi)\,|\chi|^2}{(2N+1)^2}$.

4. $1 - |\chi|^2 \geq D(\sigma_+, \sigma_-)^2$, *with equality when σ is pure, where D is the trace distance.*

Proof. Item 1 is Theorem 31 with eigenvalue differences ± 2 and Corollary 32. Item 2 follows from Eq. (72) by multiplying out 2×2 matrices.

Item 3 uses Paper B's two quadrature bounds, valid for a normal (density-operator) drive state of finite occupation and $[Q, P] = 2i$ [26]: with $G(c) = \arccos^2 c$,

$$\langle Q^2\rangle \geq \frac{G(\mathrm{Re}\,\chi)}{4\lambda^2}, \qquad (\Delta P)^2 \geq \frac{4\lambda^2|\chi|^2}{1 - |\chi|^2}.$$

The first is spectral Jensen; the second is the unitary uncertainty bound. Since $4N + 2 = \langle Q^2 + P^2\rangle$,

$$\frac{4\lambda^2|\chi|^2}{1 - |\chi|^2} \leq \langle P^2\rangle = 4N + 2 - \langle Q^2\rangle \leq 4N + 2 - \frac{G(\mathrm{Re}\,\chi)}{4\lambda^2}.$$

With $u = 4\lambda^2$, $a = G(\mathrm{Re}\,\chi)$ and $A = 4N + 2$, this reads $1 - |\chi|^2 \geq u^2|\chi|^2/(Au - a)$ for some $u > a/A$. The right side is minimized over u at $u = 2a/A$, where it equals $4a|\chi|^2/A^2$.

Item 4. If $\sigma = |\psi\rangle\langle\psi|$, the conditioned states are pure with overlap $\langle e^{i\lambda Q}\psi, e^{-i\lambda Q}\psi\rangle = \chi$, and $D^2 = 1 - |\chi|^2$. For mixed σ, applying $e^{\mp i\lambda Q} \otimes \mathbf{1}$ to a purification of σ gives purifications of $\sigma_\pm$ with overlap χ, so Uhlmann's theorem gives fidelity $F(\sigma_+, \sigma_-) \geq |\chi|$, and $D \leq (1 - F^2)^{1/2}$ [32, 33]. □

Corollary 34 (The energy–accuracy bound as a representation statement)**.** *For a flip with failure probability $\epsilon \in (0, \frac{1}{2})$ and $\theta_\epsilon = \arccos(2\epsilon - 1)$,*

$$N \;\geq\; \frac{\theta_\epsilon(1 - 2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}. \tag{75}$$

Proof. $\mathrm{Re}\,\chi = 2\epsilon - 1$ gives $|\chi|^2 \geq (1 - 2\epsilon)^2$, hence $1 - |\chi|^2 \leq 4\epsilon(1 - \epsilon)$. Combining with item 3 of Theorem 33 gives $4\epsilon(1 - \epsilon) \geq \theta_\epsilon^2(1 - 2\epsilon)^2/(2N + 1)^2$. □

This is exactly the occupation bound of Paper B. Read in the present terms, an accurate flip requires the coupling to represent Z almost multiplicatively, and the defect $1-|\chi|^2$ cannot vanish at finite occupation. Item 4 identifies the defect with information: for pure drives it is the squared distinguishability of the two field states that record the value of X. The field is then a fragment in the sense of the HAFF paper on measurement, with conditional overlap $|\chi|$; that paper determines which pointer effects such a fragment can implement [28], and the same overlap fixes the representation defect here. The exactly represented algebra $\{X\}'$ is the pointer algebra of the decoherence literature [37]: the control interface transports exactly the part of the system that the environment records.

For the phase-fixed half-turn squeezed family, Paper B gives occupations 2.958, 7.330 and 24.329 at $\epsilon = .05, .01, .001$, against bound values 2.28, 6.74 and 23.80. Its separately optimized finite quadrature-grid values are variational minima, not certified global optima; the continuous supporting ground-state argument has its own assumptions. Mixtures of coherent states require occupation of order $\pi^2/(16\epsilon)$ [26].

C.1 Same clock as mode matching

The requirement that control and environment share a clock, so that their operational windows overlap, has a precise form in the field-coupled flip. The switch g defines a normalized mode function f and the mode operator $b_f = \int \overline{f(\omega)}\, a_\omega\, d\omega$ with $Q_g = \lambda(b_f + b_f^\dagger)$ [26]. A drive is prepared in some normalized mode f', by a source with its own clock.

Proposition 35 (Mode overlap)**.** *Let the drive occupy mode f' with mean occupation N', all modes orthogonal to f' being in the vacuum, and let $c = \int \overline{f(\omega)} f'(\omega)\, d\omega$. Then the bit channel equals the channel of Theorem 33 for the reduced state of mode f, which is the output of a pure-loss channel of transmissivity $|c|^2$ and has mean occupation $|c|^2 N'$. A flip with failure ϵ therefore needs*

$$N' \;\geq\; \frac{1}{|c|^2}\left[\frac{\theta_\epsilon(1 - 2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}\right]. \tag{76}$$

If $f'(\omega) = f(\omega)e^{i\omega\Delta}$, a copy of f delayed by Δ, then $c = \int |f(\omega)|^2 e^{i\omega\Delta} d\omega$. For a positive-frequency field this integral is over $\omega > 0$. Extending a narrowband Gaussian spectral density of width Ω to the full real line gives $|c| \simeq e^{-\Omega^2\Delta^2/2}$ and an occupation penalty $e^{\Omega^2\Delta^2}$ in the range where this approximation controls the overlap. The exact positive-frequency integral, not this Gaussian approximation, determines the cost at long delays.

Proof. Write $f = \bar{c}f' + r$ with $\int \overline{f'}\, r\, d\omega = 0$ and $\|r\|^2 = 1 - |c|^2$. Since $g \mapsto b_g$ is antilinear, $b_f = c\, b_{f'} + b_r$, where $[b_r, b_r^\dagger] = 1 - |c|^2$ and b_r commutes with $b_{f'}$ and $b_{f'}^\dagger$. The mode $r/\|r\|$ is orthogonal to f' and in the vacuum, independent of mode f', so the state of mode f is that of a beam splitter of transmissivity $|c|^2$ with vacuum in the other port, and its occupation is $|c|^2 N'$. Q_g involves only mode f, so the bit channel depends only on this reduced state, and Corollary 34 applies to it. Translation gives the exact Fourier overlap. The Gaussian specialization follows only after the stated extension of the frequency support. □

The prepared drive has energy $N'\bar{\omega}'$, while only the fraction $|c|^2$ of its occupation reaches the selected mode. This is a charged preparation resource, not automatically dissipated heat. The source/switch clock offset therefore enters Paper B's feedback acceptance as a mode-matching loss.

D A representation generated by propagation

The situation is different when the "environment" is part of the same closed dynamics. In the local propagation model of Paper A, a particle with internal space $\mathbb{C}^s$ moves on a lattice Λ by a unitary step $W = \sum_{h\in N} T_h \otimes A_h$, where T_h translates by h [25]. The internal algebra is represented by

$$\phi : M_s(\mathbb{C}) \to B\big(\ell^2(\Lambda) \otimes \mathbb{C}^s\big), \qquad \phi(A) = \mathbf{1} \otimes A, \tag{77}$$

a faithful unital $*$-homomorphism, and $[\phi(A), W] = \sum_h T_h \otimes [A, A_h]$ is nonzero unless A commutes with every A_h. The requirement of Definition 28 together with nontrivial coupling is met exactly, because nothing is traced out: the translational degrees of freedom and the internal ones evolve under one unitary. The contrast with Section C is the content of Corollary 32: exactness is lost when the controlling degrees of freedom are an environment in a state with continuous spectral support.

The propagation also supplies the quadratic-space morphism that Section A found missing. Near a band point the step has a first-order symbol $H(q) = \sum_i q_i\alpha_i + \mu\beta$ with Hermitian coefficients [25].

Proposition 36 (The propagation symbol as a quadratic-space morphism)**.** *Let $j(q) = \sum_i q_i\alpha_i$ with $\alpha_1, \ldots, \alpha_d$ linearly independent. Then $j(q)^2 = Q(q)\mathbf{1}$ for a quadratic form Q if and only if, for $\mu = 0$, every eigenvalue of $H(q)$ lies on one shell $\omega^2 = Q(q)$. In that case Q is positive definite, j is an isometry from $(\mathbb{R}^d, Q)$ into $\mathrm{Herm}(s)$ with the normalized Hilbert–Schmidt form, and j extends to a unital $*$-homomorphism $Cl(\mathbb{R}^d, Q) \to M_s(\mathbb{C})$. If moreover $\mu \neq 0$, $\beta^2 = \mathbf{1}$ and $\{\alpha_i, \beta\} = 0$, then*

$$J(\omega, q) = \omega\beta - \sum_i q_i\beta\alpha_i \tag{78}$$

satisfies $J(\omega,q)^2 = (\omega^2 - Q(q))\mathbf{1}$ *and extends to a unital homomorphism* $Cl(\mathbb{R}^{1+d},\eta) \to M_s(\mathbb{C})$ *with* $\eta(\omega,q) = \omega^2 - Q(q)$*, which preserves the involution in which* β *is Hermitian and the* $\beta\alpha_i$ *are anti-Hermitian.*

Proof. The first statement is the single-shell theorem of Paper A [25] with $\mu = 0$; isometry follows from $\frac{1}{s}\operatorname{tr} j(q)^2 = Q(q)$, and the extension from the universal property of Clifford algebras. For J, expand J^2 using $\beta^2 = \mathbf{1}$ and $\beta\alpha_i\beta = -\alpha_i$: the cross terms $\omega q_i\{\beta, \beta\alpha_i\}$ vanish and $\sum q_i q_j \frac{1}{2}\{\beta\alpha_i, \beta\alpha_j\} = -Q(q)$. Hermiticity of β and anti-Hermiticity of $\beta\alpha_i = -\alpha_i\beta$ follow from $(\beta\alpha_i)^* = \alpha_i\beta$. □

J is the map $j : (V,q) \to (W,Q)$ of Section A with $(V,q) = (\mathbb{R}^{1+d},\eta)$, a Lorentzian quadratic space, and W the real span of β and the $\beta\alpha_i$ in $M_s(\mathbb{C})$, with the form defined by $Y^2 = Q_W(Y)\mathbf{1}$. Here it is supplied by the dynamics, and its existence is equivalent to an observable property of the propagation: no birefringence.

On the lattice the map is exact only at leading order. For $s = 2$ and $W(k) \in SU(2)$, the effective generator $H_{\text{eff}}(k) = (i/\tau)\operatorname{Log} W(k)$ is traceless, so $H_{\text{eff}}(k)^2 = \omega(k)^2\mathbf{1}$ holds at every k; what fails is linearity and isotropy of $k \mapsto H_{\text{eff}}(k)$. In the Weyl automaton of Paper A, $H_{\text{eff}}(k) = c\,\sigma\cdot k$ up to corrections of relative order $|k|\ell$, which include a parity-odd anisotropic term. In the split-step walk the quadratic identity $\omega^2 = c_{\text{eff}}^2 q^2 + \mu^2$ fails at order $(qa)^4$, with coefficient $-\theta^2/45 + O(\theta^4)$ in lattice units [27]. These orders are the precise sense in which a lattice propagation gives an approximate representation.

E A finite conserved control core

A finite prescribed circuit, energy supply and thermal ports can share a time-independent Hamiltonian. We combine energy-ladder control [20] with a finite clock [22], using complete energy sectors and an exactly solvable history clock. Register, gate permissions and initial resources are supplied inputs.

E.1 A guarded battery for the whole circuit

Let $H_D = \delta\sum_a q_a|a\rangle\langle a|$, with integers $0 \le q_a \le Q$, and let $H_B = \delta\sum_{n=0}^{K} n|n\rangle\langle n|$, $K \ge 2Q$. The register D includes its program, retained system and all thermal ports; an inert reference can be appended. Define

$$V(U)|a, E - q_a\rangle = \sum_b U_{ba}|b, E - q_b\rangle, \qquad Q \le E \le K, \tag{79}$$

for any unitary U, and take the identity on incomplete energy sectors.

Proposition 37 (Finite energy-conserving circuit lift)**.** *The operator* $V(U)$ *is unitary, commutes with* $H_0 = H_D + H_B$*, and* $V(U)V(W) = V(UW)$*. Put* $K = 2Q + L - 1$ *and prepare*

$$|\beta_L\rangle = \sqrt{\frac{2}{L+1}}\sum_{j=1}^{L}\sin\frac{\pi j}{L+1}|Q + j - 1\rangle, \qquad e_B = 2\sin\frac{\pi}{2(L+1)}.$$

For any input of D entangled with a reference and initially independent of this battery, the joint half trace distance between the lifted circuit $V(U_m)\cdots V(U_1)$ and $U_m\cdots U_1\otimes I_B$ is at most $\min\{1, 2Qe_B\}$. The comparison retains the battery and all correlations.

Proof. Every complete energy sector carries a copy of the entire data space; the others carry the identity. This proves unitarity, conservation and composition. Extend the battery vector by zero to a bilateral ladder with shift S. Its adjacent overlap is $\cos[\pi/(L+1)]$, so $\|(S^k - I)\beta_L\| \le |k|e_B$. With $D_q = \sum_a |a\rangle\langle a|\otimes S^{-q_a}$, $V(U) = D_q(U\otimes I)D_q^*$ on the guarded inputs. Each dressing difference has vector norm at most Qe_B, even with a retained reference. Their sum bounds the joint state distance after purification. The guards make this bilateral calculation agree with the finite-sector action; no shift wraps through an endpoint. □

The bound concerns the whole compiled circuit at a fixed register energy range. Fresh thermal units in a growing protocol enlarge Q. Battery coherence, mean energy and preparation remain resources; the battery is not an exact catalyst.

E.2 Autonomous timing and a usable output window

Let $V_j = V(U_j)$ and add clock states $|0\rangle, \dots, |m\rangle$. For $J > 0$ and $\Lambda = Jm$, set

$$H = H_0 + \Lambda I + \sum_{j=1}^{m} J\sqrt{j(m-j+1)}\big(|j\rangle\langle j-1|\otimes V_j + \text{h.c.}\big). \tag{80}$$

The clock-dependent interaction is part of the apparatus energy.

Proposition 38 (Positive autonomous execution)**.** *The Hamiltonian is nonnegative and commutes with H_0. From clock state $|0\rangle$, at $T = \pi/(2J)$ it gives*

$$e^{-iHT}(|0\rangle\otimes|\psi\rangle) = e^{-i\Lambda T}(-i)^m|m\rangle\otimes e^{-iH_0T}V_m\cdots V_1|\psi\rangle.$$

If only the first $r \le m$ edges perform gates and the rest are identities, then throughout $[T/2, 3T/2]$ the data's half trace distance from the completed lifted circuit, including its free evolution, is at most

$$P_{m,r} = \Pr\{\mathrm{Bin}(m, 1/2) < r\}. \tag{81}$$

The joint data–clock distance to the completed data and the normalized completed-tail clock state is at most $\sqrt{P_{m,r}}$.

Proof. Put $W_j = V_j\cdots V_1$, $W_0 = I$, and $\mathcal{W} = \sum_j |j\rangle\langle j|\otimes W_j$. Conjugation by $\mathcal{W}^*$ reduces the hopping term to the spin-$m/2$ chain $2JJ_x$, commuting with H_0. Its spectrum lies in $[-Jm, Jm]$; the shifted block lies in $[0, 2\Lambda]$. The spin rotation transfers 0 to m at T, as in perfect-transfer chains [21]. Its position law is $\mathrm{Bin}(m, \sin^2 Jt)$. Every prefix $j \ge r$ has the same completed data. Tracing the clock leaves at most the incomplete probability as error. Projection onto the completed tail has squared overlap one minus this probability, giving the square-root joint bound. On the stated window $\sin^2 Jt \ge 1/2$. □

The joint target uses the normalized tail, not the unchanged full clock wavefunction; the latter comparison instead has bound $2\sqrt{P_{m,r}}$. For $r = 10$, $m = 64$ and $\Lambda = .4$, $P_{m,r} \approx 1.77111\times10^{-9}$ and $T \approx 251.327$ in the declared energy units. The clock has 65 states. This finite window is not permanent halting: the closed clock eventually returns and reverses its gates. Its initialization, idle states and any output receiver remain charged resources.

E.3 Two cycles through the same thermal contact

The core has seven qubits: detector S, retained body Q, contact F, output cells E_1, E_2 and dephasing cells D_1, D_2. The body and dephasing cells have zero bare gap. The other four have gap δ, so the register has maximum charge 4. Prepare S, F, E_1, E_2 independently in $\tau = \mathrm{diag}(1-\epsilon, \epsilon)$, $\beta\delta = \ln[(1-\epsilon)/\epsilon]$, and each D_i in its zero-gap Gibbs state $I/2$. The body may be entangled with a retained reference and is never re-prepared.

Cycle i consists, in order, of

$$e^{-i\theta_i X_S P_Q}, \quad P_Q = |1\rangle\langle 1|_Q, \quad \mathrm{CZ}_{S,D_i}, \quad \mathrm{CX}_{S,Q}, \quad \mathrm{SWAP}_{S,F}, \quad \mathrm{SWAP}_{F,E_i}\,.$$

The dephased detector directly controls the body operation. Each gate is lifted and placed on an edge of Eq. (80); no external switching times enter. The first swap transfers the old detector and all its correlations to F, putting an independent Gibbs state in S. The second moves the old state to E_i and restores F from the unused Gibbs cell. Both conserve bare energy. Replacement is exact against every retained reference, including the battery, and returns a thermal state rather than a pure blank.

The first recorded bit has probability ϵ when the body is $|0\rangle$ and $\epsilon + (1-2\epsilon)\sin^2\theta_1$ when it is $|1\rangle$. This is a genuine body reading. The command acts on the already retained body and can damage its initial coherence; it is not a coherence-preservation claim. For $\epsilon = .1$, $\delta = .1$, $\theta_1 = \pi/3$ and $\theta_2 = \pi/2$, with the body initially half of a Bell pair, output populations are $.4, .26$. Net heat $Q_E = .046$ equals the battery's mean energy loss. The joint output obeys

$$\beta Q_E = S(\rho_{E_1E_2}) - 2S(\tau) + D(\rho_{E_1E_2}\|\tau\otimes\tau). \tag{82}$$

This follows from $-\ln\tau = \beta H_E + \ln Z$. Other ports, the body, reference, clock and battery remain in the full entropy account.

At $L = 31$ the battery has 39 levels and mean initial energy 1.9. The two sensing rotations give a joint error bound $(\sin\theta_1 + \sin\theta_2)e_B = .183123$; the full purification gives the smaller upper bound .070347. For $\Lambda = .4$, the complete Hamiltonian has cutoff 5.0 and initial mean 2.34. At $L = 127$, the purification upper bound is .017610, mean energy 7.14 and cutoff 14.6. These are executions retaining the complete correlated state.

The cells hold a final energy .066, including .02 initially present; net heat .046 is not automatically emitted radiation. Spatial transport, release recoil and a stationary thermal outlet are further operations. HAFF D gives a material-support branch compatible with the retained core's mean energy [23]. It changes the support's dilaton coupling; the spin cells are not identified with the preceding continuum.

F A local outlet with an autonomous clock

Heat stored in a cell can leave through a local Hamiltonian. Let the output cell be site 0 of a semi-infinite fermion chain,

$$H_{\mathrm{lead}} = \omega\sum_{j\geq 0} n_j + g\sum_{j\geq 0}(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j), \qquad \omega > 2g > 0. \tag{83}$$

Finite open chains have a nonnegative many-body Hamiltonian and spatially nearest-neighbour coupling. Equivalently they are open spin chains with XY exchange under the Jordan–Wigner map; the reference-uniform cell channel below uses this tensor-product realization. The infinite formula denotes its local dynamics, not a finite total-energy thermal bath. Initially site 0 has population p and may be correlated with old references. The β-KMS bath on $j \geq 1$ is initially independent of that entire cell–reference state. Use as comparison the same bath and the site population $\epsilon = (1 + e^{\beta\omega})^{-1}$.

Proposition 39 (Spatial export and reduced thermalization)**.** *Write $w_j(t)$ for the one-particle propagator from site 0. The excess covariance and on-site energies are*

$$\Delta\Gamma(t) = (p - \epsilon)|w(t)\rangle\langle w(t)|, \qquad \Delta E_j = \omega(p - \epsilon)|w_j(t)|^2. \tag{84}$$

For the homogeneous chain the excess hopping energy is zero, and the energy beyond the cell is $\omega(p - \epsilon)(1 - |w_0|^2)$. On the semi-infinite chain,

$$\alpha(t) := w_0(t) = e^{-i\omega t}\frac{J_1(2gt)}{gt}. \tag{85}$$

Let $b(t)$ be the population produced at site 0 from the bath alone and $n_{\text{base}}(t) = |\alpha|^2\epsilon + b(t)$. Against every old external reference, the reduced cell's half trace distance from its Gibbs replacement is at most

$$\min\{1,\ |\alpha|^2 + |\alpha|/2 + |n_{\text{base}}(t) - \epsilon|\}. \tag{86}$$

The retained lead and its initial purification are excluded from that product replacement claim.

Proof. Quadratic fermion evolution sends a covariance to $\Gamma(t) = U(t)\Gamma(0)U(t)^*$, giving the rank-one difference. On a bipartite chain successive components of w have relative phase $-i$ after removing $e^{-i\omega t}$, so $\mathrm{Re}(w_j^* w_{j+1}) = 0$. The surface spectral density is semicircular; its Fourier transform gives Eq. (85). This is standard tight-binding propagation with non-Markovian returns [19, Sec. III]. The thermal bath is even in fermion parity. The cell channel therefore sends population p to $|\alpha|^2 p + b$ and off-diagonal entry ρ_{01} to $\alpha^*\rho_{01}$. For a state with reference blocks ρ_{ab}, the diagonal difference from replacement at population n_{base} is $|\alpha|^2[(1 - \epsilon)\rho_{11} - \epsilon\rho_{00}]$, of trace norm at most $|\alpha|^2$. Positivity gives $\|\rho_{01}\|_1 \leq 1/2$. Adding the change from n_{base} to ϵ proves the bound. □

The lead need not be exactly Gibbs for the bare cell gap. The bath band has width $4g$, and the derivative of the Fermi function is bounded by $\beta/4$, so its occupation differs from ϵ by at most $\beta g/2$ throughout that band. Narrowing g controls this mismatch but lengthens transport. Equation (85) decays as $t^{-3/2}$ with returns and zeros; energy export is not monotone. Finite leads eventually recur. For a chain with sites $0, \ldots, N$, the finite and infinite wavevectors agree through Taylor order N. For $N + 1 > 2gt$, a norm bound on their difference is

$$\frac{2(2gt)^{N+1}}{(N+1)!}\left(1 - \frac{2gt}{N+2}\right)^{-1}.$$

This bounds propagation, not a global distance between finite and infinite thermal states.

For the two output cells with populations $.4, .26$, use two identical leads with $\omega = .1$, $g = .01$, $\beta = \ln 9/.1$. Their total initial excess is .046. At the first surface-amplitude zero, $t = 191.5853$, this excess is outside the cells. A finite calculation with 256 bath sites per lead gives $n_{\text{base}} = .10142988$. At $t = 2000$, excess energy .04599817 is in the leads and .04422294 is beyond site 16. Each bath initially contains energy 2.64854457. These values are transport after circuit completion; they do not discard the output correlations or identify a lattice energy profile with a null CFT stress tensor.

Reduced thermalization also differs from the exact SWAP of Appendix E. A two-site passive exchange at its perfect-transfer time is an iSWAP. Acting on $|+\rangle\langle+| \otimes \operatorname{diag}(.9, .1)$, it gives the cell exactly that Gibbs marginal, but the joint half trace distance from the product of its two marginals is .18. The bath cannot be called fresh for later cell–bath encounters.

F.1 The outlet can remain on during computation

Let H_{hist} be Eq. (80), with only its first r edges active, and let $H_B \geq 0$ be a finite bath Hamiltonian. For a local outlet interaction V_o with $\|V_o\| \leq v$, take $H_{\text{tot}} = H_{\text{hist}} + H_B + V_o + vI$. It is nonnegative and time independent. The offset v is part of its mass and energy account.

Proposition 40 (Always-on outlet bound)**.** *Put $T = \pi/(2J)$, choose $0 < t_0 < T$, and define*

$$P_0 = \Pr\{\text{Bin}(m, \sin^2 Jt_0) \leq r\}, \qquad A_r = J \sum_{j=1}^{r} \sqrt{j(m-j+1)}.$$

Compare the actual joint state at $t \in [t_0, T]$ with completed lifted data, including free evolution to t_0, and the free clock wavefunction at t_0, subsequently evolved with the outlet and identity clock edges. Keeping every reference, their half trace distance is at most

$$\min\{1,\ vt_0 + 2\sqrt{P_0} + 4A_r(t - t_0)\sqrt{P_0}\}. \tag{87}$$

Proof. Removing the always-on outlet only up to t_0 costs at most vt_0 by Duhamel's formula. Replacing all prefixes by the completed data costs at most $2\sqrt{P_0}$ in purification norm. After t_0, compare with the same outlet and free clock; only the first r history edges differ. The free-clock probability of positions $\leq r$ is at most P_0 through T. The norm of the differing edges on that state is at most $4A_r\sqrt{P_0}$. A second Duhamel bound gives the last term. No independence of the output cells or later fresh-bath composition is used. The index $\leq r$ includes the last active edge. □

For any finite transport window the bound can be made small by increasing the idle clock and its energy. For example $r = 10$, $m = 10^{10}$, $J = .001$, $\sin^2 Jt_0 = 128/m$ and two outlets with $v = .02$ give $t_0 = .1131371$, $T = 1570.7963$ and the sufficient joint bound .002262742. This uses $10^{10} + 1$ clock states and clock mean energy 10^7; it proves finite compatibility, not an efficient clock. A direct smaller Hamiltonian with 17 clock states, five battery levels and outlet norm 10^{-5} gives joint purification distance .00015225 below its certificate .00025067. Thus spatial export need not be externally switched, but a fixed-size indefinitely operating exhaust is not obtained.

G Thermal sources and their preparation

A conformal fermion defect gives a local relativistic alternative to the lattice outlet. The real-time defect construction and its thermal current are established results [17, Sections 2, 3 and 5]. Take two independent copies of the Majorana construction, so each port is a complex chiral fermion with $c = 1$. In the folded description, a constant contact angle acts on the two fields at the same contact time:

$$\begin{pmatrix}\psi_1^{\mathrm{out}}\\ \psi_2^{\mathrm{out}}\end{pmatrix} = S_\theta \begin{pmatrix}\psi_1^{\mathrm{in}}\\ \psi_2^{\mathrm{in}}\end{pmatrix}, \qquad S_\theta = \begin{pmatrix}\cos\theta & -i\sin\theta\\ -i\sin\theta & \cos\theta\end{pmatrix}, \quad \chi = \sin^2\theta. \tag{88}$$

The inputs are jointly independent, zero-charge quasifree thermal streams in one specified time coordinate, at $T_H < T_o$. This assumes a prepared reservoir, rather than an already prepared outgoing field.

Proposition 41 (Thermal transfer and its retained correlations)**.** *For $\omega > 0$ put $n_T = (1 + e^{\omega/T})^{-1}$. The source-port occupation and transferred energy current are*

$$n_\chi = (1-\chi)n_{T_H} + \chi n_{T_o}, \qquad q_\chi = \frac{\pi\chi}{12}(T_o^2 - T_H^2). \tag{89}$$

Its cross covariance with the return port has modulus $\sqrt{\chi(1-\chi)}(n_{T_o} - n_{T_H})$. For $0 < \chi < 1$ the source is not a KMS state at any single temperature; at $\chi = 1$ it is exactly the supplied hot state. Defining its stationary marginal entropy-current excess, with $h(x) = -x\ln x - (1-x)\ln(1-x)$, by

$$s_\chi = \frac{1}{\pi}\int_0^\infty [h(n_\chi) - h(n_{T_H})]d\omega,$$

one has

$$q_\chi \geq T_H s_\chi + \frac{3s_\chi^2}{\pi}, \tag{90}$$

strictly for $0 < \chi < 1$.

Proof. Rotate the full input covariance with S_θ. The factor $1/\pi$ in the integrals includes particles and holes of a complex chiral fermion; $\pi^{-1}\int \omega n_T d\omega = \pi T^2/12$. At high frequency $n_\chi \sim \chi e^{-\omega/T_o}$ for $\chi > 0$, whereas a single-temperature Fermi occupation has tail coefficient one. For fixed energy the Gibbs variational principle maximizes entropy at $T_{\mathrm{eff}} = \sqrt{T_H^2 + 12q_\chi/\pi}$. Hence $s_\chi \leq \pi(T_{\mathrm{eff}} - T_H)/6$, with equality only for that thermal state. Eliminating T_{eff} gives the bound and its strictness. □

The return occupation is $\chi n_{T_H} + (1-\chi)n_{T_o}$. Both outputs must be kept when accounting for entropy and correlations. This is not thermal replacement of an arbitrary qubit against every retained reference. If the inputs have a cross covariance, even their thermal marginals do not imply Eq. (89). A finite matched-frequency check uses the positive Hamiltonian $H = \omega(n_1 + n_2) + g(c_1^\dagger c_2 + c_2^\dagger c_1)$, $0 < g < \omega$. It commutes with the free energy and transfers equal energy out of the reservoir and into the source. This mode check does not establish a continuum limit for a microscopic moving contact.

Changing the contact has a cost in the same output fields. For a smooth prescribed $\theta(t)$ the full unequal-time covariance is $C_{\mathrm{out}}(t,t') = S(t)\,\mathrm{diag}(C_H, C_o)(t-t')S(t')^\dagger$. The

short-distance kernel is $C_T(s) = iT/[2\sinh(\pi T s)]$. At $t \pm s/2$, its common vacuum term is multiplied by $\cos[\theta(t+s/2) - \theta(t-s/2)]$. Point splitting therefore gives an added current $\dot\theta^2/(4\pi)$ in each output, so the total work supplied to the fields is

$$W_{\text{pump}} = \frac{1}{2\pi}\int \dot\theta^2 dt \geq \frac{(\Delta\theta)^2}{2\pi\Delta t}. \tag{91}$$

This agrees with the energy-shift dissipation form in adiabatic scattering [18, Eq. (8)]; here point splitting derives it exactly for the specified energy-independent contact. The inequality is Cauchy–Schwarz. A cosine ramp gives $W_{\text{pump}} = \pi(\Delta\theta)^2/(16\Delta t)$. Both ports' pump energy is charged; it is separate from the static net thermal transfer. A small angle error alone is no source certificate: $\theta = \theta_0 + \alpha\sin\Omega t$ has mean total pump current $\alpha^2\Omega^2/(4\pi)$. Instantaneous thermal entropy formulas do not apply during this switch. The controller that supplies this work and the contact's material stress still require their own realization. The input preparation has the following finite-window alternative.

G.1 Thermal correlations from a finite-energy preparation

Fix a bounded interval I of length L and $T > 0$. The geometric KMS construction is locally a vacuum pullback by the exponential map [10, Section 2.8, Proposition 2.4 and Theorem 2.5]. Its compact completion permits a finite global preparation. Use the state convention

$$C_\rho(x,y) = \frac{i\sqrt{\rho'(x)\rho'(y)}}{2\pi[\rho(x) - \rho(y) - i0]}, \qquad \langle T(x)\rangle_\rho = -\frac{c}{24\pi}\{\rho, x\}.$$

Here $c = 1$ for each complex-fermion stream, and the inverse diffeomorphism implementer acts on the vacuum.

Proposition 42 (A thermal restriction with finite global energy)**.** *There is a smooth increasing $\rho : \mathbb{R} \to \mathbb{R}$, equal to the identity outside a compact interval, whose vacuum-orbit state has the complete zero-charge thermal restriction on I. Put $h = \log\rho'$ and $g(y) = \rho'(\rho^{-1}(y))$. Its energy and variance are*

$$\begin{aligned} E_\rho &= \frac{c}{48\pi}\int h'(x)^2\,dx, \\ \operatorname{Var}_\rho(H_f) &= \frac{c}{48\pi^2}\int_0^\infty \omega^3 |\widehat{g-1}(\omega)|^2\,d\omega < \infty, \end{aligned} \tag{92}$$

where $\widehat{k}(\omega) = \int e^{-i\omega y}k(y)dy$. The global state is pure, not globally KMS. Its outside correlations purify the thermal restriction.

Proof. Set $\alpha = 2\pi T$ and prescribe $\rho = e^{\alpha x}$ on a neighborhood of I. Extend $\log\rho'$ smoothly with compact support. A disjoint bump can adjust $\int(e^h - 1)$ to zero: a positive bump has unbounded positive integral, while a negative bump on sufficiently large support has negative range exceeding any fixed surplus. Then $\rho(x) = x + \int_{-\infty}^x (e^h - 1)$ is a compact completion, up to an irrelevant translation on I. Its covariance on I is $iT/[2\sinh\pi T(x - y - i0)]$. Wick's rule identifies all local fermion correlations; the cited geometric construction gives the

corresponding observable-net restriction. Integrating the Schwarzian gives the first moment in Eq. (92). Conjugating H_f into the vacuum gives $T(g) + E_\rho$. Since $H_f\Omega = 0$, the variance is the vacuum second moment of $T(g-1)$, with the normalization of [11, Eq. (4.11)]. The test $g - 1$ is smooth and compact, so that moment is finite. □

Proposition 43 (Preparation cost in the vacuum orbit)**.** *Every compact-diffeomorphism vacuum preparation with this thermal restriction obeys*

$$E_\rho \geq \frac{\pi c T^2 L}{12} - \frac{cT}{6} \tanh \frac{\pi T L}{2}. \tag{93}$$

The right side is the infimum in this class when outside support is unrestricted, and is not attained for $T, L > 0$.

Proof. Thermal stress gives $\{\rho, x\} = -(2\pi T)^2/2$ on I. Thus $w = h'$ satisfies $w' = (w^2 - \alpha^2)/2$. The branches $w = \pm\alpha$ and the nonsingular coth branches have $w^2 \geq \alpha^2$. The remaining branches are $-\alpha \tanh[\alpha(x-s)/2]$; their derivative-square integral on I is minimized when s is its midpoint. That integral is $\alpha^2 L - 4\alpha \tanh(\alpha L/4)$. For its compact completions, flatten the endpoint derivatives in collars of width ϵ, at cost $O(\epsilon)$, and join the endpoint h-values to zero over length D, at cost $O(D^{-1})$. A disjoint bump of scale D corrects the displacement integral with bounded amplitude and cost $O(D^{-1})$. Taking $\epsilon \to 0$ and then $D \to \infty$ reaches the infimum. Zero exterior cost would force $h' = 0$ outside I, incompatible with its endpoint derivatives. □

This is a preparation-class bound, not a minimum energy for all states or complete devices. Its asymptotic slope is the thermal current. Local negative stress outside I is included; counting only the positive plateau gives a different quantity. For $L = 8$ and $T = 1/(2\pi), .31894305$, explicit smooth completions cost $.11257242, .60868371$, above the respective infima $.02748002, .15992956$. Increasing support or sharpening a boundary can approach the infimum without keeping the drive or rms resources uniformly bounded.

For a regular increasing clock $\eta(v)$ on a finite interval, complete $\rho(v) = \exp[2\pi T \eta(v)]$ instead. The entire local covariance is the thermal η covariance pulled back by that clock, and

$$\langle T(v) \rangle_\rho = \eta'(v)^2 \frac{\pi c T^2}{12} - \frac{c}{24\pi} \{\eta, v\}. \tag{94}$$

Two independently prepared copies therefore supply the two-port input correlations on the operating interval, including the output cross covariance and switching stress. Their global purity and inside–outside correlations remain in the ledger. Thermal local correlations alone do not establish a finite-window erasure bound. The cost in Eq. (92) uses the actual physical coordinate; Eq. (93) cannot be applied to a rapidity interval without its clock conversion. The local diffeomorphism has a driven stress-unitary preparation path. Its autonomous bounded approximation is developed in Appendix I.1.

G.2 A record-selected source is not its average covariance

Suppose a retained classical program R has probabilities p_r, is jointly independent of the two prepared incoming streams, and selects θ_r. Conditional on r, the preceding contact law holds.

For a finite frequency regulator with independent input mode pairs, write $\rho_{r,N} = \bigotimes_{j=1}^{N} \rho_{r,j}$ for the reduced source and $\bar{\rho}_N = \sum_r p_r \rho_{r,N}$ for its unconditional state. The return stream and program are retained in the joint process.

Proposition 44 (The source carries the program's relations)**.** *For distinct frequencies j, k, with $\Delta n_j = n_o(\omega_j) - n_H(\omega_j)$,*

$$\mathrm{Cov}(n_j, n_k) = \mathrm{Var}_p(\chi_r)\, \Delta n_j \Delta n_k. \tag{95}$$

Furthermore,

$$S(\bar{\rho}_N) = \sum_r p_r S(\rho_{r,N}) + \mathcal{I}_N, \qquad 0 \le \mathcal{I}_N \le H(R). \tag{96}$$

Here $\mathcal{I}_N = I(R : \text{source})$ in the classical–quantum record state. The product state with the same one-mode occupations has entropy $\sum_j h(\sum_r p_r n_{r,j})$, generally larger. If $H(R) = o(\mathcal{T})$ on a growing observation window and the conditional stationary source entropy rates exist, the unconditional excess rate is $\bar{s} = \sum_r p_r s_{\chi_r}$. Its mean current obeys

$$\bar{q} \ge T_H \bar{s} + \frac{3\bar{s}^2}{\pi}.$$

This limit assumes the same program is retained across the window, not a fresh independently randomized choice at each mode or time.

Proof. Conditional occupations are independent and have means $n_{r,j} = n_H(\omega_j) + \chi_r \Delta n_j$. Conditional covariance therefore vanishes; covariance of their means gives Eq. (95). The entropy identity is the classical–quantum mutual information; its upper bound is the classical program entropy. Subadditivity gives the product-marginal upper bound. Divide the identity by the window length. Averaging Eq. (90) and using $\sum_r p_r s_r^2 \ge \bar{s}^2$ proves the current bound. □

A single persistent program bit can thus create correlations across an arbitrarily long source without contributing an extensive new entropy of its own. Replacing that state by an independent Gaussian state at its averaged covariance destroys those relations. If a new program is acquired at every interval, its entire retained sequence belongs in R; $H(R)$ can then be extensive. Angle switches have the separate two-port work of Eq. (91), and neither entropy identity makes those switches free. These distinctions apply to a gravitational source selected by internal records: mean stress alone does not specify the source state.

There is an exact finite autonomous realization of the static record-controlled exchange. For matched frequencies let $K_j = c_{H,j}^\dagger c_{o,j} + c_{o,j}^\dagger c_{H,j}$ and use

$$V = \sum_r |r\rangle\langle r| \otimes \prod_j e^{-i\theta_r K_j}, \qquad H_0 = \sum_j \omega_j (n_{H,j} + n_{o,j}).$$

It commutes with H_0. Put V on one edge of the positive history clock and identities on the remaining edges. The full Hamiltonian $H_0 + H_{\mathrm{hist}}$ is nonnegative, time independent and conserves H_0. Its endpoint evolution executes the exchange uniformly over all retained references. This is a finite-mode control realization; its clock initialization and finite thermal preparations are inputs. It does not implement the moving conformal contact's material stress or remove its continuum switching work.

Proposition 45 (Two reservoirs and the complete controller ledger)**.** *Let finite reservoirs start at Gibbs states τ_H, τ_o with inverse temperatures $\beta_H > \beta_o$, jointly independent of a complete zero-bare-gap controller M. Its internal correlations and retained references are included in M, but not purifiers of the reservoirs: those would violate the stated initial independence. Under any joint unitary conserving reservoir bare energy, put $Q = \Delta E_H = -\Delta E_o$, $D_i = D(\rho_i' \| \tau_i)$ and $I' = S(\rho_H') + S(\rho_o') + S(\rho_M') - S(\rho_{HoM}')$. Then*

$$(\beta_H - \beta_o)Q = D_H + D_o + I' - \Delta S_M. \tag{97}$$

If the complete controller entropy is unchanged, the right side is $D_H + D_o + I' \geq 0$. Returning its marginal does not mean eliminating its output correlations.

Proof. The Gibbs identity gives $D_i = \beta_i \Delta E_i - \Delta S_i$. Initial independence and unitary invariance of total entropy give $\Delta S_H + \Delta S_o + \Delta S_M = I'$. Add the Gibbs identities and use energy conservation. This is the multi-reservoir form of the relative-entropy ledger underlying the sharpened Landauer principle [16, Theorem 3]. □

At frequencies $1, 2$, temperatures $.7, 1.4$, program probability .37 and angles $.2, 1.1$, direct joint evolution transfers .13174972 energy into the colder source. Its cross-frequency covariance is .00249817. Source entropy is .86881190, conditional entropy .84854079, and the independent state at averaged occupations has entropy .86900230. A five-state clock has mean energy .12 and interaction cutoff .24; its endpoint matrix and the thermal ledger agree with direct evolution. Both reservoirs have total bare cutoff 6 in these units. None of these cutoffs describes an infinite KMS field.

An operational error also needs an energy qualification before it certifies a source. For $0 \leq H_E \leq E_{\mathrm{cut}} I$, $|\operatorname{tr} H_E(\rho - \sigma)| \leq E_{\mathrm{cut}} D(\rho, \sigma)$. Without a common energy constraint, states at trace distance ϵ can differ by fixed or unbounded energy: mix the vacuum with probability ϵ in occupation $N = 1/\epsilon$. A joint clock or gate-distance certificate alone therefore does not certify stress of an unbounded field. The continuum source and its support require energy-weighted control as well as retained records.

H A physical contact: reaction, source and recoil

A prescribed scattering angle charges its work but leaves its actuator outside the model. Give the angle θ a momentum and inertia $I > 0$, and couple it to a second coordinate y with inertia $J > 0$ and potential

$$V(\theta, y) = \frac{k}{2}(\theta - y)^2 + \frac{K}{2} y^2, \qquad k, K > 0. \tag{98}$$

The coordinates are noncompact; the potential pins them. In the σ_x basis the two-port contact has charges $q_1 = 1, q_2 = -1$. For a smooth, nonnegative profile f_a of integral one, use the current Hamiltonian

$$H = \frac{p_\theta^2}{2I} + \frac{p_y^2}{2J} + V(\theta, y) + \pi \sum_{i=1}^{2} \int : \left(j_i(x) + \frac{q_i \theta f_a(x)}{2\pi} \right)^2 : \, dx. \tag{99}$$

The currents obey $[j_i(x), j_l(x')] = -i\delta_{il}\delta'(x-x')/(2\pi)$. At oscillator cutoffs, normal ordering uses the positive completed-square Hamiltonian. Its θ^2 counterterm must not be dropped. The continuum identities below apply on a domain where the current and stress products and their regulated limits exist. The positive form constructions below supply both a neutral-current Hamiltonian and the contact on the full fermion carrier. A static current displacement can still have infinite particle number; its dressing and asymptotic state need separate representation control. An oscillator cutoff is not a certificate for the full two-temperature fermion channel. The anomaly-induced reaction is the mechanism of the fermion–rotor model [13, Sections 2.1–2.2]. Here it is applied to the pinned two-coordinate actuator and its stored energy.

Proposition 46 (Current reaction with finite contact width)**.** *Set* $J_i = j_i + q_i\theta f_a/(2\pi)$. *Hamiltonian* (99) *gives*

$$(\partial_t + \partial_x)J_i = \frac{q_i}{2\pi}\dot\theta f_a, \qquad I\ddot\theta + V_\theta = -\int f_a \sum_i q_i J_i\,dx, \qquad J\ddot y + V_y = 0. \tag{100}$$

With the free initial currents defined at $t = 0$*, elimination of the field gives*

$$I\ddot\theta + V_\theta + \int_0^t \Gamma_a(t-s)\dot\theta(s)\,ds = -\xi_a(t) - \Gamma_a(t)\theta(0), \tag{101}$$
$$\Gamma_a(s) = \frac{1}{\pi}\int f_a(x) f_a(x-s)\,dx,$$
$$\xi_a(t) = \int f_a(x)\sum_i q_i j_i(0, x-t)\,dx.$$

The integrated retarded coefficient is $\gamma = 1/(2\pi)$*. For a Gaussian profile of standard deviation* a*,*

$$\Gamma_a(s) = \frac{\gamma}{\sqrt{\pi}a}e^{-s^2/(4a^2)}, \qquad \operatorname{Re}\widetilde\Gamma_a(\Omega) = \gamma e^{-a^2\Omega^2}. \tag{102}$$

Proof. The current commutator gives the first equation in (100); canonical commutators give the other two. Solve the transport equation along its characteristics. The initial $q_i\theta(0)f_a/(2\pi)$ term gives the initial slip in (101), and the source integral gives its convolution. The autocorrelation of f_a is even and has integral one on the whole line, hence $\int_0^\infty \Gamma_a(s)\,ds = 1/(2\pi)$. Fourier transformation gives (102). □

Thus damping is derived from the current algebra, not appended to an externally specified pump. For a velocity history extended into the past with bounded derivative,

$$|(\Gamma_a * v)(t) - \gamma v(t)| \le \|v'\|_\infty \int_0^\infty s\Gamma_a(s)\,ds. \tag{103}$$

The last integral is $2\gamma a/\sqrt{\pi}$ for the Gaussian. If f_a is supported in $[-a, a]$, it is at most $2\gamma a$. This controls smooth histories, not the ultraviolet error of an entire quantum channel. The initial slip is also physical: setting the mean angle to zero does not set its operator fluctuations to zero.

For independent zero-charge thermal inputs with Fermi occupations n_H, n_o, Wick contraction in the original port basis gives the unsymmetrized incoming force spectrum

$$S_\xi(\Omega) = \frac{|\widehat{f}_a(\Omega)|^2}{2\pi} \int_{\mathbb{R}} [n_H(E)(1 - n_o(E+\Omega)) + n_o(E)(1 - n_H(E+\Omega))]\, dE. \tag{104}$$

Changing variables in $S_\xi(-\Omega)$ yields

$$S_\xi(\Omega) - S_\xi(-\Omega) = 2\gamma\Omega|\widehat{f}_a(\Omega)|^2. \tag{105}$$

Indeed each occupation integrates $n(E) - n(E+\Omega)$ to Ω over the filled sea. The retarded response is temperature independent; its noise is not. Only at equal temperature T does the spectrum reduce to $\Omega|\widehat{f}_a|^2/[\pi(1 - e^{-\Omega/T})]$ and obey the single KMS fluctuation–dissipation relation. Different-temperature inputs are generally correlated in the charge basis and cannot be replaced by two independent thermal boson baths.

In a point-contact limit with well-defined renormalized products and the symmetric contact prescription, the total field power and mechanical energy satisfy

$$P_{\text{field}} = \tfrac{1}{2}\{\dot\theta, \xi\} + \gamma\dot\theta^2, \qquad \dot E_{\text{mech}} = -P_{\text{field}}. \tag{106}$$

This follows by substituting the incoming–outgoing current jump into the Sugawara stress and using the canonical mechanical equation. At finite width the full Hamiltonian supplies the energy identity, including retained current energy. The positive prescribed-angle pump term alone is not the power of a quantum actuator. Its velocity correlates with the incoming current. In a stationary interacting Gibbs state $\langle\dot\theta^2\rangle$ can be positive while the net energy transfer is zero.

Proposition 47 (A mechanical record changes the actual field source)**.** *Take a common initial state with zero first moments of the two mechanical coordinates, their momenta and the currents, and finite stress expectations as above. In branch R, displace y by A_R and leave the other first moments unchanged. Relative to the undisplaced evolution under the same Hamiltonian, let $v_R(t)$ be the deterministic mean angle velocity and define*

$$h_R(t,x) = \int_0^t v_R(s) f_a(x - t + s)\, ds, \qquad \Delta\langle T_{\text{field}}(t,x)\rangle = \gamma h_R(t,x)^2. \tag{107}$$

Then

$$\Delta E_{\text{mech}}(t) + \gamma \int h_R(t,x)^2\, dx = \frac{k+K}{2} A_R^2. \tag{108}$$

The actual excess stress scales as A_R^2 for fixed hardware.

Proof. The Heisenberg equations are linear. Displacing the initial y adds the classical solution to the undisplaced operator solution; it leaves every centered correlation unchanged. Characteristics give $\Delta\langle J_i\rangle = q_i h_R/(2\pi)$. The zero first moments remove the cross term in the stress difference, giving (107). Differentiating its spatial integral gives $v_R(\Gamma_a * v_R)$, the negative derivative of the mechanical displacement energy. The initial displacement cost is $(k+K)A_R^2/2$. □

This cancellation compares displaced and undisplaced evolutions; it does not discard the quantum noise term in (106). The common baseline stress still has to be counted. Energy retained inside the contact profile is not yet energy that has reached an external outlet. An acquired record can select A_R through a paid preparation. That preparation, its storage and its physical copying are additional operations, not free changes of a target potential.

For $I = 10, J = 20, k = 1, K = .1, a = .2$, a 192-mode positive oscillator regulator initially displaced by $A_R = 1.3$ has excess energy .9295. At $t = 80$, .5103950 is in its shifted field coordinates and .4191050 remains mechanical. Direct time-domain integration of the continuum memory equation and separate spatial integration give .5103953 and .4191052, with total energy residual 4.8×10^{-7} at step .005. This is a check of the continuum energy identity, not a cutoff channel certificate. The Gaussian profile has spatial tails. In the regulator's displaced interacting Gibbs state at $T = .02$, $t = 8$, the two branches $A_R = .5, 1.3$ give $\langle \sin^2 \theta \rangle = .426644, .716636$. These are coordinate observables, not transmission probabilities for the two-temperature fermion problem. A finite regulated bath eventually recurs.

H.1 Installing a program on the running carrier

Proposition 48 (Autonomous installation on an always-running carrier)**.** *Let $0 \le H_{\mathrm{phys}} \le E_{\mathrm{cut}} I$ be a finite actuator–field regulator, and let a prefix of r gates prepare its mechanical state from an already acquired record. Use a positive history clock with $m \ge r$, strength J, transfer endpoint $T = \pi/(2J)$ and identity edges after the prefix. The common, always-on Hamiltonian*

$$H = H_{\mathrm{hist}} + I_{\mathrm{clock}} \otimes H_{\mathrm{phys}} \ge 0 \tag{109}$$

has, for $t_0 \le t \le T$, joint distance from the completed-prefix state freely continued under H_{phys} for $t - t_0$ at most

$$d_{\mathrm{joint}} \le \min\{1, E_{\mathrm{cut}} t_0 + 2\sqrt{P_0} + 4A_r(t - t_0)\sqrt{P_0}\}, \tag{110}$$

where $P_0 = \Pr\{\mathrm{Bin}(m, \sin^2 Jt_0) \le r\}$ and $A_r = J \sum_{j=1}^{r} \sqrt{j(m - j + 1)}$. All prior references are retained. For each finite continuation window and specified error, finite sufficiently large clock resources attain that error.

Proof. During the prefix, Duhamel comparison with H_{hist} gives $E_{\mathrm{cut}} t_0$. Completing the prefix on its clock tail costs $2\sqrt{P_0}$. Thereafter compare with the identity-edge clock and the same H_{phys}. Their reference state factors across clock and data, so the nonidentity edges have norm at most $4A_r\sqrt{P_0}$; Duhamel gives the last term. The binomial lower tail decreases up to T. Set $\sin^2 Jt_0 = \mu/m$ with μ a sufficiently large multiple of $\log m$. Then $t_0 \to 0$ and the tail beats the polynomial growth of A_r. Choose T to contain the specified continuation window. □

For a bounded physical source observable $0 \le B \le B_{\mathrm{cut}} I$, the expectation error is at most $B_{\mathrm{cut}} d_{\mathrm{joint}}$. Thus a finite-cutoff record can install a mechanical program while the same carrier and field continue to evolve; no external switch at t_0 is required. A three-level actuator and three-level field, with the old record reference retained, give joint distance .00204248 below

the bound .04082928 in one small-clock check. The clock mean energy is 960, so that check is not an installation inside the mass-10 gravitational body. Larger clocks improve the guarantee at larger energy and dimension. The continuum displacement, infrared sector, spatial transport and matching to a moving action need separate control.

The upper energy cutoff is unnecessary for a class of permitted gates. The relevant condition concerns how a gate changes energy, rather than the spectrum of the carrier itself.

Proposition 49 (Energy-regular return on an unbounded carrier)**.** *Let $H_p \geq 0$ be self-adjoint, and let the finite prefix unitaries $W_j = V_j \cdots V_1$, $W_0 = I$, preserve* $\operatorname{Dom} H_p$*. Suppose*

$$B_j = W_j^* H_p W_j - H_p, \qquad b = \max_{j \leq r} \|B_j\|, \qquad c = \max_{j \leq r} \|[H_p, B_j]\|$$

extend to bounded operators, with B_j preserving $\operatorname{Dom} H_p$*. Let ψ be the normalized full input purification, including prior references, with $E = \|H_p \psi\| < \infty$. Use the same positive history clock as in Proposition 48. For $t_0 \leq t \leq T$, the joint vectors under $H_{\mathrm{hist}} + I \otimes H_p$ and the reference completed-prefix evolution differ by at most*

$$\varepsilon_m(t) = (E + b)t_0 + 2\sqrt{P_0} + 4A_r(t - t_0)\sqrt{P_0}. \tag{111}$$

Their physical rms energies obey

$$\|H_p \Psi_a(t)\| \leq M_a = E + tc + b, \qquad \|H_p \Psi_i(t)\| \leq M_i = E + b. \tag{112}$$

These bounds do not grow with the padded clock. If a self-adjoint source observable Q has $\operatorname{Dom} H_p \subseteq \operatorname{Dom} Q$ *and $\|Q\zeta\| \leq a\|H_p\zeta\| + d\|\zeta\|$, then*

$$|\langle Q \rangle_a - \langle Q \rangle_i| \leq \varepsilon_m(t)\,[a(M_a + M_i) + 2d]. \tag{113}$$

In particular, this certifies the unbounded physical energy $Q = H_p$. For any fixed finite continuation window, $\varepsilon_m \to 0$ with sufficient clock padding.

Proof. The ideal history branches are orthogonal in the clock, and $\|H_p W_j \psi\| = \|(H_p + B_j)\psi\| \leq E + b$. The early Duhamel comparison therefore costs $(E + b)t_0$; the prefix tail and late comparison are as in Proposition 48. This gives a vector bound on the retained purification, not only a distance between reduced states.

Put $G = \sum_j |j\rangle\langle j| \otimes W_{\min(j,r)}$ and write the ungated clock as K_m. In the history gauge the total Hamiltonian is

$$G^* H_{\mathrm{tot}} G = K_m \otimes I + I \otimes H_p + \sum_j |j\rangle\langle j| \otimes B_{\min(j,r)}.$$

The last term and its commutator with H_p are bounded by b, c. The propagator commutator consequently has norm at most tc, so the gauge-state rms energy is at most $E + tc$. Transforming back adds at most b. The same domain-preserving commutator argument applies to the reference. Finally write the expectation difference as $\langle \Psi_a - \Psi_i, Q\Psi_a \rangle + \langle Q\Psi_i, \Psi_a - \Psi_i \rangle$ and apply Cauchy–Schwarz. The large-clock limit follows from the same binomial tail choice. □

A positive quadratic-form source needs a weaker estimate. If $0 \le Q \le aH_p + dI$ as forms, Cauchy–Schwarz in the form norm gives

$$|\Delta\langle Q\rangle| \le (\sqrt{aM_a + d} + \sqrt{aM_i + d})\sqrt{a\varepsilon_m(M_a + M_i) + d\varepsilon_m^2}. \tag{114}$$

Here $\|H_p^{1/2}(\Psi_a - \Psi_i)\|^2 \le \|\Psi_a - \Psi_i\|\,\|H_p(\Psi_a - \Psi_i)\|$ proves the bound. A normally ordered local stress is not automatically a positive form or H_p-bounded; its relative estimate must be supplied.

Proposition 50 (A smooth chiral stress certificate)**.** *Consider a positive-energy chiral CFT satisfying the Fewster–Hollands hypotheses, with $H_f = \int T(x)\,dx$ and central charge c_{ch}. Let $0 \le f \in C_c^\infty$ and $\Lambda_f > \|f\|_\infty$. Define*

$$\begin{aligned} q_f &= \frac{c_{\mathrm{ch}}}{12\pi}\int |(\sqrt{f})'|^2\,dx, \\ q_{\mathrm{comp}} &= \frac{c_{\mathrm{ch}}}{48\pi}\int \frac{f'(x)^2}{\Lambda_f - f(x)}\,dx, \qquad d_f = q_f + q_{\mathrm{comp}}. \end{aligned} \tag{115}$$

Then, as quadratic forms,

$$0 \le T(f) + q_f I \le \Lambda_f H_f + d_f I. \tag{116}$$

If $H_f \le a_0 H_p + b_0 I$ as forms for the physical Hamiltonian of Proposition 49, its expectation certificate (114) *applies to the actual local stress $T(f)$ with $a = \Lambda_f a_0$ and $d = d_f + \Lambda_f b_0$. The constant shift cancels between the two states.*

Proof. The chiral quantum energy inequality gives $T(f) \ge -q_f I$ [12, Theorem 4.1]. To obtain the upper bound, one cannot apply that theorem directly to the non-Schwartz function $\Lambda_f - f$. Choose $k \in C_c^\infty$, $0 \le k \le 1$, equal to one on $[-1, 1]$, and set $\chi_L(x) = k(x/L)^2$. For L enclosing $\operatorname{supp} f$, the admissible test $g_L = \chi_L(\Lambda_f - f)$ has QEI constant

$$q(g_L) = q_{\mathrm{comp}} + \frac{c_{\mathrm{ch}}\Lambda_f}{12\pi L}\int |k'|^2.$$

Its two derivative supports are disjoint. On the common CFT core, $T(\chi_L)$ tends in expectation to H_f, using the integrable stress and the translation charge. Thus $T(g_L) = \Lambda_f T(\chi_L) - T(f) \ge -q(g_L)I$ gives the upper bound. The positive form then extends continuously to $\operatorname{Dom} H_f^{1/2}$. Combining it with the stated physical form inequality and (114) proves the certificate. □

For the inertial contact there is a useful conditional transfer of this field bound to the interacting carrier. In its positive completed-square regulator, write the shifted field energy as $\sum_l \omega_l(a_l + s_l\theta)^*(a_l + s_l\theta)$ and put $d_a = \sum_l \omega_l|s_l|^2 = k_{\mathrm{cur}}\int f_a^2/(4\pi)$, where $k_{\mathrm{cur}} = 2$. Since $V \ge kK\theta^2/[2(k + K)]$, the elementary square inequality gives

$$H_f \le 2H_{\mathrm{shifted}} + 2d_a\theta^2 \le \left[2 + \frac{4d_a(k + K)}{kK}\right] H_p. \tag{117}$$

In the neutral current Fock representation, that positive Hamiltonian can be constructed without a static dressing unitary. Set $g_i(\omega) = q_i\sqrt{\omega}\,\widehat{f_a}(\omega)/(2\pi)$, so $\sum_i \int |g_i|^2/\omega = d_a$. On the form domain of $H_0 = H_{\rm mech} + H_f$, define

$$\mathfrak{q}_H[\psi] = \mathfrak{q}_{\rm mech}[\psi] + \sum_i \int_0^\infty \left\| \sqrt{\omega}\, a_i(\omega)\psi + \frac{g_i(\omega)}{\sqrt{\omega}}\theta\psi \right\|^2 d\omega. \tag{118}$$

This includes the counterterm. Put $A_a = 2+4d_a(k+K)/(kK)$. The same square inequalities give $\mathfrak{q}_0 \le (1+A_a)\mathfrak{q}_H$ and $\mathfrak{q}_H \le A_a\mathfrak{q}_0$. Thus the two form norms, with $\|\psi\|^2$ added, are equivalent. The closed positive form defines a self-adjoint Hamiltonian, and (117) holds for it. Coupling cutoffs converging in $L^2(d\omega/\omega)$ converge in the common form norm and hence in norm resolvent on this field space. This does not require $g_i/\omega \in L^2$: when $\widehat{f_a}(0) \neq 0$, the static dressing can fail to be a Fock unitary even though the Hamiltonian exists.

For a compact contact profile, a stress test outside its support reads the physical outgoing current stress itself. A Gaussian profile has tails and is not covered by that exact exterior assertion. The gates must be energy-regular for this interacting H_p; free-field regularity alone does not establish that condition when the gate overlaps the contact. The exterior case admits an exact transfer below. This neutral-current construction does not establish a dressed ground state or asymptotic charged representation. The full fermion carrier and its prepared thermal inputs can instead be treated directly. A uniform zero-width limit does not follow from the form inequality.

The contact on the full fermion carrier. The two complex fermions need not be replaced by two neutral current Fock spaces. Work in their full Dirac-sea CAR representation, with positive translation generator H_f. Smooth local gauge implementers and the current anomaly act on this whole space, including its charged sectors [45, Sections 3.2–3.3 and Appendices A–B]. In the σ_x basis write $J(f) = \sum_i q_i j_i(f)$, $\kappa_c = \sum_i q_i^2 = 2$ and $d_f = \kappa_c\|f\|_2^2/(4\pi)$.

Proposition 51 (A positive contact on the full CAR carrier)**.** *For real $f \in C_c^\infty(\mathbb{R})$ and the pinned mechanical potential* (98)*, the form*

$$\mathfrak{q}_p = \mathfrak{q}_{\rm mech} + \mathfrak{q}_f + \langle\theta J(f)\rangle + d_f\langle\theta^2\rangle$$

is closed and nonnegative on the form domain of $H_0 = H_{\rm mech} + H_f$. It defines the single-contact self-adjoint H_p on the full fermion space. Put $a_0 = kK/[2(k+K)]$ and $L = \max\{2, 1 + 2d_f/a_0\}$. Then $L^{-1}\mathfrak{q}_0 \le \mathfrak{q}_p \le L\mathfrak{q}_0$. The assertion does not require a static dressing unitary.

For $b \in C_c^\infty((1,2))$ of integral one, set $b_D(x) = D^{-1}b(x/D)$ and $f_D = f - (\int f)b_D$, with disjoint supports. The corresponding compact-gauge contacts converge in norm resolvent to H_p. If $d_b = (\int f)^2\kappa_c\|b\|_2^2/(4\pi D)$ and $r_D = \sqrt{d_b/a_0} + d_b/a_0$, then, for $Lr_D < 1$,

$$\|(H_{p,D}+1)^{-1} - (H_p+1)^{-1}\| \le \frac{Lr_D}{1-Lr_D}.$$

Proof. First let $f = F'$ with $F \in C_c^\infty$. Multiplication by $e^{\pm i\lambda F}$ has a Hilbert–Schmidt commutator with the one-particle Hardy projection: its squared norm is $\iint |g(x)-g(y)|^2/(x-$

$y)^2\,dxdy/(4\pi^2)$, finite by smoothness and compact support of $g-1$. Thus $W_\lambda = e^{-i\lambda J(F)}$ is implemented on the full CAR space. On the common smooth core, $[J(F), H_f] = iJ(F')$ and $[J(F), J(F')] = i\kappa_c \int F'^2/(2\pi)$ give

$$W_\lambda H_f W_\lambda^* = H_f + \lambda J(f) + d_f\lambda^2 \geq 0.$$

For general f, use f_D above. On a finite smooth particle–hole core, $J(b_D) \to 0$ strongly: its vacuum norm is $O(D^{-1})$, and commuting it past each fermion changes the test function by multiplication with b_D, of supremum norm $O(D^{-1})$. Meanwhile $d_{f_D} \to d_f$. Passing to the limit and minimizing over λ gives the form inequality

$$|\langle\psi, J(f)\psi\rangle| \leq 2\sqrt{d_f}\,\|\psi\|\,\|H_f^{1/2}\psi\|.$$

Polarization extends the form to $\operatorname{Dom} H_f^{1/2}$. For each mechanical coordinate the shifted field form $A_f(\theta) = H_f + \theta J(f) + d_f\theta^2$ is nonnegative and obeys $A_f \leq 2H_f + 2d_f\theta^2$ and $H_f \leq 2A_f + 2d_f\theta^2$. Integrate over the mechanical coordinates and use $V \geq a_0\theta^2$. The displayed equivalent form norms establish closedness and positivity.

For the compensator, the same current bound and Cauchy–Schwarz give

$$|\langle\theta J((\int f)b_D)\rangle| \leq \sqrt{d_b/a_0}\,\mathfrak{q}_0.$$

The counterterm difference is $d_b\langle\theta^2\rangle$. Hence

$$|\mathfrak{q}_{p,D} - \mathfrak{q}_p| \leq r_D\mathfrak{q}_0 \leq Lr_D\mathfrak{q}_p.$$

The resolvent identity in the $(H_p+1)^{1/2}$ form scale gives the stated estimate. □

The finite-rms preparations of Proposition 42 now belong to the same fermion carrier as the inertial contact. Their outside purifications remain present. A constant flavor rotation preserves the vacuum but generally gives off-diagonal covariance for unequal temperatures; the rotated currents are not two independent KMS baths. The force law therefore uses the original quasifree input correlations, not a substitute product of bosonic thermal states. Finite form energy does not by itself supply the graph regularity required by the autonomous return theorem. On the stated smooth input core, exterior even local gates have the same free and interacting commutators by CAR locality, and the local-source certificate uses $H_f \leq LH_p$.

The compensator is a proof and approximation device, not a free heat outlet. For $\int f = 1$, $\operatorname{supp} f \subset [-a, a]$ and $0 \leq s \leq T < D - a$, its retarded kernel differs from the single contact's by at most $2d_b$. For every fixed D, however, $\int f_D = 0$ makes the integrated kernel vanish. Its dc response is zero, whereas the single contact's is $1/(2\pi)$. Norm-resolvent convergence is consequently not a uniform infinite-time heat-transport statement. The single-contact Hamiltonian itself only couples in $\operatorname{supp} f$; it does not require a physical compensator extending to infinity.

These estimates certify smooth, finite-window source charges. To use a sharp or unbounded gravitational kernel, one must prove the relevant source support or tail control.

An accurate field gate now has a route to an accurate local source; the source state, its correlations and its gravitational realization still have their own inputs.

The gate class is nonempty inside an accessible algebra. Let $A = \int h(s) e^{isH_p} C e^{-isH_p} ds$ for a bounded self-adjoint C and a real $h \in C_c^\infty$. Its first two commutator norms are bounded by $k_1 = \|h'\|_1 \|C\|$ and $k_2 = \|h''\|_1 \|C\|$. For $V = e^{i\lambda A}$, the energy change $B = V^* H_p V - H_p$ satisfies

$$\|B\| \le |\lambda| k_1, \qquad \|[H_p, B]\| \le |\lambda| k_2 + \lambda^2 k_1^2.$$

Both follow by differentiating $B = \int_0^\lambda e^{-isA} i[H_p, A] e^{isA} ds$. Products retain the bounded-commutator property. Consciousness II's bounded source preparation and shared-pulse result supply this smoothed gate class in a selected shared algebra for its stated translation generator and vacuum [15]. The exterior case below transfers it to the interacting current Hamiltonian. Record-controlled versions act in the record algebra tensored with that selected algebra. Thus an acquired record can install a permitted source without truncating the whole physical spectrum. An exact Weyl displacement need not satisfy the bounded energy-change condition; it is not covered by this assertion. Nor does an infinite KMS field have the finite global rms energy required here.

Proposition 52 (An exterior source gate for the interacting contact)**.** *Use the positive current Hamiltonian* (118) *or the full fermion contact of Proposition 51 with compact profile f_a. Let C be bounded, self-adjoint and localized in a bounded current interval I. Assume it and the free translates $\alpha_s^f(C)$ for $s \in \operatorname{supp} h$ belong to the selected accessible algebra, where $h \in C_c^\infty$ is real. Suppose their localization intervals are disjoint from* $\operatorname{supp} f_a$*. For $A = \int h(s) \alpha_s^f(C) ds$ and $V = e^{i\lambda A}$,*

$$[H_p, A] = [H_f, A], \qquad [H_p, [H_p, A]] = [H_f, [H_f, A]].$$

Both commutators are bounded, with the k_1, k_2 estimates above. The gate is therefore energy-regular for H_p, not only for H_f. Its bounded energy change is the same for both Hamiltonians. Record-controlled versions also meet the autonomous return conditions.

Proof. Write the interaction form as $H_p - H_f = H_{\mathrm{mech}} + \theta \sum_i q_i j_i(f_a) + d_a \theta^2$. Disjointly supported current smearings have zero symplectic pairing; their Weyl algebras commute. Thus A, and the first two derivative smearings defining its free commutators, strongly commute with every $j_i(f_a)$ and act trivially on the mechanical coordinates. The displayed equalities first hold as polarized forms on the common form domain. Free translation smoothing gives bounded commutators; these identities extend through the closed form to $\operatorname{Dom} H_p$ and imply its invariance under V. Integrating the first identity along e^{isA} gives $V^* H_p V - H_p = V^* H_f V - H_f$. The second identity gives the required bounded commutator of this energy change. A finite record control is a block sum of these same identities with a zero-gap record register. □

This class contains a genuine source preparation. Choose a non-scalar localized C in a common accessible algebra, with the local vacuum separating, and a narrow nonnegative h of integral one whose translates stay exterior. Consciousness II's bounded-source proposition then gives $V\Omega$ finite, strictly positive free-field energy for $0 < |\lambda| \|A\| < \pi$. On a finite-rms mechanical state tensored with the current vacuum, that energy is also the change of H_p:

the gate commutes with the contact interaction. With the usual local stress domain and integrability assumptions, the source perturbation vanishes outside the translated interval, and a smooth test equal to one there reads its full positive energy. The autonomous return and QEI bounds certify this preparation on the actual carrier while retaining the acquired record and prior references. A non-scalar shared algebra by itself need not contain such an exterior localized observable; the localization and permission assumptions are part of this construction. It supplies a finite source, not a thermal reservoir, a covariant local history clock or a rule that forms its own permissions.

For an explicit return, take $H_p = N$ on an infinite oscillator and retain three degenerate bits: a held source bit, its acquired record, and an output meter. A controlled NOT first acquires the source bit; that record then rotates oscillator levels $0, 1$ by .2 or .9; a final controlled NOT reads level 1 into the output meter. All higher oscillator levels form a reducing sector. This is an exact unbounded model, not a cutoff approximation. With initial source probabilities $.4, .6$ and its old purification retained, a 513-state clock of mean energy 512 gives vector error .118943 below .199530, and source energy .3839746172 against the reference .3839484296. The energy difference is 2.62×10^{-5}. The actual output-record probability is .3839746172. Clock energy decreases by the source gain; total-energy residual is below 4×10^{-12} in this check. A separate two-level block calculation agrees with the full-register evolution to 2.2×10^{-15} for the smaller clocks. Spatial transport, thermal preparation and the moving support remain distinct from this acquisition–preparation–reading realization.

H.2 The contact position can recoil

The contact profile need not follow a trajectory specified from outside. Let Y be its quantum position, $P_Y = -i\partial_Y$, and $M > 0$ its mass. On the full fermion carrier set

$$U_f(a) = e^{-iaH_f}, \qquad J_Y = U_f(Y) J(f) U_f(Y)^*.$$

The mechanical angle and storage coordinate remain internal to the body. This field translation exists on the full CAR space; it is not the static infrared dressing of the current.

Proposition 53 (Finite-mass contact and momentum conservation)**.** *The form*

$$\mathfrak{q}_{\mathrm{mov}} = \frac{\|P_Y \Psi\|^2}{2M} + \mathfrak{q}_{\mathrm{mech}} + \mathfrak{q}_f + \langle \theta J_Y \rangle + d_f \langle \theta^2 \rangle$$

is closed and nonnegative on the form domain of $P_Y^2/(2M) + H_{\mathrm{mech}} + H_f$. Its self-adjoint Hamiltonian H_{mov} strongly commutes with $P_{\mathrm{tot}} = P_Y + H_f$. A finite anchor Z of mass M_0 can be added with $P_Z^2/(2M_0) + k_s(Y - Z)^2/2$, $k_s > 0$; the conserved momentum then includes P_Z.

Proof. For almost every Y, field translation preserves the current form bound and H_f of Proposition 51. Integration over Y gives the same equivalent form norms after the positive position kinetic energy is added. The relative-anchor form is positive and closed on its natural domain as well. Simultaneous body and field translation sends J_Y first to J_{Y-a} and then to J_Y. All remaining terms, including the relative-anchor potential, are invariant. Uniqueness of the closed form realization implies commutation of H_{mov} with this translation group and therefore strong commutation with its generator. □

Thus the entire total-momentum distribution is conserved. These Hamiltonians measure energy above the body and anchor rest energies; those constant costs must be added in an apparatus ledger. The translation law is unchanged. On a common differentiable core the reaction is

$$\dot{P}_Y = -k_s(Y-Z) - \theta\partial_Y J_Y, \qquad \dot{P}_Z = k_s(Y-Z), \qquad \dot{H}_f = \theta\partial_Y J_Y.$$

The field-energy change is balanced by the body and anchor. Its instantaneous sign is not a certificate of exported heat. A positive record-controlled internal spring can change this dynamics without breaking translation symmetry; its preparation and gate regularity remain separate requirements.

The direct-integral unitary $\mathcal{W}(Y) = e^{-iYH_f}$ gives $\mathcal{W}^* P_Y \mathcal{W} = P_Y - H_f$ and $\mathcal{W}^* J_Y \mathcal{W} = J(f)$. With no anchor, the fixed total-momentum fiber is

$$H_M(p) = H_p + \frac{(p-H_f)^2}{2M}. \tag{119}$$

This is the recoil term of a mobile-field canonical transformation, as in the polaron setting [49]; the current form here is the full-CAR contact above. For a finite anchor put $M_t = M + M_0$, $\mu = MM_0/M_t$ and $r = Y - Z$. Its kinetic terms in a fiber become

$$\frac{(p-H_f)^2}{2M_t} + \frac{[p_r - (M_0/M_t)H_f]^2}{2\mu}.$$

Dropping either recoil square changes the model.

For fixed p, Eq. (119) converges in strong resolvent to H_p as $M \to \infty$. Its positive forms decrease to $\mathfrak{q}_p$ on $\operatorname{Dom} H_p^{1/2} \cap \operatorname{Dom} H_f$. This intersection contains a core for $\mathfrak{q}_p$, by equivalence with the free mechanical/field form. Resolvent variational minimizers have the closed-form lower bound; approximating a target vector by that core gives a recovery sequence with recoil penalty tending to zero. Strict convexity gives norm convergence of the resolvent vectors. This is not an operator-norm smallness statement. For a static orbit with the indicated finite fourth energy moment, Duhamel gives

$$\|(e^{-itH_M(p)} - e^{-itH_p})\psi\| \le \frac{1}{2M}\int_0^t \|(p-H_f)^2 e^{-isH_p}\psi\|\, ds.$$

Finite mean field energy alone does not meet this orbit condition.

For the unbounded harmonic coordinates used here, there is a sharper limit on a prescribed constant-velocity replacement. Take $p = Mv$ and remove the free kinetic energy $Mv^2/2$. The fiber is $H_p - vH_f + H_f^2/(2M)$, not H_p. Put $v_* = a_0/(a_0 + d_f)$. For $v < v_*$, the form of $H_p - vH_f$ is closed and nonnegative. For $v_* < v < 1$, it is unbounded below on the smooth core. To see both directions, complete the current form with coefficient $1 - v$:

$$H_p - vH_f \;\ge\; H_{\text{mech,kin}} + [a_0 + d_f - d_f/(1-v)]\theta^2.$$

For the reverse statement use the compensated compact gauge of Proposition 51. Its coherent state has $\langle J(f)\rangle = 2d_f\lambda$ and $\langle H_f\rangle = (d_f + d_b)\lambda^2$. Minimizing over λ gives mechanical leading

coefficient $a_0+d_f-d_f^2/[(1-v)(d_f+d_b)]$. First choose finite D large enough to make it negative, then send the mechanical packet center to infinity, with $y = k\theta/(k+K)$. Fixed packet widths and the conditional gauge derivative have bounded kinetic cost independent of the center. The energy tends to minus infinity. At $v = v_*$ the core form is nonnegative, but the previous equivalent form norm is lost. The full finite-mass H_{mov} remains positive: its recoil square prevents treating body kinetic energy as an unlimited constant-velocity source.

Quantum position also changes the locality requirement for the next thermal preparation. A position wavepacket has tails; its classical center lying in an interval does not localize the whole protocol. Choose a smooth position cutoff χ whose support keeps the full contact profile in a bounded interval I. Let the past interval A contain all free translates of I over the operating time, together with the other field couplings in the interaction picture. Replacing its interaction by $\chi\theta J_Y + d_f\chi^2\theta^2$ defines a positive local comparison Hamiltonian H_A. On a common operator-domain trajectory put

$$\delta_{\mathrm{loc}} = \int_0^t \|(H_{\mathrm{mov}} - H_A)e^{-isH_{\mathrm{mov}}}\Psi\|\, ds.$$

The retained-reference vector error is at most δ_{loc}. The thermal comparison in Appendix I.2 with the *actual* record distribution then has trace error at most $2\delta_{\mathrm{loc}}+\sqrt{I_0/2}$: one locality error changes the joint state, the other changes its record marginal. For source second-moment norms M_a, M_l, M_{pl}, M_{pa} of the actual, local, local-product and actual-product outputs, respectively, its source error is at most

$$\delta_{\mathrm{loc}}(M_a + M_l + M_{pl} + M_{pa}) + \sqrt{I_0}(M_l + M_{pl}).$$

All interaction-picture field couplings of the comparison protocol must lie in A; position-tail probability alone does not bound the generator integral. The body has genuine quantum recoil, but its Schrödinger kinematics does not supply a relativistic causal support, a clock or a dilaton stress tensor.

H.3 Physical locality after unfolding

The preceding H_f translates two co-propagating unfolded fields. Writing physical fields as $\xi_R(x) = \psi_1(x)$ and $\xi_L(x) = \psi_2(-x)$ changes the bulk momentum to H_1-H_2. The finite-width flavor current before diagonalization becomes

$$\int f(x)[\xi_R^\dagger(x)\xi_L(-x) + \xi_L^\dagger(-x)\xi_R(x)]\, dx.$$

It is a mirror coupling, not a local bulk density. The CAR give $[J(f), \xi_L(y)] = -f(-y)\xi_R(-y)$, allowing an immediate response between different positions. Point contact avoids that spatial kernel, but a profile $f_a(x) = a^{-1}f(x/a)$ at fixed integral has $d_{f_a} = d_f/a$. The existing energy bounds are not uniform in that limit. Neither folding nor shrinking supplies a finite local material body for free.

A local current alternative uses two flavors in each physical direction, with charges $+1, -1$ after a within-direction flavor rotation. Let $H_R, H_L \geq 0$, $H = H_R + H_L$, $P = H_R - H_L$, and $d_s = \kappa_s\|f\|_2^2/(4\pi)$, $\kappa_R = \kappa_L = 2$. Both currents $J_R(x), J_L(x)$ now act at the same physical x.

Proposition 54 (Bidirectional current and physical recoil)**.** *The form*

$$H_p^{\mathrm{bi}} = H_{\mathrm{mech}} + H + \theta[J_R(f) + J_L(f)] + d\theta^2, \qquad d = d_R + d_L,$$

is closed and nonnegative on the free form domain. Translating its current by e^{-iYP} and adding $P_Y^2/(2M)$ gives a nonnegative Hamiltonian conserving $P_Y + P$. Its fixed-momentum fiber is $H_p^{\mathrm{bi}} + (p-P)^2/(2M)$. For the harmonic internal coordinates above and equal $d_R = d_L = d/2$, $H_p^{\mathrm{bi}} - vP$ has a positive closed form for $|v| < \sqrt{a_0/(a_0+d)}$ and is unbounded below on the smooth core for $\sqrt{a_0/(a_0+d)} < |v| < 1$.

Proof. In physical coordinates the left current has the opposite signs in both the current commutator and its commutator with H_L. For $f = F'$, $G = J_R(F) - J_L(F)$ therefore satisfies

$$e^{-i\theta G} H e^{i\theta G} = H + \theta(J_R(f) + J_L(f)) + d\theta^2.$$

Compact compensation in each direction handles arbitrary compact f, as in Proposition 51. The same equivalent form norms and translation proof then apply, with P replacing H_f. Completing the two current forms separately gives

$$H_p^{\mathrm{bi}} - vP \geq H_{\mathrm{mech,kin}} + \left[a_0 + d - \frac{d_R}{1-v} - \frac{d_L}{1+v}\right]\theta^2.$$

The bracket is $a_0 - dv^2/(1-v^2)$ in the balanced case. For necessity, use independent compact compensated coherent states in the two directions. Their minimized leading coefficient is $a_0 + d - d_R^2/[(1-v)(d_R + d_{b,R})] - d_L^2/[(1+v)(d_L + d_{b,L})]$. It is negative for fixed sufficiently spread compensators above the stated threshold. Harmonic packet centers tending to infinity then give the negative core sequence, with bounded kinetic cost. □

At equality only core nonnegativity is asserted. The finite-mass Hamiltonian remains positive with its recoil retained. This alternative has four complex chiral fields, $c_R = c_L = 2$; each actual outlet must count the fields routed through it. Four input preparations, their purifiers and the additional control permissions are real resources. For interval lengths a, b and separation $g > 0$ in the product free vacuum, Appendix I.2 uses $I_0 = \frac{2}{3}\log[1 + ab/(g(g + a + b))]$ for these four fields, twice the two-field value. Both the conditional preparation error and the source-moment certificate must use this larger I_0. The balanced free carrier and its physical recoil do not yet quantize a propagating material controller or its complete stress tensor.

I Local preparation and correlated-window renewal

I.1 A local preparation can be executed autonomously

The compact preparation unitary V of Proposition 42 is local, but its energy change need not be bounded. A preparation bit permits a bounded regularization that controls the field state in energy norm as well as Hilbert norm. Let H act trivially on that zero-gap bit and set

$$C = |1\rangle\langle 0| \otimes V + |0\rangle\langle 1| \otimes V^*, \qquad u_0 = |0\rangle\Omega, \quad u_1 = Cu_0.$$

Then $C = C^*$, $C^2 = I$ and $e^{-i\pi C/2}u_0 = -iu_1$. For a nonnegative even $h \in C_c^\infty(-1,1)$ of integral one, put $\mu_2 = \int s^2 h(s)ds$ and

$$C_\delta = \int h(s)e^{i\delta sH}Ce^{-i\delta sH}ds, \qquad W_\delta = e^{-i\tau C_\delta}, \quad \tau = \pi/2.$$

Proposition 55 (Local preparation in the energy graph norm)**.** *Suppose $F_l(t) = e^{itH}Ce^{-itH}u_l$, $l = 0, 1$, is twice continuously differentiable in the H graph norm for $|t| \le \delta_0$. Let $L_2 = \max_{l,t} \|F_l''(t)\|$ and $G_2 = \max_{l,t} \|HF_l''(t)\|$ there. For $0 < \delta \le \delta_0$ define*

$$e_\delta = \mu_2\delta^2 L_2/\sqrt{2}, \qquad g_\delta = \mu_2\delta^2 G_2/\sqrt{2}, \qquad k_1 = \|h'\|_1/\delta, \quad k_2 = \|h''\|_1/\delta^2.$$

The bounded local gate W_δ has energy-regular constants $b \le \tau k_1$ and $c \le \tau k_2 + \tau^2 k_1^2$. Moreover,

$$\begin{aligned} \|(W_\delta - e^{-i\tau C})u_0\| &\le \tau e_\delta, \\ \|H(W_\delta - e^{-i\tau C})u_0\| &\le \tau g_\delta + \tfrac{1}{2}\tau^2 k_1 e_\delta. \end{aligned} \tag{120}$$

For a compact diffeomorphism of the vacuum Virasoro module the stated graph condition holds for free translation $H = H_f$. For every finite source tolerance, a finite positive autonomous history clock can therefore execute this preparation while approximating its local thermal restriction and each specified smooth stress test.

Proof. Time smoothing gives $\|[H, C_\delta]\| \le k_1$ and $\|[H, [H, C_\delta]]\| \le k_2$, hence the gate constants by Proposition 49. Evenness of h cancels the first Taylor term. The second-order remainder bounds $(C_\delta - C)u_l$ by $\mu_2\delta^2 L_2/2$ and its H norm by $\mu_2\delta^2 G_2/2$. The ideal gate path lies in $\operatorname{span}\{u_0, u_1\}$ with coefficient sum at most $\sqrt{2}$. Duhamel gives the first estimate. For $\zeta \in \operatorname{Dom} H$,

$$\|He^{-itC_\delta}\zeta\| \le \|H\zeta\| + tk_1\|\zeta\|.$$

Inserting this in the same Duhamel integral gives the second estimate, which tends to zero as $O(\delta)$. The smooth diffeomorphism representation preserves $\bigcap_n \operatorname{Dom} L_0^n$, and its action is smooth there [12, Sections 5.1–5.2, Proposition 5.1]. Möbius translation generators act continuously on this domain; Ω, $V\Omega$ and the required orbit therefore have the stated graph differentiability. First make the preparation error small by choosing δ, then, for fixed δ, apply the autonomous return and smooth-stress certificates. The required clock is finite at each tolerance, although its resources need not stay bounded as that tolerance decreases. □

No global vacuum or energy projector enters C_δ. Its support is the preparation interval enlarged by the short translations. If this interval stays outside the pinned-current contact, the commutator equalities above make W_δ energy regular for the same interacting H_p. Graph convergence also needs its H_p-graph condition on the input; commutator equality alone does not give it. Schwartz mechanical vectors tensored with the smooth vacuum core supply an explicit class: the mechanical polynomials, smeared currents and twice-differentiated exterior orbit act continuously on that core. Overlapping intervals and more general mechanical states need separate domain control.

In the free-field execution, field-energy gain is exactly the loss of the clock-interaction energy $\langle H_{\text{hist}}\rangle$; their sum is conserved. The preparation bit, initial vacuum, allowed local

gate, clock coherence and waveform are supplied resources. Locality here is locality of a permitted continuum-algebra interaction, not a derived laboratory actuator or a covariant moving clock. A finite record can select among these gates while retaining its coherences and references. Online prediction of its changing thermal clock and physical renewal remain further requirements.

I.2 A retained record can select the next thermal window

A later interval of the same field is not an independent reservoir merely because it has not yet met the contact. The vacuum correlates disjoint regions; a record made in one can steer the conditional state in the other. Separation nevertheless gives a quantitative renewal condition without discarding those correlations.

Proposition 56 (Record-selected local thermal preparation)**.** *Consider c_f independent complex chiral fermions in their vacuum. Let A, B be split-separated intervals of lengths a, b and gap $d > 0$. The complete past field protocol and acquired record R are a normal local channel on A, with initial ancillas independent of the field. Here R denotes the commuting readout algebra of the retained register, not an extra physical dephasing. For each result r, let an even $V_r \in \mathcal{A}(B)$ prepare a specified thermal restriction τ_r on $J \subset B$ from the vacuum. Put*

$$I_0 = \frac{c_f}{6} \log \left(1 + \frac{ab}{d(d+a+b)} \right).$$

After record-controlled preparation,

$$\frac{1}{2} \left\| \rho'_{RJ} - \sum_r p_r |r\rangle\langle r| \otimes \tau_r \right\| \leq \sqrt{I_0/2}. \tag{121}$$

The norm is that of normal states on the local algebra. The bound is a probability-weighted conditional error, not a uniform bound for arbitrarily rare results.

Let Q_r be a smooth source test on J and $Q = \bigoplus_r V_r^* Q_r V_r$. On the split algebra set $\alpha = \rho_{RB}$, $\beta = \rho_R \otimes \omega_B$ and suppose $M_a^2 = \alpha(Q^2)$, $M_i^2 = \beta(Q^2)$ are finite. Then

$$|\alpha(Q) - \beta(Q)| \leq \sqrt{I_0}(M_a + M_i). \tag{122}$$

Neither assertion replaces the actual outside purification by a product state.

Proof. The past local channel leaves the B marginal equal to ω_B. Relative-entropy data processing gives $S(\alpha\|\beta) = I(R : B) \leq I_{\mathrm{vac}}(A : B)$. For free complex chiral fermions, the multi-interval vacuum entropy formula gives the displayed I_0 [46, Section 4.1, Eq. (63)]. For commuting even observable restrictions it supplies an upper bound by data processing; the graded fermion fields are not being replaced by bosonic tensor factors. Measuring an arbitrary binary effect and applying classical Pinsker gives $\|\alpha - \beta\|/2 \leq \sqrt{I_0/2}$. Controlled preparation and restriction are contractions, proving Eq. (121).

For the source bound use the natural-cone representatives ξ_α, ξ_β. Spectral Jensen for the relative modular operator gives [47, Section 2, Eqs. (2.1)–(2.6)]

$$S(\alpha\|\beta) \geq -2 \log \langle \xi_\alpha, \xi_\beta \rangle \geq \|\xi_\alpha - \xi_\beta\|^2.$$

Faithful regularization gives the same bound for finite-entropy normal states. Expanding the two expectations of Q and applying Cauchy–Schwarz gives Eq. (122). □

The estimate can also retain more than the classical record. If the past protocol outputs a larger controller algebra $\mathcal{K}$, the same data-processing argument gives $S(\rho_{\mathcal{K}B}\|\rho_{\mathcal{K}}\otimes\omega_B) \le I_0$. The controller may include its mechanics and references initially independent of the field. Any subsequent normal joint channel on $\mathcal{K}$ and B preserves the local trace bound. Its output source obeys Eq. (122) with the actual and comparison output moments. This also controls a later joint contact, provided its participating algebra is genuinely contained in this input. It does not include the entire field complement as an independent reference.

The source moments are explicit inputs. For the record-only estimate, if the past instrument has local effects $E_r \in \mathcal{A}(A)$ and $m_r^2 = \omega((V_r^* Q_r V_r)^2)$, then locality and $0 \le E_r \le I$ give $M_a^2 \le \sum_r m_r^2$ and $M_i^2 = \sum_r p_r m_r^2$. Smooth compact preparations and stress tests have finite m_r. A trace-distance bound alone would not control an unbounded source.

For a desired local trace error $\epsilon > 0$, a sufficient separation is

$$d \ge \frac{\sqrt{(a+b)^2 + 4ab/(e^{12\epsilon^2/c_f} - 1)} - (a+b)}{2}.$$

This is the gap required by the vacuum-information certificate, not a necessary distance for every instrument: an uninformative record can already be independent at a smaller gap.

For a specified finite sequence, use the actual hull of the past field operations at each step and a separated preparation support for the next window. The estimate applies to that actual past, without resetting it. The local gates can be smoothed and executed by the positive history clock under the graph-domain conditions above; its additional gate and source errors are charged separately. All prior registers and field purifications remain present. The gap consumes spatial reach and transport time. The record must reach the preparer through a permitted physical channel; a controlled unitary does not itself construct that transport. The growing supports, storage and clock therefore define a finite operating family, not a fixed-capacity perpetual reservoir. This local certificate also does not prove that the whole future field and every old quantum reference are jointly a product state. Nor does it alone certify the entropy rate of a correlated released field.

J Actual diagnostic return and terminal reading

J.1 The reading itself supplies the repair and contact program

The program need not be a label given to the apparatus. Use three zero-gap pointer bits for world, sensor and record frame, and three meter bits. Start the routine meter blank and the other two meters in independent diagonal states with bit probability .1. Three controlled NOTs copy body parity into the routine meter; two copy sensor and frame values into their respective noisy meters. All six bits and the actual held body remain in the core.

The acquired meter word o controls a fixed repair table $a(o)$. The table is the first two-step decision of Consciousness III's continuing-body model [14], with independent coordinate fault

probability .05. It depends on the acquired word and known model, not on the inaccessible true cause. On one history edge use the unitary

$$V = \sum_{o} |o\rangle\langle o| \otimes X_{a(o)} \otimes S_{\theta \mathbf{1}_{a(o)\neq 0}}, \tag{123}$$

where $X_0 = I$ and $X_{e_j} = X_j$. The exchange S acts on two matched-frequency fermion cells initially prepared at temperatures $.7, 1.4$, with $\omega = 1$ and $\theta = .7$. Thus the actual physical repair permission and actual energy transfer are performed by one program-controlled history edge. Whether the scorer later calls the repair correct has no effect on the interaction.

Put the five meter gates and this gate on the first six edges of an eight-edge clock, followed by two identities. Every gate commutes with reservoir bare energy. The joint positive conserved Hamiltonian has eight qubit registers, nine clock states and dimension 2304. At $J = .06$, clock mean energy is .48, interaction cutoff .96, and the total spectral cutoff is 2.96. Initial mean energy is 1.00197392. Direct evolution retaining 128 purification columns agrees with the full gate product to vector error 2.7×10^{-14}; bare energy conservation residual is 4.5×10^{-16} in this check. Actual restoration failure is .02891, repair probability .1355, and cold-port energy gain is .00761031. The latter equals $\Pr(a \neq 0)\sin^2\theta(n_o - n_H)$, not a probability of incorrect repair supplied as a source command. The contact angle is a hardware choice: setting it to zero leaves the logical repair intact. This construction connects actual repair to an actual source; it does not make that heat transfer a universal minimum cost of repairing a pointer. The complete six-bit controller entropy is unchanged. Reservoir relative entropies are .0001838782 and .0001317734, and joint output total correlation is .0051202854. They obey Eq. (97) with residual below 3×10^{-16}. Reservoir purification columns are retained for the vector certificate, not folded into its initially independent controller in the entropy ledger.

This is a first-interval conserved realization of acquisition returning to repair and a source. Initial meter purity, biased noise, reservoir preparation, reference labels and hardware are supplied and charged separately from the finite Hamiltonian's energy. The two cells store transferred energy; spatial release and a covariant supported contact are not implied by this finite calculation. Its value is that reading, decision, repair and source no longer need mutually incompatible clocks or an external selector within the demonstrated interval.

Proposition 57 (A finite continuing diagnostic return)**.** *For every specified horizon L, the rational pointer model with $h = 1/20$, $\epsilon = 1/10$ and the two-step decision admits one finite, time-independent, nonnegative Hamiltonian. Its completed circuit has the joint law of all L logical intervals: the same damaged and repaired body, all actual meter and action records, exact retained posterior weights, and one record-selected thermal exchange per interval. No inaccessible fault bit enters the decision circuit. With a fresh matched-frequency cell pair of gap ω per interval,*

$$Q_L = \omega \sin^2\theta(n_o - n_H) \sum_{t=1}^{L} \Pr(a_t \neq 0). \tag{124}$$

For a circuit with m padded history edges, clock mean energy is Jm, the total energy cutoff is $2L\omega + 2Jm$, and its exact completion time is $\pi/(2J)$ in units $\hbar = 1$.

Proof. Prepare $3L$ independent diagonal fault ancillas and let each control the corresponding held-body flip. Prepare $3L$ meter bits as above. The observer's reversible decision uses only the meter words, retained integer weights and known earlier actions. The rational update has denominator 800000 per interval; all weights and comparisons have finite integer size. A reversible Boolean circuit computes each update and action, keeping its inputs and garbage. The action controls the held-body flip and exchange with the next prepared thermal pair. Every gate commutes with the sum H_0 of cell energies, because the logic is degenerate and each exchange has matched gaps. Put the gates on a positive history clock. The prefix-unitary conjugation gives $H_0 + JmI + 2JJ_x$, proving positivity, bare-energy conservation and full-reference endpoint completion. Each unused pair is still jointly independent of the acquired program at its contact. Conditional energy transfer is zero or $\omega \sin^2 \theta(n_o - n_H)$, which proves Eq. (124). □

For the first three logical intervals, the expected number of repairs is .44635607 and $Q_3 = .02506944$ at the stated cell settings. There are nine readings, nine prepared fault bits and six thermal cells. Keeping all preceding weight vectors requires at most 984 bits in this case, before records, arithmetic workspace and reversible garbage. The general saved-weight cost grows quadratically with L. This construction removes the need for a new externally chosen circuit at every diagnostic return. Its logical intervals are not equally spaced proper-time samples of a continuously acting bath: only the completed joint law is asserted. All ancillas and cells are prepared before the finite run, all outputs remain in its ledger, and the clock can reverse after completion. It proves neither a fixed-capacity cyclic apparatus nor a covariant moving contact. These are distinct renewal and material implementation questions, not failures of the finite return.

J.2 An actual terminal record and probe replacement

Let probe P retain any old reference, record R start in $|0\rangle$, and a paid blank B start in $|+\rangle$. Write $C_Y = P_-^Y \otimes I_R + P_+^Y \otimes X_R$, then apply the true SWAP_{PB}. The record-one probability is $\mathrm{tr}(P_+^Y \rho_P)$. The final probe is $|+\rangle$, independent of all old references and the record; its old correlations have moved into B and R, not disappeared. Positive generators $gP_+^Y \otimes (I - X_R)$ and $g(I - \mathrm{SWAP}_{PB})$ implement these gates at duration $\pi/(2g)$. A fermionic exchange is not a substitute for this true SWAP.

The physical probe may continue to acquire a phase while the meter executes. Move the r nonidentity gates of the clock in Appendix E to its last r edges. All earlier edges are identities. Put $T = \pi/(2J)$, choose $t_0 < T$ and define

$$P_0 = \Pr\{\mathrm{Bin}(m, \sin^2 Jt_0) \geq m - r\}, \qquad A = J \sum_{j=m-r+1}^{m} \sqrt{j(m-j+1)}.$$

Proposition 58 (Terminal read under a retained phase coupling)**.** *Let the actual probe coupling be $\lambda d(t) Z_P$, with $|d(t)| \leq D$. Relative to writing the probe's actual phase at t_0 and replacing it at T, the full joint half trace distance obeys*

$$d_{\mathrm{joint}} \leq 2At_0\sqrt{P_0} + 2\sqrt{P_0} + \lambda D(T - t_0). \tag{125}$$

All old references, the blank, record and clock are included. For any specified finite trajectory and positive error tolerance, finite m and $t_0 < T$ suffice.

Proof. Before t_0, compare with the identity-edge clock and the actual free probe phase. Only the last r edges differ. Their action on that reference is bounded by $2A\sqrt{P_0}$, since the binomial upper tail increases up to T. Duhamel gives the first term. At t_0, dressing the reference with the gate prefixes changes its joint state by at most $2\sqrt{P_0}$. From there to T, omit only the bounded phase coupling, at cost $\lambda D(T-t_0)$; the compiled clock completes the meter and SWAP exactly. First choose t_0 near T, then increase m to suppress the tail. □

The positive clock has mean energy Jm, not zero. Add the probe/blank offset $2\epsilon_P I$, with $2\epsilon_P > \lambda D$, to keep the complete internal Hamiltonian positive. For $m = 10^5$, $r = 2$, $T = 80$, $t_0 = 78$ and $\lambda D = .00032$, a sufficient error is below .000641; clock mean energy is 1963.4954. This cannot be inserted into a mass-10 support without changing that support. A direct 17-state clock check gives joint distance .0034887 below its certificate .0046290. The construction renews the probe over a finite window. It consumes a coherent blank and a record slot and does not reset the entire reader or its clock. Its phase frame and allowed gate family remain inputs. HAFF D also solves a smaller, 129-state meter clock jointly with a source-matched mean moving support [23]. Its energy is included in the initial supported mass, and the actual record depends on the dilaton trajectory. That mean-field calculation does not include the contact reservoir's microscopic degrees or stress. For the specified C_Y, SWAP and coherent blank, an antiunitary symmetry makes the mean probe force vanish throughout the read, although the force operator and its fluctuations remain nonzero.

K Local linear-current control of stress flows

Consider the neutral current on a circle of length 2π, with

$$J_n = \sqrt{n}\, a_n, \quad J_{-n} = \sqrt{n}\, a_n^*, \quad J_0 = 0, \qquad H = L_0 = \sum_{n\geq 1} n a_n^* a_n.$$

Our Fourier convention is $f_n = (2\pi)^{-1}\int_0^{2\pi} f(\theta)e^{-in\theta}\,d\theta$. The same convention applies to the real current smearing h. The primitive interaction is $-Bj(h)$, where $j(h) = \sum_{n\geq 1}\sqrt{n}(h_{-n}a_n + h_n a_n^*)$ and B is a finite controller coefficient. Fix $h \in C_c^\infty(I,\mathbb{R})$ on a proper arc I, with $h_n \neq 0$ for $n \geq 1$. Such an h exists: in the Fréchet space of real smooth functions supported in a fixed compact subarc, each equation $h_n = 0$ defines a proper closed subspace, so the intersection of their open dense complements is nonempty. For a prescribed finite cutoff N, a nonnegative sufficiently narrow bump already gives $h_n \neq 0$ for $1 \leq n \leq N$; this alternative makes the probe depend on N. No uniform lower bound on $|h_n|$ is assumed.

Proposition 59 (A neutral-current stress compiler)**.** *Suppose signed local linear-current coupling, finite degenerate-helper rotations and record phase gates, calibrated clock preparation and finite readout are available, with coherent control intervals. For every real smooth f, $t \in \mathbb{R}$, $E < \infty$ and $\delta > 0$, a finite positive autonomous program implements $e^{itT(f)}$ within joint half-trace distance δ, uniformly over inputs with $\mathrm{tr}(H\rho) \leq E$ and arbitrary retained*

references. Its helpers and original clock are included in the comparison with their designated final states. Here

$$T(f) = \frac{1}{2} \sum_{p,q \in \mathbb{Z}} f_{-p-q} : J_p J_q :$$

is the $c = 1$ Sugawara stress tensor. The program uses no field reset, heralding or primitive quadratic coupling.

Proof. We first include the field's free propagation. For a finite quadrature $Q(v) = \sum_{n \le N} (v_n a_n^* + \overline{v_n} a_n)$ put

$$G_n = \frac{v_n}{\sqrt{n} h_n}, \qquad g_v(u) = \frac{1}{2\pi} \sum_{n \le N} (G_n e^{-inu} + \overline{G_n} e^{inu}), \quad 0 \le u \le 2\pi.$$

The Hamiltonian

$$H - s g_v(u) A j(h) + s^2 g_v(u)^2 C_h, \qquad C_h = \sum_{n \ge 1} |h_n|^2,$$

is positive by completion of the oscillator squares, with shifts $s g_v(u) A h_n / \sqrt{n}$ and $A = X$ or Y. In the interaction picture $[j_u(h), j_w(h)]$ is a scalar. Thus the Magnus series terminates after its scalar second term, and $e^{-2\pi i H} = 1$ gives the endpoint gate $e^{i\phi_s} e^{isAQ(v)}$. The phase ϕ_s is computed from the scalar commutator and the completion integral. Subsequent pulses start at integer periods and therefore act on the same $Q(v)$. Changing s to $-s$ pays for the inverse without reversing H. A Y pulse may instead use a paid Z rotation conjugating the X coupling. For coherent record branches, the known phases are removed by record phase gates.

The exact trigonometric comparator does not have smooth compact-time switching. Multiplying it by an all-order-flat endpoint taper χ_η makes the physical coupling smooth and gives $\|(1 - \chi_\eta) g_v\|_1 \le 2\eta \|g_v\|_\infty$. Writing $C_1 = \sum n |h_n|^2$ and $D_h = \sum n^2 |h_n|^2$, we have

$$\|j(h)\psi\| \le 2\sqrt{C_h} \|H^{1/2}\psi\| + \sqrt{C_1} \|\psi\|, \qquad \sqrt{\langle H \rangle_u} \le \sqrt{E} + \sqrt{D_h} |s| \|g_v\|_1.$$

The second bound follows directly from the controlled displacement. It includes the transient excitation of all modes, not just the selected endpoint modes. Duhamel's formula then makes each taper error arbitrarily small in joint vector norm, including old references. For a finite word of such pulses the energy bound is $(\sqrt{E} + \sqrt{D_h} L)^2$, where $L = \sum_{\text{slots}} |s| \|g_v\|_1$.

Next use the current-factor cutoff

$$T_N(f) = \frac{1}{2} \sum_{0 < |p|, |q| \le N} f_{-p-q} : J_p J_q : .$$

This is a selfadjoint finite oscillator quadratic. Finite occupation vectors are analytic for a finite quadratic, hence form an essential selfadjointness core. Let $P_M = 1_{[0,M]}(H)$, M an integer. It has finite rank. For $N > M$, the omitted number-preserving terms have annihilator index $m \le M$ and creator index $n > N$. Their norm on P_M is bounded by

$\sqrt{nM}|f_{n-m}|$. Omitted pair annihilators vanish; pair creators have norm at most $M+2$. Consequently

$$\|[T(f)-T_N(f)]P_M\| \le b_N(M) \longrightarrow 0,$$
$$b_N(M) = \sqrt{M}\sum_{m=1}^{M}\sum_{n>N}\sqrt{n}|f_{n-m}| + \frac{M+2}{4}\sum_{k>N}k^2|f_k|.$$

For the pair term we used $\sum_{n=1}^{k-1}\sqrt{n(k-n)} \le k^2/2$. These absolutely convergent sums identify the limit on the finite-energy core as the Sugawara $T(f)$. Carpi–Weiner's essential selfadjointness theorem for real smooth stress smearings [2, Theorem 4.4] supplies the self-adjoint closure of the limit; it does not substitute for the current-factor tail estimate. On this common core, the resolvent identity and density of $(T(f)-z)D_{\mathrm{fin}}$ give strong resolvent convergence. Hence $e^{itT_N(f)} \to e^{itT(f)}$ strongly, and

$$a_N(M,t) = \|[e^{itT_N(f)} - e^{itT(f)}]P_M\| \longrightarrow 0.$$

The uniform reference estimate is obtained only once, at the input of the whole program. For isometries V, W, purification and the P_M decomposition give

$$D_{\mathrm{joint}}(V\rho V^*, W\rho W^*) \le \|(V-W)P_M\| + 2\sqrt{\frac{E}{M+1}}.$$

It is independent of reference dimension. Thus the factor-cutoff unitaries converge uniformly over the stated mean-energy class. The core-tail bound is not itself a rate for $a_N(M,t)$, since a stress-evolved vector need not stay in P_M.

Write $r = (q_1, p_1, \dots, q_N, p_N)$. A real symmetric matrix K_N gives

$$T_N = \tfrac{1}{2}r^T K_N r - c_N, \quad c_N = \tfrac{f_0}{2}\sum_{n\le N} n, \quad T_N = \tfrac{1}{2}\sum_{\ell\le 2N}\kappa_\ell Q_\ell^2 - c_N$$

after real orthogonal diagonalization. This is an algebraic decomposition, not a free optical transformation: each Q_ℓ is selected directly by the local pulse above. Finite quadratic Trotter products converge on oscillator Schwartz vectors and hence strongly, including after restriction to P_M. Each square exponential is compiled with the same retained helper by four Pauli pulses. For an exponent of magnitude $|\alpha|$, n blocks and $s = \sqrt{|\alpha|/(2n)}$, the second-moment estimate is

$$D_{\mathrm{joint}} \le (13\sqrt{2})^{1/3}\sqrt{|\alpha| m_2}\, n^{-1/6}, \qquad m_2 = \mathrm{tr}(\rho Q_\ell^2) \le 4\,\mathrm{tr}(H\rho) + 1.$$

To see the moment bound, on a spectral value q set $C_\vartheta = e^{i\vartheta X}e^{i\vartheta Y}e^{-i\vartheta X}e^{-i\vartheta Y}$. Taylor's integral remainder gives $\|C_\vartheta - e^{-2i\vartheta^2 Z}\| \le 13|\vartheta|^3$: the third derivative of the four-factor product has norm at most 64, and the quadratic exponential contributes at most $2\vartheta^4$ for $|\vartheta| \le 1$; outside that interval the unitary distance is at most 2. Telescoping n blocks yields a spectral error $\min\{2, A|q|^3\}$, with $A = 13|\alpha|^{3/2}/(2\sqrt{2n})$. The inequality $\min\{4, A^2q^6\} \le 4^{2/3}A^{2/3}q^2$ and the spectral theorem give the displayed joint bound, retaining the helper and reference. The sign is chosen by the Pauli order; all inverse pulses are paid. Moments for

a finite word are evaluated on its ideal prefixes, whose oscillator moments are finite on P_M. Telescoping the complete word against those prefixes avoids both fresh-helper resets and a circular block-count estimate.

Finally approximate the smooth finite schedule in L^1 by constant slots with rational history-clock thresholds. Every slot has the positive field Hamiltonian $H - B_j j(h) + C_h B_j^2$. Install the finite rotations and record phase gates on the original clock path; these gates commute with the degenerate data bare energy. The resulting sum of the positive history clock, data energy and completed field squares is a positive autonomous Hamiltonian on the common field-energy domain. For $G = \max_j \|B_j\|$ and $D_3 = \sum n^3 |h_n|^2$, its field rms obeys

$$D^+ \|H\psi_u\| \le 2G\sqrt{D_h}\sqrt{\|H\psi_u\|} + G\sqrt{D_3}.$$

The bound is independent of clock size. The finite-boundary history-clock estimate therefore makes the complete endpoint error arbitrarily small on P_M, with the same original clock and all record branches retained. Choosing successively M, N, Trotter length, square-block counts, taper/slot precision and clock size allocates any prescribed total δ. □

The resources can grow sharply. With $\Gamma(v) = \|g_v\|_1 \le 2\sum_{n\le N} |v_n|/(\sqrt{n}|h_n|)$, a square exponent of size $|\alpha|$ uses field duration $8\pi n$, area $2\sqrt{2n|\alpha|}\Gamma(v)$ and positive completion integral

$$\frac{2C_h|\alpha|}{\pi} \sum_{n\le N} \frac{|v_n|^2}{n|h_n|^2}.$$

Helper rotations, phase gates and readout have their own finite budgets; rotation slots can last 2π so the field drift is identity. For an autonomous clock of duration T and size $m+1$, $J = \pi/(2T)$, its initial mean energy is mJ, norm $2mJ$ and largest edge coefficient at most $J(m+1)/2$. Preparation, drive bandwidth and retained storage are additional resources. These are sufficient finite bounds, not optimal costs or net-heat identities.

The uniform channel statement assumes only mean energy. Expectations of unbounded stress require separate graph-norm control: for smooth g, $\|T(g)\psi\| \le C_g\|(H+1)\psi\|$. Small joint channel error alone does not certify source convergence on the whole mean-energy class. The finite program's field moments also do not include a covariant stress tensor for clock and support.

The target preserves the Virasoro vacuum subspace, although the individual current pulses and finite T_N approximants can leave it; the endpoint leakage is part of the error above. The compiler concerns the neutral current sector. Its controls are not invariant under the $SO(3)$ action in the $SU(2)_1$ carrier, so an always-on selecting noise needs a separate control/noise certificate. The circle, calibrated port, coherent interval and controller preparation remain physical inputs.

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